



MP3718

4-Cell to 24-Cell, High-Accuracy Battery Monitor and Protector with Coulomb Counting

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

DESCRIPTION

The MP3718 is a robust battery management device, providing a complete analog front-end (AFE) monitoring and protection solution. It is designed for multiple-cell series battery management systems (BMS). This device can support I²C or SPI communication. It supports connections for 4-cell to 24-cell series battery packs, with an absolute voltage exceeding 140V on particular pins.

The MP3718 integrates two separate analog-to-digital converters (ADCs). The first ADC measures each channel's differential cell voltages (up to 24 channels), die temperature, and 10-channel temperature via an external NTC thermistor. The second ADC measures the charge/discharge current via an external current-sense resistor.

When paired with an MPF4278x fuel gauge, the MP3718 can achieve a state-of-charge (SOC) error to within 2%.

The device has high-side drivers and features robust protection functions. Two discharge MOSFET drives, the CDSG provides the discharging fault protection in the charging circuit of the split port application, the DSG provides the fault protection in the discharging circuit. In addition to MOSFETs based protection, it also has a secondary protection to burn out the three-terminal fuse.

The device can operate in standalone mode without a microcontroller.

It also integrates the balancing MOSFETs to equalize mismatched cells, with the option to drive the external balancing MOSFETs.

It has an optimized baseline current consumption that depends on the operating mode.

For applications requiring compliance with IPC 2221B or UL 60950-1, conformal coating may be required depending on the battery pack voltage.

The MP3718 is available in a LQFP-64 (7mmx7mm) package.

FEATURES

- Supports 4 to 24 Stacked Cells
- I²C or SPI Interface, Support 8-Bit CRC
- **Analog Front-End (AFE) Monitor:**
 - Two Separate ADCs Provide Cell and Pack Voltage Strictly Synchronized with Current
 - 15-Bit ADC with $\leq 3\text{mV}$ Total Cell Measurement Error at 25°C
 - 16-Bit ADC with $\pm 0.3\%$ (Pack Current and Coulomb Counting) Measurement Error at 25°C via SRP-SRN
- **Hardware Configurable Protections:**
 - Charge/Discharge OCP and SCP
 - Cell UVP and OVP
 - Pack UVP and OVP
 - Cell Low/High-Temperature Protection
 - Die High-Temperature Protection
- **Integrated High-Side MOSFET Driver**
 - Capable of Driving Multiple Paralleled N-Channel MOSFETs
 - Current Loop Regulation during DSG MOSFET Soft Start
 - GPIO or Register Control MOSFET
 - CDSG driver for charging port
- **Automatic or Manual Cell-Balancing**
- **Standalone Mode without Microcontroller**
- **Delicate Functions:**
 - Reduced Current Standby Mode
 - 5V and 3.3V LDO
 - Load/Charger Plug-In Detection
 - Open-Wire Detection
 - Permanent Dead Battery Flag
 - Lockable Programmable Memory for Protection Configuration
- **Random Cell Connection Tolerant**
- Available in a LQFP-64 (7mmx7mm) Package



APPLICATIONS

- Power and Gardening Tools
- Energy Storage System (ESS)
- E-Bikes, E-Scooters
- Light Electric Vehicles (LEVs)
- Battery Backups and UPS

All MPS parts are lead-free, halogen-free, and adhere to the RoHS directive. For MPS green status, please visit the MPS website under Quality Assurance. "MPS", the MPS logo, and "Simple, Easy Solutions" are trademarks of Monolithic Power Systems, Inc. or its subsidiaries.

TYPICAL APPLICATION

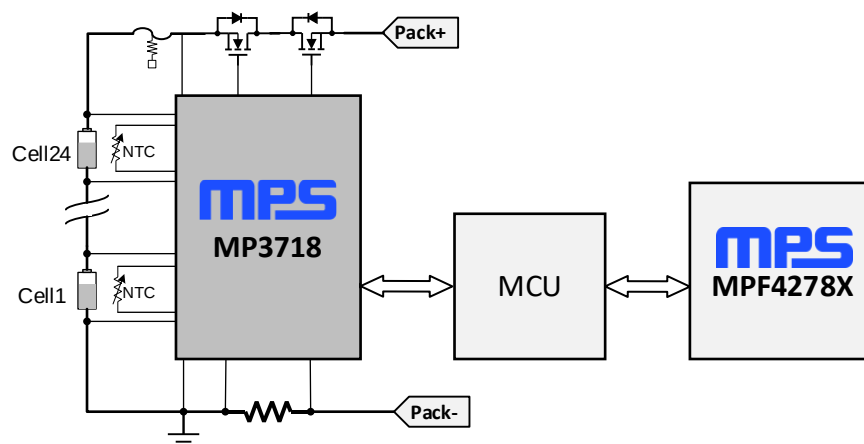


Figure 1: MP3718 one port application

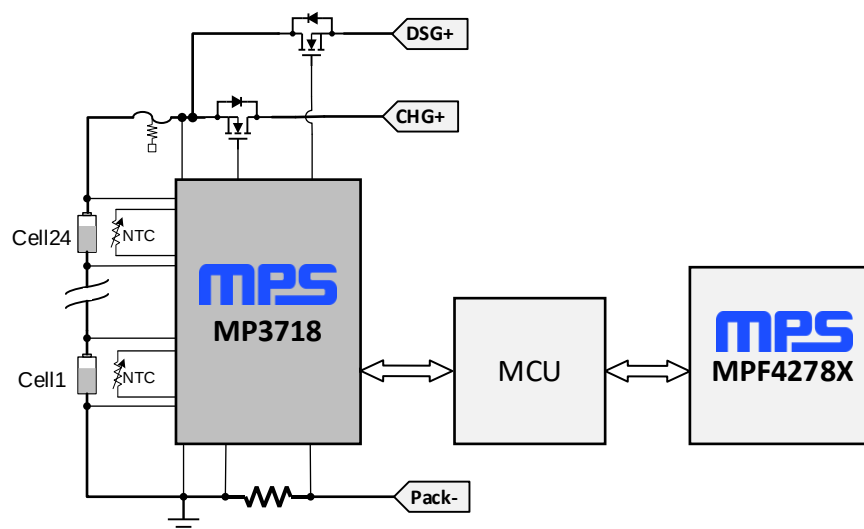


Figure 2: MP3718 separate port application



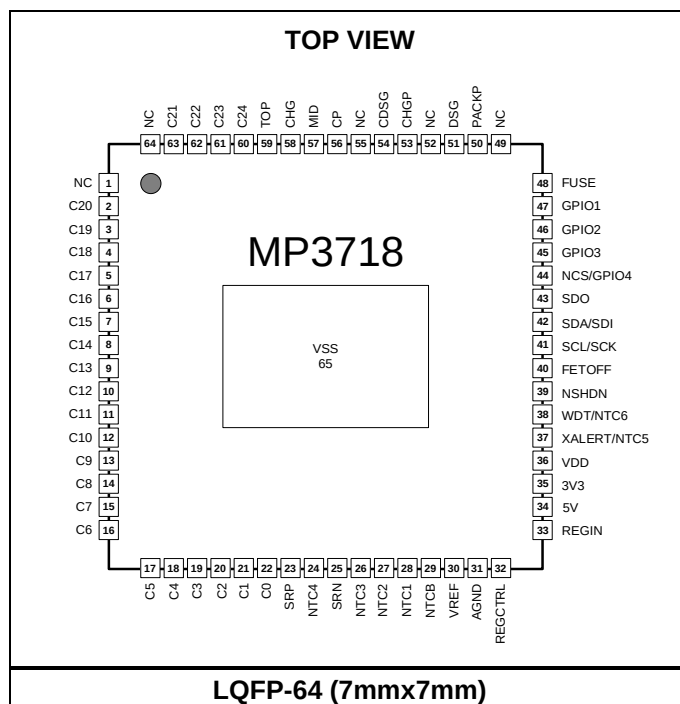
ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating	Comment
MP3718DFN-xxxx**	LQFP-64 (7mmx7mm)	See Below	3	
MP3718DFN-0000	LQFP-64 (7mmx7mm)	See Below	3	I ² C, LDO on in shutdown
MP3718DFN-0002	LQFP-64 (7mmx7mm)	See Below	3	SPI, LDO on in shutdown

* For Tray, add suffix -T (e.g. MP3718DFN-xxxx-T).

** “-xxxx” is the configuration code identifier for the register settings. Each “x” can be a hexadecimal value between 0 and F. Contact an MPS FAE to create this unique number.

PACKAGE REFERENCE



**PIN FUNCTIONS**

Pin #	Name	Type	Description
1	NC	-	Not connected.
2	C20	I	Connect to the positive pin of cell 20.
3	C19	I	Connect to the positive pin of cell 19.
4	C18	I	Connect to the positive pin of cell 18.
5	C17	I	Connect to the positive pin of cell 17.
6	C16	I	Connect to the positive pin of cell 16.
7	C15	I	Connect to the positive pin of cell 15.
8	C14	I	Connect to the positive pin of cell 14.
9	C13	I	Connect to the positive pin of cell 13.
10	C12	I	Connect to the positive pin of cell 12.
11	C11	I	Connect to the positive pin of cell 11.
12	C10	I	Connect to the positive pin of cell 10.
13	C9	I	Connect to the positive pin of cell 9.
14	C8	I	Connect to the positive pin of cell 8.
15	C7	I	Connect to the positive pin of cell 7.
16	C6	I	Connect to the positive pin of cell 6.
17	C5	I	Connect to the positive pin of cell 5.
18	C4	I	Connect to the positive pin of cell 4.
19	C3	I	Connect to the positive pin of cell 3.
20	C2	I	Connect to the positive pin of cell 2.
21	C1	I	Connect to the positive pin of cell 1.
22	C0	I	Connect to the negative pin of cell 1.
23	SRP	I	Positive sense pin.
24	NTC4	I	Thermistor 4 terminal.
25	SRN	I	Negative sense pin.
26	NTC3	I	Thermistor 3 terminal.
27	NTC2	I	Thermistor 2 terminal.
28	NTC1	I	Thermistor 1 terminal.
29	NTCB	O	NTC bias.
30	VREF	P	ADC reference voltage.
31	AGND	P	Ground. Bypass capacitor of VREF and NTCB are recommended to connect to AGND.
32	REGCTRL	P	Turn-on control for external BJT low-dropout (LDO) regulator.
33	REGIN	P	The external 5V and 3.3V LDO regulator input. Connect an external 3.3μF bypass capacitor from REGIN to VSS
34	5V	P	5V voltage output to drive external peripherals. Bypass 5V with an external 1μF capacitor to VSS.
35	3V3	P	3.3V voltage output to drive external peripherals. Bypass 3V3 with an external 1μF capacitor to VSS.
36	VDD	P	4.8V rail for internal use. Connect a 1μF bypass capacitor from VDD to VSS
37	XALERT/NTC5	I/O	Multi-function pin. This pin can be set as Interrupt alert output or NTC5.
38	WDT/NTC6	I/O	Multi-function pin. This pin can be set as watchdog timer pin or NTC6.
39	NSHDN	I	Active-low shutdown input signal.
40	FETOFF	I	MOSFET forced off pin.



PIN FUNCTIONS (continued)

Pin #	Name	Type	Description
41	SCL/SCK	I	Multi-function pin. This pin can be set as the I ² C interface clock or the SPI interface clock. This pin is open drain mode in I ² C version.
42	SDA/SDI	I/O	Multi-function pin. This pin can be set as the I ² C interface data or the SPI serial data input. This pin is open drain mode in I ² C version.
43	SDO	O	SPI serial data output. If this is I ² C version, this pin is high impedance mode.
44	NCS/GPIO4	I/O	Multi-function pin. This pin can be set as General-purpose pin 4 (GPIO4), NTC10, auto cell balancing activation or SPI cable selection.
45	GPIO3	I/O	Multi-function pin. This pin can be set as General-purpose pin 3 (GPIO3), NTC9, or fault indication.
46	GPIO2	I/O	Multi-function pin. This pin can be set as GPIO2, NTC8, high side DSG driver indication, or low side DSG output pin.
47	GPIO1	I/O	Multi-function pin. This pin can be set as GPIO1, NTC7, high side CHG driver indication, or low side CHG output pin.
48	FUSE	I/O	Secondary fuse sense and driver.
49	NC	-	Not connected.
50	PACKP	I/O	Pack sensing voltage (load side). DSG driver reference voltage.
51	DSG	O	Discharge MOSFET driver.
52	NC	-	Not connected.
53	CHGP	I/O	Connect to charger input port. CDSG driver reference voltage.
54	CDSG	O	Discharge MOSFET driver in charge path.
55	NC	-	Not connected.
56	CP	P	Charge pump regulated voltage. Connect a 47nF capacitor from CP to MID, then adjust the value based on the number of parallel DSG and CHG MOSFETs.
57	MID	P	Low-current positive supply pin. Used as the charge pump supply input.
58	CHG	O	Charge MOSFET driver.
59	TOP	I	Battery-side pack-sensing voltage. TOP must be connected to the top of the battery stack, which is the highest positive voltage in the battery pack. CHG driver reference voltage.
60	C24	I	Connect to the positive pin of cell 24.
61	C23	I	Connect to the positive pin of cell 23.
62	C22	I	Connect to the positive pin of cell 22.
63	C21	I	Connect to the positive pin of cell 21.
64	NC	-	Not connected.
65	VSS	P	Ground. Power GND is recommended to connect to VSS. Connect VSS near the positive connection (SRP) of the low-side sense resistor.

**ABSOLUTE MAXIMUM RATINGS** ⁽¹⁾

CHG, CP, DSG to AGND	-0.3V to +140V
TOP, MID to AGND	-0.3V to +140V
CHGP, CDSG to AGND	-0.3V to +140V
PACKP to AGND	-0.3V to +140V
Cn - C(n-1) (n: 1 to 24)	-0.3V to +20V
Cn to AGND (n: 0 to 24)	- 0.3V to +140V
SRP, SRN to AGND	-0.5V to +6V
CP to MID	-0.3V to +20V
REGCTRL, FUSE to AGND	-0.3V to +15V
REGIN to AGND	-0.3V to +7V
NSHDN to AGND	-0.3V to +6V
All other pins to AGND	-0.3V to +6V
Junction temperature	150°C
Lead temperature	260°C
Storage temperature	- 65°C to +150°C

Recommended Operating Conditions ⁽⁴⁾

TOP to AGND	10V to 120V
Cx - C(x-1) (x: 1 to 24) ⁽⁵⁾	-0.2V to 5V
C0 to AGND	-0.25V to +0.3V
Operating temperature (T _j)	-40°C to +125°C
SRP to SRN	-100mV to +100mV

Thermal Metrics ⁽⁶⁾

Junction-to-ambient (R _{θJA})	46.6°C/W
Junction-to-case (top) (R _{θJC(TOP)})	14.5°C/W
Junction-to-board (top) (R _{θJC(TOP)})	27.1°C/W

Notes:

- 1) Exceeding these ratings may damage the device.
- 2) Tested per ANSI/ESDA/JEDEC JS-001.
- 3) Tested per ANSI/ESDA/JEDEC JS-002.
- 4) The device is not guaranteed to function outside of its operating conditions.
- 5) The stack voltage should be greater than 10V
- 6) Metrics provided using set-up conditions compliant with EIA/JESD51-2, 7, and 8.



ELECTRICAL CHARACTERISTICS

Connected Cells=24, V_{cell}=3.6V, V_{MID}=86.4V, V_{REGIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
Supply Current and Leakage						
Total shutdown current	I _{TOT_SHDN}	MID + REGIN average current, NSHDN pin low, 3.3V and 5V off, T _A = 25°C		2		μA
		MID + REGIN average current, NSHDN pin low, 3.3V and 5V off, T _A = -40°C to +105°C			TBD	μA
		MID + REGIN average current, 3.3V and 5V on, NSHDN pin low, T _A = 25°C		13.5		μA
		MID + REGIN current, 3.3V and 5V on, NSHDN pin low, T _A = -40°C to +105°C			TBD	μA
Total safe state current	I _{TOT_SAFE}	MID + REGIN average current, IC in safe state (all MOSFETs off), hardware monitoring off, plug-in detection disabled, no HR ADC conversion, T _A = 25°C		23		μA
		MID + REGIN average current, IC in safe state (all MOSFETs off), hardware monitoring off, plug-in detection disabled, no HR ADC conversion, T _A = -40°C to +105°C			TBD	μA
Total sleep state current	I _{TOT_SLEEP}	MID + REGIN current, IC in sleep state (all MOSFETs off), hardware monitoring off, TOP voltage comparator enable, no HR ADC conversion, T _A = 25°C		30		μA
		MID + REGIN current, IC in sleep state (all MOSFETs off), hardware monitoring off, TOP voltage comparator enable, no HR ADC conversion, T _A = -40°C to 105°C			TBD	μA
Total standby state current	I _{TOT_STANDBY}	MID + REGIN average current, IC in standby state (N-channel MOSFETs on), hardware monitoring interval is 7.5s, no HR ADC conversion, T _A = 25°C		90		μA
		MID + REGIN average current, IC in standby state (N-channel MOSFETs on), hardware monitoring interval is 7.5s, no HR ADC conversion, T _A = -40°C to 105°C			TBD	μA
Total active state current	I _{TOT_ACTIVE}	MID + REGIN average current, IC in active state (N-channel MOSFETs on), hardware monitoring interval is 240ms, no HR ADC conversion, T _A = 25°C		240		μA
		MID + REGIN average current, IC in active state (N-channel MOSFETs on), hardware monitoring interval is 240ms, no HR ADC conversion, T _A = -40°C to 105°C			260	μA

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, VMID=86.4V, REGIN=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
Total full speed HR ADC update mode current	I _{TOT_FSUM}	MID + REGIN average current, IC in pure ADC mode, all voltage ADC channels are converted every 20ms, no Coulomb Counting, T _A = 25°C		1000		μA
		MID + REGIN average current, IC in pure ADC mode, all voltage ADC channels are converted every 20ms, no Coulomb Counting, T _A = -40°C to 105°C			TBD	μA
Cell leakage	I _{Cx_LEAK}	T _A = 25°C	-200		+200	nA
		T _A = -40°C to +105°C	-600		+600	nA
Supported Series Cells						
Supported cell number	N _{CELL} ⁽⁷⁾		4		24	
MID input under-voltage lockout (UVLO) threshold	V _{IN_UVLO}	Falling edge, T _A = -40°C to +105°C			13.2	V
		Rising edge, T _A = -40°C to +105°C			14.2	V
Current Sense						
Current analog-to-digital converter (ADC) conversion time	t _{iADC} ⁽⁷⁾	16 bits (15 bits + sign)		384		μs
Current ADC measurement range	V _{SRPN}	T _A = -40°C to +105°C	-100		+100	mV
Current ADC measurement error	I _{ADC_ERR}	V _{SRN_SRP} = 0mV, T _A = 25°C	-21		21	μV
		V _{SRN_SRP} = ±10mV, T _A = 25°C	TBD		TBD	μV
		V _{SRN_SRP} = ±90mV, T _A = 25°C	-291		291	μV
Current ADC measurement error	I _{ADC_ERR}	V _{SRN_SRP} = 0mV, T _A = -40°C to 105°C	TBD		TBD	μV
		V _{SRN_SRP} = ±10mV, T _A = -40°C to 105°C	TBD		TBD	μV
		V _{SRN_SRP} = ±90mV, T _A = -40°C to 105°C	-561		561	μV
Current ADC measurement gain error ⁽⁸⁾	I _{ADC_GAIN_ERR}	FSR = 100mV, SRP-N common mode: +300mV to -125mV from AGND, T _A = 25°C	-0.3		+0.3	%
		FSR = 100mV, SRP-N common mode: +300mV to -125mV from AGND, T _A = -40°C to 105°C	-0.6		+0.6	%
Current sense offset	V _{IADC_ACC_OFF SET}	SRP - SRN = 0V, SRP-N common mode: +300mV to -125mV from AGND, T _A = -40°C to +105°C	TBD		TBD	μV
SRP and SRN leakage	I _{SRPN_LEAK}	IADC not running, T _A = -40°C to +105°C	-500		+500	nA

Note:

7) Guaranteed by design.

8) Gain error does not include offset error, e.g., 10mV input, the gain error is 10mV×0.3%=30μV, offset error is 21μV, so the total error is 30μV +21μV=51μV. The error can be further reduced by post-PCB assembly calibration. Contact an MPS FAE for the related application note

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{mid}=86.4V, R_{GIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
SRP and SRN differential input current	I _{SRPN_DIFF}	IADC running, SPR - SRN = 100mV, absolute value, T _A = -40°C to +105°C		0.5		μA
		IADC running, SPR - SRN < 5mV, T _A = -40°C to +105°C		50		nA
Sigma Delta Voltage ADC						
Voltage ADC conversion time	t _{VADC} ⁽⁷⁾	16 bits (15 bits + sign), without current ADC concurrency, T _A = -40°C to +105°C		384		μs
		16 bits (15 bits + sign), with current ADC concurrency, T _A = -40°C to +105°C		768		μs
Cell Voltage Measurement						
Cell ADC measurement range	V _{CELL_MEA_RANGE}	T _A = -40°C to +105°C	-0.2		5	V
Total cell measurement error	V _{CELL_ERR}	V _{CELL} = 1V to 4.5V, T _A = 25°C	-3		+3	mV
		V _{CELL} = 1V to 4.5V, T _A = -20°C to 65°C	TBD		TBD	mV
		V _{CELL} = 1V to 4.5V, T _A = -40°C to +105°C	TBD		TBD	mV
		V _{CELL} = -0.2 V to 5V, T _A = -40°C to +105°C	TBD		TBD	mV
C0 to AGND voltage	V _{C0_TO_AGND}	C0 to AGND voltage drop resulting in less than 10mV error on cell 1, T _A = -40°C to +105°C	TBD			V
Cell ADC input current	I _{IN_CELL_CONV}	Input current during ADC conversion when V _{CELL} = 5V, T _A = 25°C, cell 1 is measured by the C1 to C0 voltage		10		μA
Die Temperature						
Die temperature operating range	T _{DIE}		-40		+125	°C
Die Temp. Total Measurement Error	T _{DIE_ERR}	T _J = -40°C to +125°C	TBD		TBD	°C
Over-temperature (OT) analog shutdown threshold	T _{DIE_OTSD}			140	155	°C
Negative Temperature Coefficient (NTC) Temperature Measurement						
NTC voltage measurement range	V _{NTC_RANGE}	Nominal NTC range within a percentage of NTCB, T _A = 25°C	1		100	%
NTC total measurement error	V _{NTC_ERR}	15-bit conversion, with NTCB = 3.3V, T _A = -40°C to 105°C	-55		+55	LSB
NTCx input leakage	I _{NTC_IN_LEAK}	ADC off, on each NTC pin, T _A = -40°C to +105°C			250	nA
NTCx input leakage ADC on	I _{NTC_LEAK_CONV}	ADC converting NTCx, on each NTC pin, T _A = 25°C			250	nA

Note:

7) Guaranteed by design.

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{mid}=86.4V, R_{GIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
VTOP Measurement						
TOP measurement range	V _{TOP_MEAS_RANGE}	T _A = -40°C to +105°C	10		120	V
TOP total measurement error	V _{TOP_ERR}	15-bit ADC reading, T _A = -40°C to +105°C, 20Ω filtering resistor	TBD		TBD	mV
PACKP Measurement						
Pack measurement range	V _{PACKP_MEAS_RANGE}	T _A = -40°C to +105°C	0		120	V
Pack total measurement error	V _{PACKP_ERR}	15-bit ADC reading, T _A = -40°C to +105°C, 100Ω filtering resistor	TBD		TBD	mV
GPIO Measurement						
GPIO measurement range	V _{GPIO}	T _A = 25°C	0		3.3	V
GPIO total measurement error	V _{GPIO_ERR}	15-bit ADC reading, T _A = 25°C	-5		+5	mV
		15-bit ADC reading, T _A = -40°C to +105°C	TBD		TBD	mV
GPIO input current during ADC conversion	I _{GPIO_CONV_IN}	GPIO pins input current during ADC conversion, T _A = -40°C to 105°C		1		μA
Regulators Measurements						
5V total measurement error	V _{5V_MEAS_ERR}	ADC reading, T _A = 25°C	-15		+15	mV
		ADC reading, T _A = -40°C to +105°C	TBD		TBD	mV
3V3 total measurement error	V _{3V3_MEAS_ERR}	ADC reading, T _A = 25°C	-15		+15	mV
		ADC reading, T _A = -40°C to +105°C	TBD		TBD	mV
Hardware Protection						
Cell OV/SOV steps	V _{OVSOV_STEP} ⁽⁷⁾			4.88		mV
Cell OV/SOV accuracy	V _{OVSOV_ACC}	V _{CELL} = 2V to 4.5V, T _A = -20°C to 65°C	-9.76		+9.76	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +105°C	-14.64		+14.64	mV
Cell UV steps	V _{UV_STEP} ⁽⁷⁾			19.53		mV
Cell UV accuracy	V _{UVTH_ACC}	V _{CELL} = 2V to 4.5V, T _A = -20°C to 65°C	TBD		TBD	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +105°C	TBD		TBD	mV
Pack OV/UV steps	V _{POUTH_STEP} ⁽⁷⁾			29.3		mV
Pack OV/UV accuracy	V _{POUTH_ACC}	T _A = -40°C to +105°C	TBD		TBD	
Pack SOV/SUV steps	V _{PSOUTH_STEP} ⁽⁷⁾			29.3		mV
Pack SOV/SUV accuracy	V _{PSOUTH_ACC}	T _A = -40°C to +105°C	TBD		TBD	
Deep discharge steps	V _{DEPTH_STEP} ⁽⁷⁾			19.5		mV

Note:

7) Guaranteed by design.

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, VMID=86.4V, REGIN=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
Deep discharge accuracy	V _{DEPTH_ACC}	V _{CELL} = 2V to 4.5V, -20°C to 65°C	-19.5		+19.5	mV
		V _{CELL} = 1V to 5V, T _A = -40°C to +105°C	-39		+39	mV
OC fastest detection time	t _{OC_DGL_MIN}	Fastest deglitch setting and latency time, T _A = -40°C to +105°C			120	μs
OCL discharge FSR value	V _{OCL_DSG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		38.4		mV
OCL discharge accuracy	V _{OCL_DSG_ACC}	Setting from 1.2mV to 12mV, T _A = -40°C to +105°C	TBD		TBD	mV
		Setting from 12mV to 115.2mV, T _A = -40°C to +105°C	TBD		TBD	%
OCL discharge FSR value	V _{OCL_DSG_FSR_3X}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		115.2		mV
OCH discharge FSR value	V _{OCH_DSG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		38.4		mV
OCH discharge accuracy	V _{OCH_DSG_ACC}	Setting from 1.2mV to 12mV, T _A = -40°C to +105°C	TBD		TBD	mV
		Setting from 12mV to 115.2mV, T _A = -40°C to +105°C	TBD		TBD	%
OCH discharge FSR value	V _{OCH_DSG_FSR_3X}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		115.2		mV
OC charge FSR value	V _{OC_CHG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		38.4		mV
OC charge accuracy	V _{OC_CHG_ACC}	Setting from 1.2mV to 12mV, T _A = -40°C to +105°C	TBD		TBD	mV
		Setting from 12mV to 115.2mV, T _A = -40°C to +105°C	TBD		TBD	%
OC charge FSR value	V _{OC_CHG_FSR_3X}	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		115.2		mV
SC fastest detection time	t _{SC_DGL_MIN}	Fastest deglitch setting and latency time, T _A = -40°C to +105°C			TBD	μs
SC discharge FSR value	V _{SC_DSG_FSR_1X}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = 25°C		176		mV
SC discharge full scale accuracy	V _{SC_DSG_FSR_1X_ACC}	Range and LSB selector = 0 (1x), FSR (max setting), T _A = -40°C to +105°C	TBD		TBD	%

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{MID}=86.4V, R_{GIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
SC discharge offset	VSC_DSG_FS R_1X_OFF	Range and LSB selector = 0 (1x), offset on smallest setting, T _A = -40°C to +105°C	TBD		TBD	mV
SC discharge FSR value	VSC_DSG_FS R_3X	Range and LSB selector = 1 (3x), FSR (max setting), T _A = 25°C		528		mV
SC discharge full scale accuracy	VSC_DSG_FS R_3X_ACC	Range and LSB selector = 1 (3x), FSR (max setting), T _A = -40°C to +105°C	TBD		TBD	%
Cell Balancing						
R _{DS(ON)} balancing MOSFET	R _{DS(ON)_BAL_FET}	T _A = 25°C		28		Ω
		T _A = -40°C to +105°C	12		50	Ω
Low-Dropout Regulator (LDO) Supply and References						
REGCTRL output voltage	V _{REGCTRL}	V _{MID} = 10V to 120V, with REGCTRL load current = 0mA and 1mA, T _A = - 40°C to +105°C	TBD	6.44	TBD	V
REGIN loaded output voltage	V _{REGIN} ⁽⁷⁾	With external REGIN BJT, V _{MID} = 10V to 120V, with load current = 1mA and 50mA, T _A = -40°C to +105°C	TBD	5.5	TBD	V
5V nominal voltage	V _{5V}	REGIN=5.5V, T _A = 25°C		5		V
5V output accuracy	V _{5V_ACC}	REGIN=5.5V, load current=0mA to 45mA, T _A = -40°C to +105°C	4.9		5.1	V
5V analog UV	V _{5V_UV}	Falling edge		4.5		V
5V short-circuit current	I _{5V_EFET}	REGIN=5.5V, T _A = -40°C to +105°C	TBD	55	70	mA
3V3 nominal voltage	V _{3V3}	REGIN=5.5V, T _A = 25°C		3.3		V
3V3 output accuracy	V _{3V3_ACC}	REGIN=5.5V, load current=0mA to 45mA, T _A = -40°C to +105°C	3.135		3.465	V
3V3 short-circuit current	I _{3V3_EFET}	REGIN=5.5V, T _A = -40°C to +105°C	45	55	70	mA
3V3 analog UV	V _{3V3_UV}	Falling edge		2.85		V
VDD output voltage	V _{DD}	REGIN=5.5V, NSHDN=High, T _A = 25°C		4.8		V
		REGIN=5.5V, NSHDN=low (IC in shutdown mode), 5V and 3V3 off, T _A = 25°C		4.5		V
VDD short-circuit current in shutdown	I _{VDD_LIMIT}	NSHDN=low (IC in shutdown mode), 5V and 3V3 off, T _A = 25°C	TBD	2.5		mA

Note:

7) Guaranteed by design.

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{mid}=86.4V, R_{GIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
Reference voltage	V _{REF}	T _A = 25°C		3.3		V
NTCB pull-up voltage	V _{NTCB}	NTCB enabled, no load, T _A = 25°C		3.3		V
NTCB load	I _{NTCB_MAXLOAD}	NTCB enabled, max load, T _A = 25°C		4		mA
		NTCB enabled, max load, T _A = -40°C to +105°C	TBD		TBD	mA
GPIO						
GPIO input voltage (high)	V _{IH-GPIO}	T _A = -40°C to +105°C	2.5		V _H	V
GPIO input voltage (low)	V _{IL-GPIO}	T _A = -40°C to +105°C			0.8	V
GPIO output voltage (high)	V _{OH-GPIO}	V _H = 3.3V or 5V, T _A = -40°C to +105°C	V _H - 0.5		V _H + 0.5	V
GPIO output voltage (low)	V _{OL-GPIO}	V _H = 3.3V or 5V, T _A = -40°C to +105°C	0		0.6	V
GPIO push-pull resistor (up)	R _{GPIO_PUP}	GPIO high voltage = 3.3V		430		Ω
GPIO push-pull resistor (down)	R _{GPIO_PDOWN}	GPIO low voltage = 0V		110	200	Ω
GPIO pull-up mode	R _{GPIO_WEAK_PUP}			17		kΩ
FUSE pin						
FUSE max push-pull current	I _{FUSE_PUP}			11		mA
FUSE output voltage (high)	V _{OH-FUSE}	V _{TOP} = 10V to 120V, Load current=2mA, T _A = -40°C to +105°C	TBD	6.5	TBD	V
FUSE input voltage (high)	V _{IH-FUSE}	T _A = -40°C to +105°C	2.5		7	V
FUSE input voltage (Low)	V _{IL-FUSE}	T _A = -40°C to +105°C	0		0.9	V
NSHDN Pin						
NSHDN internal pull down	R _{SHDN_PD}	NSHDN=0V to 5V		5		MΩ
NSHDN deglitch to enter shutdown	t _{NSHDN_DGL_ENTER}	T _A = 25°C		2		ms
NSHDN deglitch to exit shutdown	t _{NSHDN_DGL_EXIT}	T _A = 25°C		1		ms
NSHDN falling threshold	V _{NSHDN_FALL}			1		V
NSHDN rising threshold	V _{NSHDN_RISE}			2.4	TBD	V
WDT Pin						
Reset pulse length with WDT reset enable	t _{WDT_RSTPULSE_LEN_RST_ENABLE}			25		ms
Reset pulse length with WDT reset disable	t _{WDT_RSTPULSE_LEN_RST_DISABLE}			50		ms

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{MID}=86.4V, REG_{IN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
High-Side MOSFET (HS-MOSFET) Drive						
CHG gate drive on voltage	V _{CHG_ON}	C _{LOAD} = 80nF, V _{MID} = 15V to 120V, static value at CHG driver on, 12.5V selected, the CHG voltage base on V _{TOP} , T _A = -40°C to +105°C	11.1		13.4	V
		C _{LOAD} = 80nF, V _{MID} = 10V to 120V, static value at CHG driver on, 5V selected, the CHG voltage base on V _{TOP} , T _A = -40°C to +105°C	4.5		5.5	V
CHG gate drive off voltage	V _{CHG_OFF}	C _{LOAD} = 80nF, V _{MID} = 10V to 120V, static value at CHG driver off, the CHG voltage base on V _{TOP} , T _A = -40°C to +105°C	TBD		TBD	V
CHG driver on threshold	V _{CHG_ON_TH}	12.5V selected, the CHG voltage base on V _{TOP} , rising edge, T _A = -40°C to +105°C	2		6.6	V
		5V selected, the CHG voltage base on V _{TOP} , rising edge, T _A = -40°C to +105°C	1		3.5	V
CHG driver off threshold	V _{CHG_OFF_TH}	CHG driver off state, falling edge, T _A = -40°C to +105°C	0.5		2.2	V
CHG gate drive turn-on resistance	R _{CHG_TON}	FET drive turned on, PACKP = V _{TOP} , T _A = 25°C		2600		Ω
CHG gate driver turn-off resistance	R _{CHG_TOFF}	MOSFET driver turned off, PACKP = V _{TOP} , T _A = 25°C		840		Ω
DSG gate driver on voltage	V _{DSG_ON}	C _{LOAD} = 80nF, V _{MID} = 15V to 120V, 12.5V selected, static value at DSG driver on, the DSG voltage base on V _{PACKP} , T _A = -40°C to +105°C	9.3		13.4	V
		C _{LOAD} = 80nF, V _{MID} = 10V to 120V, 5V selected, static value at DSG driver on, the DSG voltage base on V _{PACKP} , T _A = -40°C to +105°C	4.5		5.5	V
DSG gate drive off voltage	V _{DSG_OFF}	C _{LOAD} = 80nF, V _{MID} = 10V to 120V, static value at DSG driver off, the DSG voltage base on V _{PACKP} , T _A = -40°C to +105°C	TBD		-1	V
DSG driver on threshold	V _{DSG_ON_TH}	12.5V selected, the DSG voltage base on V _{PACKP} , rising edge, T _A = -40°C to +105°C	2		6.6	V
		5V selected, the DSG voltage base on V _{PACKP} , rising edge, T _A = -40°C to +105°C	1		3.5	V
DSG driver off threshold	V _{DSG_OFF_TH}	DSG driver off state, falling edge, T _A = -40°C to +105°C	0.5		2.1	V

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{MID}=86.4V, R_{GIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
DSG gate driver turn-on resistance	R _{DSG_ON}	FET driver turned on, PACKP = V _{TOP} , T _A = 25°C		2400		Ω
DSG gate driver turn-off resistance	R _{DSG_OFF}	MOSFET driver turned off, PACKP = V _{TOP} , DSG = PACKP + 5V, T _A = 25°C		600		Ω
DSG constant current regulation accuracy	I _{DSG_CC}	Setting to the maximum value, T _A = -40°C to +105°C	TBD		TBD	%
CDSG gate driver on voltage	V _{CDSG_ON}	C _{LOAD} = 80nF, V _{MID} = 15V to 120V, 12.5V selected, static value at CDSG driver on, the CDSG voltage base on V _{CHGP} , T _A = -40°C to +105°C	9.3		14.1	V
		C _{LOAD} = 80nF, V _{TOP} = 10V to 120V, 5V selected, static value at CDSG driver on, the CDSG voltage base on V _{CHGP} , T _A = -40°C to +105°C	4.5		5.5	V
CDSG gate driver off voltage	V _{CDSG_OFF}	C _{LOAD} = 80nF, V _{MID} = 10V to 120V, static value at CDSG driver off, the CDSG voltage base on V _{CHGP} , T _A = -40°C to +105°C	TBD		TBD	V
CDSG driver on threshold	V _{CDSG_ON_TH}	12.5V selected, the CDSG voltage base on V _{CHGP} , rising edge, T _A = -40°C to +105°C	2		6	V
		5V selected, the CDSG voltage base on V _{CHGP} , rising edge, T _A = -40°C to +105°C	1		3.4	V
CDSG driver off threshold	V _{CDSG_OFF_TH}	CDSG driver off state, falling edge, T _A = -40°C to +105°C	0.5		1.65	V
CDSG gate driver turn-on resistance	R _{CDSG_ON}	FET driver turned on, CHGP = V _{TOP} , T _A = 25°C		2400		Ω
CDSG gate driver turn-off resistance	R _{CDSG_OFF}	FET driver turned off, CHGP = V _{TOP} , CDSG = CHGP + 5V, T _A = 25°C		600		Ω
Charge Pump						
Charge pump output voltage	V _{CP}	V _{MID} = 15V to 120V, 12.5V driver level selected, T _A = 25°C		14.5		V
		V _{MID} = 10V to 120V, 5V driver level selected, T _A = 25°C		7.5		V
Charge pump output voltage accuracy	V _{CP_ACC}	T _A = -40°C to +105°C	TBD		TBD	V
Charge pump turn-on time	t _{CP_TON}	V _{CP} - AGND transition from V _{MID} to V _{MID} + V _{CP} , 47nF capacitor between CP and MID, T _A = -40°C to +105°C			10	ms

**ELECTRICAL CHARACTERISTICS (continued)**Connected Cells=24, V_{cell}=3.6V, V_{MID}=86.4V, V_{REGIN}=5.5V, T_A = 25°C, unless otherwise noted.

Parameter Name	Symbol	Condition	Min	Typ	Max	Units
Pull-Up and Comparators						
PACKP pre-charge	I _{PACKP_PUP}	V _{TOP} = V _{MID} = 10V to 120V, T _A = 25°C		750		μA
CHGP pre-charge	I _{CHGP_PUP}	V _{TOP} = V _{MID} = 10V to 120V, T _A = 25°C		750		μA
PACKP/CHGP higher than V _{TOP}	V _{HGR_VTOP}	V _{TOP} = V _{MID} = 10V to 120V, T _A = 25°C, rising edge		280		mV
PACKP/CHGP lower than V _{TOP}	V _{LWR_TOP}	V _{TOP} = V _{MID} = 10V to 120V, T _A = 25°C, falling edge		-1.77		V
Standby current comparator threshold	V _{SBY_COMP_TH}	T _A = -40°C to +105°C	TBD	250	TBD	μV
Standby current comparator detect time	T _{SBY_COMP}	T _A = 25°C			10	ms
PACKP short recovery threshold	V _{PACKP_SCOC_REC_TH}	T _A = 25°C		2		V
		T _A = -40°C to +105°C	1.5		2.5	V
Short recovery pull-up current	I _{SCOC_PUP}	T _A = -40°C to +105°C	TBD	250	TBD	μA
Internal Clock						
Internal clock frequency	f _{2M_CLOCK}	V _{MID} = 10V to 120V, T _A = 25°C		2		MHz
		V _{MID} = 10V to 120V, T _A = -40°C to +105°C	TBD	2	TBD	MHz
I ² C Communication Interface						
I ² C clock frequency	f _{I2C}	T _A = 25°C			400	kHz
SCL, SDA input voltage low	V _{ILOW}	T _A = -40°C to +105°C			1	V
SCL, SDA input voltage high	V _{IHIGH}	T _A = -40°C to +105°C	2.35			V
SCL, SDA input hysteresis	V _{IHYST}	T _A = -40°C to +105°C		0.66		V
SPI Communication Interface						
SPI clock frequency	f _{SPI}	T _A = 25°C			1.0	MHz
SPI input low	V _{SPI_IN_LOW}	SDI, CLK, T _A = -40°C to +105°C			1	V
SPI input high	V _{SPI_IN_HIGH}	SDI, CLK, T _A = -40°C to +105°C	2.35			V



FUNCTIONAL BLOCK DIAGRAM

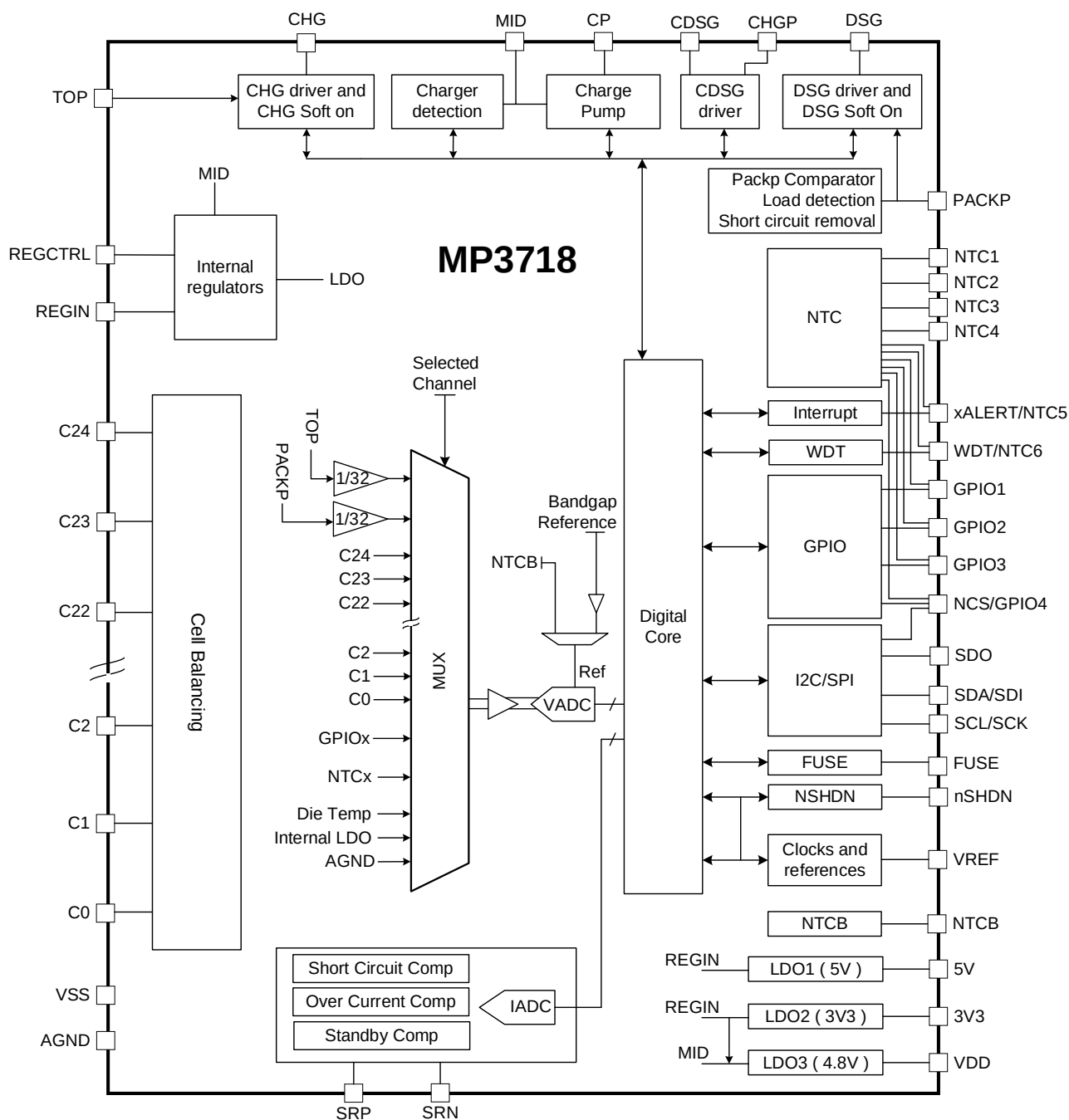


Figure 3: Functional Block Diagram



APPLICATION INFORMATION

MP3718 is a high-accuracy, 4-cell to 24-cell, high-accuracy monitor and protector.

The MP3718 includes high-side MOSFET (HS-FET) drivers for charge and discharge independent control and protection. It can automatically scan the cell voltage and temperature at regular intervals, and continuously monitor the current, and then compare the results with the configured thresholds. If the results exceed the thresholds, the device can automatically turn off the MOSFET or burn down the fuse according to the fault type and initialization configuration.

The host microcontroller can obtain cells' voltage, current and temperature from this device via I²C or SPI commands. Based on this information, the host controls the external devices to protect the cells, calculates the pack SOC, and balances the mismatched cell.

MP3718 integrates self-diagnostics functions, which include the power rails monitor, ADC self-test, non-volatile memory (NVM) check, and watchdog to enhance the system robustness.

REGULATORS AND POWER DOMAINS

The MP3718 is powered from VMID and REGIN.

There are three main power rails:

- VDD (4.8V): internal use only.
- 5V: 5V voltage output to drive external peripherals.
- 3.3V: 3.3V voltage output to drive external peripherals.

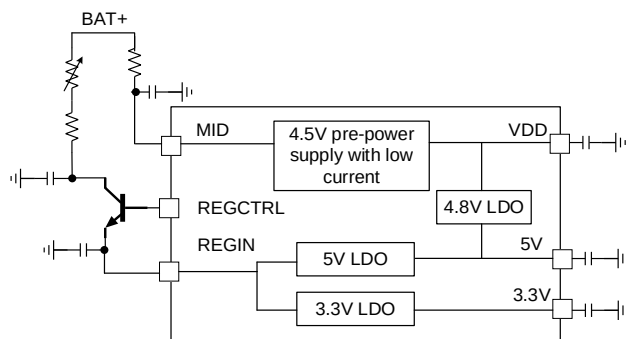


Figure 4: Power Rails

The REGIN is connected to the emitter of the external BJT transistor and REGCTRL is connected to the base of the BJT. The

REGCTRL outputs a stable 6.2V voltage which passes through the BE junction of the transistor to provide a voltage of approximately 5.6V to the REGIN. The REGIN can also be powered directly from the external 5V DC source.

The 5V and 3.3V rails are generated by the internal LDOs. The 5V and 3.3V rails can be configured to be on or off in shutdown mode by V5_SHDN_EN and V3P3_SHDN_EN.

These power rails are monitored internally, and the device can provide an interrupt or fault if the voltage falls below thresholds.

NON-VOLATILE MEMORY (NVM) CONFIGURATION

The default values of most registers can be configured through the non-volatile memory (NVM). Some registers are one-time programmable (OTP), while others are multiple-time programmable (MTP). The MTP registers can be programmed up to two times. These values can be locked, so that the same version of the IC can be adjusted for similar projects with minor differences.

To configure the MTP, follow the steps below:

1. Apply VDD=5.5V.
2. Set store_cmd_access_code = 0xBAF4.
3. Make sure the register has been set to the expected value.
4. Set store_nvm_cmd = 1.
5. Read store_in_progress, if it equal to 0, it means MTP program is done.
6. Read the otp_nvm_status to confirm the programming.

MAIN MODES

The MP3718's main operation modes are listed and described below. Figure 5 shows the main states transition diagram.

Shutdown Mode

Shutdown mode minimizes the amount of leakage from the battery pack, which extends the battery pack's shelf storage life.

In shutdown mode, the communication interface is unavailable. VDD is on, and the 5V and 3.3V

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

can be configured on. If 3V3 is enable, 5V cannot be disable. If any of the following situations occur, the device will enter shutdown mode.

- NSHDN pin is low.
- V_{MID} is lower than the V_{IN_UVLO} .
- The die temperature exceeds the T_{DIE_OTSD} .

Sleep Mode

In order to reduce the current consumption when the remaining capacity of battery pack is small, MP3718 provides a low voltage threshold of TOP. When the voltage of TOP below than this threshold, the current is not charging state, and the voltage of MID is not higher than TOP, IC into sleep mode.

Sleep mode is characterized by the following conditions:

- The protection MOSFETs are forced OFF.
- Hardware autonomous protection is disable.
- HR ADC, Coulomb counter, cell balancing is disable.
- The communication interface is enabled.

The MID vs. PACKP comparator is enabled for the wake up.

The DSG MOSFET dead and CHG MOSFET dead detection can be enable in sleep mode.

If `SAFE_WAKEUP_EN` (OTP) is 1 and `TOP_SLEEPUV_STS` is 0, IC will go to safe mode from shutdown mode.

If `SAFE_WAKEUP_EN` (OTP) is 0, it will go to safe mode until V_{MID} is higher than V_{TOP} or a write command of `FORCE_SLEEP_EXIT=1`.

Once a fuse fault occurs or the TOP voltage falls below a configurable threshold, the device can enter sleep mode. This mode can be used when the host MCU fails, and the battery pack needs to save power.

Safe Mode

In safe mode, no fault conditions exist and both MOSFETs are disabled. The device can continue taking high resolution ADC measurements. Power consumption can be reduced by disabling or slowing down the autonomous hardware monitoring. Safe mode is characterized by the following conditions:

- The protection MOSFETs are turned off.
- The hardware autonomous protection state machine is disabled, unless it is forcibly enabled.
- The communication interface is enabled.
- The highly resolution (HR) ADC readings can be updated manually or automatically.
- The cell balancing can be performed manually or automatically.

Disable autonomous hardware monitoring by commanding `SAFE_MONITOR_EN = 0`. Adjust the hardware monitoring rate by setting the value of `STBY_MONITOR_CFG`.

Turn on one FET or both FETs, the device goes to active mode.

Active Mode

In active mode, at least there is one MOSFET on, and the battery pack system can be powered or be charged by a downstream system through the CHG and DSG N-channel MOSFETs.

Hardware autonomous monitoring is automatically enabled in active mode. The default update rate is 240ms but can be changed to 120ms.

Standby Mode

In standby mode, no fault conditions exist, and one or both MOSFETs are enabled. However, the discharge or charge current is very low. Standby mode supports low current consumption with the ability to transition quickly to active mode.

In standby mode, the interval time each hardware monitor slot can be independently lengthened via the `STBY_MONITOR_CFG` register. This reduces the average current consumption.

During standby a coulomb counting will be monitoring the current with less than 100% duty cycle.

To enter standby mode, `STBY_STATE_EN` should be set to 1, and the current must be below the standby current threshold for a configurable time.



If the pack current exceeds the standby current (STBY_CUR_TH), the MP3718 will immediately go to active mode.

Fault Mode

In fault mode, either one MOSFET driver off or both the CHG and DSG MOSFET drivers off in response to a fault event. Hardware protection monitoring is still enabled during a fault.

Fault mode can be cleared manually or through automatic fault recovery, depending on the configuration.

Pure ADC Mode (ESS)

As an alternative to safe and active mode, pure ADC mode skips the hardware monitor slot to provide a faster rate of HR ADC updates. Note that self-diagnostics, fault indication and other hardware monitoring is not available in pure ADC mode. Set the ESS_MODE_EN=1 in normal mode, the device goes to pure ADC mode.

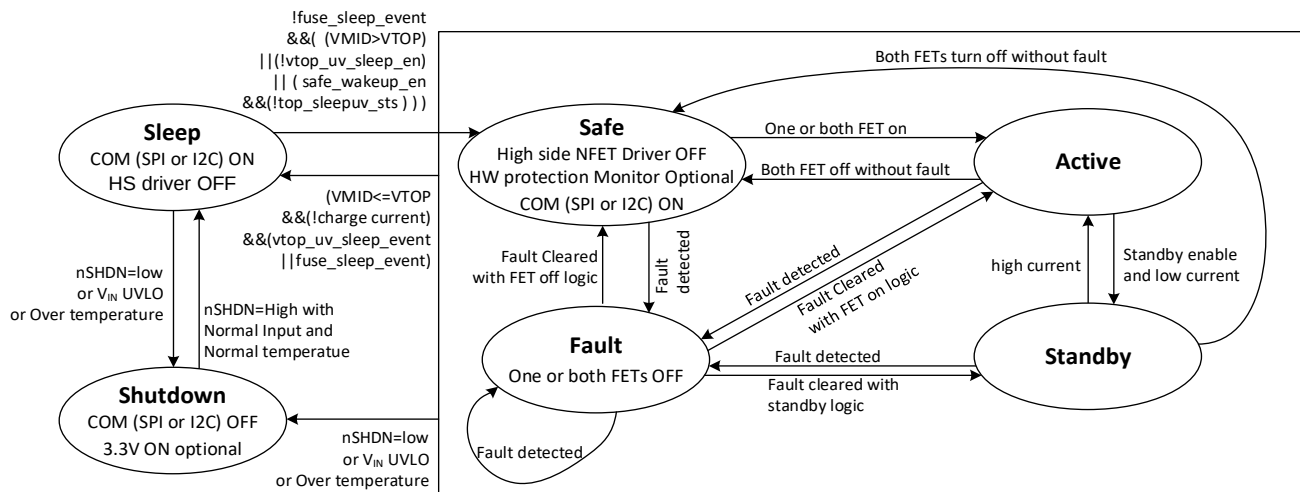


Figure 5: Main States Diagram with monitor protection

REGISTERS (DEFAULT AND LOCK)

Most of the configuration settings and options have a configurable default value. Certain registers can be configured independently, or they can be locked in read-only mode. This prevents critical safety features from being changed.

OPERATING SCHEDULE

When device in safe mode with SAFE_MONITOR_EN = 1, or in fault mode, or in active mode, or in standby mode with STBY_MONITOR_EN = 1, the timeslots are sequenced as: A - B - B - B - ... - B - A - B - B - ..., where A is the hardware protection timeslot and B is the intermediate timeslot.

The slot-A (hardware protection timeslot) is used to monitor the voltage and temperature channels programmed to be monitored. This timeslot has a fixed length of 20ms. The slot-A is also used to detect the cell list needs to auto-balance and the ADC temperature compensation update. The

hardware voltage monitors and cell balancing list detection can be completed in one slot-A.

The slot-B (intermediate timeslot) is used to execute the functions of auto-open wire detection, high-resolution (HR) ADC scan, manual open wire detection, and cell balancing, which in order of priority. This slot-B has a fixed length of 20ms. Any function starts on the rising edge of the slot-B, and if this function cannot be completed in one slot-B, it will continue in the next slot-B if there is no higher-priority function request.

The Coulomb counter can be executed in both slot-A and slot-B and can be run concurrently with other functions.

The number of slot-B that occur between slot-A depends on ACTIVE_MONITOR_CFG, STBY_MONITOR_CFG, and the operation mode of the device.

When device in pure ADC (ESS) mode with DISABLE_HWP_SLOT = 1, the slot-A is disabled and 2ms temperature compensation re-



calculated once every 15 slot-B. Since the slot-A is disabled, the voltage hardware monitor and auto-balance are ineffective.

Note that some register configurations are modified by serial communication, the register

value is loaded by the chip's internal logic only at slot-A. The register ADC_CTRL (A8h) shows the slot status.

Figure 6 shows the typical sequenced Diagram

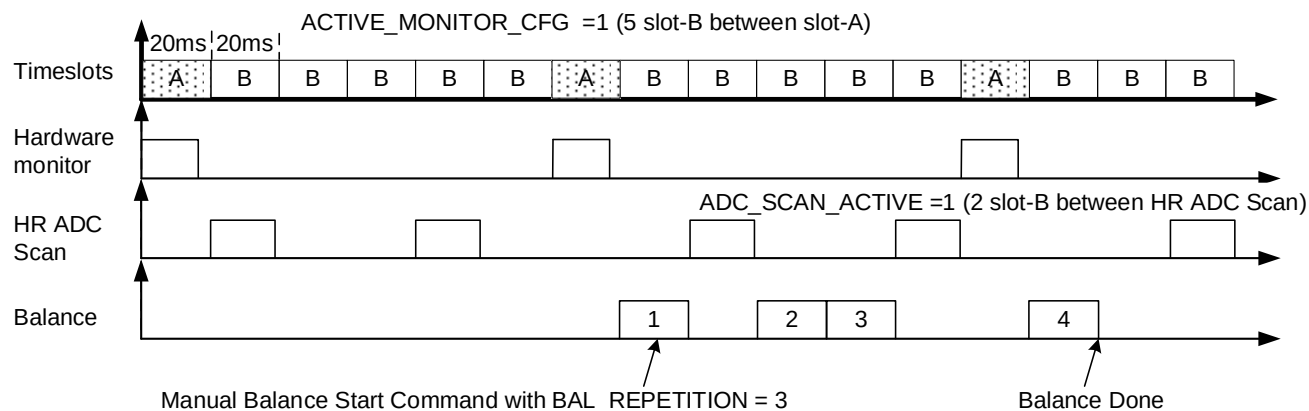


Figure 6: The Typical sequence diagram

High-Resolution Voltage ADC Scan

A voltage scan can convert the following classes of inputs:

- Die temperature voltage
- NTC voltages
- GPIO voltages
- Cell voltages
- TOP to AGND voltage
- PACKP to AGND voltage

If a class of inputs is not enabled, the scan skips to the next class. Control bits are provided to enable individual channels in each class (refer to HR_SCAN1, HR_SCAN2, HR_SCAN3 for more information).

The default run time for each channel is 78μs settling time and 384μs conversion time, and when the current ADC and voltage ADC are running together (cell sync, TOP sync, or Coulomb counter enable), the time for each channel is 384μs settling time and 384μs conversion time. So, in one slot-B the device can measure all channels without Coulomb counting and without cells synchronous conversion, or maximum 23 channels with coulomb counting, or maximum 26 channels without coulomb counting and with cell synchronous conversion enable.

The MP3718 allows for cell inputs to be shorted or connected across corresponding bus bars. The cell measurement range is -0.2V to 5V. Figure 7 shows the bus bar connection.

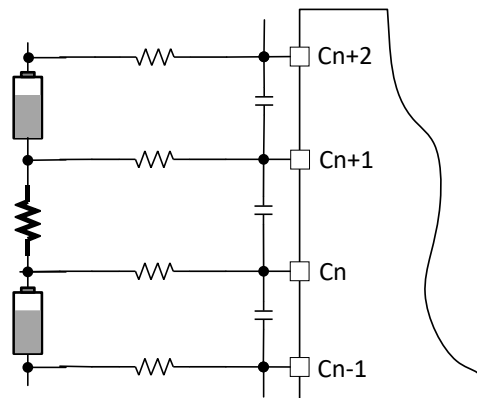


Figure 7: Bus Bar Connection

All NTC ADC channels are a ratio metric conversion. When the ADC converts the NTC channel, the ADC reference is switched to NTCB. In this scenario, all effects related to NTCB drifts due to the temperature and operating conditions are removed from the reading. Figure 8 shows the architecture for NTC ADC conversion.

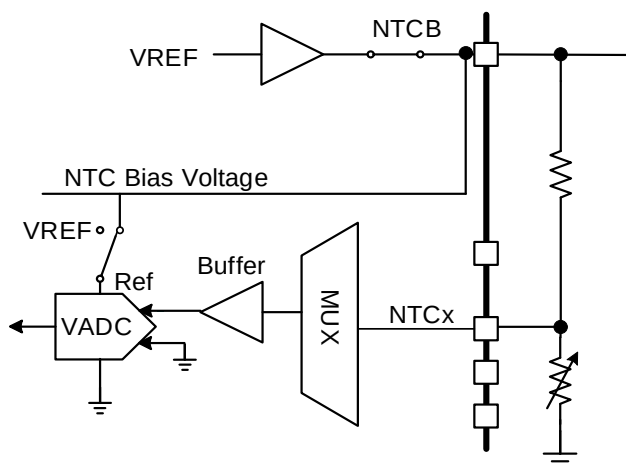


Figure 8: Architecture for NTC ADC Conversion

Manual HR ADC Scan

The manual HR ADC scan means that the HR ADC reading refresh needs the special communication command from host MCU each time.

The sequence below provides the typical steps to handle the HR ADC scan manually:

1. Initialize enable the channels and select whether cell sync or TOP sync in HR_SCAN1, HR_SCAN2, HR_SCAN3, enable the die temperature monitor by DIE_TEMP_DIG_EN =1 to allow the ADC temperature compensation, and, set ADC_SCAN_CONT_EN = 0 to enable manual HR ADC Scan, then set the SAFE_MONITOR_EN = 1 to load these configurations.
2. Set ADC_SCAN_GO =1 to start the HR ADC scan.
3. Once SCAN_DONE_STS =1 means the HR ADC scan is done.
4. Read the HR ADC reading from 6Ah to A7h.
5. To start the next HR ADC scan, set ADC_SCAN_GO =0 and then set ADC_SCAN_GO = 1.

Automatic HR ADC Scan in active mode

The automatic HR ADC scan means that the HR ADC reading can automatically refresh by the AFE itself.

The sequence below provides the typical steps to start the continuous and automatic HR ADC scan.

1. Initialize the configuration to enable the channel and select whether cell sync or TOP

sync in HR_SCAN1, HR_SCAN2, and HR_SCAN3, enable the die temperature monitor by DIE_TEMP_DIG_EN =1 to allow the ADC temperature compensation, and then set the SAFE_MONITOR_EN = 1 to load these configurations.

2. Configure the HR ADC scan rate in ADC_SCAN_SAFE (the rate in standby) and ADC_SCAN_ACTIVE.
3. Set ADC_SCAN_CONT_EN = 1 to start the continuous HR ADC scan in active mode.
4. To stop the continuous ADC scan, set ADC_SCAN_CONT_EN = 0.

If the automatic HR ADC also required in safe mode, set ADC_SCAN_CONT_SAFE_EN = 1 in step 3. Note that the current consumption will increase if the automatic HR ADC enabled in safe mode.

Automatic HR ADC Scan in ESS Mode

The sequence below provides the typical steps to start the continuous and automatic HR ADC scan in pure ADC mode.

1. Initialize the configuration to enable the channel and select whether cell sync or TOP sync in HR_SCAN1, HR_SCAN2, and HR_SCAN3, enable the die temperature monitor by DIE_TEMP_DIG_EN =1 to allow the ADC temperature compensation, and then set the SAFE_MONITOR_EN = 1 to load these configurations.
2. Configure the HR ADC scan rate in ADC_SCAN_ACTIVE.
3. Set ESS_MODE_EN=1 and DISABLE_HWP_SLOT =1 to enter the pure ADC mode.
4. Set ADC_SCAN_CONT_EN = 1 to enable the continuous HR ADC scan in pure ADC mode.
5. To stop the continuous ADC scan, set ADC_SCAN_CONT_EN = 0.

The ADC_SCAN_GO is invalid for the continuous ADC scan. When the continuous and automatic HR ADC scan is required, the ADC complete state is invalid.

Synchronized Voltage and Current Reading

For cell and pack voltages, the exact, synchronized current readings can be read in



dedicated registers paired with the matching voltage reading. Synchronous current readings can be enabled using CELL_SYNC_EN and VTOP_SYNC_EN (refer to HR_SCAN2). ADC results can be got from 6Ch to 9Bh.

The synchronized voltage and current reading can be requested concurrently with Coulomb counting monitoring.

Cell-Balancing

The battery pack reaches its end of life (EOL) when the capacity of the weakest cell is too low to be usable. Unbalanced cells yield the same effect, with one cell at a higher SOC, and another at a lower SOC. To reduce the effects of the issue, it is critical to balance lithium cells.

Unbalanced cells can be caused by cell temperature differences, different self-discharge rates, and general production tolerances that affect cell chemistry and alter aging rates among strings of parallel cells. Cell-balancing can extend the usable life of the battery pack by keeping the voltage of each cell within the defined safe operating area (SOA).

Internal MOSFET Balancing

The MP3718 incorporates internal MOSFETs that support direct cell balancing, without the use of external MOSFETs or BJTs, for currents up to about 58mA (with a 4V cell voltage). The balance current for each cell is dictated by the combined filter resistor and the on resistance of the internal balancing MOSFET ($R_{(DS)ON}$).

External Balancing

A balancing current exceeding 58mA can be achieved by adding external balancing MOSFETs or BJTs. Board thermals and the wiring resistance limit this balancing current. For external MOSFETs, it is also limited by the MOSFETs' $R_{(DS)ON}$ at the appropriate gate voltage (V_{GS}). For external BJTs, it may be limited by the DC current gain (h_{FE}). For external MOSFETs, a minimum gate threshold voltage of 1.8V is required (assuming a 4.2V lithium cell) to turn on the balancing MOSFETs.

Cell Balancer Control

The MP3718 will only conduct cell balancing on valid inputs as configured by the CELL_S_CTRL and CELL_MONITOR_SKIP registers.

The MP3718 can simultaneously balance only even or odd cells. Internal logic ensures that simultaneously odd/even balancing is not allowed. Figure 9 shows the internal logic of the cell balancing execution sequence.

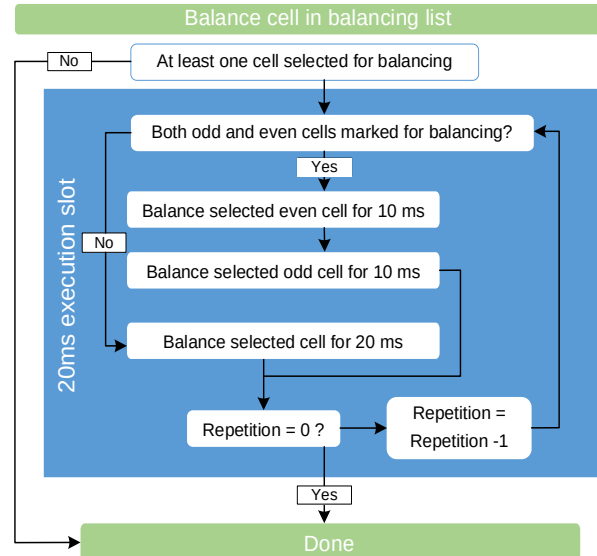


Figure 9: Executing Balancing for Cell in Balancing List

Since cell1/3/5/.../21/23 are fixed as odd cell channels and cell2/4/6/.../22/24 are fixed as even cells, if one channel is shorted or busbar connected by CELL_MONITOR_SKIP, adjacent cells may be balanced at the same time, causing the external balancing circuit of the upper cell to fail to conduct. For example, cell9 set as skips, and cell8, cell10 are set as balancing together, the external balancing circuit of cell 10 can't conduct.

To avoid the above scenario, it is recommended to skip two adjacent channels for automatic balancing, as shown in Figure 3. For manual balancing, it can be avoided by skipping two adjacent channels, or the host MCU can avoid writing adjacent cells to the balancing list at the same time.

Manual Cell-Balancing

The cells that need to be balanced are manually marked by writing to the appropriate register (refer to BAL_LIST and BAL_LISTH).

The balancing time for marked cells can be configured via BAL_REPETITION, with 0 to 31 repetitions. When BAL_REPETITION is set to 0, there is only one 20ms execution (refer to BAL_CFG).



Set `BALANCE_GO = 1` to start the manual cell-balancing. To start the next balancing once the balancing is completed, set `BALANCE_GO = 0` and then set `BALANCE_GO = 1`.

The host MCU should periodically read the AFE die temperature to ensure it is within the operating range, as an excessive temperature can lead to OT (Over Temperature) shutdown.

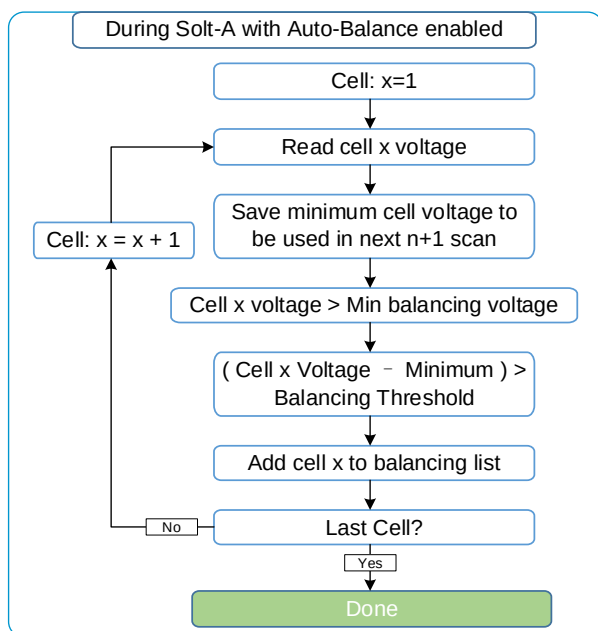


Figure 10: Automatic Cell-Balancing List Update

Automatic Cell-Balancing

Some or all of the cell balancing functions can be automated by the MP3718, by using automatic cell-balancing. Unlike manual balancing, there is no need to configure the balancing list when using automatic balancing.

The following configurations are available for automatic cell-balancing:

1. Enable automatic cell balancing by setting the register `BALANCE_MODE_REG = 1` (refer to `BAL_CFG` (41h)).
2. Set the balancing threshold via the register `BAL_MSM_TH` (refer to `BAL_THR` (42h)). This is the minimum voltage difference between the current cell voltage and the lowest cell voltage to be eligible for cell balancing. This value ranges between 9.765mV and 78.12mV, with 9.765mV steps.
3. Set the minimum balancing voltage via the register `CELL_BAL_MIN` (refer to `BAL_THR`

(42h)). This is the minimum cell voltage for a cell to be eligible for balancing. This value ranges from 2500mV to 4961mV with 39mV steps.

4. Set `ABAL_ON_SAFE` (refer to `BAL_THR` (42h)) to enable or disable automatic cell-balancing in safe mode. Set `ABAL_ON_CHARGE` to enable or disable automatic cell-balancing when the state of the current is in charge mode. Set `ABAL_ON_STBY` to enable or disable automatic cell-balancing when the state of the current is in standby mode.
5. Configure the method for iterations automatic cell-balancing via the register `BALANCE_MODE_CTRL` (refer to `BAL_CFG` (41h)). The device can be configured to use register control `BALANCE_GO` (refer to `BAL_CTRL` (40h)) control or `GPIO4` control to initiate automatic cell-balancing.
6. When `AUTO_BAL_ALWAYS = 0` (refer to `BAL_CFG` (41h)), automatic cell balancing will not operate continuously. In this configuration, automatic cell-balancing is initiated via register or via `GPIO3` control. The register `BALANCE_MODE_CTRL` configures the method of control (register or `GPIO4`). Once initiated, automatic cell-balancing will execute for a specific number of iterations, based on the value in the register `BAL_REPETITION`. Automatic cell balancing should then be re-initiated as needed (via register or `GPIO4`).
7. When `AUTO_BAL_ALWAYS = 1`, the device will continuously balance the cells whenever imbalance is detected. With automatic cell-balancing, there is no need for host MCU intervention, and the cell-balancing repetition number is ignored. Disable constant automatic balancing by setting register `AUTO_BAL_ALWAYS = 0`. The `BALANCE_GO`, `BAL_DONE_STS`, `BALANCING_COUNT_STS` and `AUTO_BAL_SKIPPED_XXX` are invalid under the constant automatic balancing mode. If the balancing condition is not satisfied, like die over temperature or open wire detected or discharging current, the balancing will be suspended. The balancing done interrupt won't be triggered under this mode.

Figure 10 shows the cell-balancing list update for automatic cell-balancing. If



CELL_xx_MONITOR_SKIP =1, this cell channel voltage won't affect the automatic cell-balancing.

Cell-Balancing and HR ADC Concurrency

In general, HR ADC scanning has a higher priority than cell balancing, so if both functions are requested at the same time, HR ADC will run first. To detect whether the cell-balancing circuit is working properly, the device provides HR ADC scan and cell-balancing concurrent modes. The sequence below provides the typical steps to handle the cell-balancing circuit check.

1. Start the HR ADC scan without balancing.
2. Set BAL_ODD_ON_HRSCAN=1 and then start the HR ADC scan, the internal balancing MOSFET of the odd cell is turned on during the HR ADC scan.
3. Set BAL_EVEN_ON_HRSCAN=1, then start HR ADC scan, even cell internal equalization MOSFET turns on during HR ADC scan.
4. Comparing the two HR ADC results above, the host MCU determines whether the cell balancing circuit is normal or not.

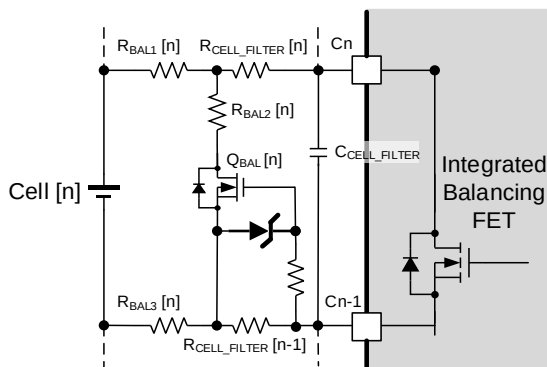


Figure 11: External Balance Diagnostic Circuit

Note that the BAL_ODD_ON_HRSCAN and BAL_EVEN_ON_HRSCAN shouldn't be set to 1 together. To detect the external balancing circuit, the balancing resistor needs to be split into 3. Figure 11 shows the circuit.

Coulomb Counter

The MP3718 supports the continuous accumulation of current over a period, and in the case of dynamic current, the average current over the period can be obtained.

The period length can be set from 24.576ms to 1572.864ms. current continuous sample times can be set from 64 to 4096. A single current ADC

conversion time is 384μs, and when set to 24.576ms, the current is sampled 64 times continuously. When the accumulation is complete, the value is stored in the CC_ACC_LSBS (lower 2 bytes) and CC_ACC_MSBS (upper 2 bytes) registers for MCU reading, and an interrupt is generated.

There are three operating modes for the Coulomb counter.

- **Single-shot mode:** the Coulomb counter is executed only once. The next start requires writing 0 to the enable bit before writing 1. The corresponding register configuration is CC_B2B_ACC_CTRL = 0 and CC_SAMPLE_MODE = 0.
- **Intermittent mode:** the Coulomb counter works as an active-wait- active mode. When CC_SAMPLE_MODE = 1 and CC_B2B_ACC_CTRL = 1, the Coulomb counting runs for CC_INT_CNT time, then waits for CC_SAMPLE_TIME_SEL time, and then continues.
- **Back-to-back mode:** the Coulomb counter works as a continuous back to back mode. When CC_SAMPLE_MODE = 0 and CC_B2B_ACC_CTRL = 1, a new Coulomb counting conversion automatically restarts as soon as the current count is completed.

The sequence below shows the typical steps to handle a Coulomb counting conversion:

1. Initialize enable the Coulomb counter function by CC_EN_SAFE=1 and CC_EN_FAULT= 1 to allow the Coulomb counter function in safe mode and fault mode.
2. Configure the Coulomb counter length via CC_INT_CNT.
3. Configure the CC_SAMPLE_MODE and CC_SAMPLE_MODE to set the Coulomb counter operating mode. If intermittent mode used, the CC_SAMPLE_TIME_SEL should be configured.
4. Set CC_EN to enable starting the conversion.
5. Accumulation reaches the amount specified in CC_INT_CNT. CC_ACC_LSBS and CC_ACC_MSBS are updated to the values reported in CC_RT_ACC_LSBS and CC_RT_ACC_MSBS.

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

6. When in single-shot mode, the CC_DONE flag can be checked to verify whether accumulation has been completed. Readings are available in CC_ACC_LSB and CC_ACC_MSB.
7. If CC_B2B_ACC_CTRL=1, the CC_RT_ACC_xSBS registers are automatically cleared and Coulomb counting restarts. The integrated value for completed accumulation can be read from CC_ACC_xSBS. The CC_DONE status is invalid in this mode. If CC_ACC_INT_EN is enabled, a Coulomb counting interrupt is triggered when CC_ACC_xSBS updates.
8. Set CC_EN = 0 to stop Coulomb counting.

If an CC_ACC_INT_STS interrupt is detected, it should be cleared using the CC_ACC_STS_INT_CLEAR command.

Hardware Voltage Monitor

An autonomous hardware state machine periodically schedules the conversion for all relevant channels during slot-A, and the results are internally checked and compared to the limits.

The deglitch filters described refers to the number of consecutive results during which the relevant channel exceeds its thresholds. The interval between results is depending on the operation mode, ACTIVE_MONITOR_CFG, and STBY_MONITOR_CFG.

The results from monitoring can be reported to the XALERT, GPIO1, GPIO2, GPIO3, or FUSE pin or directly action on the MOSFET driver when the result exceeds its threshold and keep deglitch time if the interrupt, fault or fuse fault is enabled. More details information refers to the hardware monitor section.

Open Wire Detection

The open wire can detect the opened cell (or opened wire).

It has manual open wire and automatic open wire.

Manual Open Wire Detection

The host MCU can trigger open-wire detection by writing OPEN_WIRE_GO =1. Once all cells have been detected, the flag OPEN_WIRE_DONE reports to 1. For next new open-wire detection set OPEN_WIRE_GO = 1 after set the OPEN_WIRE_GO = 0.

The detected wires depend on the CELL_S_CTRL register (see CELLS_CTRL (00h)) and CELL_x_MONITOR_SKIP. The all used cells' up wire and down wire can be detected.

The manual open wire will execute in slot-B with higher priority than the balancing and lower priority than HR ADC Scan.

Automatic Open Wire Detection

Set AOPW_EN_DSG =1, during discharge or standby current the device will automatically execute open wire detection when cell under-voltage (UV), over-voltage (OV), mismatch (MSMT), dead, or severe over-voltage (SOV) is detected.

Set AOPW_EN_CHG =1, during charge current the device will automatically execute open wire detection when cell UV/OV/MSMT/DEAD/SOV is detected.

The detected wires are the upper and lower wire of which cell UV/OV/MSMT/DEAD/SOV is detected.

The automatic open wire will execute after the hardware monitor with higher priority than the HR ADC scan in slot-B. Therefore, the HR ADC reading update will be affect if automatic open wire triggered.

Once the upper or lower wire of this cell has been detected as open, this cell will not be reported as UV/OV/MSMT/DEAD/SOV.

PROTECTION MOSFET ENABLE CONTROL

The MP3718 can leave safe mode and enter active mode via pin control or register control, though only one control method can be selected at a time. The FET_SRC register and can be set either use the register (ACTIVE_CTRL register) or pin (GPIO1/2) to control the MOSFETs.

There are two control strategies that can be selected via the FET_CFG register: simple or direct. For simple mode, only using the GPIO1 or ACTIVE_CTRL[0] to control both CHG and DSG MOSFET. For direct mode, using GPIO1 or ACTIVE_CTRL[0] to control the DSG MOSFET, and using GPIO2 or ACTIVE_CTRL[1] to control CHG MOSFET.

The MOSFET turn on signal can be configured as active level or active on rising edge via the FT_STATE_SEL register. If it is set as rising edge,



IC will return to safe mode once the fault with both FETs off has been cleared. Must pull down GPIO or set the ACTIVE_CTRL to 0 firstly and then pull up GPIO or set ACTIVE_CTRL to 1 to turn on the MOSFET. If it is set as the level, and the MOSFET control signal is still on, IC will return to active with MOSFET on once the fault has been cleared. If FT_STATE_SEL = 0 and only one FET turn off in fault mode, when all fault cleared, IC goes to active mode with only one FET turn on.

The DSG/CHG driver fault recovery, DOC1/DOC2/COC fault recovery and fault exit by CHG or DSG current force the MOSFET to be turned on again even the FT_STATE_SEL=0, and the MOSFET will keep turn on if the fault is cleared.

CHG and DSG MOSFET Structure

Depending on the charging and discharging ports respectively, the MP3718 provides two high side MOSFET architectures.

One port structure: charging port and the discharge port is the same port. Connect the PACKP to charging and discharging ports, and connect the CHGP and CDSG to VSS.

Separate port structure: the charging port and the discharge port are the different port. Connect the PACKP to the discharging port and connect the CHGP to the charging port. The DSG is the discharging MOSFET driver in discharging port, the CDSG is the discharging MOSFET driver in charging port. During charging port conduction, the discharging-related fault can still actively control the CDSG driver to prevent accidental over discharging due to charger damaged. Note that the CDSG and DSG can't turn on together.

The switch operates from discharging port to charging port by host MCU is following:

1. Turn off the DSG FET
2. Set the CDSG_DRIVER_CTRL to 1.
3. Delay 500us.
4. turn on the CDSG FET.

MOSFET Driving Ability

The HS-FET driver can drive multiple DSG/CHG FETs in parallel. The CP capacitor value must be increased, depending on the number of parallel MOSFETs in the application.

Considering that when the MID voltage is less than 18V, the CP will operate at a lower voltage.

So, for the application of low series battery packs, it is recommended to set the driver voltage low via FET_DRV_LVL. If it is equal to 10h or 11h, the CP voltage would regulate in lower mode, which is about 50% of the normal CP voltage.

The driver also has a strong turn off capability, which is equivalent to connecting a resistor (RCHG_OFF or RDSG_OFF in the electrical characteristics table) between the gate and source of the MOSFETs during turn-off.

There is a driver timer out to monitor the start of CHG and DSG MOSFET and it is paused during the soft start. Once the timer out with the MOSFET turn on fail, it will report the driver fault.

Short-Circuit Detection prior to DSG Start

MP3718 provides the short-circuit detection option prior to the DSG MOSFET start to make sure the DSG MOSFET starts up in a safe condition.

For the small capacitance on PACK+, set PACKP_PRE_CHECK_ENABLE = 1, FORCE_DSG_SOFTSTART_BELOW_2V = 0 and PACKP_2V_TIMER_ENABLE = 0, a 250uA current source to attempt to charge PACKP above 2V in the required time before the DSG start. If the PACKP can't be charged above 2V in the required time, the DSG won't start, and report as short-circuit fault and discharge over current fault.

For the large capacitance on PACK+, set PACKP_PRE_CHECK_ENABLE = 0, FORCE_DSG_SOFTSTART_BELOW_2V = 1 and PACKP_2V_TIMER_ENABLE = 1, the device will use the DSG FET and softstart control as a stronger current source to charge the PACKP above 2V. If the PACKP can't be charged above 2V in the required time, the DSG will turn off, and report as DSG driver fault, short-circuit fault and discharge over current fault.

DSG MOSFET Soft Start (DSS)

The DSG MOSFET driver supports limited current loop regulation when DSG MOSFET soft start is enabled via DSG_SOFTON_EN = 1. The limited current loop is realized by monitoring the differential voltages of SRP and SRN to control the DSG output. There are 4-Stage limited current depending on the voltage difference between VTOP and PACKP. Table 1 shows the constant current at different PACKP voltage. VSS_REF is set by DSGSS_CURRENT, which is equal to DSGSS_CURRENT. DSGSS_CURRENT has

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

two ranges set by DSGSS_CURRENT_RNG, one is from 50 μ V to 800 μ V with 50 μ V LSB, the other is from 250 μ V to 4000 μ V with 250 μ V LSB.

Table 1: 4-level constant current soft start

V _{PACKP} /V _{TOP} Ratio	Soft-Start Reference
> 87%	$\leq 8 * VSS_REF$
75%-87%	$\leq 4 * VSS_REF$
50%-75%	$\leq 2 * VSS_REF$
< 50%	$\leq VSS_REF$

During the DSG soft start process, the current output capability of the DSG pin is limited, which the maximum output capability is (VSS_REF-VSRP_SRN)*gm. VSRP_SRN is the voltage between SRP and SRN of AFE. gm is a voltage scaling factor and typical value is 0.02. For the application circuit design, it is necessary to consider that the leakage current of the peripheral circuits of DSG must be less than the maximum output capability of the DSG pin, otherwise the DSG will not be able to start ⁽¹¹⁾.

(11) Please contact MPS FAE for the related application note.

CHG MOSFET Soft Start (CSS)

The CHG MOSFET driver supports a configurable soft start (CSS) using the CHG_SOFTON_PUP register to set the CHG driver pull-up current values (ranging from 3 μ A to 10 μ A). When CHG_SOFTON_EN is enabled, the MOSFET drive circuit automatically uses CHG SS to control the output current of the CHG pin, so that VGS rises slowly. During CSS, the threshold of the charge overcurrent protection is CHG_SOFTON_OC_LIM, and which deglitch time is limited to 10ms. Note that when set CHG OC deglitch (OC_CHG_DG) less than 10ms, don't enable CHG_SOFTON_OC_EN. Once the voltage between SRP and SRN exceeds CHG_SOFTON_OC_LIM and keep 10ms, the CHG MOSFET will turn off and report the CHG OC fault.

MOSFET Start Retry

Once the MOSFET driver fault report, the MOSFET driver can be auto-recovery turn on by enable the recovery function.

After the MOSFET cool time, the MOSFET will retry to turn on, which process is the same as the normal process, which means if the soft start is enabled, it will still enter the soft start sequence.

The retry times can be set to 1 time, 2 times, 3times or always retry.

MOSFET dead detection

The DSG MOSFET dead detection is enabled by dsgefet_dead_en=1. The DSG MOSFET dead can be detected in the following conditions:

- When the DSG FET is turn off, and the dsgefet_drv = 0, if the discharge current is higher than dsgefet_dead_curr_lim and keep for dsgefet_dead_dgl time
- When the DSG FET is turn off, but the dsgefet_drv = 1, and the dsgefet_drv keep report to 1 until the dsgefet_dead_dgl timer out.

The CHG MOSFET dead detection is enabled by chgefet_dead_en=1. The CHG MOSFET dead can be detected in the following conditions:

- When the CHG FET is turn off, and the chgefet_drv = 0, if the charge current is higher than chgefet_dead_oc and keep for chgefet_dead_dg time
- When the CHG FET is turn off, but the chgefet_drv = 1, and the chgefet_drv keep report to 1 until the chgefet_dead_dgl timer out.

FUNCTIONAL PINS**GPIO Pins**

The GPIO pins can be directly controlled by the MCU through registers, or they can be assigned to special functions. The GPIO pins have the following functions:

- Push-pull output: Set GPIOx_IO = 0, the GPIOx_O is the control bit for high or low output. The x suffix after GPIO, with values 1/2/3/4, represents the 4 GPIO pins respectively
- Pull-up capability: Connect a 20k Ω pull-up resistor to 3V3 or 5V, the register GPIOx_IO = 1 and GPIOx_PUP = 1
- Digital input: Set GPIOx_IO = 1 and GPIOx_TYPE = 0, the GPIOx is the state of this GPIO.
- Analog input: Set GPIOx_IO = 1 and GPIOx_TYPE = 1, and GPIOx_NTCy = 0, this GPIO pin set as a buffered ADC input with a 0V to 3.3V range.



- NTC analog input: Set GPIOx_IO = 1 and GPIOx_TYPE = 1, and GPIOx_NTCy = 1, this GPIO pin set as NTC ADC input.

If GPIO1 set as push-pull output with GPIO1_FSEL = 1 and SBD_GPIO = 1 (refer to FT_REC_CFG1 (64h)), the GPIO1 will output high when cell UV detected. When GPIO1_FSEL = 1 and SBD_GPIO = 0, the GPIO1 as the DSG FET driver state indicator.

If GPIO2 set as push-pull output with GPIO2_FSEL = 1 and SBD_GPIO = 1, the GPIO2 will output high when cell OV or open wire detected. When GPIO2_FSEL = 1 and SBD_GPIO = 0, the GPIO2 as the CHG FET driver state indicator.

In addition to the GPIO functions, the GPIO1 and GPIO2 pins can also be configured as the MOSFET control pins (input).

If GPIO3 set as push-pull output with GPIO3_FSEL = 1, the GPIO3 will output high when any fault detected. The output is latched by the xx_FAULT_STS = 1, and can be cleared by writing xx_FAULT_CLEAR to 1.

In addition to the GPIO capability, the GPIO4 digital input can be configured to initiate automatic cell-balancing.

FUSE Pin

The FUSE pin can be used the secondary fuse sense and driver. When the MP3718 detects a secondary failure, FUSE pin will output to burn down the fuse to protect the battery pack.

Once FUSE is triggered, after a configurable timer out, IC will go to sleep mode and been locked in sleep mode until MID or REGIN voltage drop below the UVLO and repower.

WDT Pin

The WDT pin is controlled by the watchdog timer (WDT), and provides the following functions:

- Toggles 25ms high pulse when a watchdog event occurs to reset an external IC (e.g. the host MCU).
- Triggers a self-reset to the IC by bringing the device back to its default values.
- Can be pulled high externally to reset the device.

XALERT pin

The XALERT pin can be configured to be an active high (3.3V) or active low interrupt pin. When set to active low, XALERT goes low if there is a pending interrupt. When set to active high, this pin goes high if there is a pending interrupt.

Hardware monitor and other events trigger this pin, but certain bits must be enabled to ensure that only the relevant sources generate an interrupt. The output is latched by the xx_INT_STATUS = 1, and can be cleared by writing xx_INT_CLEAR to 1.

PLUGIN AND REMOVAL DETECTION

There are two analog comparators, one comparing the voltages between PACKP and TOP, the other comparing the voltages between CHGP and TOP. When CDSG_DRIVER_CTRL = 0, the VTOP vs. PACKP comparator is selected. When CDSG_DRIVER_CTRL = 1, the VTOP vs. CHGP comparator is selected.

The user can get the port status by querying the register VTOP_COMP_STS. It can show V_{PACKP(CHGP)} higher or lower or equal to V_{TOP}, which is used to detect the state of charger and load.

MP3718 also provides the below port detection.

Charger insertion detection

Set CHG_DET_EN = 1, the device will turn on the CHGP voltage comparator and wait for CHGP > VTOP. When CHGP > VTOP, the MP3718 will consider that a charger is connected, and CHG_DET_DONE_STS = 1, CHG_IN = 1.

If CONN_DET_INT_EN = 1, the charger insert event can be output to XALERT pin.

Charger removal detection

Once the charging-related fault happens, etc. the cell OV, the system may into current save mode until the charger removal. So a charger removal detection is need for the system.

Once the charger is already present, set VTOP_CMP_EN = 1 to enable the PACKP voltage comparator. If PACKP_V_INT_EN = 1, when the PACKP comparator changes state it will flag an interrupt. And then check the CHG_IN or VTOP_COMP_STS to confirm if the charger removed.

Consider that when the CHG FET is off, some chargers' output is a periodically battery pack full



voltage pulse, this device provides an additional 1s deglitch time for charger removal detection.

Load insert detection

Set LD_PLUGIN_DET_EN=1, PACKP will output 250uA try to charge the PACK+, try to charge it to close the TOP. During this time, the LD_SETTLING=1.

If the PACK+ can't be charged to the voltage equal to VTOP in T_PLUGIN_PRECHARGE_MAX time, it considers that a load is already present.

If the PACK+ is charged to VTOP, the LD_SETTLING =0 and LDDDET_ENGAGED =1. It will wait the PACKP < VTOP state to trigger the load insert event. If CONN_DET_INT_EN= 1, the load insert event can be output to XALERT pin.

For the large capacitance on PACK+, suggest soft start the DSG FET firstly, and then set LD_PLUGIN_DET_EN =1 and turn off the DSG FETs.

Load removal detection

Once the discharging-related fault happens, such as DSG SC, the system may check at intervals to see if the load has been removed.

For the small capacitance on PACK+, set LD_RML_DET_EN = 1 and PACKP_2V_TIMER_ENABLE = 0, a 250uA current source to attempt to charge PACKP above 2V in the required time. If the PACKP can be charged above 2V in the required time. The device considers the load is removed.

For the large capacitance on PACK+, set LD_RML_DET_EN = 1, PACKP_PRE_CHECK_ENABLE = 0, and PACKP_2V_TIMER_ENABLE = 1, the device will use the DSG FET and softstart control as a stronger current source to charge the PACKP above 2V. Both DSG FET and 250uA current source will turn on until PACKP is above 2V. Once PACKP is above 2V, the 250uA current source will hold on and the DSG FET will turn off. If PACKP remains above 2V for the required time the load is considered removed.

If CONN_DET_INT_EN= 1, the load removal detected done event can be output to XALERT pin. The LOAD_CHARGER_STATUS should be check to whether the load has been removed.

HARDWARE MONITOR

The hardware monitor can automatically monitor the voltage, current, and temperature of the cell pack without the intervention of a host MCU. It can also monitor the power status of the device itself.

Voltage Monitor

Voltage monitor is implemented by an autonomous hardware state machine that periodically schedules the conversion for the relevant channels and compares the results with their thresholds. The SAFE_MONITOR_EN should be set to 1 to enable the voltage monitor in safe mode.

If less than 24 cells are used, only the cells enabled by the CELL_S_CTRL register are monitored. The intermediate cell channels can be configured to skip the hardware protection scan via CELL_x_MONITOR_SKIP when the intermediate cell channel is connected to a bus bar.

Cell Under-Voltage (UV)

Set CELL_UV_EN =1 to enable cell UV comparisons. The threshold can be set by CELL_UV from 0mV to 4980.5mV with 19.53mV steps. The deglitch time can be set by CELL_UV_DG from 1 to 16 times with 1 time steps. The cell UV reports when the cell voltage is below CELL_UV hold CELL_UV_DG number of consecutive slots-A.

Cell UV status can be output to GPIO1 pin when GPIO1_IO = 0, GPIO1_FSEL = 1 and SBD_GPIO = 1.

Cell UV status can trigger the DSG FET or both CHG and DSG FETs off when CELL_UV_FAULT_EN = 1.

Cell UV status can be output to XALERT pin when CELL_UV_INT_EN = 1, and INT_ALERT_CTRL = 1.

The CELL_x_UV_RT_STS registers can be queried to determine which cell is in UV state. Note that this status register is the real-time status.

Cell Over-Voltage (OV)

Set CELL_OV_EN =1 to enable cell OV comparisons. The threshold can be set by CELL_OV from 0mV to 4995.1mV with 4.88mV steps. The deglitch time can be set by CELL_OV_DG from 1 to 16 times with 1 time steps.



The cell OV reports when the cell voltage is above CELL_OV hold CELL_OV_DG number of consecutive slots-A.

Cell OV status can be output to GPIO2 pin when GPIO2_IO = 0, GPIO2_FSEL = 1 and SBD_GPIO = 1.

Cell OV status can trigger the CHG FET or both CHG and DSG FETs off when CELL_OV_FAULT_EN = 1.

Cell OV status can be output to XALERT pin when CELL_OV_INT_EN = 1, and INT_ALERT_CTRL = 1.

The CELL_x_OV_RT_STS registers can be queried to determine which cell is in OV state. Note that this status register is the real-time status.

Cell Mismatched (MSMT)

Set CELL_MSMT_EN = 1 to enable cell mismatch detection. The mismatch threshold can be set by CELL_MSMT_TH from 0mV to 1210.9mV with 39.06mV steps. The minimum voltage threshold can be set by CELL_MSMT_MIN from 2500mV to 4960.9mV with 39.06mV steps. The deglitch time can be set by CELL_MSMT_DG from 1 to 16 times with 1 time steps. Cell MSMT reports when the cell voltage is higher than CELL_MSMT_MIN and the difference to the minimum cell voltage exceeds CELL_MSMT_TH hold CELL_MSMT_DG number of consecutive slots-A. The minimum cell voltage is reported in the CELL_MSMT_LOWER register. Cell MSMT status can trigger both CHG and DSG FETs off when CELL_MSMT_FUSE_EN = 1.

Cell MSMT status can be output to XALERT pin when CELL_MISMATCH_INT_EN = 1, and INT_ALERT_CTRL = 1.

Cell dead status can be output to FUSE pin to burn down the external fuse when CELL_MSMT_FUSE_EN = 1.

The CELL_x_MSMT_RT_STS registers can be queried to determine which cell is mismatched. Note that this status register is the real-time status.

Once an MSMT cell is detected, the CELL_MSMT_LOG_STS flag is set to 1. This flag remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the CELL_MSMT_DET_CLEAR command, power reset or WDT reset.

Cell Dead

Set CELL_DEAD_EN = 1 to enable cell dead comparisons. The threshold can be set by CELL_DEAD_LIMIT from 0mV to 2480.5mV with 19.53mV steps. The deglitch time can be set by CELL_DEAD_DG from 1 to 16 times with 1 time steps. Cell dead reports when the cell voltage is below CELL_DEAD_LIMIT hold CELL_DEAD_DG number of consecutive slots-A.

Cell dead status can trigger both CHG and DSG FETs off when CELL_DEAD_FAULT_EN = 1.

Cell dead status can be output to XALERT pin when CELL_DEAD_INT_EN = 1, and INT_ALERT_CTRL = 1.

Cell dead status can be output to FUSE pin when CELL_DEAD_FUSE_EN = 1.

The CELL_x_DEAD_RT_STS registers can be queried to determine which cell is in dead state. Note that this status register is the real-time status.

Once a dead cell is detected, the CELL_DEAD_LOG_STS flag is set to 1. This flag remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the CELL_DEAD_DET_CLEAR command, power reset or WDT reset.

Cells Severe Over-Voltage (SOV)

Set CELL_SOV_EN = 1 to enable cell SOV comparisons. The threshold can be set by CELL_SOV from 0mV to 4995.1mV with 4.88mV steps. The deglitch time can be set by CELL_SOV_DG from 1 to 16 times with 1 time steps. The cell SOV reports when the cell voltage is above CELL_SOV hold CELL_SOV_DG number of consecutive slots-A.

Cell SOV status can trigger both CHG and DSG FETs off when CELL_SOV_FAULT_EN = 1.

Cell SOV status can be output to XALERT pin when CELL_SOV_INT_EN = 1, and INT_ALERT_CTRL = 1.

Cell SOV status can be output to FUSE pin when CELL_SOV_FUSE_EN = 1.

The CELL_x_SOV_RT_STS registers can be queried to determine which cell is in SOV state. Note that this status register is the real-time status.

Once a SOV cell is detected, the CELL_SOV_LOG_STS flag is set to 1. This flag

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the CELL_SOV_DET_CLEAR command, power reset or WDT reset.

Pack UV

Set VTOP_UV_EN =1 to enable battery pack UV comparisons. The threshold can be set by VTOP_UV_LIMIT from 0mV to 119971.215mV with 29.297mV steps. The deglitch time can be set by VTOP_UV_DG from 1 to 16 times with 1 time steps. The PACK UV reports when the VTOP to AGND voltage is below VTOP_UV_LIMIT hold VTOP_UV_DG number of consecutive slots-A.

PACK UV status can trigger DSG FET or both CHG and DSG FETs off when VTOP_UV_FAULT_EN_CTRL = 1.

PACK UV status can be output to XALERT pin when VTOP_UV_INT_EN = 1, and INT_ALERT_CTRL = 1.

The VTOP_UV_RT_STS registers can be queried to determine if the battery pack is in UV state. Note that this status register is the real-time status.

Pack OV

Set VTOP_OV_EN =1 to enable battery pack OV comparisons. The threshold can be set by VTOP_OV_LIMIT from 0mV to 119971.215mV with 29.297mV steps. The deglitch time can be set by VTOP_OV_DG from 1 to 16 times with 1 time steps. The PACK OV reports when the VTOP to AGND voltage is above VTOP_OV_LIMIT hold VTOP_OV_DG number of consecutive slots-A.

PACK OV status can trigger CHG FET or both CHG and DSG FETs off when VTOP_OV_FAULT_EN_CTRL = 1.

PACK OV status can be output to XALERT pin when VTOP_OV_INT_EN = 1, and INT_ALERT_CTRL = 1.

The VTOP_OV_RT_STS registers can be queried to determine if the battery pack is in OV state. Note that this status register is the real-time status.

Pack SUV

Set VTOP_SUV_EN =1 to enable battery pack SUV comparisons. The threshold can be set by VTOP_SUV_LIMIT from 0mV to 79980.5mV with 29.297mV steps. The deglitch time can be set by VTOP_SUV_DG from 1 to 16 times with 1 time steps. The PACK SUV reports when the VTOP to

AGND voltage is below VTOP_SUV_LIMIT hold VTOP_SUV_DG number of consecutive slots-A.

PACK SUV status can trigger both CHG and DSG FETs off when VTOP_SUV_FAULT_EN = 1.

PACK SUV status can be output to XALERT pin when VTOP_SUV_INT_EN = 1, and INT_ALERT_CTRL = 1.

PACK SUV status can be output to FUSE pin when VTOP_SUV_FUSE_EN = 1.

The VTOP_SUV_RT_STS registers can be queried to determine if the battery pack is in SUV state. Note that this status register is the real-time status.

Once the pack SUV is detected, the VTOP_SUV_LOG_STS flag is set to 1. This flag remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the VTOP_SUV_DET_CLEAR command, power reset or WDT reset.

Pack SOV

Set VTOP_SOV_EN =1 to enable battery pack SOV comparisons. The threshold can be set by VTOP_SOV_LIMIT from 0mV to 119971.215mV with 29.297mV steps. The deglitch time can be set by VTOP_SOV_DG from 1 to 16 times with 1 time steps. The PACK SOV reports when the VTOP to AGND voltage is above VTOP_SOV_LIMIT hold VTOP_SOV_DG number of consecutive slots-A.

PACK SOV status can trigger both CHG and DSG FETs off when VTOP_SOV_FAULT_EN = 1.

PACK SOV status can be output to XALERT pin when VTOP_SOV_INT_EN = 1, and INT_ALERT_CTRL = 1.

PACK SOV status can be output to FUSE pin when VTOP_SOV_FUSE_EN = 1.

The VTOP_SOV_RT_STS registers can be queried to determine if the battery pack is in SOV state. Note that this status register is the real-time status.

Once the pack SOV is detected, the VTOP_SOV_LOG_STS flag is set to 1. This flag remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the VTOP_SOV_DET_CLEAR command or power reset or WDT reset.

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE****Temperature Monitor**

The cell temperature is monitored through the negative temperature coefficient (NTC) resistor close to the cell. The PCB temperature is monitored through the NTC resistor on PCB. The chip die temperature is monitored through internal analog signal.

An autonomous hardware state machine that periodically schedules the conversion for the temperature channels and compares the results with their thresholds implements temperature monitor.

The SAFE_MONITOR_EN should be set to 1 to enable the temperature monitor in safe mode.

There are four thresholds for the cell temperature.

Cell Over-Temperature (OT)

Cell over-temperature (OT) is set by NTC_CELL_OT from 0% to 99.906% with 0.09766% steps. Percentage means the voltage across the NTC resistor as a proportion of the NTCB voltage. The cell OT occurs when NTC ratio is less than NTC_CELL_OT hold deglitch number. The cell OT status can be output to XALERT pin when NTC_OT_INT_EN = 1, and INT_ALERT_CTRL = 1.

The cell OT status with charging current can trigger the CHG FET or both CHG and DSG FETs off when NTC_CELL_HOT_CHG_FAULT_EN = 1.

Cell Under-Temperature (UT)

Cell under-temperature (UT) is set by NTC_CELL_UT from 0% to 99.906% with 0.09766% steps. The cell UT occurs when NTC ratio is greater than NTC_CELL_UT hold deglitch number.

The cell UT status can be output to XALERT pin when NTC_UT_INT_EN = 1, and INT_ALERT_CTRL = 1.

The cell UT status with charging current can trigger the CHG FET or both CHG and DSG FETs off when NTC_CELL_COLD_CHG_FAULT_EN = 1.

Cell Severe Over-Temperature (SOT)

Cell severe over-temperature (SOT) is set by NTC_CELL_SOT from 0% to 99.906% with 0.09766% steps. The cell SOT occurs when NTC ratio is less than NTC_CELL_SOT hold deglitch number.

The cell SOT status can be output to XALERT pin when NTC_SOT_INT_EN = 1, and INT_ALERT_CTRL = 1.

The cell SOT status can trigger the DSG FET or both CHG and DSG FETs off when NTC_CELL_HOT_DSG_FAULT_EN = 1.

Cell Severe Under-Temperature (SUT)

Cell severe under-temperature (SUT) is set by NTC_CELL_SUT from 0% to 99.906% with 0.09766% steps. The cell SUT occurs when NTC ratio is greater than NTC_CELL_SUT hold deglitch number.

The cell SUT status can be output to XALERT pin when NTC_SUT_INT_EN = 1, and INT_ALERT_CTRL = 1.

The cell SUT status can trigger the DSG FET or both CHG and DSG FETs off when NTC_CELL_COLD_DSG_FAULT_EN = 1.

Cell Temperature Monitor

All 10 NTC channels can be set to cell temperature. For the NTC1/2/3/4, set NTCx_EN = 1 and NTCx_TYPE_SEL = 0 to enable the cell temperature monitor. For the NTC5/6/7/8/9/10, set NTCx_CELL_EN = 1 to enable the cell temperature, and meanwhile this pin should set as NTC input mode (For the details refer to GPIO pins section).

The cell UT and OT share the same deglitch time which can be set by NTC_CELL_UTOT_DG from 1 to 8 times with 1-time steps. The cell SUT and SOT share the same deglitch time which can be set by NTC_CELL_SUTOT_DG from 1 to 8 times with 1-time steps.

The NTCx_CELL_xx_RT_STS can be queried to determine if any cell temperature exceeds the cell operating temperature range. Note that this status register is the read-time status.

Generally, the cell UT corresponds to charging under temperature, the cell OT corresponds to charging over temperature, the cell SUT corresponds to discharging under temperature, the cell SOT corresponds to discharging over temperature. Therefore, $NTC_CELL_SOT < NTC_CELL_OT < NTC_CELL_UT < NTC_CELL_SUT$. If the temperature range of charging is wider than that of discharging, set NTC_CHG_COLDER and NTC_CHG_HOTTER to 1.



PCB OT

The NTC1/2/3/4, can be set to PCB OT comparisons by `NTCx_EN = 1` and `NTCx_TYPE_SEL = 1`. The threshold can be set by `PCB_MNTR_HOT` from 0% to 99.906% with 0.09766% steps. The deglitch time can be set by `PCB_MNTR_DG` from 1 to 8 times with 1 time steps. The PCB OT occurs when NTC ratio is less than `PCB_MNTR_HOT` hold `pcb_mntr_dg` number of consecutive slots-A.

The PCB OT status can be output to XALERT pin when `INT_ALERT_CTRL = 1`, and `PCB_MNTR_HOT_INT_EN = 1`.

The PCB OT status can trigger both CHG and DSG FETs off when `PCB_MNTR_FAULT_EN = 1`.

The `NTCx_PCB_MNTR_HOT_STS` registers can be queried to determine if the PCB is in OT state. Note that this status register is the read-time status.

Die Over-Temperature (OT)

Set `DIE_TEMP_DIG_EN = 1` to enable Die temperature comparisons. The threshold can be set by `DIE_TEMP_HOT` from 70°C and 120°C with 0.474°C steps. The die OT reports when the die temperature exceeds the `DIE_TEMP_HOT` in a slot-A.

Die OT status can trigger both CHG and DSG FETs off when `DIE_TEMP_DIG_FAULT_EN = 1`.

Die OT status can be output to XALERT pin when `VTOP_SUV_INT_EN = 1`, and `INT_ALERT_CTRL = 1`.

PACK SUV status can be output to FUSE pin when `VTOP_SUV_FUSE_EN = 1`, and `DIE_TEMP_DIG_FAULT_EN = 1`.

The `DIE_TEMP_HOT_RT_STS` registers can be queried to determine if the die is in OT state. Note that this status register is the read-time status.

Additionally, there is an analog comparator for the die temperature, which forces the device to shut down at 140°C. The digital die temperature threshold is typically used as an early warning, since the analog threshold forces the IC to shut down. Until the analog OT condition is resolved, the IC remains in a forced shutdown state, and the communication interface is unavailable.

Current monitor

There are three analog current comparators for pack current monitoring, the standby current comparator, the overcurrent comparator, and the short-circuit current comparator. The `SAFE_SCOC_EN` should be set to 1 to enable the current monitor in safe mode. Unlike voltage and temperature monitors, current monitoring runs all the time.

There is one digital current comparator for small standby current detection.

Standby Current Monitor

The standby current threshold can be set by the `STBY_CUR_TH` register, there are four levels of 250uV, 375uV, 500uV and 625uV can be selected. When the voltage generated by the charging current across the sense resistor is greater than `STBY_CUR_TH`, the device reports the current state is charging (`PACK_CURRENT_STATUS = 2'b11`). And when the voltage generated by the discharging current across the sense resistor is greater than `STBY_CUR_TH`, it reports the current state is discharge (`PACK_CURRENT_STATUS = 2'b01`).

The current state change event can be output to XALERT pin when `pack_current_int_en = 1`, and `INT_ALERT_CTRL = 1`.

Digital Current Monitor

When `DSTBC_EN=1`, and the hardware monitoring is enabled (in active/standby/fault mode or in safe mode with `SAFE_MONITOR_EN = 1`), the current ADC will run one time during each hardware slot, the value is compared to the `DSTBC_TH` threshold. If current value > `DSTBC_TH`, the `ICHG_GT_STBY = 1` and `IDSG_GT_STBY=0`. If current value < -`DSTBC_TH`, the `IDSG_GT_STBY = 1` and `ICHG_GT_STBY = 0`. If current value < `DSTBC_TH` and current value > -`DSTBC_TH`, the `ICHG_GT_STBY = 0` and `IDSG_GT_STBY=0`. If `DSTBC_EN=0`, the `ICHG_GT_STBY = 0` and `IDSG_GT_STBY=0`. If `PACK_CURRENT_INT_EN = 1`, and `ICHG_GT_STBY` from 0 to 1 or `IDSG_GT_STBY` from 0 to 1, the `PACK_CURRENT_INT_STS` should be report to 1.

Discharge Over-Current (DOC) Monitor

There are two level DOC monitor. The OC1 should be set to low threshold with long deglitch time. The



OC2 should be set to high threshold with short deglitch time. Two possible range (RNG) levels are available for each threshold.

Set OC1_DCHG_EN_CTRL = 1 to enable DOC1 monitor. The threshold can be set by OC1_DCHG_LIM from 1.2mV to 38.4mV with 1.2mV steps if OC1_DCHG_RNG = 0, or from 3.6mV to 115.2mV with 3.6mV if OC1_DCHG_RNG = 1. The deglitch time can be set by OC1_DCHG_DGL from 0ms to 315ms with 5ms steps if OC1_DCHG_DGL_RNG = 0, or from 0ms to 2520ms with 40ms steps if OC1_DCHG_DGL_RNG = 1. When the deglitch time set to 0, the actual deglitch time is about 60us. The DOC1 event reports when the voltage generated by the discharging current across the sense resistor is greater than OC1_DCHG_LIM hold OC1_DCHG_DGL time.

Set OC2_DCHG_EN_CTRL = 1 to enable DOC2 monitor. The threshold can be set by OC2_DCHG_LIM from 1.2mV to 38.4mV with 1.2mV steps if OC2_DCHG_RNG = 0, or from 3.6mV to 115.2mV with 3.6mV if OC2_DCHG_RNG = 1. The deglitch time can be set by OC2_DCHG_DGL from 0ms to 315ms with 5ms steps if OC2_DCHG_DGL_RNG = 0, or from 0ms to 2520ms with 40ms steps if OC2_DCHG_DGL_RNG = 1. When the deglitch time set to 0, the actual deglitch time is about 60us. The DOC2 event reports when the voltage generated by the discharging current across the sense resistor is greater than OC2_DCHG_LIM hold OC2_DCHG_DGL time.

The DOC1 event can be output to XALERT pin when OC1_DCHG_INT_EN = 1, and INT_ALERT_CTRL = 1.

The DOC1 event can trigger the DSG FET or both CHG and DSG FET off when OC1_DCHG_FAULT_EN = 1.

The DOC2 event can be output to XALERT pin when OC2_DCHG_INT_EN = 1, and INT_ALERT_CTRL = 1.

The DOC2 event can trigger the DSG FET or both CHG and DSG FET off when OC2_DCHG_FAULT_EN = 1.

The OC1_DCHG_RT_STS and OC2_DCHG_RT_STS can be queried to determine if any discharge over current event

happened. Note that these status registers are the read-time status.

Charge Over-Current (COC) Monitor

Set OC_CHG_EN_CTRL = 1 to enable COC monitor. The threshold can be set by OC_CHG_LIM from 1.2mV to 38.4mV with 1.2mV steps if OC_CHG_RNG = 0, or from 3.6mV to 115.2mV with 3.6mV if OC_CHG_RNG = 1. The deglitch time can be set by OC_CHG_DG from 0ms to 315ms with 5ms steps if OC_CHG_DGL_RNG = 0, or from 0ms to 2520ms with 40ms steps if OC_CHG_DGL_RNG = 1. When the deglitch time set to 0, the actual deglitch time is about 60us. The COC event reports when the voltage generated by the charging current across the sense resistor is greater than OC_CHG_LIM hold OC_CHG_DG time.

The COC event can be output to XALERT pin when OC_CHG_INT_EN = 1, and INT_ALERT_CTRL = 1.

The COC event can trigger the CHG FET or both CHG and DSG FET off when OC_CHG_FAULT_EN = 1.

The OC_CHG_RT_STS register can be queried to determine if charge over current event happened. Note that this status register is the read-time status.

DSG Short-Circuit Current (DSC) Monitor

The DSC monitoring is performed on the short-circuit current analog comparator. Set SC_DCHG_EN_CTRL = 1 to enable DSC monitor. The threshold can be set by SC_DCHG_LIM from 5.5mV to 176mV with 5.5mV steps if SC_DCHG_RNG = 0, or from 16.5mV to 528mV with 16.5mV if SC_DCHG_RNG = 1. The deglitch time can be set by SC_DCHG_DG from 0μs to 317.5μs with 2.5μs steps. When the deglitch time set to 0, there is no deglitch time, the detection time is about 4μs. The DSC event reports when the voltage generated by the discharging current across the sense resistor is greater than SC_DCHG_LIM hold SC_DCHG_DG time.

The DSC event can be output to XALERT pin when SHORT_CURR_INT_EN = 1, and INT_ALERT_CTRL = 1.

The DSC event can trigger the DSG FET or both CHG and DSG FET off when OC1_DCHG_FAULT_EN = 1.

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

The SC_DCHG_RT_STS register can be queried to determine if the short-circuit event happened. Note that this status register is the read-time status.

Diagnostics

The device has the following self-diagnostic features, which can enhance the robustness of the system. To enable the LDO monitoring and ADC self-diagnostics, the SAFE_MONITOR_EN should be set to 1 in safe mode.

LDO (VDD, 5V, 3.3V) Rail Monitoring

An autonomous hardware state machine that periodically schedules the conversion for LDO (VDD, 5V and 3V3) channels and compares the results with their thresholds implements the LDO rail monitor.

The VDD UV event can be output to XALERT pin when VDD_INT_EN = 1, and INT_ALERT_CTRL = 1.

Set V5_EN = 1 to enable the 5V monitoring. The threshold can be set by V5_UV_LIMIT with 12.89mV steps and is recommended to be set 6587.11mV. The 5V UV reports when the 5V voltage less than V5_UV_LIMIT in a slot-A. The 5V UV event can be output to XALERT pin when V5_INT_EN = 1, and INT_ALERT_CTRL = 1. The real time status V5_UV_RT_STS can be queried to determine if the 5V UV event happened.

Set V3V3_EN = 1 to enable the 3V3 monitoring. The threshold can be set by V3V3_UV_LIMIT with 12.89mV steps and is recommended to be set 3287.11mV. The 3V3 UV reports when the 3V3 voltage less than V3V3_UV_LIMIT in a slot-A. The 3V3 UV event can be output to XALERT pin when V3V3_INT_EN = 1, and INT_ALERT_CTRL = 1. The real time status V3V3_UV_RT_STS can be queried to determine if the 3V3 UV event happened.

3V3, 5V, and VDD also provide the analog comparators monitor, which the threshold is 90% of normal voltage.

3V3, 5V, and VDD UV status can trigger both CHG and DSG FETs off when LDO_FAULT_EN = 1.

3V3, 5V, and VDD UV status can be output to FUSE pin when LDO_UV_FUSE_EN = 1.

Once the 3V3, 5V, or VDD UV is detected, the LDO_UV_LOG_STS flag is set to 1. This flag

remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the LDO_UV_DET_CLEAR command, power reset or WDT reset.

ADC Self-Diagnostics

The device will alternately ADC convert two known voltage and ensures the conversion results are switch alternately. One is 1.2V LDO, and the other is AGND. Both ADC results should be in a predefined windows that accounts for tolerance and temperature shifts. The current slot-A monitors the internal 1.2V LDO, the next slot-A monitors the AGND and so on alternately.

Set ADC_SELF_TEST_EN = 1 to enable this monitor. If the results do not match the expected, the ADC fail event detected. This ADC fail event also cover the MUX fail issue.

The ADC fail event can be output to XALERT pin when ADC_SELF_TEST_EN = 1, and INT_ALERT_CTRL = 1.

The ADC fail event can be output to GPIO3 pin when GPIO3_IO = 0, GPIO3_FSEL = 1 and ADC_FAIL_FAULT_EN = 1.

The ADC fail event can be output to FUSE pin when ADC_FAIL_FUSE_EN = 1.

The SELF_TEST_STS_UV and SELF_TEST_STS_OV registers can be queried to determine if the ADC fail event happened. Note that these status registers are the read-time status.

Once the ADC fail is detected, the ADC_FAIL_LOG_STS flag is set to 1. This flag remains at 1 even the device enters and exits shutdown mode. This flag can be cleared by the ADC_FAIL_DET_CLEAR command, power reset or WDT reset.

NVM Cyclic Redundancy Check (CRC)

The NVM CRC detects whether the value of the non-volatile memory in the device is normal or not, which is executed when the device exits shutdown mode. It also can be executed by the host communication command. If the NVM CRC fails, the NVM_CRC_OUTCOME reports the result. The NCM event can be output to XALERT or GPIO3.

Watchdog Timer

The watchdog timer monitors communication with the I²C or SPI interface. If there is no write to the



WDT_RST register within a certain interval, the timer is triggered. The watchdog follows has a bark-bite style.

The bark notifies the device when the MCU has failed to clear the watchdog timer. This results in a bark timeout, which can be configured to trigger the alert pin. The WDT bark counter settings are 25ms LSB, with up to 3.2 seconds maximum.

After a bite timeout, the device's registers reset to the default values. The WDT bite counter settings are 25ms LSB, with up to 3.2 seconds maximum.

Using the bidirectional, configurable WDT, it is possible to set up the IC to self-reset and/or to reset an external IC if its reset trigger is connected to the WDT pin.

Communication CRC

Typically, the CRC detects errors in exchanged data. The MP3718 supports CRC for exchanged data. When this functionality is enabled, the following is possible:

- Write a transaction to 1 register (2 bytes) to append a 1-byte CRC. A correct CRC is required for a successful write transaction.
- Read a transaction for 1 register (2 bytes) to append a 1-byte CRC.
- The read transaction length can be temporarily increased to 126 bytes using a dedicated register. The CRC is appended at the end of the transaction.

The communication CRC is implemented by the CRC-8 algorithm. The CRC can detect 1 incorrect bit. However, all CRC techniques encounter limitations when there are more incorrect bits.

If more than 1 bit is corrupted, the error detection depends on the specific pattern. Statistically, the detection can occur, but it is not guaranteed. In general, the ability to detect errors decreases when there is more data to check. For more details, see the I²C Read section and SPI Read section.

INTERRUPTS

Interrupt statuses are split between 3 register addresses (RD_INT0, RD_INT1 and RD_INT2). Each interrupt source has a matching enable bit and a dedicated clear bit to clear the interrupt. Table 2, Table 3 and Table 4 interrupts.



Table 2: RD_INT0 Interrupts

Condition	Flag	Description	Related Status Register(s)
Cell OV	CELL_OV_INT_STS	Unified flag for cell OV conditions. The related register reports each cell's OV flag, which is checked to determine which cell is generating the interrupt.	Flags in address, RD_CELL_OV
Cell UV	CELL_UV_INT_STS	Unified flag for cell UV conditions. The related register reports each cell's UV flag, which is checked to determine which cell is generating the interrupt.	Flags in address, RD_CELL_UV
VTOP pack OV	VTOP_OV_INT_STS	Dedicated flag for battery pack OV conditions detected on the TOP to AGND.	
VTOP pack UV	VTOP_UV_INT_STS	Dedicated flag for battery pack UV conditions detected on the TOP to AGND. This is unrelated to the VTOP UV analog threshold.	
DSG FET dead	DSGFET_DEAD_INT_STATUS	Dedicated flag for DSGFET dead conditions	
CHG FET dead	CHGFET_DEAD_INT_STATUS	Dedicated flag for DSGFET dead conditions	
NTC SUT	NTC_SUT_INT_STS	Unified flag for NTC cold conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTCX_CELL_COLD_DSG_STS,
NTC SOT	NTC_SOT_INT_STS	Unified flag for NTC hot conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTCX_CELL_SOT_RT_STS
Watchdog interrupt	WDT_INT_STS	Notification for either a bite or bark event from the communication watchdog timer.	WDT_BARKED, WDT_BITE
AFE mode change	AFE_MODE_CHANGE_INT_STS	The AFE has changed states (e.g. safe mode, active mode, fault mode, standby mode).	PWR_STATE
Scan complete	VSCAN_DONE_INT_STS	Notification that the high-resolution VADC scan has finished converting of all the channels on its list.	
CC done	CC_DONE_INT_STS	Flag reporting that a new Coulomb counting accumulation value has been updated to CC_ACC_LSB and CC_ACC_MSB.	
Plug-in detection	CONN_DET_INT_STS	The plug-in detection logic reports that either a device (capacitive load or charger) has been connected, or the detection cannot be successfully complete.	LD_IN, CHG_IN, CHGDET_FAIL, LDDET_FAIL
Pack current	PACK_CURRENT_INT_STS	Interrupt indicating a change in the battery pack current range (discharge, standby, or charge).	PACK_CURRENT_STATUS
COMM CRC fail	COMM_CRC_FAIL_INT_STS	Notification that a wrong CRC write even has been occurred. The previous write command is invalid.	COMM_CRC_BAD_WRITE

**Table 3: RD_INT1 Interrupts**

Name	Flag	Description	Related Status Register(s)
NVM CRC	NVM_CRC_EVENT_INT_STS	Reports whether an interrupt related to an NVM memory CRC fault has been detected.	
5V	V5_INT_STS	The 5V rail is below the defined UV threshold.	
3V3	V3V3_INT_STS	The 3.3V rail is below the defined UV threshold.	
VDD	VDD_INT_STS	The VDD rail is below the defined UV threshold.	
Open wire	OPEN_WIRE_INT_STS	Notification that there is one or more wire has been detected from no wire open status.	Flags in address, RD_OPENL
Dead cell	CELL_DEAD_INT_STS	The dead cell threshold has been reached.	Flags in address, RD_CELL_DEAD
Mismatched cells	CELL_MISMATCH_INT_STS	The voltage difference between cells is too great.	Flags in address, RD_CELL_MSMT
Die temperature	DIE_TEMP_INT_STS	The die temperature is too high; reported by the digital die temperature check.	
PCB temperature hot	PCB_MNTR_HOT_INT_STS	The NTC monitoring the PCB temperature is too hot. Unified flag for NTC hot conditions for all NTC channels configured in PCB mode. To identify which NTC is the source of the interrupt, the individual flags should be checked.	NTC1_PCB_MNTR_HOT_STS, NTC2_PCB_MNTR_HOT_STS, NTC3_PCB_MNTR_HOT_STS, NTC4_PCB_MNTR_HOT_STS
Self-test fail	SELF_TEST_INT_STS	The ADC is failing its self-diagnostic test and/or the conversion of this value is outside the specified boundaries.	SELF_TEST_STS_OV, SELF_TEST_STS_UV
Balancing complete	BAL_DONE_INT_STS	Balancing has been completed or prevented.	Flags in address, BAL_STS
PACKP voltage	PACKP_V_INT_STS	The voltage on PACKP/MID node has changed compared to VTOP.	PACKP_COMP_STS
FET driver	FET_DRIVER_INT_STS	Reports a MOSFET driver issue	CHG_DRIVER_FAULT_STS DSG_DRIVER_FAULT_STS
NTC UT	NTC_UT_INT_STS	Unified flag for NTC cold conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTCX_CELL_UT_RT_STS
NTC OT	NTC_OT_INT_STS	Unified flag for NTC hot conditions for all NTC channels configured in cell mode. To identify which NTC is the source of the interrupt, the additional individual flags should be checked.	NTCX_CELL_OT_RT_STS

**Table 4: RD_INT2 Interrupts**

Name	Flag	Description	Related Status Register(s)
Cell SOV	CELL_SOV_INT_STAT US	Unified flag for cell SOV conditions. The related register reports each cell's SOV flag, which is checked to determine which cell is generating the interrupt.	Flags in address, RD_CELL_SOV
VTOP pack SOV	VTOP_SOV_INT_STAT US	Dedicated flag for battery pack SOV conditions detected on the TOPS to AGND.	
VTOP pack SUV	VTOP_SUV_INT_STAT US	Dedicated flag for battery pack SUV conditions detected on the TOPS to AGND. This is unrelated to the VMID UVLO analog threshold.	
OC	OVER_CURR_ INT_STS	OC interrupt for charge, discharge low, and discharge high.	OCL_DCHG_STS, OCH_DCHG_STS, OC_CHG_STS
SC	SHORT_CURR_ INT_STS	Short-circuit interrupt for discharge short-circuit current or the short-circuit removal detection before DSG MOSFET turn on	SC_CHG_STS,
FUSE failure			Flags in address, DEAD/SOV/TOP_SOV/TOP_S UV/LDO_UV/SELF_ADC/DSG FAIL/CHG FAIL

The interrupt source can be configured using a TYPE selector. Depending on the interrupt, the TYPE selector provides a few options:

- Level (high)
- Rising edge
- Falling edge
- Rising and falling edge

Each interrupt that is related to a protection flag follows the processes described above.

Some interrupts provide the mask setting, which can release the interrupt pin, while the register status still retain.

MOSFET FAULT

In fault mode, some fault case an option is available to turn off only one FET or turn off both discharge and charge MOSFETs. Faults can be cleared either by an automatic recovery or manual MCU clearing. The fault enable control bit for each fault has a configurable default value, and the option to permanently lock the register in read-only mode. Table 6 lists certain faults and their recovery methods. Note that the fault cannot be manual cleared if the real time status remains 1. For example, if $CELL_x_OV_RT_STS = 1$ and $OV\ threshold > VCELL > OV\ threshold - hysteresis$, the cell OV

fault can't be manual cleared. In this case, set $CELL_OV_FAULT_EN = 0$ to force the fault to be cleared, note that set $CELL_OV_FAULT_EN$ should be set back to 1 after this clearing. Other faults can be forced to be cleared by the similar way.

There is a recovery logic option, if discharge current $>$ standby when in FAULT with only CHG FET off, return on CHG FET to prevent overheating FET component. If this Fault still exists and the current isn't discharge, turn off the CHG Fault again. Refer table 5 for the details.

There is a recovery logic option, if charge current $>$ standby when in FAULT with only DSG FET off, return on DSG FET to prevent overheating FET component. If this Fault still exist and the current isn't charge, trigger the DSG Fault again.

**Table 5: MOSFET behavior and fault status when a reverse current is detected**

FET_BD_EN	xx_EXIT_DSG /CHG_CURR	MOSFET Behavior	Fault Condition and Status
0	0	Direct turn on	Fault remained until condition is removed when auto recovery enabled
0	1	Direct turn on	Voltage-type faults will be cleared at the next SlotA. Current-type faults will be cleared immediately.
1	0	OFF until fault cleared	Fault remained until condition is removed when auto recovery enabled
1	1	OFF until fault cleared	Voltage-type faults will be cleared at the next SlotA. Current-type faults will be cleared immediately.

Table 6: Fault and Recovery Management (Part I)

Fault	Enable Control	Behavior in Fault	Recovery Method(s)
Cell OV	CELL_OV_FAULT_EN	Configurable by: CELL_OV_FAULT_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (CELL_OV_FAULT_CLR = 1) - Automatic recovery: This method is enabled via CELL_OV_REC = 1. Real time status of cell UV/dead/SOV is still present: The fault will auto clear when Vcell < OV threshold - hysteresis hold CELL_OV_DG + 16 slotA times. Real time status of cell UV/dead/SOV is not present: The fault will auto clear when Vcell < OV threshold - hysteresis hold CELL_OV_DG time. An additional recovery logic option is available via CELL_OV_EXIT_DSG_CURR. If it is set to 1, automatic recovery is enabled and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear after one slot-A.
Cell UV	CELL_UV_FAULT_EN	Configurable by CELL_UV_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (CELL_UV_FAULT_CLR = 1) - Automatic recovery: This method is enabled via CELL_UV_REC = 1. Real time status of cell OV/dead/SOV is still present: The fault will auto clear when Vcell > UV threshold + hysteresis hold CELL_UV_DG + 16 slotA times. Real time status of cell OV/dead/SOV is not present: The fault will auto clear when Vcell > UV threshold + hysteresis hold CELL_UV_DG times. An additional recovery logic option is available via CELL_UV_EXIT_CHG_CURR. If it is set to 1, automatic recovery is enabled and set as DSG FET off, the DSG FET will automatically turn on once a charge current is detected, and the fault status will auto clear after one slot-A.
Cell dead voltage	CELL_DEAD_FAULT_EN	Turn off CHG and DSG FET	The host MCU writes to the fault clear command (CELL_DEAD_FAULT_CLR). CELL_DEAD_LOG_STS can only be cleared by CELL_DEAD_DET_CLEAR.
Cell mismatch	CELL_MSMT_FAULT_EN	Turn off CHG and DSG FET	The host MCU writes to CELL_MSMT_FAULT_CLR. CELL_MSMT_LOG_STS can only be cleared by CELL_MSMT_DET_CLEAR.



Open wire	OPEN_WIRE_FAULT_EN	Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (OPEN_WIRE_FAULT_CLR.) - Automatic recovery: This method is enabled with OPEN_WIRE_FAULT_REC.
VTOP OV	VTOP_OV_FAULT_EN_CTRL	Configurable by VTOP_OV_FAULT_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (VTOP_OV_FAULT_CLR.) - Automatic recovery: This method is enabled via VTOP_OV_REC. The fault will auto clear when VTOP < OV threshold – hysteresis hold VTOP_OV_DG + 16 slotA times. - An additional recovery logic option is available via VTOP_OV_EXIT_DSG_CURR. If it is set to 1, automatic recovery is enabled and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear after one slot-A.
VTOP UV	VTOP_UV_FAULT_EN_CTRL	Configurable by VTOP_UV_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (VTOP_UV_FAULT_CLR.) - Automatic recovery: This method is enabled with VTOP_UV_FAULT_REC. The fault will auto clear when VTOP > UV threshold + hysteresis hold VTOP_UV_DG + 16 slotA times. - An additional recovery logic option is available via VTOP_UV_EXIT_CHG_CURR. If it is set to 1, automatic recovery is enabled and set as DSG FET off, the DSG FET will automatically turn on once a charge current is detected, and the fault status will auto clear after one slot-A.
Cell NTC too hot discharge	NTC_CELL_HOT_DSG_FAULT_EN	Configurable by NTC_CELL_DSG_HOT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The MCU writes to the clear bit (NTC_CELL_DSG_HOT_FAULT_CLR) - Automatic mode: This method is enabled via NTC_CELL_DSG_HOT_REC = 1. The fault will auto clear when the NTC voltage (V_{NTC}) exceeds (hot discharge threshold + hysteresis) hold NTC_CELL_SUTOT_DG + 16 slotA times.
Cell NTC too cold discharge	NTC_CELL_COLD_DSG_FAULT_EN	Configurable by NTC_CELL_DSG_COLD_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The MCU writes to the clear bit (NTC_CELL_DSG_COLD_FAULT_CLR) - Automatic mode: This method is enabled via NTC_CELL_DSG_COLD_REC = 1. The fault will auto clear when $V_{NTC} < (\text{cold discharge threshold} - \text{hysteresis})$ hold NTC_CELL_SUTOT_DG + 16 slotA times.



Cell NTC too hot charge	NTC_CELL_HOT_CHG_FAULT_EN	Configurable by NTC_CELL_CHG_HOT_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The MCU writes to the clear bit (NTC_CELL_CHG_HOT_FAULT_CLR) - Automatic recovery: This method is enabled via NTC_CELL_CHG_HOT_REC = 1. The fault will auto clear when V_{NTC} exceeds (hot charge threshold + hysteresis) hold NTC_CELL_UTOT_DG + 16 slotA times. An additional recovery logic option is available via NTC_HOT_EXIT_BY_CURR. If it is set to 1, automatic recovery is enabled and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear after one slot-A.
Cell NTC too cold charge	NTC_CELL_COLD_CHG_FAULT_EN	Configurable by NTC_CELL_CHG_COLD_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The MCU writes to the clear bit (NTC_CELL_CHG_COLD_FAULT_CLR) - Automatic recovery: This method is enabled via NTC_CELL_CHG_COLD_REC = 1. The fault will auto clear when V_{NTC} falls below (cold charge threshold - hysteresis) hold NTC_CELL_UTOT_DG + 16 slotA times. An additional recovery logic option is available via NTC_COLD_EXIT_BY_CURR. If it is set to 1, automatic recovery is enabled and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear after one slot-A.
Pack OC discharge I	OC1_DCHG_FAULT_EN	Configurable by OC1_DCHG_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (OC1_DCHG_FAULT_CLR) - Automatic recovery: Enabled with OC1_DCHG_REC. This device tries to turn on the MOSFETs to check if the OC condition is removed. If removed, the device auto-clears the fault status and keep MOSFET on. If the condition is not removed within the time set by OC1_DCHG_RETRY, then the device reports to OC_DCHG_RETRY_FAILED. An additional recovery logic option is available via DSG_OC1_EXIT_CHG_CURR. If it is set to 1, OC_DCHG_RETRY_FAILED = 1 and set as DSG FET off, the DSG FET will automatically turn on once a charge current is detected, and the fault status will auto clear. An additional recovery logic option is available via DSG_OC1_EXIT_VMID. If it is set to 1, OC_DCHG_RETRY_FAILED = 1 and set as both FETs off, the FETs will automatically turn on once PACKP/MID > TOP is detected, and the fault status will auto clear.



Pack OC discharge II	OC2_DCHG_FAULT_EN	Configurable by OC2_DCHG_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (OC2_DCHG_FAULT_CLR) - Automatic recovery: Enabled with OC2_DCHG_REC. This device tries to turn on the MOSFETs to check if the OC condition is removed. If removed, the device auto-clears the fault status and keep MOSFET on. If the condition is not removed within the time set by OC2_DCHG_RETRY, then the device reports to OC_DCHG_RETRY_FAILED. An additional recovery logic option is available via DSG_OC2_EXIT_CHG_CURR. If it is set to 1, OC_DCHG_RETRY_FAILED = 1 and set as DSG FET off, the DSG FET will automatically turn on once a charge current is detected, and the fault status will auto clear. An additional recovery logic option is available via DSG_OC2_EXIT_VMID. If it is set to 1, OC_DCHG_RETRY_FAILED = 1 and set as both FETs off, the FETs will automatically turn on once PACKP/MID > TOP is detected, and the fault status will auto clear.
Pack OC charge	OC_CHG_FAULT_EN	Configurable by OC_CHG_FAULT_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (OC_CHG_FAULT_CLR) - Automatic recovery: Enabled with OC_CHG_REC. This device tries to turn on the MOSFETs to check if the OC condition is removed. If removed, the device auto-clears the fault status and keep MOSFET on. If the condition is not removed within the time set by OC_CHG_RETRY, then the device reports to OC_CHG_RECOVERY_FAILED. An additional recovery logic option is available via CHG_OC_EXIT_DSG_CURR. If it is set to 1, OC_CHG_RECOVERY_FAILED= 1 and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear. An additional recovery logic option is available via CHG_OC_EXIT_DSG_VPACKP. If it is set to 1, OC_CHG_RECOVERY_FAILED= 1 and set as both FETs off, the FETs will automatically turn on once PACKP < TOP is detected, and the fault status will auto clear.



Pack short-circuit current discharge	SC_DCHG_FAULT_EN	Configurable by SC_DSG_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (SC_DCHG_FAULT_CLR) - Automatic recovery: Enabled with SC_DCHG_REC. The device monitors the PACKP voltage to see if it rises to 2V. If PACKP > 2V, the fault will auto clear. If this condition is not met by SC_PUP_RETRY_N, then the device reports to SC_DCHG_RECOVERY_FAILED. An additional recovery logic option is available via DSG_SC_EXIT_CHG_CURR. If it is set to 1, SC_DCHG_RECOVERY_FAILED = 1 and set as DSG FET off, the DSG FET will automatically turn on once a charge current is detected, and the fault status will auto clear. An additional recovery logic option is available via DSG_SC_EXIT_CHG_VMID. If it is set to 1, SC_DCHG_RECOVERY_FAILED = 1 and set as both FETs off, the FETs will automatically turn on once PACKP/MID > TOP is detected, and the fault status will auto clear.
LDO UV	LDO_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (LDO_FAULT_CLR).
ADC self-test failure	ADC_ST_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (ADC_FAIL_FAULT_CLEAR).
PCB monitor too hot	PCB_MNTR_FAULT_EN	Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (PCB_MNTR_FAULT_CLR) - Automatic recovery: This method is enabled via PCB_MNTR_REC = 1. The fault will auto clear when the NTC voltage exceeds (monitor hot threshold + hysteresis) hold PCB_MNTR_DG + 16 slotA times.
Die too hot digital	DIE_TEMP_DIG_FAULT_EN	Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery: The host MCU writes to the fault clear command (DIE_TEMP_FAULT_CLR) - Automatic recovery: The device waits for the die temperature to drop below (digital temperature threshold - hysteresis). This method is enabled with DIE_TEMP_FAULT_REC
Die too hot analog	N/A; hardcoded enable	Turn off CHG and DSG FET	The IC returns to safe mode from shutdown mode once the die temperature drops below (analog temperature threshold - hysteresis).
DSG Driver turn on fail	TURNON_TIME_OUT_FAULT	Configurable by DSG_DRIVER_FAULT_ACTION 0: Turn off DSG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery only. The host MCU writes to the fault clear command (DSG_DRIVER_FAULT_CLR). - Automatic recovery: Enabled with DSG_DRIVER_REC. This device tries to turn on the MOSFETs. If the MOSFETs can turn on, the device auto-clears the fault status and keep MOSFET on. If the MOSFETs can't turn on within the time set by DSG_DRIVER_RETRY, then the device reports to DSG_FET_DRV_RECOVERY_FAILED. An additional recovery logic option is available via DSG_DRIVER_FAIL_EXIT_CHG_CURR. If it is set to 1, DSG_FET_DRV_RECOVERY_FAILED = 1 and set as DSG FET off, the DSG FET will automatically turn



			on once a charge current is detected, and the fault status will auto clear.
CHG Driver turn on fail	TURNON_TIME_OUT_FAULT	Configurable by CHG_DRIVER_FAULT_ACTION 0: Turn off CHG FETs 1: Turn off CHG and DSG FET	Configurable for manual or automatic recovery. - Manual recovery only. The host MCU writes to the fault clear command (CHG_DRIVER_FAULT_CLR). - Automatic recovery: Enabled with CHG_DRIVER_REC. This device tries to turn on the MOSFETs. If the MOSFETs can turn on, the device auto-clears the fault status and keep MOSFET on. If the MOSFETs can't turn on within the time set by CHG_DRIVER_RETRY, then the device reports to CHG_FET_DRV_RECOVERY_FAILED. An additional recovery logic option is available via CHG_DRIVER_FAIL_EXIT_DSG_CURR. If it is set to 1, CHG_FET_DRV_RECOVERY_FAILED = 1 and set as CHG FET off, the CHG FET will automatically turn on once a discharge current is detected, and the fault status will auto clear.
Cell severe OV	CELL_SOV_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (CELL_SOV_FAULT_CLEAR).
VTOP severe OV	VTOP_SOV_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (VTOP_SOV_FAULT_CLEAR)
VTOP severe UV	VTOP_SUV_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (VTOP_SUV_FAULT_CLEAR)
NVM CRC	NVM_FAULT_EN	Turn off CHG and DSG FET	Manual recovery only. The host MCU writes to the fault clear command (NVM_CRC_CLR)



FUSE FAULT

When the IC in active/standby/fault mode, the FUSE fault can be enabled. There is a configuration bit to enable it in safe mode. Once the secondary failure detected, FUSE pin will output a high level to blow the fuse to protect the battery pack.

For the standalone application, we don't suggest enabling the cell serious OV and cell dead, since when the battery is assembling, the cell overvoltage and undervoltage is easy to be detected by the disconnection.

Table 7 shows the fuse fault items.

Table 7: Fuse Fault and Recovery Management

Fault	Enable Control	Behavior in Fault	Recovery Method(s)
Cell dead voltage	CELL_DEAD_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the CELL_DEAD_FUSE_EN firstly and then send the clear command CELL_DEAD_FAULT_CLR = 1.
Cell mismatch	CELL_MSMT_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the CELL_MSMT_FUSE_EN firstly and then send the clear command CELL_MSMT_FAULT_CLR = 1.
LDO UV	LDO_UV_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the LDO_UV_FUSE_EN firstly and then send the clear command LDO_FAULT_CLR
ADC self-test failure	ADC_ST_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the ADC_FAIL_FUSE_EN firstly and then send the clear command ADC_FAIL_FAULT_CLEAR = 1.
Cell serious OV	CELL_SOV_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the CELL_SOV_FUSE_EN firstly and then send the clear command CELL_SOV_FAULT_CLEAR = 1.
VTOP serious OV	TOP_SOV_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the TOP_SOV_FUSE_EN firstly and then send the clear command VTOP_SOV_FAULT_CLEAR = 1.
VTOP serious UV	TOP_SUV_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the TOP_SUV_FUSE_EN firstly and then send the clear command VTOP_SUV_FAULT_CLEAR = 1.
DSG MOSFET fail	DSGFET_DEAD_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the DSGFET_DEAD_FUSE_EN firstly and then send the clear command DSGFET_DEAD_FAULT_CLEAR = 1.
CHG MOSFET fail	CHGFET_DEAD_FUSE_EN	Fuse driver pull high and turn off both FETs	Manual recovery only. Disable the CHGFET_DEAD_FUSE_EN firstly and then send the clear command CHGFET_DEAD_FAULT_CLEAR = 1.



I²C INTERFACE

The MP3718 can use an I²C interface to flexibly set parameters and report device statuses instantaneously. The I²C is a two-wire serial interface with two bus lines: a serial data line (SDA) and a serial clock line (SCL). Both SDA and SCL are open drains, and they must be connected to the positive supply voltage via a pull-up resistor.

The IC operates as a slave device that receives control inputs from the master device, such as a microcontroller (MCU). SCL is always driven by the master device. The I²C interface supports both standard mode (up to 100kbit/s) and fast mode (up to 400kbit/s).

All transactions begin with a start (S) command and are terminated by a stop (P) command. Start and stop commands are always generated by the master. A start command is defined as a high-to-low transition on the SDA while the SCL is high. A stop command is defined as a low-to-high transition on the SDA while the SCL is high (see Figure 12).

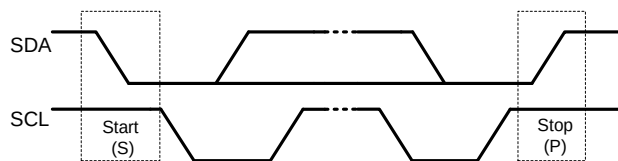


Figure 12: Start and Stop Commands

For data validity, the data on the SDA must be stable during the high period of the clock. The high or low state of the SDA can only change when the clock signal on the SCL is low (see Figure 13).

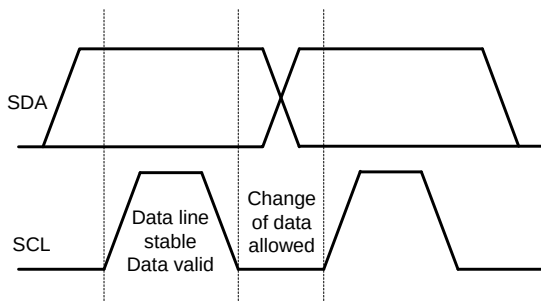


Figure 13: Bit Transfer on the I2C Bus

Every byte on the SDA must be 8 bits long. The number of bytes that can be transmitted per transfer is unrestricted. Data is transferred with the most significant bit (MSB) first.

Each byte must be followed by an acknowledge (ACK) bit, which is generated by the receiver to signal to the transmitter that the byte was successfully received.

The ACK signal occurs when the transmitter releases the SDA line during the acknowledge clock pulse. This allows the receiver to pull the SDA line low. The SDA line stays low during the high period of the ninth clock.

If the SDA line is high during the 9th clock pulse, this is defined as a not acknowledge (NACK) signal. The master can then generate either a stop command to abort the transfer, or a repeated start (Sr) command to start a new transfer.

After the start command, a slave address is sent. This address is 7 bits long, followed by an 8th data direction bit (R/W). A 0 indicates a transmission (write), and a 1 indicates a request for data (read). Figure 14 shows the address bit arrangement.

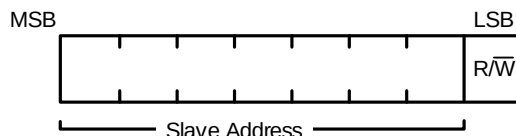


Figure 14: 7-Bit Address

Figure 15 shows a data transfer on the I²C bus.

When CRC is enabled, the payload is limited to 2 bytes by default. This value can be increased to 63 words (126 bytes) using the XFR_NUM_RD_WORDS register. XFR_NUM_RD_WORDS retains the new value only for the next transaction, so ensure that the next transaction uses the increased payload count (see Figure 16).

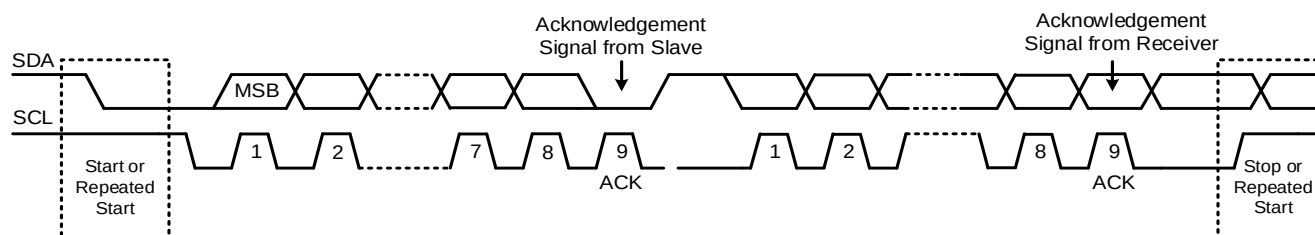


Figure 15: Data Transfer on the I²C Bus Read

CRC in an I²C Read Transaction

After completing the request for a read transaction, the host MCU must ensure that the CRC matches the value provided in the read transaction to confirm that no bits were corrupted during the transmission.

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes, following the order in which the bytes are transmitted or received. If it were bit-wise, the MSB would be processed first in each byte. An example of this sequence is shown below:

1. Slave address byte + write (02h)
2. Register address byte (00h)
3. Slave address byte + read (03h)
4. Register address byte (00h)
5. Word 1 - byte 1 / bits[7:0] (14h)
6. Word 1 - byte 2 / bits[15:8] (00h)

This sequence results in CRC = 0xE9 being appended as the last CRC byte.

Although this byte of the fourth sequence does not exist in the actual read sequence, it must be added in the CRC calculation.

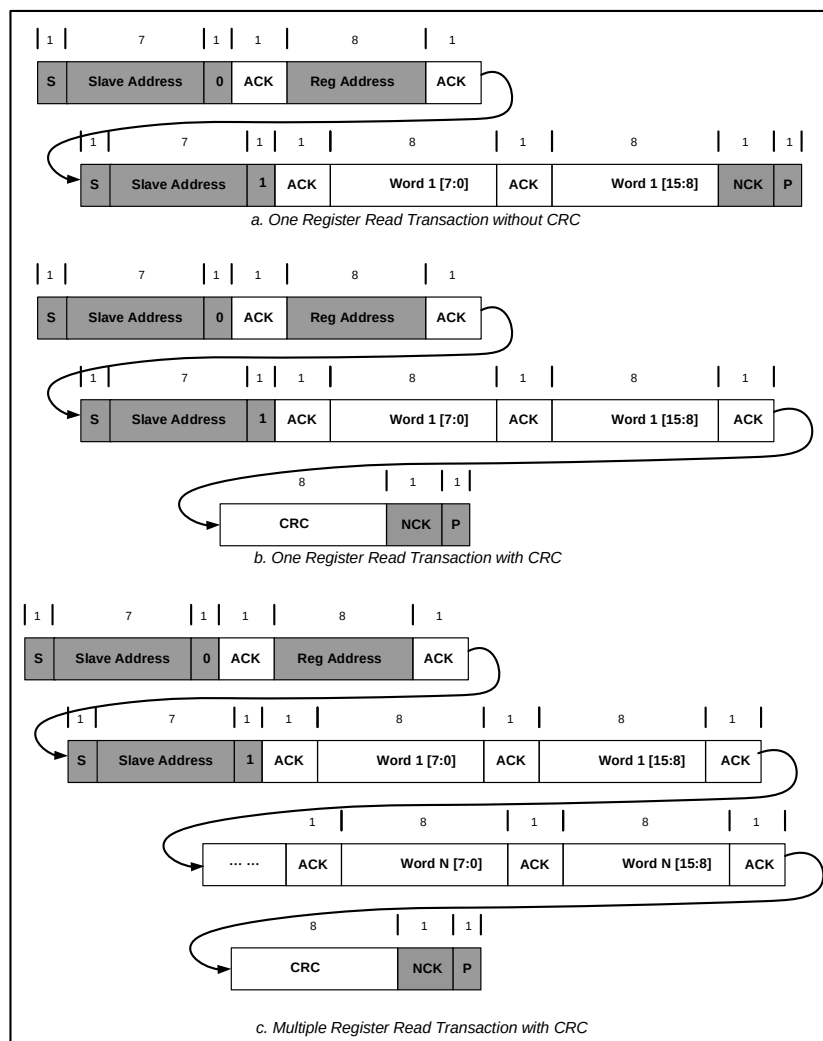


Figure 16: I²C Read Transaction



I²C Write

Figure 17 shows how the byte is ordered in a write transaction. When CRC is enabled, the targeted register address is modified according to payload only if the CRC results match.

CRC in an I²C Write Transaction

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes following the order in which they are transmitted or received. If it were bit-wise, the MSB would

first be processed in each byte. An example of this sequence is shown below:

1. Slave address byte + write (02h)
2. Register address byte (00h)
3. Word 1 - byte 1 / bits[7:0] (14h)
4. Word 1 - byte 2 / bits[15:8] (00h)

This sequence results in CRC = 0x5B, which the host MCU should append as the last CRC byte to ensure a successful transaction.

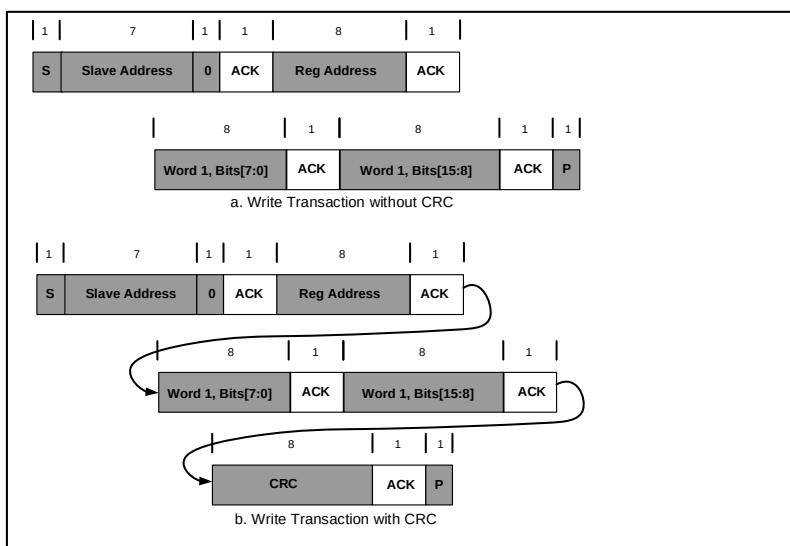


Figure 17: I²C Write Transaction

SERIAL PERIPHERAL INTERFACE (SPI)

The MP3718 has an interface that is compatible with serial peripheral interfaces (SPIs). This interface is configured to operate using clock phase (CPHA) = 1 and clock polarity (CPOL) = 1. Consequently, data on SDI must be stable during SCK's rising edge. Words are transferred with most significant bit (MSB) first.

During a write transaction, the data value on SDI is latched to the device on SCK's rising edge. During a read transaction, the bit stream is presented on SDO with the MSB first, and is valid during SCK's rising edge, while the SDO transitions on SCK's falling edge.

nCS must remain low for the entire command sequence duration, including the time between the command byte and subsequent data. During a write command, data is latched on nCS's rising edge (see Figure 18).

SPI Data Protocol

For a successful SPI transaction, nCS must go low and there must be a successful match between the transaction slave address and the internally configured slave address (located in register DEVICE_ADD). The MP3718's SPI requires a specific transaction structure.

SPI Read

Read transactions should have the fields arranged with a matching order for the slave address, read bit, register address, and data payload (see Figure 19).

When CRC is enabled, the payload is limited to two bytes by default. This value can be increased to 63 words (126 bytes) using the XFR_NUM_RD_WORDS register. XFR_NUM_RD_WORDS retains the new value only for the next transaction, so ensure that the



next transaction uses the increased payload count.

CRC in an SPI Read Transaction

After completing the request for a read transaction, the host MCU must ensure that the CRC matches the value provided in the read transaction to confirm that no bits were corrupted during the transmission.

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes, following the order in which bytes are transmitted or received. If it were bit-wise, the MSB would first be processed in each byte. An example of this sequence is shown below:

1. Slave address byte + read (03h)
2. Register address byte (00h)
3. Word 1 - byte 1 / bits[7:0] (14h)
4. Word 1 - byte 2 / bits[15:8] (00h)

This sequence results in CRC = 0x4D being appended as the last CRC byte.

SPI Write

Write transactions should have the fields arranged with a slave address, write bit, register address, and data payload (see Figure 20). When CRC is enabled, the SPI write payload is limited to 2 bytes and targeted register addresses are modified according to the payload only if the CRC matches.

CRC in an SPI Write Transaction

The CRC-8 algorithm follows the polynomial $(1 + x^1 + x^2 + x^8)$, and is applied byte-wise to the bytes following the order in which they are transmitted or received. If it were bit-wise, the MSB would first be processed in each byte. An example of this sequence is shown below:

1. Slave address byte + write (02h)
2. Register address byte (00h)
3. Word 1 - byte 1 / bits[7:0] (14h)
4. Word 1 - byte 2 / bits[15:8] (00h)

This sequence results in CRC = 0x5B, which the host MCU should append as the last CRC byte to ensure a successful transaction.

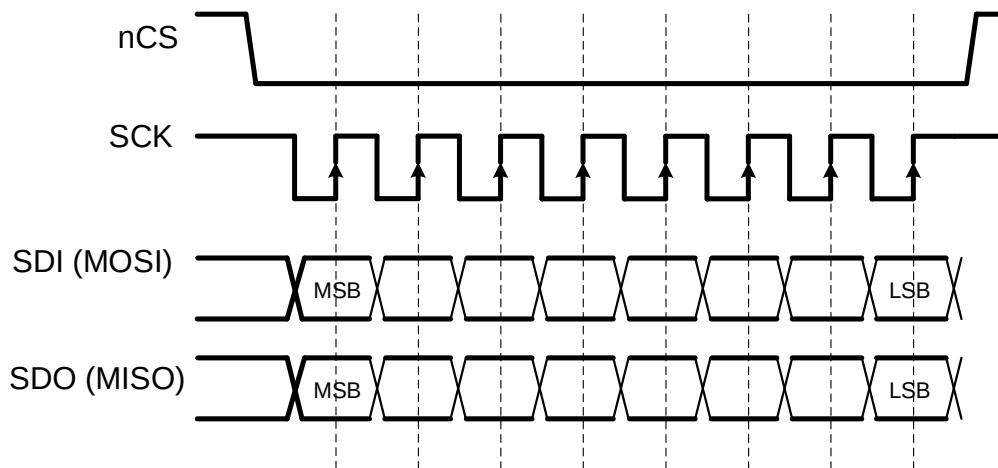
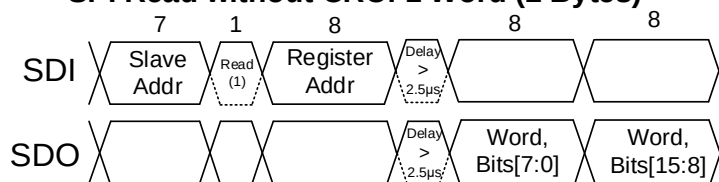
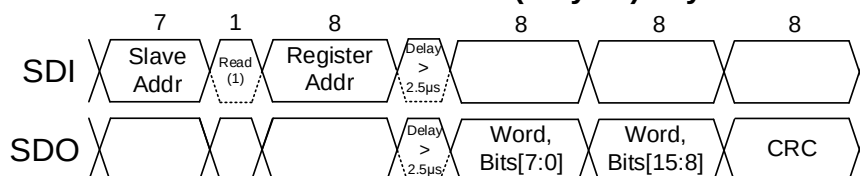
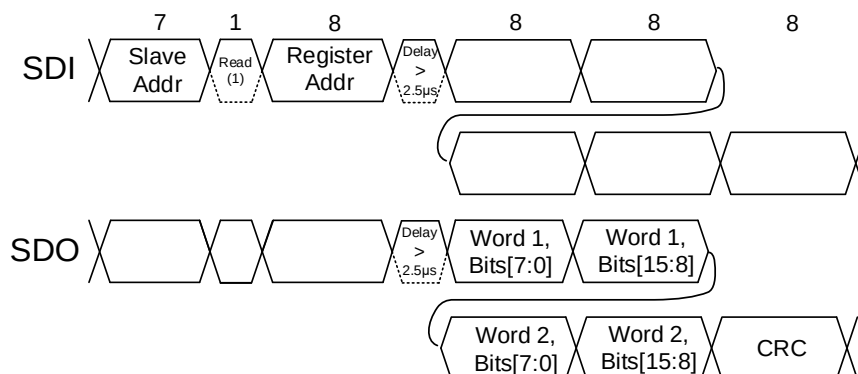
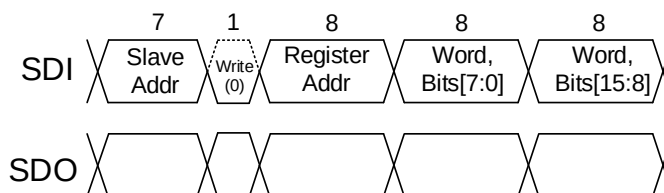
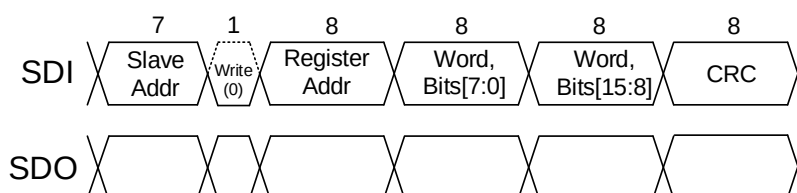


Figure 18: SPI Signal Sequencing

**SPI Read without CRC: 1 Word (2 Bytes)****SPI Read with CRC: 1 Word (2 Bytes) Payload****SPI Read with CRC: 2 Words (4 Bytes) Payload****Figure 19: SPI Read Transaction****SPI Write without CRC: 1 Word (2 Bytes)****SPI Write with CRC: 1 Word (2 Bytes)****Figure 20: SPI Write Transaction**



SUPPORTED COMMANDS

Command Code	Command Name	Type	Bytes
00h	CELLS_CTRL	RW	2
01h	CELL_MONITOR_SKIP	RW	2
02h	AFE_CFG_STATUS	R	2
03h	LOAD_CHARGER_STATUS	R	2
04h	LOAD_CHARGER_CFG	RW	2
05h	PWR_CFG	RW	2
06h	STB_CFG	RW	2
07h	FET_MODE	RW	2
08h	FET_CFG	RW	2
09h	WDT_CFG	RW	2
0Ah	GPIO_WDT_STATUS	R	2
0Bh	GPIO_OUT	RW	2
0Ch	GPIO_CFG	RW	2
0Dh	PINS_CFG	RW	2
0Eh	PRO_EN_CTRL1	RW	2
0Fh	PRO_EN_CTRL2	RW	2
10h	RD_INT0	R	2
11h	RD_INT1	R	2
12h	RD_INT2	R	2
13h	INT0_CLR	W	2
14h	INT1_CLR	W	2
15h	INT2_CLR	W	2
16h	INT0_EN	RW	2
17h	INT1_EN	RW	2
18h	INT2_EN	RW	2
19h	INT_TYPE0	RW	2
1Ah	INT_TYPE1	RW	2
1Bh	INT_TYPE2	RW	2
1Ch	MASK_INT0	RW	2
1Dh	MASK_INT1	RW	2
1Eh	FAULT_EN1	RW	2
1Fh	FAULT_EN2	RW	2
20h	FET_DEAD_LIM	RW	2
21h	FET_DEAD_DEG	RW	2
22h	FUSE_CFG	RW	2
23h	VTOP_UV_SLEP_CFG	RW	2
24h	DSGOC_LIM	RW	2
25h	DSGOC_DEG	RW	2
26h	CHGOC_DEG	RW	2
27h	DSGSC_CFG	RW	2
28h	RD_CELL_UV	R	2
29h	RD_CELL_OV	R	2



SUPPORTED COMMANDS (continued)

Command Code	Command Name	Type	Bytes
2Ah	RD_OVUV_STS	R	2
2Bh	RD_CELL_SOV	R	2
2Ch	RD_CELL_DEAD	R	2
2Dh	RD_SOVDEAD_STS	R	2
2Eh	RD_CELL_MSMT	R	2
2Fh	RD_CELLH_MSMT	R	2
30h	RD_RT_STS	R	2
31h	CELL_MSMT_STS	R	2
32h	PACK_SOV	RW	2
33h	PACK_SUV	RW	2
34h	CELL_SOV	RW	2
35h	CELL_HYST	RW	2
36h	PACK_UV_OV	RW	2
37h	CELL_UV	RW	2
38h	CELL_OV	RW	2
39h	PACK_UV	RW	2
3Ah	PACK_OV	RW	2
3Bh	CELL_DEAD_THR	RW	2
3Ch	CELL_MSMT	RW	2
3Dh	LOG_STS	R	2
3Eh	LOG_STS_CLR	W	2
3Fh	BAL_STS	R	2
40h	BAL_CTRL	RW	2
41h	BAL_CFG	RW	2
42h	BAL_THR	RW	2
43h	BAL_LIST	RW	2
44h	BAL_LISTH	RW	2
45h	RD_NTC_SUT	R	2
46h	RD_NTC_SOT	R	2
47h	RD_NTC_UT	R	2
48h	RD_NTC_OT	R	2
49h	NTC_CFG	RW	2
4Ah	NTCC_SOTHR	RW	2
4Bh	NTCC_SUTHR	RW	2
4Ch	NTCC_OTHR	RW	2
4Dh	NTCC_UTHR	RW	2
4Eh	NTCM_OTHR	RW	2
4Fh	DIE_OT	RW	2
51h	RD_VA3P3	R	2
52h	RD_VA5	R	2
53h	RD_VASELF	R	2
54h	RD_OPENL	R	2

**SUPPORTED COMMANDS** *(continued)*

Command Code	Command Name	Type	Bytes
55h	RD_OPENH_CTL	RW	2
56h	OPW_GO	R	2
57h	OPEN_CFG	RW	2
58h	NVM_CRC_GO	R	2
59h	V3P3_UV	RW	2
5Bh	V5_UV	RW	2
5Ch	FT_STS0	R	2
5Dh	FT_STS1	R	2
5Eh	FT_STS2	R	2
5Fh	FT_CLR	W	2
60h	FT2_CLR	W	2
61h	FT_ACT_CFG	RW	2
62h	FT_REC_EN	RW	2
63h	FT_REC_CFG0	RW	2
64h	FT_REC_CFG1	RW	2
65h	FT_REC_CFG2	RW	2
66h	RD_CCIRQL	R	2
67h	RD_CCIRQH	R	2
68h	RD_CCACCQL	R	2
69h	RD_CCACCQH	R	2
6Ah	RD_VTOP	R	2
6Bh	RD_ITOP	R	2
6Ch	RD_VCELL1	R	2
6Dh	RD_ICELL1	R	2
6Eh	RD_VCELL2	R	2
6Fh	RD_ICELL2	R	2
70h	RD_VCELL3	R	2
71h	RD_ICELL3	R	2
72h	RD_VCELL4	R	2
73h	RD_ICELL4	R	2
74h	RD_VCELL5	R	2
75h	RD_ICELL5	R	2
76h	RD_VCELL6	R	2
77h	RD_ICELL6	R	2
78h	RD_VCELL7	R	2
79h	RD_ICELL7	R	2
7Ah	RD_VCELL8	R	2
7Bh	RD_ICELL8	R	2
7Ch	RD_VCELL9	R	2
7Dh	RD_ICELL9	R	2
7Eh	RD_VCELL10	R	2
7Fh	RD_ICELL10	R	2



SUPPORTED COMMANDS (continued)

Command Code	Command Name	Type	Bytes
80h	RD_VCELL11	R	2
81h	RD_ICELL11	R	2
82h	RD_VCELL12	R	2
83h	RD_ICELL12	R	2
84h	RD_VCELL13	R	2
85h	RD_ICELL13	R	2
86h	RD_VCELL14	R	2
87h	RD_ICELL14	R	2
88h	RD_VCELL15	R	2
89h	RD_ICELL15	R	2
8Ah	RD_VCELL16	R	2
8Bh	RD_ICELL16	R	2
8Ch	RD_VCELL17	R	2
8Dh	RD_ICELL17	R	2
8Eh	RD_VCELL18	R	2
8Fh	RD_ICELL18	R	2
90h	RD_VCELL19	R	2
91h	RD_ICELL19	R	2
92h	RD_VCELL20	R	2
93h	RD_ICELL20	R	2
94h	RD_VCELL21	R	2
95h	RD_ICELL21	R	2
96h	RD_VCELL22	R	2
97h	RD_ICELL22	R	2
98h	RD_VCELL23	R	2
99h	RD_ICELL23	R	2
9Ah	RD_VCELL24	R	2
9Bh	RD_ICELL24	R	2
9Ch	RD_VNTC1	R	2
9Dh	RD_VNTC2	R	2
9Eh	RD_VNTC3	R	2
9Fh	RD_VNTC4	R	2
A0h	RD_VNTC5	R	2
A1h	RD_VNTC6	R	2
A2h	RD_VGPIO1	R	2
A3h	RD_VGPIO2	R	2
A4h	RD_VGPIO3	R	2
A5h	RD_VGPIO4	R	2
A6h	RD_TDIE	R	2
A7h	RD_VPACKP	R	2
A8h	ADC_CTRL	R	2
A9h	ADC_CFG	RW	2

**SUPPORTED COMMANDS** *(continued)*

Command Code	Command Name	Type	Bytes
AAh	PCB_GAIN_CAL	RW	2
ABh	TRIMG_IPCB	RW	2
ACH	HR_SCAN1	RW	2
ADh	HR_SCAN2	RW	2
AEnh	HR_SCAN3	RW	2
AFh	CC_CTRL	R	2
B0h	CC_CFG	RW	2
B1h	COMM_CFG	RW	2
B2h	COMM_CTRL	R	2
B5h	SILC_INFO2	RW	2
BEh	NVM_STATUS	R	2
BFh	NVM_CRC_STATUS	R	2
C0h	OTP_STORE_CMD	R	2
C1h	STORE_CMD_ACCESS_CODE	RW	2
C2h	OTP_READ_CMD	RW	2
EDh	AUTO_BAL_LIST	R	2
EEh	AUTO_BALH_LIST	R	2

**REGISTER MAP****CELLS_CTRL (00h)****Format:** Unsigned binary

The CELLS_CTRL command configures the number of stacked cells in use, and the skipped cell channels (Cell18-23) during hardware monitoring.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	CELL_23_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 23 channel will be skipped during hardware monitoring. 1'b 0: Cell 23 not skip during hardware monitoring 1'b 1: Cell 23 skip during hardware monitoring This bit allows MTP
12	R/W (can lock to read-only)	CELL_22_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 22 channel will be skipped during hardware monitoring. 1'b 0: Cell 22 not skip during hardware monitoring 1'b 1: Cell 22 skip during hardware monitoring This bit allows MTP
11	R/W (can lock to read-only)	CELL_21_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 21 channel will be skipped during hardware monitoring. 1'b 0: Cell 21 not skip during hardware monitoring 1'b 1: Cell 21 skip during hardware monitoring This bit allows MTP
10	R/W (can lock to read-only)	CELL_20_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 20 channel will be skipped during hardware monitoring. 1'b 0: Cell 20 not skip during hardware monitoring 1'b 1: Cell 20 skip during hardware monitoring This bit allows MTP
9	R/W (can lock to read-only)	CELL_19_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 19 channel will be skipped during hardware monitoring. 1'b 0: Cell 19 not skip during hardware monitoring 1'b 1: Cell 19 skip during hardware monitoring This bit allows MTP
8	R/W (can lock to read-only)	CELL_18_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 18 channel will be skipped during hardware monitoring. 1'b 0: Cell 18 not skip during hardware monitoring 1'b 1: Cell 18 skip during hardware monitoring This bit allows MTP
7:5	R	RESERVED	N/A	Reserved. Do not change this register value.



4:0	R/W (can lock to read-only)	CELL_S_CTRL	5'b 10100	Sets the number of stacked cells in use. 5'b 00000: Cells 4~1 enabled 5'b 00001: Cells 5~1 enabled 5'b 00010: Cells 6~1 enabled 5'b 00011: Cells 7~1 enabled 5'b 00100: Cells 8~1 enabled 5'b 00101: Cells 9~1 enabled 5'b 00110: Cells 10~1 enabled 5'b 00111: Cells 11~1 enabled 5'b 01000: Cells 12~1 enabled 5'b 01001: Cells 13~1 enabled 5'b 01010: Cells 14~1 enabled 5'b 01011: Cells 15~1 enabled 5'b 01100: Cells 16~1 enabled 5'b 01101: Cells 17~1 enabled 5'b 01110: Cells 18~1 enabled 5'b 01111: Cells 19~1 enabled 5'b 10000: Cells 20~1 enabled 5'b 10001: Cells 21~1 enabled 5'b 10010: Cells 22~1 enabled 5'b 10011: Cells 23~1 enabled 5'b 10100~11111: Cells 24~1 enabled These bits allow MTP
-----	--------------------------------	-------------	-----------	--

CELL_MONITOR_SKIP (01h)**Format:** Unsigned binary

The CELL_MONITOR_SKIP command configures the skipped cell channels (Cell2-17) during hardware monitoring.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	CELL_17_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 17 channel will be skipped during hardware monitoring. 1'b 0: Cell 17 not skip during hardware monitoring 1'b 1: Cell 17 skip during hardware monitoring This bit allows MTP
14	R/W (can lock to read-only)	CELL_16_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 16 channel will be skipped during hardware monitoring. 1'b 0: Cell 16 not skip during hardware monitoring 1'b 1: Cell 16 skip during hardware monitoring This bit allows MTP
13	R/W (can lock to read-only)	CELL_15_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 15 channel will be skipped during hardware monitoring. 1'b 0: Cell 15 not skip during hardware monitoring 1'b 1: Cell 15 skip during hardware monitoring This bit allows MTP
12	R/W (can lock to read-only)	CELL_14_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 14 channel will be skipped during hardware monitoring. 1'b 0: Cell 14 not skip during hardware monitoring 1'b 1: Cell 14 skip during hardware monitoring This bit allows MTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

11	R/W (can lock to read-only)	CELL_13_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 13 channel will be skipped during hardware monitoring. 1'b 0: Cell 13 not skip during hardware monitoring 1'b 1: Cell 13 skip during hardware monitoring This bit allows MTP
10	R/W (can lock to read-only)	CELL_12_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 12 channel will be skipped during hardware monitoring. 1'b 0: Cell 12 not skip during hardware monitoring 1'b 1: Cell 12 skip during hardware monitoring This bit allows MTP
9	R/W (can lock to read-only)	CELL_11_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 11 channel will be skipped during hardware monitoring. 1'b 0: Cell 11 not skip during hardware monitoring 1'b 1: Cell 11 skip during hardware monitoring This bit allows MTP
8	R/W (can lock to read-only)	CELL_10_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 10 channel will be skipped during hardware monitoring. 1'b 0: Cell 10 not skip during hardware monitoring 1'b 1: Cell 10 skip during hardware monitoring This bit allows MTP
7	R/W (can lock to read-only)	CELL_9_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 9 channel will be skipped during hardware monitoring. 1'b 0: Cell 9 not skip during hardware monitoring 1'b 1: Cell 9 skip during hardware monitoring This bit allows MTP
6	R/W (can lock to read-only)	CELL_8_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 8 channel will be skipped during hardware monitoring. 1'b 0: Cell 8 not skip during hardware monitoring 1'b 1: Cell 8 skip during hardware monitoring This bit allows MTP
5	R/W (can lock to read-only)	CELL_7_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 7 channel will be skipped during hardware monitoring. 1'b 0: Cell 7 not skip during hardware monitoring 1'b 1: Cell 7 skip during hardware monitoring This bit allows MTP
4	R/W (can lock to read-only)	CELL_6_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 6 channel will be skipped during hardware monitoring. 1'b 0: Cell 6 not skip during hardware monitoring 1'b 1: Cell 6 skip during hardware monitoring This bit allows MTP
3	R/W (can lock to read-only)	CELL_5_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 5 channel will be skipped during hardware monitoring. 1'b 0: Cell 5 not skip during hardware monitoring 1'b 1: Cell 5 skip during hardware monitoring This bit allows MTP



2	R/W (can lock to read-only)	CELL_4_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 4 channel will be skipped during hardware monitoring. 1'b 0: Cell 4 not skip during hardware monitoring 1'b 1: Cell 4 skip during hardware monitoring This bit allows MTP
1	R/W (can lock to read-only)	CELL_3_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 3 channel will be skipped during hardware monitoring. 1'b 0: Cell 3 not skip during hardware monitoring 1'b 1: Cell 3 skip during hardware monitoring This bit allows MTP
0	R/W (can lock to read-only)	CELL_2_MONITOR_SKIP	1'b 0	If it is set to 1, the cell 2 channel will be skipped during hardware monitoring. 1'b 0: Cell 2 not skip during hardware monitoring 1'b 1: Cell 2 skip during hardware monitoring This bit allows MTP

AFE_CFG_STATUS (02h)**Format:** Unsigned binary

The AFE_CFG_STATUS command controls CHG and DSG MOSFETs' on or off and controls selecting load or charge ports. It also reports power state, pack current, charge-pump, CHG and DSG driver statuses, and MOSFETs' turn-on timeout.

Bits	Access	Bit Name	Default	Description
15	R	CP_STS	1'b 0	Indicates the charge pump's status. 1'b 0: Off 1'b 1: On
14	R	DSG_DRV_TRANS	1'b 0	When CDSG_DRIVER_CTRL = 0, reports whether the DSG driver is changing. 1'b 0: The DSG driver is settled (off or at its target voltage) 1'b 1: The DSG driver is changing states When CDSG_DRIVER_CTRL = 1, reports whether the CDSG driver is changing. 1'b 0: The CDSG driver is settled (off or at its target voltage) 1'b 1: The CDSG driver is changing states
13	R	CHG_DRV_TRANS	1'b 0	Reports whether the CHG driver is changing. 1'b 0: The CHG driver is settled (off or at its target voltage) 1'b 1: The CHG driver is changing states
12	R	DSG_DRV	1'b 0	When CDSG_DRIVER_CTRL = 0, reports whether the DSG driver is on or off. 1'b 0: Off 1'b 1: On When CDSG_DRIVER_CTRL = 1, reports whether the CDSG driver is on or off. 1'b 0: Off 1'b 1: On
11	R	CHG_DRV	1'b 0	Reports whether the CHG driver is on or off. 1'b 0: Off 1'b 1: On



10	R	FET_TIMEOUT	1'b 0	When true, an issue was detected during the latest turn-on attempt. 1'b 0: False 1'b 1: True
9:8	R/W (can lock to read-only)	ACTIVE_CTRL	2'b 00	With FET_SRC = Register Control and FET_CFG = Simple Mode: 2'b 00: All MOSFETs are OFF 2'b 01: Turn on CHG and DSG 2'b 10: All MOSFETs are OFF 2'b 11: Turn on CHG and DSG With FET_SRC = Register Control and FET_CFG = Direct Mode: 2'b 00: All MOSFETs are OFF 2'b 01: Turn on DSG 2'b 10: Turn on CHG 2'b 11: Turn on CHG and DSG These bits allow OTP
7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W	CDSG_DRIVER_CTRL	1'b 0	Controls the selection between load and charge ports. 1'b 0: Load port selected. 1'b 1: Charger port selected. Following procedure applied while changing the ports 1. FETs will be turned off. (if already turned-on). 2. charge OC (over-current), discharge short-circuit (SC), discharge OC, discharge FET driver fault and charge FET driver faults will be cleared. 3. Resets the associated faults; retry counter and terminates the associated hiccup timer
4:3	R	PACK_CURRENT_STATUS	2'b 00	Returns the pack current status. 2'b 01: Battery pack current is discharging 2'b 10: Battery pack current is in standby range, meaning it is within $\pm$ STBY_CUR_TH 2'b 11: Battery pack current is charging Others: Invalid commands
2:0	R	PWR_STATE	3'b 000	Returns the power state status. 3'b 001: Safe 3'b 010: Standby 3'b 011: ESS 3'b 100: Active 3'b 101: Fault 3'b 110: Sleep Others: Invalid commands

LOAD_CHARGER_STATUS (03h)**Format:** Unsigned binary

The LOAD_CHARGER_STATUS command reports load removal detection status: busy, failed, completed, or detected. It also reports load/charger plug-in detection results, including the status of PACKP detection pre-bias and pre-charging, the completed load/charger plug-in detection, and the detected load/charger plug-in. Besides, it reports VTOP voltage comparison result.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15	R	LD_RML_DONE	1'b 0	When true, load plug-in detection is complete. This bit can be cleared by writing 0 to LD_RML_DET_EN. 1'b 0: False 1'b 1: True
14	R	LD_RML_FAIL	1'b 0	When true, load plug-in detection has failed. 1'b 0: False 1'b 1: True
13	R	LD_RML_STS	1'b 0	Reports the detected/not detected load removal event. This bit can be cleared by writing 0 to LD_RML_DET_EN. 1'b 0: Not detected 1'b 1: Detected
12	R	LD_RML_DET_BUSY	1'b 0	When true, load removal detection is busy after load removal detection is manually enabled or an OC or SC protection has occurred. 1'b 0: False 1'b 1: True
11	R	RESERVED	N/A	Reserved. Do not change this register value.
10	R	LD_DET_DONE_STS	1'b 0	When true, load plug-in detection is complete. This bit can be cleared by writing 0 to LD_PLUGIN_DET_EN. 1'b 0: False 1'b 1: True
9	R	LD_IN	1'b 0	Reports the detected/not detected load plug-in event. This bit can be cleared by writing 0 to LD_PLUGIN_DET_EN. 1'b 0: Not detected 1'b 1: Detected
8	R	LDDET_ENGAGED	1'b 0	Reports the status of PACKP detection pre-bias. When true, the PACKP voltage has reached the VTOP voltage and is waiting to detect a plug-in. The state machine for load plug-in detection is enabled, and the current source/sink is enabled. When false, detection is ineffective. 1'b 0: False 1'b 1: True
7	R	LD_SETTLING	1'b 0	Reports the status of PACKP detection pre-charging. When true, the PACKP voltage is approaching the VTOP voltage, but has not reached it. The state machine for load plug-in detection is enabled, and the current source/sink is enabled. When false, detection is in a different state. 1'b 0: False 1'b 1: True
6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R	CHG_DET_DONE_STS	1'b 0	When true, charger detection is complete. This bit can be cleared by writing 0 to CHG_DET_EN. 1'b 0: False 1'b 1: True
4	R	CHG_IN	1'b 0	Reports the detected/not detected charger plug-in event. This bit can be cleared by writing 0 to CH_DET_EN. 1'b 0: Not detected 1'b 1: Detected
3	R	RESERVED	N/A	Reserved. Do not change this register value.



2:0	R	VTOP_COMP_STS	3'b 000	Reports the VTOP voltage comparison result. The exact voltage level for the transition can be slightly different based on the direction of the transition and the applied hysteresis. When CDSG_DRIVER_CTRL=0 3'b 0xx: Comparison result not available 3'b 100: VTOP +160mV >= VPACKP >= VTOP -1.6V 3'b 101: VPACKP < VTOP -1.6V 3'b 110: VPACKP > VTOP +160mV 3'b 111: Comparison result not available When CDSG_DRIVER_CTRL=1 3'b 0xx: Comparison result not available 3'b 100: VTOP +160mV >= VCHGP >= VTOP -1.6V 3'b 101: VCHGP < VTOP -1.6V 3'b 110: VCHGP > VTOP +160mV 3'b 111: Comparison result not available
-----	---	---------------	---------	---

LOAD_CHARGER_CFG (04h)**Format:** Unsigned binary

The LOAD_CHARGER_CFG command configures the PACKP pre-charging expiration timer and the delay time, and it also enables the VTOP voltage comparator.

Bits	Access	Bit Name	Default	Description
15:14	R/W (can lock to read-only)	LD_RML_RETRY_N	2'b 00	Sets how many times load removal detection retries. 2'b 00: Keep trying 2'b 01: 1 try 2'b 10: 2 tries 2'b 11: 4 tries These bits allow MTP
13:11	R/W (can lock to read-only)	LD_RML_RETRY_DELAY	3'b 000	Sets the delay between load removal detection and a retry. 3'b 000: 1s 3'b 001: 2s 3'b 010: 4s 3'b 011: 6s 3'b 100: 10s 3'b 101: 15s 3'b 110: 20s 3'b 111: 25s These bits allow MTP
10:9	R/W (can lock to read-only)	SHORT_CIRCUIT_REMOVAL_DETECT_TIME	2'b 00	Sets the time for short circuit removal detection. 2'b 00: 125ms 2'b 01: 250ms 2'b 10: 500ms 2'b 11: 1s These bits allow MTP
8	R/W (can lock to read-only)	PACKP_2V_TIMER_ENABLE	1'b 1	Enables the PACKP to be charged to more than 2V by discharge soft start. This bit should only be used when PACKP_PRE_CHECK_ENABLE=0. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



7:5	R/W (can lock to read-only)	T_PLUGIN_PRECHARGE_MAX	3'b 101	Selects the PACKP pre-charging expiration timer and the delay time in turning off the biasing after reaching PACKP = VTOP. 3'b 000: The timer is 0.2s and the delay time is 84ms 3'b 001: The timer is 0.4s and the delay time is 165ms 3'b 010: The timer is 0.8s and the delay time is 334ms 3'b 011: The timer is 1.6s and the delay time is 334ms 3'b 100: The timer is 3.2s and the delay time is 334ms 3'b 101: The timer is 6.4s and the delay time is 672ms 3'b 110: The timer is 12s and the delay time is 1332ms 3'b 111: The timer is 24s and the delay time is 2700ms These bits allow OTP
4	R/W (can lock to read-only)	MID_CMP_DG_EN	1'b 1	Enables the deglitch time for PACKP or CHGP not exceeds VTOP detection. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W	VTOP_CMP_EN	1'b 1	Manually enables the VTOP comparator to compare the voltage of VTOP with VMID and PACKP. When disabled, the comparator can still be internally enabled for other functions, such as terminal detection or fault recovery. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W	LD_RML_DET_EN	1'b 0	Enables load removal detection. If this detection is re-enabled, this bit must be written to 0 and then written to 1 again. Write 0 to this bit to clear the load removal detection function status bits. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	LD_PLUGIN_DET_EN	1'b 0	Enables load plug-in detection. If this detection is re-enabled, this bit must be written to 0 and then written to 1 again. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	CHG_DET_EN	1'b 0	Enables charger removal and plug-in detection. If this detection is re-enabled, this bit must be written to 0 and then written to 1 again. Write 0 to this bit to clear the charger detection function status bits. 1'b 0: Disabled 1'b 1: Enabled

PWR_CFG (05h)

Format: Unsigned binary

The PWR_CFG(FET_CFG1) command controls enable/disables of sleep and shutdown mode, including forced exit from sleep mode, 3V3/5V powered-on in shutdown mode, whether to enter safe mode first, and IC goes to sleep condition. It also controls the 3V3 and 5V LDO regulator in shutdown mode. Enable voltage and current protection monitoring during safe mode. It also configures the voltage protection monitoring refreshing intervals in active and fault mode, and configures the CHG and DSG gate driver voltages. Besides, it reports 3V3 and 5V regulator in shutdown mode.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:14	R/W (can lock to read-only)	FET_DRV_LVL	2'b 00	Defines VGS for the CHG and DSG MOSFETs. 2'b00: 5V with low CP 2'b01: 7.5V with low CP 2'b10: 10V with high CP 2'b11: 12.5V with high CP These bits allow MTP
13	R	RESERVED	N/A	Reserved. Do not change this register value.
12	R/W	FORCE_SLEEP_EXIT	1'b 0	Enables or disables forced exit from sleep mode. When enabled, force IC exit from sleep mode. 1'b 0: Disabled 1'b 1: Enabled
11	R	RESERVED	N/A	Reserved. Do not change this register value.
10	R/W (can lock to read-only)	V5_SHDN_EN	1'b 1	Enables the 5V LDO regulator in shutdown mode. When V3P3_SHDN_EN=1, V5_SHDN_EN =0 is invalid. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
9	R	V5_SHOFF_STS	1'b 0	Reports the internal enable or disable control for the 5V regulator during shutdown mode. When enabled, the 5V rails stay on even in shutdown mode. 1'b 0: Disabled 1'b 1: Enabled
8	R	V3P3_SHOFF_STS	1'b 0	Reports the internal enable or disable control for the 3.3V regulator during shutdown mode. When enabled, the 3.3V rails stay on even in shutdown mode. 1'b 0: Disabled 1'b 1: Enabled
7	R/W (can lock to read-only)	V3P3_SHDN_EN	1'b 1	Enables the 3.3V LDO regulator during shutdown mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
6	R/W (can lock to read-only)	FORCE_SLEEP_BY_FUSE	1'b 0	Enables or disables the IC goes to sleep mode if the FUSE pin is high after FUSE_SLEEP_DELAY time. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	VTOP_UV_SLEEP_EN	1'b 0	Enables or disables sleep mode. When enabled, IC goes to sleep mode when VTOP is equal to or below VTOP_UV_SLEEP_THR. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	SAFE_WAKEUP_EN	1'b 1	Enables whether to enter safe mode first. When enabled, force IC goes to safe firstly when IC from shutdown to sleep. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



3	R/W (can lock to read-only)	ACTIVE_MONITOR_CFG	1'b 0	Selects the interval used for voltage protection monitoring to refresh the ADC reading in active mode or fault mode. 1'b 0: Voltage protection readings are refreshed every 120ms 1'b 1: Voltage protection readings are refreshed every 240ms This bit allows OTP
2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R/W (can lock to read-only)	SAFE_SCOC_EN	1'b 0	Enables short-circuit (SC) and over-current (OC) monitoring, even in safe mode. This function is disabled by default, since the MOSFETs are disabled in safe mode. If enabled, current consumption increases. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
0	R/W (can lock to read-only)	SAFE_MONITOR_EN	1'b 0	Enables or disables voltage protection monitoring during safe mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

STB_CFG (06h)**Format:** Unsigned binary

The STB_CFG command configures the standby comparator current threshold, the voltage protection refreshing interval in standby or safe mode, and the deglitch time from active to standby mode. It also controls enable/disable of the standby state, the digital hysteresis on the standby current comparator, voltage protection monitoring during standby mode. Besides, it reports the on/off setting for standby mode.

Bits	Access	Bit Name	Default	Description
15	R	STBY_STATE	1'b 0	Reports the on/off setting for standby mode. If off, the AFE does not transition to standby mode from active mode, regardless of whether the current level is in the standby mode range. 1'b 0: Off 1'b 1: On
14:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12:8	R/W (can lock to read-only)	STBY_ENTER_DGL	5'b 11000	Sets the deglitch time from active to standby. 5'b 00000: No deglitch delay 5'b 00001: The current below standby keep 1s, IC from active to standby 5'b 00010: The current below standby keep 2s, IC from active to standby ... 5'b 11111: The current below standby keep 31s, IC from active to standby These bits allow OTP
7	R/W (can lock to read-only)	FET_BD_EN	1'b 0	Enable the MOSFET body diode when a reserve current detected. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6:4	R/W (can lock to)	STBY_MONITOR_CFG	3'b 011	Selects the interval used for voltage protection monitoring to refresh the ADC reading while in standby or safe mode.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

	read-only)			3'b 000: Voltage protection readings are refreshed every 240ms 3'b 001: Voltage protection readings are refreshed every 500ms 3'b 010: Voltage protection readings are refreshed every 1000ms 3'b 011: Voltage protection readings are refreshed every 2000ms 3'b 100: Voltage protection readings are refreshed every 3760ms 3'b 101: Voltage protection readings are refreshed every 7500ms 3'b 110: Voltage protection readings are refreshed every 15000ms 3'b 111: Voltage protection readings are refreshed every 30000ms These bits allow OTP
3	R/W	STBY_HYS	1'b 0	Enables the digital hysteresis on the standby current comparator. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2:1	R/W	STBY_CUR_TH	2'b 01	Sets the standby comparator current threshold. 2'b 00: 250μV 2'b 01: 375μV 2'b 10: 500μV 2'b 11: 625μV These bits allow OTP
0	R/W (can lock to read-only)	STBY_STATE_EN	1'b 0	Enables or disables standby mode. When disabled, the device cannot enter standby mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

**FET_MODE (07h)****Format:** Unsigned binary

The FET_CFG command configures CHG/DSG MOSFETs SS: short-circuit detection sequence, forced DSG SS, MOSFETs turn-on timeout delay, CHG SS timeout, OC threshold and OC deglitch time during CHG SS. It also control enable/disable of driver turn-on timeout report. Besides, it configures MOSFET control: the high/low side MOSFET driver mode, control mode, control logic, control source.

Bits	Access	Bit Name	Default	Description
15:13	R/W	CHG_TURN_ON_TIMEOUT	3'b 000	Sets the CHG MOSFET turn-on timeout delay. Applies to CHG MOSFET. This function is paused during soft start (SS). LSB: 20ms Offset: 40ms Range: 40ms to 180ms Value (ms) = Setting x 20 + 40 These bits allow OTP
12	R/W (can lock to read-only)	LOW_SIDE_FET_MODE	1'b 0	Sets the MOSFET driver as high or low side mode. When it is set to low side, the PACKP comparator and the high side driver circuit are disabled. 1'b 0: High side FET mode 1'b 1: Low side FET mode This bit allows OTP
11	R/W	CHG_SOFTON_OC_LIM	1'b 0	Sets the OC threshold that is only applicable when the CHG undergoes SS. 1'b 0: 3.6mV 1'b 1: 4.8mV This bit allows OTP
10	R/W	CHG_SOFTON_OC_EN	1'b 1	Enables CHG SS OC for the CHG MOSFET driver during CHG SS. This check will temporarily force with 10ms deglitch and CHG SS OC limit threshold to monitor CHG MOSFET SOA during CHG SS. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R/W	TURNON_TIMEOUT_FAULT	1'b 0	Enables the driver fault report for the turn-on timeout. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
8	R/W	CHG_SS_TURN_ON_TIMEOUT	1'b 1	Sets the CHG SS timeout. 1'b 0: 25ms 1'b 1: 50ms This bit allows OTP
7:5	R/W	DSG_TURN_ON_TIMEOUT	3'b 000	Sets the MOSFET turn-on timeout delay. Applies to both the CHG and DSG MOSFETs. This function is paused during soft start (SS). LSB: 20ms Offset: 40ms Range: 40ms to 180ms Value (ms) = Setting x 20 + 40. These bits allow OTP



4	R/W	FORCE_DSG_SOFT_START_BELOW_2V	1'b 1	Enables or disables forced DSG SS in all conditions when the PACKP is lower than the threshold of about 2V. When enabled, force DSG MOSFET to SS. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W	PACKP_PRE_CHECK_ENABLE	1'b 1	Enables the short-circuit detection sequence prior to the MOSFET turning on. If detection is enabled and the device detects that PACKP is shorted to ground, then the discharge short-circuit fault is generated. When enabled, a 250µA current source is used to attempt to charge PACKP above the about 2V threshold. The load removal detection function will use the 250µA current source to check that the load is removed. This bit should only be used when PACKP_2V_TIMER_ENABLE=0. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W (can lock to read-only)	FT_STATE_SEL	1'b 0	When this bit is set to 0, the MOSFETs transition to start-up via a control source (depending on FET_SRC), which is disabled then enabled. When this bit is set to 1, the MOSFETs start up depending on the control source status. 1'b 0: Rising edge 1'b 1: Level This bit allows OTP
1	R/W (can lock to read-only)	FET_CFG	1'b 0	Selects the CHG and DSG MOSFET control logic. 1'b 0: Simple mode. The internal logic automatically determines the turn-on/-off sequencing. 1'b 1: Direct mode. In direct mode with pin control, GPIO1 controls the DSG MOSFET and GPIO2 controls the CHG MOSFET. When direct mode is selected and faults are enabled, the IC still uses the internal logic to turn off the MOSFETs. This bit allows OTP
0	R/W (can lock to read-only)	FET_SRC	1'b 0	Defines the source that enables the MOSFETs and switches to enter active mode. The CHG and DSG MOSFETs are controlled by the host MCU, via either register or pin. 1'b 0: Register control 1'b 1: GPIO (GPIO1 or GPIO1 and GPIO2 depending on FET_CFG) This bit allows OTP

FET_CFG (08h)**Format:** Unsigned binary

The FET_CFG command configures the DSG MOSFET SS, including initial current threshold, initial current threshold's range and LSB, timer limit. It also controls enable/disable of CHG/DSG MOSFET driver soft start and the change of MOSFET turn on/off sequence.



Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	CHG_SOFTON_EN	1'b 1	Enables soft start for the CHG MOSFET driver. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14:12	R/W (can lock to read-only)	CHG_SOFTON_PUP	3'b 001	Sets the pull-up current values during CHG MOSFET SS. LSB: 1μA Offset: 3μA Range: 3μA to 10μA Value (μA) = Setting + 3 These bits allow MTP
11	R	RESERVED	N/A	Reserved. Do not change this register value.
10	R/W (can lock to read-only)	DSG_SOFTON_EN	1'b 1	Enables soft start for the DSG MOSFET driver. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9:5	R/W (can lock to read-only)	DSG_SS_TURN_ON_TIMEOUT	5'b 00000	Sets the DSG MOSFET soft start timer limit. If the VPACKP can't reach to VTOP in this time, the DSG driver fault is generated. LSB: 40ms Offset: 40ms Range: 40ms to 1280ms Value (ms) = Setting x 40 + 40 These bits allow MTP
4	R/W (can lock to read-only)	DSGSS_CURRENT_RNG	1'b 0	Selects the range and LSB for the DSG MOSFET SS initial current threshold. 1'b 0: 50μV/ LSB, 800μV FSR 1'b 1: 250μV/ LSB, 4000μV FSR This bit allows MTP
3:0	R/W (can lock to read-only)	DSGSS_CURRENT	4'b 0000	Sets the DSG MOSFET soft start initial current (ISS) threshold. With a 1x range (DSGSS_CURRENT_RNG = 0): LSB: 50μV Offset: 50μV Range: 50μV to 800μV Value (μV) = Setting x 50 + 50 With a 5x range (DSGSS_CURRENT_RNG = 1): LSB: 250μV Offset: 250μV Range: 250μV to 4000μV Value (μV) = Setting x 250 + 250 When VPACKP < 50% x VTOP, DSG soft start current is ISS When 50% x VTOP ≤ VPACKP < 75% x VTOP, DSG soft start current is 2 x ISS When 75% x VTOP ≤ VPACKP < 87% x VTOP, DSG soft start current is 4 x ISS When VPACKP ≥ 87% x VTOP, DSG soft start current is 8 x ISS These bits allow MTP

WDT_CFG (09h)**Format:** Unsigned binary

The WDT_CFG command configures watchdog bark and bite timeout. It also controls enable/disable of the communication watchdog.



Bits	Access	Bit Name	Default	Description
15:9	R/W (can lock to read-only)	WDT_BARK_CFG	7'b 0110110	Configures the watchdog bark timeout. The bark setting defines the delay from the last watchdog reset to the bark. LSB: 25ms Offset: 25ms Range: 25ms to 3200ms Value (ms) = Setting x 25 + 25 These bits allow OTP
8:2	R/W (can lock to read-only)	WDT_BITE_CFG	7'b 1011110	Configures the watchdog bite timeout. The bite setting defines the delay from the bark to the bite. LSB: 25ms Offset: 25ms Range: 25ms to 3200ms Value (ms) = Setting x 25 + 25 These bits allow OTP
1	R	RESERVED	N/A	Reserved. Do not change this register value.
0	R/W (can lock to read-only)	WDT_COM_CTRL	1'b 0	Enables watchdog communication. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

GPIO_WDT_STATUS (0Ah)**Format:** Unsigned binary

The GPIO_WDT_STATUS command controls the reset of Watchdog. It also reports watchdog bite and bark status, GPIO1-4 and fuse status.

Bits	Access	Bit Name	Default	Description
15:11	R	RESERVED	N/A	Reserved. Do not change this register value.
10	W	WDT_RST	1'b 0	Writing 1 to this bit resets the watchdog timer counter, and clears WDT_BITE and WDT_BARKED. This is a self-clearing register.
9	R	WDT_BITE	1'b 0	Indicates whether the bite timer has expired, and a bite event has occurred. This event toggles the WDT pin, if WDT_RPT is set to enable. 1'b 0: False 1'b 1: True
8	R	WDT_BARKED	1'b 0	Indicates whether the bark timer has expired and a bark event has occurred. 1'b 0: False 1'b 1: True
7:5	R	RESERVED	N/A	Reserved. Do not change this register value.
4	R	FUSE_STS	1'b 0	Reports whether the fuse fault has been detected or the FUSE pin has detected high voltage, which is the latched status. 1'b 0: Not detected 1'b 1: Detected
3	R	GPIO4	1'b 0	Reports the GPIO4 pin status. 1'b 0: Low 1'b 1: High
2	R	GPIO3	1'b 0	Reports the GPIO3 pin status. 1'b 0: Low 1'b 1: High



1	R	GPIO2	1'b 0	Reports the GPIO2 pin status. 1'b 0: Low 1'b 1: High
0	R	GPIO1	1'b 0	Reports the GPIO1 pin status. 1'b 0: Low 1'b 1: High

GPIO_OUT (0Bh)**Format:** Unsigned binary

The GPIO_OUT command configures GPIO1-4's output level and digital ADC standby current comparator threshold. It also controls enable/disable of the digital ADC standby current comparator. Besides, it reports digital standby current status (charge and discharge).

Bits	Access	Bit Name	Default	Description
15	R/W	DSTBC_EN	1'b 0	Enables the digital ADC standby current comparator. 1'b 0: Disabled 1'b 1: Enabled
14	R	IDSG_GT_STBY	1'b 0	Returns the digital standby current status. 1'b 0: Battery pack discharge current is in digital standby range 1'b 1: Battery pack discharge current is out digital standby range, meaning it exceeded DSTBC_TH
13	R	ICHG_GT_STBY	1'b 0	Returns the digital standby current status. 1'b 0: Battery pack charge current is in digital standby range 1'b 1: Battery pack charge current is out digital standby range, meaning it lower than -DSTBC_TH
12:7	R/W	DSTBC_TH	6'b 000101	Sets the digital ADC standby current comparator threshold. LSB: 3.05μV Offset: 0μV Range: 0μV to 192.15μV Value (μV) = Setting x 3.05
6:4	R	RESERVED	N/A	Reserved. Do not change this register value.
3	R/W (can lock to read-only)	GPIO4_O	1'b 0	Sets the target output level for GPIO4 (high or low). This bit is effective only when GPIO4 is used as a digital output (GPIO4_IO is set to output). 1'b 0: Low 1'b 1: High This bit allows OTP
2	R/W (can lock to read-only)	GPIO3_O	1'b 0	Sets the target output level for GPIO3 (high or low). This bit is effective only when GPIO3 is used as a digital output (GPIO3_IO is set to output). 1'b 0: Low 1'b 1: High This bit allows OTP
1	R/W	GPIO2_O	1'b 0	Sets the target output level for GPIO2 (high or low). This bit is effective only when GPIO2 is used as a digital output (GPIO2_IO is set to output). 1'b 0: Low 1'b 1: High This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

0	R/W	GPIO1_O	1'b 0	Sets the target output level for GPIO1 (high or low). This bit is effective only when GPIO1 is used as a digital output (GPIO1_IO is set to output). 1'b 0: Low 1'b 1: High This bit allows OTP
---	-----	---------	-------	--

GPIO_CFG (0Ch)
Format: Unsigned binary

The GPIO_CFG command configures GPIO1-4's direction, type of input, type of ADC input, and pull-up capability.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	GPIO4_NTC10	1'b 0	The setting defines the type of ADC input for GPIO4. 1'b 0: General purpose ADC input 1'b 1: NTC10 Input, this setting is valid when gpio4_type=1 This bit allows OTP
14	R/W (can lock to read-only)	GPIO4_PUP	1'b 1	Enables the GPIO4 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO4. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	GPIO4_TYPE	1'b 0	Defines the type of input for GPIO4. 1'b 0: Digital input 1'b 1: Buffered ADC input, 3.3V range This bit allows OTP
12	R/W (can lock to read-only)	GPIO4_IO	1'b 1	Defines the direction for GPIO4. 1'b 0: Output 1'b 1: Input This bit allows OTP
11	R/W (can lock to read-only)	GPIO3_NTC9	1'b 0	The setting defines the type of ADC input for GPIO3. 1'b 0: general purpose ADC input 1'b 1: NTC9 Input, this setting is valid when gpio3_type=1 This bit allows OTP
10	R/W (can lock to read-only)	GPIO3_PUP	1'b 1	Enables the GPIO3 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO3. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R/W (can lock to read-only)	GPIO3_TYPE	1'b 0	Defines the type of input for GPIO3. 1'b 0: Digital input 1'b 1: Buffered ADC input, 3.3V range This bit allows OTP
8	R/W (can lock to read-only)	GPIO3_IO	1'b 1	Defines the direction for GPIO3. 1'b 0: Output 1'b 1: Input This bit allows OTP



7	R/W (can lock to read-only)	GPIO2_NTC8	1'b 0	The setting defines the type of ADC input for GPIO2. 1'b 0: general purpose ADC input 1'b 1: NTC8 Input, this setting is valid when gpio2_type=1 This bit allows OTP
6	R/W (can lock to read-only)	GPIO2_PUP	1'b 0	Enables the GPIO2 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO2. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	GPIO2_TYPE	1'b 0	Defines the type of input for GPIO2. 1'b 0: Digital input 1'b 1: Buffered ADC input, 3.3V range This bit allows OTP
4	R/W (can lock to read-only)	GPIO2_IO	1'b 1	Defines the direction for GPIO2. 1'b 0: Output 1'b 1: Input This bit allows OTP
3	R/W (can lock to read-only)	GPIO1_NTC7	1'b 0	The setting defines the type of ADC input for GPIO1 1'b 0: general purpose ADC input 1'b 1: NTC7 Input, this setting is valid when gpio1_type=1 This bit allows OTP
2	R/W (can lock to read-only)	GPIO1_PUP	1'b 0	Enables the GPIO1 pull-up capability. When enabled, a 20kΩ pull-up resistor is applied to GPIO1. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read-only)	GPIO1_TYPE	1'b 0	Defines the type of input for GPIO1. 1'b 0: Digital input 1'b 1: Buffered ADC input, 3.3V range This bit allows OTP
0	R/W (can lock to read-only)	GPIO1_IO	1'b 1	Defines the direction for GPIO1. 1'b 0: Output 1'b 1: Input This bit allows OTP

PINS_CFG (0Dh)**Format:** Unsigned binary

The PIN_CFG command configures GPIO1-4 and NTC5-6 function type, GPIO pull-up voltage, VREF and NTCB dynamic-on, and xALERT active type. It also controls enable/disable of watchdog bite event toggle and watchdog reset.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	NTCB_DYNAMIC_ON	1'b 1	Enables dynamic bias on NTCB during ADC conversions of NTC channels. When disabled, NTCB is always enabled, which increases current consumption. 1'b 0: Always keep on 1'b 1: Dynamic on This bit allows OTP
12:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R/W (can lock to read-only)	NTC6_CFG_WDT	1'b 0	Sets the WDT pin as the NTC6 input. 1'b 0: Watchdog input and output 1'b 1: NTC6 input This bit allows OTP
8	R/W (can lock to read-only)	GPIO_LV_CFG	1'b 0	Sets the GPIO1-4, WDT and XALERT pin pull-up voltage. 1'b 0: 3V3 1'b 1: 5V This bit allows OTP
7	R/W (can lock to read-only)	NTC5_CFG_XALERT	1'b 0	Sets xALERT pin as NTC5 input 1'b 0: xALERT output 1'b 1: NTC5 input This bit allows OTP
6	R/W (can lock to read-only)	WDT_RST_EN	1'b 1	Enables the WDT pin to reset the MP3718 back to its factory settings. When disabled, a WDT pulse caused by a watchdog bite does not trigger a reset. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	WDT_RPT	1'b 1	Enables a watchdog bite event triggers the WDT pin to toggle a high pulse. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	GPIO4_FSEL	1'b 0	Sets GPIO4 as the auto-balance controller, which is valid when GPIO4 is set as input and balance set as GPIO control. 1'b 0: GPIO 1'b 1: auto-balance controller This bit allows OTP
3	R/W (can lock to read-only)	GPIO3_FSEL	1'b 0	Set GPIO3 as the fault indicator, which is valid when GPIO3 is set to be an output. 1'b 0: GPIO 1'b 1: Fault indicator This bit allows OTP
2	R/W (can lock to read-only)	GPIO2_FSEL	1'b 0	Sets GPIO2 as the DSG indicator, which is valid when GPIO2 is set to be an output. 1'b 0: GPIO 1'b 1: DSG indicator This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

1	R/W (can lock to read-only)	GPIO1_FSEL	1'b 0	Sets GPIO1 as the CHG indicator, which is valid when GPIO1 is set to be an output. 1'b 0: GPIO 1'b 1: CHG indicator This bit allows OTP
0	R/W (can lock to read-only)	ALERT_POL	1'b 0	Sets the polarity of xALERT pin when the interrupt is pending. 1'b 0: Active high. xALERT pin goes high when an interrupt is pending 1'b 1: Active low. xALERT pin goes low when an interrupt is pending This bit allows OTP

PRO_EN_CTRL1 (0Eh)
Format: Unsigned binary

The PRO_EN_CTRL1 command controls enable/disable of voltage monitoring protection, including cell OV/UV/SOV/dead/mismatch, CHG/DSG MOSFETs dead, ADC self-diagnostic, and digital die temperature too hot. It also controls enable/disable of current monitoring protection, including discharge OC1 and 2, charge OC, and discharge SC.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	AOPW_EN_DSG	1'b 0	If enabled, when cell UV/OV/MSMT/DEAD/SOV detected in discharge or standby current, the auto open wire will run. If the one wire is detected as open, the upper cell and down cell's fault will not be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W (can lock to read-only)	AOPW_EN_CHG	1'b 0	If enabled, when cell UV/OV/MSMT/DEAD/SOV detected in charge current, the auto open wire will run. If the one wire is detected as open, the upper cell and down cell's fault will not be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	CHGFET_DEAD_EN	1'b 0	Enables the monitoring for CHG MOSFET healthy. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R/W (can lock to read-only)	DSGFET_DEAD_EN	1'b 0	Enables the monitoring for DSG MOSFET healthy. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
11	R/W (can lock to read-only)	DIE_TEMP_DIG_EN	1'b 1	Enables digital die temperature monitoring. This bit must be enabled for die temperature hot faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R/W (can lock to read-only)	ADC_SELF_TEST_EN	1'b 1	Enables the ADC self-test check that indicates whether the applied voltage is within the allowed range. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R	RESERVED	N/A	Reserved. Do not change this register value.
8	R/W (can lock to read-only)	CELL_MSMT_EN	1'b 0	Enables cell mismatch monitoring protection. This bit must be enabled for cell mismatch faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	CELL_DEAD_EN	1'b 0	Enables cell dead protection. This bit must be enabled for cell dead faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	CELL_SOV_EN	1'b 0	Enables cell serious over voltage protection (SOVP). This bit must be enabled for SOV faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	CELL_OV_EN	1'b 0	Enables cell OVP. This bit must be enabled for OV faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	CELL_UV_EN	1'b 0	Enables cell UVP. This bit must be enabled for UV faults or interrupts to be triggered. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W (can lock to read-only)	SC_DCHG_EN_CTRL	1'b 1	Enables discharge short-circuit monitoring. When disabled, this fault does not trigger an interrupt or a fault, regardless of whether those bits are enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
2	R/W (can lock to read-only)	OC_CHG_EN_CTRL	1'b 1	Enables charge OC limit monitoring. When disabled, a charge OC limit event does not trigger a fault or interrupt, regardless of whether the fault and interrupt bits are enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

1	R/W (can lock to read-only)	OC2_DCHG_EN_CTRL	1'b 1	Enables discharge OC limit 2 monitoring. When disabled, a discharge OC limit 2 event does not trigger a fault or interrupt, regardless of whether the fault and interrupt bits are enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
0	R/W (can lock to read-only)	OC1_DCHG_EN_CTRL	1'b 1	Enables discharge OC limit 1 monitoring. When disabled, a discharge OC limit 1 event does not trigger a fault or interrupt, regardless of whether the fault and interrupt bits are enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP

PRO_EN_CTRL2 (0Fh)
Format: Unsigned binary

The PRO_EN_CTRL2 command controls enable/disable of voltage monitoring protection, including VTOP OV/UV/SOV/SUV, 3V3/5V UV and NTC.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	NTC10_CELL_EN	1'b 0	Enables cell temperature protection monitoring for NTC10 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W (can lock to read-only)	NTC9_CELL_EN	1'b 0	Enables cell temperature protection monitoring for NTC9 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	NTC8_CELL_EN	1'b 1	Enables cell temperature protection monitoring for NTC8 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R/W (can lock to read-only)	NTC7_CELL_EN	1'b 1	Enables cell temperature protection monitoring for NTC7 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

11	R/W (can lock to read-only)	NTC6_CELL_EN	1'b 1	Enables cell temperature protection monitoring for NTC6 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
10	R/W (can lock to read-only)	NTC5_CELL_EN	1'b 1	Enables cell temperature protection monitoring for NTC5 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R/W (can lock to read-only)	NTC4_EN	1'b 1	Enables protection monitoring for NTC4 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
8	R/W (can lock to read-only)	NTC3_EN	1'b 1	Enables protection monitoring for NTC3 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	NTC2_EN	1'b 1	Enables protection monitoring for NTC2 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	NTC1_EN	1'b 1	Enables protection monitoring for NTC1 if monitoring is required. Monitoring is automatically required when the IC is not in safe mode, but this function can be enabled in safe mode with PROTECT_IN_SAFE_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	V3V3_EN	1'b 1	Enables the ADC 3V3 monitoring check to ensure that the 3.3V supply is above the defined UV level. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	V5_EN	1'b 1	Enables the ADC 5V monitoring check to ensure that the 5V supply is above the defined UV level. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

3	R/W (can lock to read-only)	VTOP_SUV_EN	1'b 0	Enables the monitoring for battery pack (VTOP) serious under-voltage protection (SUVP). 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W (can lock to read-only)	VTOP_SOVP_EN	1'b 0	Enables the monitoring for battery pack (VTOP) serious over-voltage protection (SOVP). 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read-only)	VTOP_OV_EN	1'b 0	Enables the monitoring for battery pack (VTOP) over-voltage protection (OVP). 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
0	R/W (can lock to read-only)	VTOP_UV_EN	1'b 0	Enables the monitoring for battery pack (VTOP) under-voltage protection (UVP). 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

RD_INT0 (10h)
Format: Unsigned binary

The RD_INT0 command reports interrupt, including cell OV/UV, VTOP OV/UV, CHG/DSG MOSFETs dead, NTC cell discharging cold/hot, detected watchdog.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R	COMM_CRC_FAIL_INT_STS	1'b 0	Reports whether an interrupt related to a wrong CRC write even has been detected. 1'b 0: Not detected 1'b 1: Detected
13	R	PACK_CURRENT_INT_STS	1'b 0	Reports whether an interrupt related to a battery pack current change has been detected. The pack current changes when switching between discharge, standby, and charge mode, which is reported in PACK_CURRENT_STATUS. 1'b 0: Not detected 1'b 1: Detected
12	R	CONN_DET_INT_STS	1'b 0	Reports whether an interrupt related to charger or load plug-in detection has been detected. 1'b 0: Not detected 1'b 1: Detected
11	R	CC_ACC_INT_STS	1'b 0	Reports whether an interrupt related a new Coulomb counting accumulation value has occurred, and updated CC_ACC_LSBS and CC_ACC_MSBS. 1'b 0: Not detected 1'b 1: Detected


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R	VSCAN_DONE_INT_STS	1'b 0	Reports whether an interrupt related to a high-resolution voltage scan has been detected. 1'b 0: Not detected 1'b 1: Detected
9	R	AFE_MODE_CHANGE_INT_STS	1'b 0	Reports whether an interrupt related to an AFE mode change has been detected. A mode change is a change between any of the operating modes (e.g. Safe, Standby, ESS, Active, Fault, and Sleep). 1'b 0: Not detected 1'b 1: Detected
8	R	WDT_INT_STS	1'b 0	Reports whether an interrupt related to watchdog communication has been detected. A watchdog bark and a watchdog bite event can both trigger a watchdog event interrupt. 1'b 0: Not detected 1'b 1: Detected
7	R	NTC_SOT_INT_STS	1'b 0	Reports whether an interrupt related to an NTC over temperature event during discharge has been detected. This signal is generated by combining the interrupt detection for all 10 NTC channels. So NTCx_CELL_SOT_RT_STS (x=1~10) should be checked to identify the NTC source of the interrupt. 1'b 0: Not detected 1'b 1: Detected
6	R	NTC_SUT_INT_STS	1'b 0	Reports whether an interrupt related to an NTC under temperature event during discharge has been detected. This signal is generated by combining the interrupt detection for all 10 NTC channels. So NTCx_CELL_SUT_RT_STS (x=1~10) should be checked to identify the NTC source of the interrupt. 1'b 0: Not detected 1'b 1: Detected
5	R	CHGFET_DEAD_INT_STATUS	1'b 0	Reports whether an interrupt related to CHG MOSFET dead has been detected. 1'b 0: Not detected 1'b 1: Detected
4	R	DSGFET_DEAD_INT_STATUS	1'b 0	Reports whether an interrupt related to DSG MOSFET dead has been detected. 1'b 0: Not detected 1'b 1: Detected
3	R	VTOP_OV_INT_STS	1'b 0	Reports whether an OV interrupt related to the battery stack positive terminal (VTOP) has been detected. 1'b 0: Not detected 1'b 1: Detected
2	R	VTOP_UV_INT_STS	1'b 0	Reports whether a UV interrupt related to the battery stack positive terminal (VTOP) has been detected. 1'b 0: Not detected 1'b 1: Detected
1	R	CELL_UV_INT_STS	1'b 0	Reports whether an interrupt related to a cell under-voltage (UV) event has been detected. The flags located in RD_CELL_UV should be checked to identify which cell caused the interrupt. 1'b 0: Not detected 1'b 1: Detected



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

0	R	CELL_OV_INT_STS	1'b 0	Reports whether an interrupt related to a cell over-voltage (OV) event has been detected. The flags located in RD_CELL_OV should be checked to identify which cell caused the interrupt. 1'b 0: Not detected 1'b 1: Detected
---	---	-----------------	-------	--

RD_INT1 (11h)**Format:** Unsigned binary

The RD_INT1 command reports interrupt including NVM CRC fault, 5V/3V3/VDD UV, open wire, cell dead, cell mismatch, die temperature too hot, NTC PCB hot, ADC self-diagnostic, completed cell-balancing, pack voltage change, MOSFET driver error, NTC cell charging cold/hot.

Bits	Access	Bit Name	Default	Description
15	R	NTC_OT_INT_STS	1'b 0	Reports whether an interrupt related to an NTC over temperature event during charge has been detected. This signal is generated by combining the interrupt detection for all 10 NTC channels. So NTCx_CELL_OT_RT_STS (x=1~10) should be checked to identify the NTC source of the interrupt. 1'b 0: Not detected 1'b 1: Detected
14	R	NTC_UT_INT_STS	1'b 0	Reports whether an interrupt related to an NTC under temperature event during charge has been detected. This signal is generated by combining the interrupt detection for all 10 NTC channels. So NTCx_CELL_UT_RT_STS (x=1~10) should be checked to identify the NTC source of the interrupt. 1'b 0: Not detected 1'b 1: Detected
13	R	FET_DRIVER_INT_STS	1'b 0	Reports whether an interrupt related to a MOSFET driver error has been detected. MOSFET driver errors cover the following conditions: MOSFET timeout An issue detected during a short-circuit or OC fault clearing sequence. An issue during CHG or DSG SS, such as a short-circuit or an OC condition. 1'b 0: Not detected 1'b 1: Detected
12	R	PACKP_V_INT_STS	1'b 0	Reports whether an interrupt related to a PACKP voltage change event has been detected. 1'b 0: Not detected 1'b 1: Detected
11	R	BAL_DONE_INT_STS	1'b 0	Reports whether an interrupt related to a cell-balancing done event has been detected. 1'b 0: Not detected 1'b 1: Detected
10	R	SELF_TEST_INT_STS	1'b 0	Reports whether an interrupt related to an ADC self-diagnostic event has been detected. The SELF_TEST_STS_OV and SELF_TEST_STS_UV flags should be checked to identify the cause of the interrupt. 1'b 0: Not detected 1'b 1: Detected
9	R	RESERVED	N/A	Reserved. Do not change this register value.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

8	R	PCB_MNTR_HOT_INT_STS	1'b 0	Reports whether an interrupt related to the PCB NTC over temperature monitor has been detected. NTC1_PCB_MNTR_HOT_STS, NTC2_PCB_MNTR_HOT_STS, NTC3_PCB_MNTR_HOT_STS, and NTC4_PCB_MNTR_HOT_STS should be checked to identify which NTC caused the interrupt. 1'b 0: Not detected 1'b 1: Detected
7	R	DIE_TEMP_INT_STS	1'b 0	Reports whether an interrupt related to a digital die temperature event has been detected. 1'b 0: Not detected 1'b 1: Detected
6	R	CELL_MISMATCH_INT_STS	1'b 0	Reports whether an interrupt related to a cell mismatch has been detected. Flags in the RD_CELL_MSMT and CELL_MSMT_LOWER registers should be read to identify the cells triggering the event. 1'b 0: Not detected 1'b 1: Detected
5	R	CELL_DEAD_INT_STS	1'b 0	Reports whether an interrupt related to a cell dead has been detected. 1'b 0: Not detected 1'b 1: Detected
4	R	OPEN_WIRE_INT_STS	1'b 0	Reports whether an interrupt related to an open-wire event has been detected. The open wire status registers (located at the RD_OPENH and RD_OPENL addresses) must be checked to determine whether there are disconnected wires. Clearing OPEN_WIRE_GO before completing the detection will also result in an Open Wire event Interrupt. 1'b 0: Not detected 1'b 1: Detected
3	R	VDD_INT_STS	1'b 0	Reports whether an interrupt related to a UV condition on VDD has been detected. A VDD UV event can be triggered by the VDD analog UV comparator. The VDD analog comparator is always enabled. 1'b 0: Not detected 1'b 1: Detected
2	R	V3V3_INT_STS	1'b 0	Reports whether an interrupt related to a UV condition on 3V3 has been detected. This is triggered by monitoring the 3V3 ADC or the 3.3V analog comparator. The 3.3V UV analog comparator is always enabled, so when 3V3_INT_EN is enabled, it can trigger an interrupt even if 3V3_EN is disabled. 1'b 0: Not detected 1'b 1: Detected
1	R	V5_INT_STS	1'b 0	Reports whether an interrupt related to a UV condition on 5V has been detected. This is triggered by monitoring the 5V ADC or the 5V analog comparator. The 5V UV analog comparator is always enabled, so when 5V_INT_EN is enabled, it can trigger an interrupt even if 5V_EN is disabled. 1'b 0: Not detected 1'b 1: Detected



0	R	NVM_CRC_EVENT_INTERRUPT_STS	1'b 0	Reports whether an interrupt related to a NVM memory CRC fault has been detected. When enabled, this interrupt can be triggered by the following conditions: The NVM CRC calculation fails while restoring the NVM. NVM CRC is manually triggered using NVM_CRC_DO. In this scenario, an interrupt event is generated when a CRC check is completed (for either passing or failing). 1'b 0: Not detected 1'b 1: Detected
---	---	-----------------------------	-------	--

RD_INT2 (12h)**Format:** Unsigned binary

The RD_INT2 command reports interrupt including cell SOV, VTOP SOV/SUV, OC, SC, discharge OC1, discharge OC2, charge OC, discharge SC, fuse driver, real-time discharge OC1, real-time discharge OC2, real-time charge OC, real-time discharge SC.

Bits	Access	Bit Name	Default	Description
15	R	FUSE_DRIVER_INTERRUPT_STS	1'b 0	Reports whether a fuse driver interrupt has been detected. 1'b 0: Not detected 1'b 1: Detected
14:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12	R	SC_DCHG_RT_STS	1'b 0	Reports whether a discharge short-circuit event has been detected in real time. 1'b 0: Not detected 1'b 1: Detected
11	R	SC_DCHG_STS	1'b 0	Reports whether a latched discharge short-circuit event has been detected. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling) selected by SC_DCHG_INT_TYPE. For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. This status bit can be cleared with SHORT_CURR_INT_CLEAR. 1'b 0: Not detected 1'b 1: Detected
10	R	OC_CHG_RT_STS	1'b 0	Reports in real time whether a charge OC limit is detected. 1'b 0: Not detected 1'b 1: Detected
9	R	OC2_DCHG_RT_STS	1'b 0	Reports in real time whether a discharge OC limit 2 is detected. 1'b 0: Not detected 1'b 1: Detected
8	R	OC1_DCHG_RT_STS	1'b 0	Reports in real time whether a discharge OC limit 1 is detected. 1'b 0: Not detected 1'b 1: Detected



7	R	OC_CHG_STS	1'b 0	Indicates the latched charge OC limit interrupt status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. The type is defined by OC_CHG_INT_TYPE. This status bit can be cleared with OVER_CURR_INT_CLEAR. 1'b 0: Not detected 1'b 1: Detected
6	R	OC2_DCHG_STS	1'b 0	Indicates the latched discharge over-current (OC) limit 2 interrupt status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. The type is defined by OC_DCHG_INT_TYPE, with the type selection being common for both limit 1 and limit 2 interrupts. This status bit can be cleared with OVER_CURR_INT_CLEAR. 1'b 0: Not detected 1'b 1: Detected
5	R	OC1_DCHG_STS	1'b 0	Indicates the latched discharge over-current (OC) limit 1 interrupt status. The latched value depends on the selected interrupt type (e.g. rising, falling, or rising and falling). For a rising edge type, the bit is set to 1 if a rising edge is detected. For a falling edge type, the bit is set to 1 if a falling edge is detected. For a rising and falling edge type, the bit is set to 1 if either edge is detected. For a level, the bit is set to 1 if the source signal is high. The type is defined by OC_DCHG_INT_TYPE, with the type selection being common for both limit 1 and limit 2 interrupts. This status bit can be cleared with OVER_CURR_INT_CLEAR. 1'b 0: Not detected 1'b 1: Detected
4	R	SHORT_CURR_INT_STS	1'b 0	Reports whether an interrupt related to a short-circuit event has been detected. This signal is generated by combining the interrupt detection for discharge short-circuit and charge short-circuit. SC_CHG_STS and SC_DCHG_STS should be checked to identify the source of the interrupt. 1'b 0: Not detected 1'b 1: Detected



3	R	OVER_CURR_INT_STS	1'b 0	Reports whether an interrupt related to an over-current (OC) event has been detected. This signal is generated by combining the interrupt detection for discharge OC and charge OC. OC1_DCHG_STS, OC2_DCHG_STS, and OC_CHG_STS should be checked to identify the source of the interrupt. 1'b 0: Not detected 1'b 1: Detected
2	R	VTOP_SUV_INT_STATUS	1'b 0	Reports whether an SUV interrupt related to the battery stack positive terminal (VTOP) has been detected. 1'b 0: Not detected 1'b 1: Detected
1	R	VTOP_SOV_INT_STATUS	1'b 0	Reports whether an SOV interrupt related to the battery stack positive terminal (VTOP) has been detected. 1'b 0: Not detected 1'b 1: Detected
0	R	CELL_SOV_INT_STATUS	1'b 0	Reports whether an interrupt related to a cell serious over voltage (SOV) event has been detected. The flags located in RD_CELL_SOV should be checked to identify which cell caused the interrupt. 1'b 0: Not detected 1'b 1: Detected

INT0_CLR (13h)**Format:** Unsigned binary

The INT0_CLR command controls the bit-clear of interrupts, including cell and VTOP OV/UV, DSG/CHG MOSFET dead, NTC cell discharging cold/hot, watchdog event, AFE mode change, completed high-resolution voltage scan, completed CC, charger/load plug-in, battery pack current change, wrong CRC write.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	W	COMM_CRC_FAIL_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a wrong CRC write interrupt. This is a self-clearing register.
13	W	PACK_CURRENT_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a battery pack current change interrupt. This is a self-clearing register.
12	W	CONN_DET_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a charger/load connection interrupt. This is a self-clearing register.
11	W	CC_ACC_STS_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a CC completed interrupt. This is a self-clearing register.
10	W	VSCAN_DONE_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a completed high-resolution voltage scan interrupt. This is a self-clearing register.
9	W	AFE_MODE_CHANGE_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear an AFE mode change interrupt. This is a self-clearing register.
8	W	WDT_INTERRUPT_CLEAR	1'b 0	Writing 1 will clear the posted watchdog event interrupt and related notifying bit WDT_STS. This is a self-clearing register.
7	W	NTC_SOT_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a cell NTC severe over temperature monitor interrupt. This is a self-clearing register.
6	W	NTC_SUT_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a cell NTC severe under temperature monitor interrupt. This is a self-clearing register.
5	W	CHGFET_DEAD_INTERRUPT_CLEAR	1'b 0	Write 1 to this bit to clear a CHG MOSFET dead interrupt. This is a self-clearing register.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

4	W	DSGFET_DEAD_INT_CLEAR	1'b 0	Write 1 to this bit to clear a DSG MOSFET dead interrupt. This is a self-clearing register.
3	W	VTOP_OV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a battery pack OV interrupt. This is a self-clearing register.
2	W	VTOP_UV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a battery pack UV interrupt. This is a self-clearing register.
1	W	CELL_UV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell UV interrupt. This is a self-clearing register.
0	W	CELL_OV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell OV interrupt. This is a self-clearing register.

INT1_CLR (14h)
Format: Unsigned binary

The INT1_CLR command controls the bit-clear of interrupts, including NVM CRC, 5V/3V3/VDD UV, open-wire, cell dead, cell mismatch, NTC PCB hot, ADC self-diagnostic, completed cell-balancing, PACKP voltage change, MOSFET driver failure, NTC cell charging cold/hot, digital die temperature too hot.

Bits	Access	Bit Name	Default	Description
15	W	NTC_OT_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell NTC over temperature monitor interrupt. This is a self-clearing register.
14	W	NTC_UT_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell NTC under temperature monitor interrupt. This is a self-clearing register.
13	W	FET_DRIVER_INT_CLEAR	1'b 0	Write 1 to this bit to clear a MOSFET driver event interrupt. This is a self-clearing register.
12	W	PACKP_V_INT_CLEAR	1'b 0	Write 1 to this bit to clear a PACKP voltage change interrupt. This is a self-clearing register.
11	W	BAL_DONE_INT_CLEAR	1'b 0	Write 1 to this bit to clear a completed balancing interrupt. This is a self-clearing register.
10	W	SELF_TEST_INT_CLEAR	1'b 0	Write 1 to this bit to clear an ADC self-diagnostic interrupt. This is a self-clearing register.
9	R	RESERVED	N/A	Reserved. Do not change this register value.
8	W	PCB_MNTR_TEMP_INT_CLEAR	1'b 0	Write 1 to this bit to clear an PCB NTC over temperature monitor interrupt. This is a self-clearing register.
7	W	DIE_TEMP_INT_CLEAR	1'b 0	Write 1 to this bit to clear a digital die temperature too hot event interrupt. This is a self-clearing register.
6	W	CELL_MISMATCH_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell mismatch interrupt. This is a self-clearing register.
5	W	CELL_DEAD_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell dead interrupt. This is a self-clearing register.
4	W	OPEN_WIRE_INT_CLEAR	1'b 0	Write 1 to this bit to clear an open wire interrupt. This is a self-clearing register.
3	W	VDD_INT_CLEAR	1'b 0	Write 1 to this bit to clear a VDD UV interrupt. This is a self-clearing register.
2	W	V3V3_INT_CLEAR	1'b 0	Write 1 to this bit to clear a 3V3 UV interrupt. This is a self-clearing register.
1	W	V5_INT_CLEAR	1'b 0	Write 1 to this bit to clear a 5V UV interrupt. This is a self-clearing register.
0	W	NVM_CRC_EVENT_INT_CLEAR	1'b 0	Write 1 to this bit to clear an NVM CRC interrupt. This is a self-clearing register.

**INT2_CLR (15h)****Format:** Unsigned binary

The INT2_CLR command controls the bit-clear of interrupts, including cell SOV, VTOP SOV/SUV, OC, SC, and fuse driver event.

Bits	Access	Bit Name	Default	Description
15	W	FUSE_DRIVER_INT_CLEAR	1'b 0	Write 1 to this bit to clear a fuse driver interrupt. This is a self-clearing register.
14:5	R	RESERVED	N/A	Reserved. Do not change this register value.
4	W	SHORT_CURR_INT_CLEAR	1'b 0	Write 1 to this bit to clear a short-circuit interrupt. This is a self-clearing register.
3	W	OVER_CURR_INT_CLEAR	1'b 0	Write 1 to this bit to clear an over-current interrupt. This is a self-clearing register.
2	W	VTOP_SUV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a battery pack SUV interrupt. This is a self-clearing register.
1	W	VTOP_SOV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a battery pack SOV interrupt. This is a self-clearing register.
0	W	CELL_SOV_INT_CLEAR	1'b 0	Write 1 to this bit to clear a cell SOV interrupt. This is a self-clearing register.

INT0_EN (16h)**Format:** Unsigned binary

The INT0_EN command controls enable/disable of interrupt, including cell OV/UV, VTOP OV/UV, CHG/DSG MOSFETs dead, NTC cell discharging cold/hot, detected watchdog communication, battery pack current direction changes, AFE mode change, completed high-resolution voltage scan, completed CC, charger/load plug-in, pack current switching status, wrong CRC write. It also controls enable/disable of interrupt reporting to xALERT pin.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	INT_ALERT_CTRL	1'b 1	Enables interrupt reporting to the xALERT pin. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W	COMM_CRC_FAIL_INT_EN	1'b 0	Enables interrupt reporting for a wrong CRC write event. 1'b 0: Disabled 1'b 1: Enabled
13	R/W	PACK_CURRENT_INT_EN	1'b 0	Enables interrupt reporting for battery pack current direction changes. 1'b 0: Disabled 1'b 1: Enabled
12	R/W	CONN_DET_INT_EN	1'b 0	Enables interrupt reporting for charger/load connection. 1'b 0: Disabled 1'b 1: Enabled
11	R/W	CC_ACC_INT_EN	1'b 0	Enables interrupt reporting for a CC completion. 1'b 0: Disabled 1'b 1: Enabled


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R/W	VSCAN_DONE_INT_EN	1'b 0	Enables interrupt reporting for High resolution VADC Scan Completed. 1'b 0: Disabled 1'b 1: Enabled
9	R/W	AFE_MODE_CHANGE_INT_EN	1'b 0	Enables interrupt reporting an AFE mode change. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	WDT_INT_EN	1'b 0	Enables interrupt reporting for watchdog communication. 1'b 0: Disabled 1'b 1: Enabled
7	R/W	NTC_SOT_INT_EN	1'b 0	Enables interrupt reporting for cell NTC severe over temperature monitoring conditions. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	NTC_SUT_INT_EN	1'b 0	Enables interrupt reporting for cell NTC severe under temperature monitoring conditions. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	CHGFET_DEAD_INT_EN	1'b 0	Enables interrupt reporting for a CHG MOSFET dead event. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	DSGFET_DEAD_INT_EN	1'b 0	Enables interrupt reporting for a DSG MOSFET dead event. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	VTOP_OV_INT_EN	1'b 0	Enables interrupt reporting for battery pack OV conditions. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	VTOP_UV_INT_EN	1'b 0	Enables interrupt reporting for battery pack UV conditions. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	CELL_UV_INT_EN	1'b 0	Enables interrupt reporting for cell UV conditions. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	CELL_OV_INT_EN	1'b 0	Enables interrupt reporting for cell OV conditions. 1'b 0: Disabled 1'b 1: Enabled

INT1_EN (17h)
Format: Unsigned binary

The INT1_EN command controls enable/disable of interrupt, including NVM CRC event, 5V/3V3/VDD UV, open-wire, cell dead, cell mismatch, digital die temperature too hot, NTC PCB hot, ADC self-diagnostic, completed cell-balancing, PACKP voltage change, MOSFET driver failure, NTC cell charging cold/hot.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15	R/W	NTC_OT_INT_EN	1'b 0	Enables interrupt reporting for cell NTC over temperature monitoring conditions. 1'b 0: Disabled 1'b 1: Enabled
14	R/W	NTC_UT_INT_EN	1'b 0	Enables interrupt reporting for cell NTC under temperature monitoring conditions. 1'b 0: Disabled 1'b 1: Enabled
13	R/W	FET_DRIVER_INT_EN	1'b 0	Enables interrupt reporting for MOSFET driver issues. 1'b 0: Disabled 1'b 1: Enabled
12	R/W	PACKP_V_INT_EN	1'b 0	Enables interrupt reporting for PACKP voltage changes. 1'b 0: Disabled 1'b 1: Enabled
11	R/W	BAL_DONE_INT_EN	1'b 0	Enables interrupt reporting for completed cell-balancing. 1'b 0: Disabled 1'b 1: Enabled
10	R/W	SELF_TEST_INT_EN	1'b 0	Enables interrupt reporting for an ADC self-diagnostic OV/UV condition. 1'b 0: Disabled 1'b 1: Enabled
9	R	RESERVED	N/A	Reserved. Do not change this register value.
8	R/W	PCB_MNTR_HOT_INT_EN	1'b 0	Enables interrupt reporting for an NTC over temperature PCB condition. 1'b 0: Disabled 1'b 1: Enabled
7	R/W	DIE_TEMP_INT_EN	1'b 0	Enables interrupt reporting for a die over-temperature event. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	CELL_MISMATCH_INT_EN	1'b 0	Enables interrupt reporting for a cell mismatch condition. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	CELL_DEAD_INT_EN	1'b 0	Enables interrupt reporting for a cell dead condition. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	OPEN_WIRE_INT_EN	1'b 0	Enables interrupt reporting for completed open-wire detection. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	VDD_INT_EN	1'b 0	Enables interrupt reporting for a VDD UV event. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	V3V3_INT_EN	1'b 0	Enables interrupt reporting for a 3V3 UV event. 1'b 0: Disabled 1'b 1: Enabled



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

1	R/W	V5_INT_EN	1'b 0	Enables interrupt reporting for a 5V UV event. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	NVM_CRC_EVENT_INT_EN	1'b 0	Enables interrupt reporting for an NVM CRC event. 1'b 0: Disabled 1'b 1: Enabled

INT2_EN (18h)**Format:** Unsigned binary

The INT2_EN command controls enable/disable of interrupt, including cell SOV, VTOP SOV/SUV, OC, SC, discharge OC1, discharge OC2, charge OC, fuse driver.

Bits	Access	Bit Name	Default	Description
15	R/W	FUSE_DRIVER_INT_EN	1'b 0	Enables interrupt reporting for fuse driver conditions. 1'b 0: Disabled 1'b 1: Enabled
14:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7	R/W	OC_CHG_INT_EN	1'b 0	Enables bit-level control for a charge OC interrupt. OVER_CURR_INT_EN must be enabled for this bit to be effective. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	OC2_DCHG_INT_EN	1'b 0	Enables bit-level control for a discharge OC2 interrupt. OVER_CURR_INT_EN must be enabled for this bit to be effective. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	OC1_DCHG_INT_EN	1'b 0	Enables bit-level control for a discharge OC1 interrupt. OVER_CURR_INT_EN must be enabled for this bit to be effective. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	SHORT_CURR_INT_EN	1'b 0	Enables interrupt reporting for short-circuit conditions. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	OVER_CURR_INT_EN	1'b 0	Enables interrupt reporting for OC conditions. Note that there are individual enable/disable control bits: OC_CHG_INT_EN, OC2_DCHG_INT_EN and OC1_DCHG_INT_EN. OVER_CURR_INT_EN acts as a master enabler, so it must be enabled for the charger and discharge interrupt to work. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	VTOP_SUV_INT_EN	1'b 0	Enables interrupt reporting for battery pack SUV conditions. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	VTOP_SOV_INT_EN	1'b 0	Enables interrupt reporting for battery pack SOV conditions. 1'b 0: Disabled 1'b 1: Enabled



0	R/W	CELL_SOV_INT_EN	1'b 0	Enables interrupt reporting for cell SOV conditions. 1'b 0: Disabled 1'b 1: Enabled
---	-----	-----------------	-------	---

INT_TYPE0 (19h)**Format:** Unsigned binary

The INT_TYPE0 command configures interrupt triggering logic: input level, rising edge, falling edge, or rising and falling edges. Interrupt includes cell and VTOP UV/OV, NTC cell discharging and charging cold/hot.

Bits	Access	Bit Name	Default	Description
15:14	R/W	CELL_OV_INT_TYPE	2'b 00	Controls the triggering logic for a cell OV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
13:12	R/W	CELL_UV_INT_TYPE	2'b 00	Controls the triggering logic for a cell UV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
11:10	R/W	NTC_OT_INT_TYPE	2'b 00	Controls the triggering logic for a cell NTC over temperature CHG interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
9:8	R/W	NTC_UT_INT_TYPE	2'b 00	Controls the triggering logic for a cell NTC under temperature CHG interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
7:6	R/W	NTC_SOT_INT_TYPE	2'b 00	Controls the triggering logic for a cell NTC severe over temperature DSG interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal



5:4	R/W	NTC_SUT_INT_TYPE	2'b 00	Controls the triggering logic for a cell NTC severe under temperature DSG interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
3:2	R/W	VTOP_OV_INT_TYPE	2'b 00	Controls the triggering logic for a battery pack OV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
1:0	R/W	VTOP_UV_INT_TYPE	2'b 00	Controls the triggering logic for a battery pack UV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal

INT_TYPE1 (1Ah)**Format:** Unsigned binary

The INT_TYPE1 command configures interrupt triggering logic: input level, rising edge, falling edge, or rising and falling edges. Interrupt includes 5V/3V3/VDD UV, charge OC, discharge OC, and discharge SC.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13:12	R/W	VDD_INT_TYPE	2'b 00	Controls the triggering logic for a VDD UV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
11:10	R/W	V3V3_INT_TYPE	2'b 00	Controls the triggering logic for a 3V3 UV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
9:8	R/W	V5_INT_TYPE	2'b 00	Controls the triggering logic for a 5V UV interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal



				2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5:4	R/W	SC_DCHG_INT_TYPE	2'b 00	Controls the triggering logic for discharge SC interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
3:2	R/W	OC_DCHG_INT_TYPE	2'b 00	Controls the triggering logic for discharge OC interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
1:0	R/W	OC_CHG_INT_TYPE	2'b 00	Controls the triggering logic for charge OC interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal

INT_TYPE2 (1Bh)**Format:** Unsigned binary

The INT_TYPE2 command configures interrupt triggering logic: input level, rising edge, falling edge, or rising and falling edges. Interrupt includes cell dead, cell mismatch, die temperature, NTC PCB hot, and ADC self-diagnostic.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11:10	R/W	SELF_TEST_INT_TYPE	2'b 00	Controls the triggering logic for an ADC self-diagnostic interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
9:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7:6	R/W	PCB_MNTR_TEMP_INT_TYPE	2'b 00	Controls the triggering logic for an PCB NTC over temperature monitor interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal



				2'b 11: Interrupt generated on the rising and falling edges of the input status signal
5:4	R/W	DIE_TEMP_INT_TYPE	2'b 00	Controls the triggering logic for a die temperature interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
3:2	R/W	CELL_MISMATCH_INT_TYPE	2'b 00	Controls the triggering logic for a cell mismatch interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal
1:0	R/W	CELL_DEAD_INT_TYPE	2'b 00	Controls the triggering logic for a cell dead interrupt. 2'b 00: Interrupt generated on input status signal high 2'b 01: Interrupt generated on the rising edge of the input status signal 2'b 10: Interrupt generated on the falling edge of the input status signal 2'b 11: Interrupt generated on the rising and falling edges of the input status signal

MASK_INT0 (1Ch)**Format:** Unsigned binary

The MASK_INT0 command controls enable/disable the mask of interrupt. Interrupt includes cell and VTOP OV/UV, OC, SC, NTC cell discharging and charging cold/hot.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R/W	NTCS_CELL_OT_MASK	1'b 0	Enables clearing the NTC cell over temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	NTCS_CELL_UT_MASK	1'b 0	Enables clearing the NTC cell under temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
7	R/W	NTCS_CELL_SOT_MASK	1'b 0	Enables clearing the NTC cell severe over temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled



6	R/W	NTCS_CELL_SUT_MASK	1'b 0	Enables clearing the NTC cell severe under temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	SC_MASK	1'b 0	Enables clearing the SC interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	OVER_CURR_MASK	1'b 0	Enables clearing the OC interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	VTOP_OV_MASK	1'b 0	Enables clearing the VTOP OV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	VTOP_UV_MASK	1'b 0	Enables clearing the VTOP UV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	CELL_UV_MASK	1'b 0	Enables clearing the cell UV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	CELL_OV_MASK	1'b 0	Enables clearing the cell OV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled

MASK_INT1 (1Dh)**Format:** Unsigned binary

The MASK_INT1 command controls enable/disable the mask of interrupt. Interrupt includes ADC self-diagnostic, 5V/3V3/VDD UV, cell mismatch, die temperature too hot, NTC PCB hot, cell SOV, VTOP SUV/SOV, and DSG/CHG MOSFET dead.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11	R/W	CHGFET_DEAD_MASK	1'b 0	Enables clearing the CHG MOSFET dead interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R/W	DSGFET_DEAD_MASK	1'b 0	Enables clearing the DSG MOSFET dead interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
9	R/W	VTOP_SOV_MASK	1'b 0	Enables clearing the VTOP SOV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	VTOP_SUV_MASK	1'b 0	Enables clearing the VTOP SUV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
7	R/W	CELL_SOV_MASK	1'b 0	Enables clearing the cell SOV interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	PCB_MNTR_HOT_MASK	1'b 0	Enables clearing the PCB NTC over temperature monitor interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	DIE_TEMP_DIG_MASK	1'b 0	Enables clearing the die temperature interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	CELL_MSMT_MASK	1'b 0	Enables clearing the cell mismatch interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	VDD_MASK	1'b 0	Enables clearing the VDD interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	V3V3_MASK	1'b 0	Enables clearing the 3V3 interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	V5_MASK	1'b 0	Enables clearing the 5V interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled



0	R/W	SELF_TEST_MASK	1'b 0	Enables clearing the ADC self-diagnostic interrupt flag and pulls the interrupt pin low, unless there are other interrupts pending. When disabled, the interrupt process can set the interrupt flag again. 1'b 0: Disabled 1'b 1: Enabled
---	-----	----------------	-------	---

FAULT_EN1 (1Eh)**Format:** Unsigned binary

The FAULT_EN1 command controls enable/disable of fault monitoring. Fault monitoring includes cell and VTOP UV/OV, cell dead, cell mismatch, NTC cell discharging and charging cold/hot, NTC PCB hot, die temperature too hot, discharge OC1 and 2, charge OC, and discharge SC.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	CELL_MSMT_FAULT_EN	1'b 0	Enables a cell mismatch fault detection and reaction. When enabled, this cell mismatch event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W (can lock to read-only)	CELL_DEAD_FAULT_EN	1'b 0	Enables a cell dead fault detection and reaction. When enabled, this cell dead event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	SC_DCHG_FAULT_EN	1'b 1	Enables a discharge SC fault detection and reaction. When enabled, this SC event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R/W (can lock to read-only)	OC_CHG_FAULT_EN	1'b 1	Enables a charge OC fault detection and reaction. When enabled, this OC event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
11	R/W (can lock to read-only)	OC2_DCHG_FAULT_EN	1'b 1	Enables a discharge OC 2 fault detection and reaction. When enabled, this OC event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
10	R/W (can lock to read-only)	OC1_DCHG_FAULT_EN	1'b 1	Enables a discharge OC 1 fault detection and reaction. When enabled, this OC event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

9	R/W (can lock to read-only)	DIE_TEMP_DIG_FAULT_EN	1'b 1	Enables a die over-temperature fault detection and reaction. When enabled, this die over-temperature event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
8	R/W (can lock to read-only)	PCB_MNTR_FAULT_EN	1'b 0	Enables a PCB NTC over temperature monitor fault detection and reaction. When enabled, this PCB NTC over temperature monitor event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	NTC_CELL_COLD_DISCHARGE_FAULT_EN	1'b 0	Enables a cell NTC under temperature discharging monitor fault detection and reaction. When enabled, this cell NTC under temperature discharging monitor event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	NTC_CELL_HOT_DISCHARGE_FAULT_EN	1'b 0	Enables a cell NTC over temperature discharging monitor fault detection and reaction. When enabled, this cell NTC over temperature discharging monitor event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	NTC_CELL_COLD_CHARGE_FAULT_EN	1'b 0	Enables a cell NTC under temperature charging monitor fault detection and reaction. When enabled, this cell NTC under temperature charging monitor event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	NTC_CELL_HOT_CHARGE_FAULT_EN	1'b 0	Enables a cell NTC over temperature charging monitor fault detection and reaction. When enabled, this cell NTC over temperature charging monitor event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W (can lock to read-only)	VTOP_OV_FAULT_ENABLE_CTRL	1'b 0	Enables a battery pack OV fault detection and reaction. When enabled, this battery pack OV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W (can lock to read-only)	VTOP_UV_FAULT_ENABLE_CTRL	1'b 0	Enables a battery pack UV fault detection and reaction. When enabled, this battery pack UV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



1	R/W (can lock to read-only)	CELL_OV_FAULT_EN	1'b 0	Enables a cell OV fault detection and reaction. When enabled, this cell OV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
0	R/W (can lock to read-only)	CELL_UV_FAULT_EN	1'b 0	Enables a cell UV fault detection and reaction. When enabled, this cell UV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

FAULT_EN2 (1Fh)**Format:** Unsigned binary

The FAULT_EN2 command controls enable/disable of fault monitoring and reaction. After fault detected, reactions are divided into two types: AFE moves into fault state or GHV/Fuse pin output high. The former reaction related fault includes cell SOV, VTOP SUV/SOV, ADC self-diagnostic, 3V3/5V UV, open wire, NVM CRC, and the latter includes cell dead, cell SOV, cell mismatch, VTOP SUV/SOV, and DSG/CHG MOSFET dead, ADC self-diagnostic, and 3V3/5V UV.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	LDO_UV_FUSE_EN	1'b 0	Enables a 3V3, 5V or VDD UV fuse reaction. When enabled, this 3V3 UV or 5V UV event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W (can lock to read-only)	ADC_FAIL_FUSE_EN	1'b 0	Enables an ADC self-diagnostic fuse reaction. When enabled, this an ADC self-diagnostic event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	CHGFET_DEAD_FUSE_EN	1'b 0	Enables a CHG MOSFET dead fuse reaction. When enabled, this CHG MOSFET dead event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R/W (can lock to read-only)	DSGFET_DEAD_FUSE_EN	1'b 0	Enables a DSG MOSFET dead fuse reaction. When enabled, this DSG MOSFET dead event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
11	R/W (can lock to read-only)	CELL_MSMT_FUSE_EN	1'b 0	Enables a cell mismatch fuse reaction. When enabled, this cell mismatch event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R/W (can lock to read-only)	VTOP_SUV_FUSE_EN	1'b 0	Enables a battery pack SUV fuse reaction. When enabled, this battery pack SUV event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R/W (can lock to read-only)	VTOP_SOV_FUSE_EN	1'b 0	Enables a battery pack SOV fuse reaction. When enabled, this battery pack SOV event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
8	R/W (can lock to read-only)	CELL_SOV_FUSE_EN	1'b 0	Enables a cell SOV fuse reaction. When enabled, this cell SOV event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	CELL_DEAD_FUSE_EN	1'b 0	Enables a cell dead fuse reaction. When enabled, this cell dead event drives the FUSE pin to high. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	NVM_FAULT_EN	1'b 0	Enables an NVM CRC fault detection and reaction. When enabled, this NVM CRC event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5	R/W (can lock to read-only)	OPEN_WIRE_FAULT_EN	1'b 0	Enables an open wire fault detection and reaction. When enabled, this open wire event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	LDO_FAULT_EN	1'b 0	Enables a 3V3, 5V, or VDD UV fault detection and reaction. 3V3, 5V, or VDD UV is triggered by monitoring its ADC or analog comparator. When enabled, 3V3, 5V or VDD UV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W (can lock to read-only)	ADC_FAIL_FAULT_EN	1'b 0	Enables an ADC self-diagnostic fault detection and reaction. When enabled, this an ADC self-diagnostic event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



2	R/W (can lock to read-only)	VTOP_SUV_FAULT_EN	1'b 0	Enables a battery pack SUV fault detection and reaction. When enabled, this battery pack SUV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read-only)	VTOP_SOV_FAULT_EN	1'b 0	Enables a battery pack SOV fault detection and reaction. When enabled, this battery pack SOV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
0	R/W (can lock to read-only)	CELL_SOV_FAULT_EN	1'b 0	Enables a cell SOV fault detection and reaction. When enabled, this cell SOV event puts the AFE in a fault state. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

FET_DEAD_LIM (20h)**Format:** Unsigned binary

The FET_DEAD_LIM command configures CHG and DSG MOSFETs dead current thresholds, and thresholds range and LSB.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13:12	R/W (can lock to read-only)	CHGFET_DEAD_CURR_RNG	2'b 00	Selects the range and LSB for the CHG MOSFET dead current threshold (CHGFET_DEAD_CURR_LIM). 2'b 00: Using 1x range, 1.2mV/ LSB, 80mV FSR 2'b 01: Using standby current range 2'b 10: Using 3x range, 3.6mV/ LSB, 240mV FSR 2'b 11: Using standby current range These bits allow MTP
11:7	R/W (can lock to read-only)	CHGFET_DEAD_CURR_LIM	5'b 10000	Sets the CHG MOSFET dead current threshold. With standby current range (CHGFET_DEAD_CURR_RNG = 1 or 3): LSB: 125µV Offset: 250µV Range: 250µV to 625µV, when CHGFET_DEAD_CURR_LIM > 3, the threshold is 625µV With 1x range (CHGFET_DEAD_CURR_RNG = 0) LSB: 1.2mV Offset: 1.2mV Range: 1.2mV to 38.4mV With 3x range (CHGFET_DEAD_CURR_RNG = 2) LSB: 3.6mV Offset: 3.6mV Range: 3.6mV to 115.2mV These bits allow MTP
6:5	R/W (can lock to read-only)	DSGFET_DEAD_CURR_RNG	2'b 00	Selects the range and LSB for the DSG MOSFET dead current threshold (DSGFET_DEAD_CURR_LIM). 2'b 00: Using 1x range, 1.2mV / LSB, 38.4mV FSR 2'b 01: Using standby current range 2'b 10: Using 3x range, 3.6mV / LSB, 115.2mV FSR



				2'b 11: Using standby current range These bits allow MTP
4:0	R/W (can lock to read-only)	DSGFET_DEAD_CURR_LIM	5'b 10000	Sets the DSG MOSFET dead current threshold. With standby current range (DSGFET_DEAD_CURR_RNG = 1 or 3): LSB: 125μV Offset: 250μV Range: 250μV to 625μV, when DSGFET_DEAD_CURR_LIM > 3, the threshold is 625μV With 1x range (DSGFET_DEAD_CURR_RNG = 0) LSB: 1.2mV Offset: 1.2mV Range: 1.2mV to 38.4mV With 3x range (DSGFET_DEAD_CURR_RNG = 2) LSB: 3.6mV Offset: 3.6mV Range: 3.6mV to 115.2mV These bits allow MTP

FET_DEAD_DEG (21h)**Format:** Unsigned binary

The FET_DEAD_DEG command configures CHG and DSG MOSFETs dead current deglitch and deglitch bit weight.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	CHGFET_DEAD_DG_RNG	1'b 0	Sets the CHG MOSFET dead current deglitch bit weight, and selects the range for CHGFET_DEAD_DGL. 1'b 0: 5ms LSB / 315ms FSR 1'b 1: 40ms LSB / 2520ms FSR This bit allows MTP
12:7	R/W (can lock to read-only)	CHGFET_DEAD_DG_L	6'b 010110	Sets the CHG MOSFET dead current deglitch value and cannot set it to 0. With CHGFET_DEAD_DG_RNG = 0: LSB: 5ms Range: 5ms to 315ms Value (ms) = Setting x 5 With CHGFET_DEAD_DG_RNG = 1: LSB: 40ms Range: 40ms to 2520ms Value (ms) = Setting x 40 These bits allow MTP
6	R/W (can lock to read-only)	DSGFET_DEAD_DG_RNG	1'b 0	Sets the DSG MOSFET dead current deglitch bit weight, and selects the range for DSGFET_DEAD_DGL. 1'b 0: 5ms LSB / 315ms FSR 1'b 1: 40ms LSB / 2520ms FSR This bit allows MTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

5:0	R/W (can lock to read-only)	DSGFET_DEAD_DGL	6'b 010110	Sets the DSG MOSFET dead current deglitch value and cannot set it to 0. With DSGFET_DEAD_DG_RNG = 0: LSB: 5ms Range: 5ms to 315ms Value (ms) = Setting x 5 With DSGFET_DEAD_DG_RNG = 1: LSB: 40ms Range: 40ms to 2520ms Value (ms) = Setting x 40 These bits allow MTP
-----	--------------------------------	-----------------	------------	--

FUSE_CFG (22h)
Format: Unsigned binary

The FUSE_CFG command sets the fuse driver pull-up keep time, and the delay time from fuse triggered to AFE sleeping.

Bits	Access	Bit Name	Default	Description
15:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7:4	R/W	FUSE_DRIVER_TIME	4'b 0000	Sets the fuse driver pull high keep time. When FUSE_DRIVER_TIME= 0, the time is disable and it always remains pull up. LSB: 4s Range: 4s to 60s Value (s) = Setting x 4 These bits allow MTP
3:0	R/W	FUSE_SLEEP_DELAY	4'b 0000	Sets the delay time from fuse triggered to sleep mode. LSB: 1s Offset: 1s Range: 1s to 16s Value (s) = Setting x 1 + 1 These bits allow OTP

VTOP_UV_SLEP_CFG (23h)
Format: Unsigned binary

The VTOP_UV_SLEP_CFG command configures the VTOP UV threshold for AFE's sleep mode.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11:0	R/W (can lock to read-only)	VTOP_UV_SLEEP_THR	12'b 000000000000	Sets VTOP UV threshold for sleep mode. Once the VTOP is lower than this threshold, IC goes into sleep mode. LSB: 19.53125mV Range: 0mV to 79980.46875mV Value (mV) = Setting x 19.53125 These bits allow MTP

DSGOC_LIM (24h)
Format: Unsigned binary

The DSGOC_LIM command configures discharge OC1 and OC2 thresholds and thresholds range and LSB.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.



13	R/W (can lock to read-only)	OC2_DCHG_RNG	1'b 1	Selects the range and LSB for the discharge OC level 2 threshold. 1'b 0: 1.2mV/ LSB, 38.4mV FSR 1'b 1: 3.6mV/ LSB, 115.2mV FSR This bit allows MTP
12:8	R/W (can lock to read-only)	OC2_DCHG_LIM	5'b 01001	Sets the discharge OC 2 threshold, which should be higher than OC 1 threshold. With a 1x range (OC2_DCHG_RNG = 0): LSB: 1.2mV Offset: 1.2mV Range: 1.2mV to 38.4mV Value (mV) = Setting x 1.2 + 1.2 With a 3x range (OC2_DCHG_RNG = 1): LSB: 3.6mV Offset: 3.6mV Range: 3.6mV to 115.2mV Value (mV) = Setting x 3.6 + 3.6 These bits allow MTP
7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W (can lock to read-only)	OC1_DCHG_RNG	1'b 0	Selects the range and LSB for the discharge OC level 1 threshold. 1'b 0: 1.2mV/ LSB, 38.4mV FSR 1'b 1: 3.6mV/ LSB, 115.2mV FSR This bit allows MTP
4:0	R/W (can lock to read-only)	OC1_DCHG_LIM	5'b 10000	Sets the discharge OC 1 threshold. With a 1x range (OC1_DCHG_RNG = 0): LSB: 1.2mV Offset: 1.2mV Range: 1.2mV to 38.4mV Value (mV) = Setting x 1.2 + 1.2 With a 3x range (OC1_DCHG_RNG = 1): LSB: 3.6mV Offset: 3.6mV Range: 3.6mV to 115.2mV Value (mV) = Setting x 3.6 + 3.6 These bits allow MTP

DSGOC_DEG (25h)**Format:** Unsigned binary

The DSGOC_DEG command configures discharge OC1 and OC2 deglitch value and deglitch bit weight.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R/W (can lock to read-only)	OC2_DCHG_DGL_RNG	1'b 0	Sets the discharge OC 2 deglitch bit weight, and selects the range for OC2_DCHG_DGL. 1'b 0: 5ms LSB / 315ms FSR 1'b 1: 40ms LSB / 2520ms FSR This bit allows MTP
13:8	R/W (can lock to read-only)	OC2_DCHG_DGL	6'b 000100	Sets the discharge OC 2 deglitch value. When OC2_DCHG_DGL = 0, there is no deglitch, regardless of the LSB. This means an OC event is flagged as soon as it is detected. The response time is about 60µs.



				With OC2_DCHG_DGL_RNG = 0: LSB: 5ms Range: 0ms to 315ms Value (ms) = Setting x 5 With OC2_DCHG_DGL_RNG = 1: LSB: 40ms Range: 0ms to 2520ms Value (ms) = Setting x 40 These bits allow MTP
7	R	RESERVED	N/A	Reserved. Do not change this register value.
6	R/W (can lock to read-only)	OC1_DCHG_DGL_RNG	1'b 0	Sets the discharge OC 1 deglitch bit weight, and selects the range for OC1_DCHG_DGL. 1'b 0: 5ms LSB / 315ms FSR 1'b 1: 40ms LSB / 2520ms FSR This bit allows MTP
5:0	R/W (can lock to read-only)	OC1_DCHG_DGL	6'b 010110	Sets the discharge OC 1 deglitch value. When OC1_DCHG_DGL = 0, there is no deglitch, regardless of the LSB. This means an OC event is flagged as soon as it is detected. The response time is about 60µs. With OC1_DCHG_DGL_RNG = 0: LSB: 5ms Range: 0ms to 315ms Value (ms) = Setting x 5 With OC1_DCHG_DGL_RNG = 1: LSB: 40ms Range: 0ms to 2520ms Value (ms) = Setting x 40 These bits allow MTP

CHGOC_DEG (26h)**Format:** Unsigned binary

The CHGOC_DEG command controls enable/disable of charge OC threshold, deglitch, threshold range and LSB, and deglitch range and LSB.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R/W (can lock to read-only)	OC_CHG_DGL_RNG	1'b 0	Sets the charge OC deglitch bit weight, and selects the range for OC_CHG_DG. 1'b 0: 5ms LSB / 315ms FSR 1'b 1: 40ms LSB / 2520ms FSR This bit allows MTP
13:8	R/W (can lock to read-only)	OC_CHG_DG	6'b 000100	Sets the charge OC deglitch value. When OC_CHG_DG = 0, there is no deglitch, regardless of the LSB. This means an OC event is flagged as soon as it is detected. The response time is about 60µs. With OC_CHG_DGL_RNG = 0: LSB: 5ms Range: 0ms to 315ms Value (ms) = Setting x 5 With OC2_DCHG_DGL_RNG = 1: LSB: 40ms Range: 0ms to 2520ms Value (ms) = Setting x 40 These bits allow MTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W (can lock to read-only)	OC_CHG_RNG	1'b 0	Selects the range and LSB selector for the charge OC threshold (OC_CHG_LIM). 1'b 0: 1.2mV/ LSB, 38.4mV FSR 1'b 1: 3.6mV/ LSB, 115.2mV FSR This bit allows MTP
4:0	R/W (can lock to read-only)	OC_CHG_LIM	5'b 10000	Sets the charge OC threshold. With a 1x range (OC_CHG_RNG = 0): LSB: 1.2mV Offset: 1.2mV Range: 1.2mV to 38.4mV Value (mV) = Setting x 1.2 + 1.2 With a 3x range (OC_CHG_RNG = 1): LSB: 3.6mV Offset: 3.6mV Range: 3.6mV to 115.2mV Value (mV) = Setting x 3.6 + 3.6 These bits allow MTP

DSGSC_CFG (27h)
Format: Unsigned binary

The DSGSC_CFG command controls enable/disable of discharge short-circuit threshold, deglitch time, and thresholds range and LSB.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:8	R/W (can lock to read-only)	SC_DCHG_DG	7'b 0000001	Configures the discharge short-circuit deglitch time. When SC_DCHG_DG = 0, there is no deglitch time. LSB: 5μs Range: 0μs to 635μs Value (μs) = Setting x 5 These bits allow MTP
7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W (can lock to read-only)	SC_DCHG_RNG	1'b 0	Selects the range and LSB for the discharge short-circuit threshold (SC_DCHG_LIM). 1'b 0: 5.5mV/ LSB, 176mV FSR 1'b 1: 16.5mV/ LSB, 528mV FSR This bit allows MTP
4:0	R/W (can lock to read-only)	SC_DCHG_LIM	5'b 10001	Configures the discharge short-circuit threshold. With a 1x range (SC_DCHG_RNG = 0): LSB: 5.5mV Offset: 5.5mV Range: 5.5mV to 176mV Value (mV) = Setting x 5.5 + 5.5 With a 3x range (SC_DCHG_RNG = 1): LSB: 16.5mV Offset: 16.5mV Range: 16.5mV to 528mV Value (mV) = Setting x 16.5 + 16.5 These bits allow MTP

**RD_CELL_UV (28h)****Format:** Unsigned binary

The RD_CELL_UV command reports whether a UV condition has been detected on cell 1-16.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 16. 1'b 0: This cell is not UV 1'b 1: This cell is UV
14	R	CELL_15_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 15. 1'b 0: This cell is not UV 1'b 1: This cell is UV
13	R	CELL_14_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 14. 1'b 0: This cell is not UV 1'b 1: This cell is UV
12	R	CELL_13_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 13. 1'b 0: This cell is not UV 1'b 1: This cell is UV
11	R	CELL_12_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 12. 1'b 0: This cell is not UV 1'b 1: This cell is UV
10	R	CELL_11_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 11. 1'b 0: This cell is not UV 1'b 1: This cell is UV
9	R	CELL_10_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 10. 1'b 0: This cell is not UV 1'b 1: This cell is UV
8	R	CELL_9_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 9. 1'b 0: This cell is not UV 1'b 1: This cell is UV
7	R	CELL_8_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 8. 1'b 0: This cell is not UV 1'b 1: This cell is UV
6	R	CELL_7_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 7. 1'b 0: This cell is not UV 1'b 1: This cell is UV
5	R	CELL_6_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 6. 1'b 0: This cell is not UV 1'b 1: This cell is UV
4	R	CELL_5_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 5. 1'b 0: This cell is not UV 1'b 1: This cell is UV
3	R	CELL_4_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 4. 1'b 0: This cell is not UV 1'b 1: This cell is UV


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

2	R	CELL_3_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 3. 1'b 0: This cell is not UV 1'b 1: This cell is UV
1	R	CELL_2_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 2. 1'b 0: This cell is not UV 1'b 1: This cell is UV
0	R	CELL_1_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 1. 1'b 0: This cell is not UV 1'b 1: This cell is UV

RD_CELL_OV (29h)
Format: Unsigned binary

The RD_CELL_OV command reports whether an OV condition has been detected on cell 1-16.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 16. 1'b 0: This cell is not OV 1'b 1: This cell is OV
14	R	CELL_15_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 15. 1'b 0: This cell is not OV 1'b 1: This cell is OV
13	R	CELL_14_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 14. 1'b 0: This cell is not OV 1'b 1: This cell is OV
12	R	CELL_13_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 13. 1'b 0: This cell is not OV 1'b 1: This cell is OV
11	R	CELL_12_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 12. 1'b 0: This cell is not OV 1'b 1: This cell is OV
10	R	CELL_11_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 11. 1'b 0: This cell is not OV 1'b 1: This cell is OV
9	R	CELL_10_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 10. 1'b 0: This cell is not OV 1'b 1: This cell is OV
8	R	CELL_9_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 9. 1'b 0: This cell is not OV 1'b 1: This cell is OV
7	R	CELL_8_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 8. 1'b 0: This cell is not OV 1'b 1: This cell is OV
6	R	CELL_7_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 7. 1'b 0: This cell is not OV 1'b 1: This cell is OV



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

5	R	CELL_6_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 6. 1'b 0: This cell is not OV 1'b 1: This cell is OV
4	R	CELL_5_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 5. 1'b 0: This cell is not OV 1'b 1: This cell is OV
3	R	CELL_4_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 4. 1'b 0: This cell is not OV 1'b 1: This cell is OV
2	R	CELL_3_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 3. 1'b 0: This cell is not OV 1'b 1: This cell is OV
1	R	CELL_2_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 2. 1'b 0: This cell is not OV 1'b 1: This cell is OV
0	R	CELL_1_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 1. 1'b 0: This cell is not OV 1'b 1: This cell is OV

RD_OVUV_STS (2Ah)**Format:** Unsigned binary

The RD_OVUV_STS command reports whether a UV/OV condition has been detected on cell 17-24.

Bits	Access	Bit Name	Default	Description
15	R	CELL_24_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 24. 1'b 0: This cell is not OV 1'b 1: This cell is OV
14	R	CELL_23_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 23. 1'b 0: This cell is not OV 1'b 1: This cell is OV
13	R	CELL_22_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 22. 1'b 0: This cell is not OV 1'b 1: This cell is OV
12	R	CELL_21_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 21. 1'b 0: This cell is not OV 1'b 1: This cell is OV
11	R	CELL_20_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 20. 1'b 0: This cell is not OV 1'b 1: This cell is OV
10	R	CELL_19_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 19. 1'b 0: This cell is not OV 1'b 1: This cell is OV
9	R	CELL_18_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 18. 1'b 0: This cell is not OV 1'b 1: This cell is OV


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

8	R	CELL_17_OV_RT_STS	1'b 0	Reports whether an OV condition has been detected on cell 17. 1'b 0: This cell is not OV 1'b 1: This cell is OV
7	R	CELL_24_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 24. 1'b 0: This cell is not UV 1'b 1: This cell is UV
6	R	CELL_23_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 23. 1'b 0: This cell is not UV 1'b 1: This cell is UV
5	R	CELL_22_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 22. 1'b 0: This cell is not UV 1'b 1: This cell is UV
4	R	CELL_21_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 21. 1'b 0: This cell is not UV 1'b 1: This cell is UV
3	R	CELL_20_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 20. 1'b 0: This cell is not UV 1'b 1: This cell is UV
2	R	CELL_19_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 19. 1'b 0: This cell is not UV 1'b 1: This cell is UV
1	R	CELL_18_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 18. 1'b 0: This cell is not UV 1'b 1: This cell is UV
0	R	CELL_17_UV_RT_STS	1'b 0	Reports whether a UV condition has been detected on cell 17. 1'b 0: This cell is not UV 1'b 1: This cell is UV

RD_CELL_SOV (2Bh)
Format: Unsigned binary

The RD_CELL_SOV command reports whether an SOV condition has been detected on cell 1-16.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 16. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
14	R	CELL_15_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 15. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
13	R	CELL_14_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 14. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
12	R	CELL_13_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 13. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

11	R	CELL_12_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 12. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
10	R	CELL_11_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 11. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
9	R	CELL_10_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 10. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
8	R	CELL_9_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 9. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
7	R	CELL_8_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 8. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
6	R	CELL_7_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 7. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
5	R	CELL_6_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 6. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
4	R	CELL_5_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 5. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
3	R	CELL_4_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 4. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
2	R	CELL_3_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 3. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
1	R	CELL_2_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 2. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
0	R	CELL_1_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 1. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV

RD_CELL_DEAD (2Ch)
Format: Unsigned binary

The RD_CELL_DEAD command reports whether a dead condition has been detected on cell 1-16.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 16. 1'b 0: This cell is not dead 1'b 1: This cell is dead


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

14	R	CELL_15_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 15. 1'b 0: This cell is not dead 1'b 1: This cell is dead
13	R	CELL_14_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 14. 1'b 0: This cell is not dead 1'b 1: This cell is dead
12	R	CELL_13_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 13. 1'b 0: This cell is not dead 1'b 1: This cell is dead
11	R	CELL_12_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 12. 1'b 0: This cell is not dead 1'b 1: This cell is dead
10	R	CELL_11_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 11. 1'b 0: This cell is not dead 1'b 1: This cell is dead
9	R	CELL_10_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 10. 1'b 0: This cell is not dead 1'b 1: This cell is dead
8	R	CELL_9_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 9. 1'b 0: This cell is not dead 1'b 1: This cell is dead
7	R	CELL_8_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 8. 1'b 0: This cell is not dead 1'b 1: This cell is dead
6	R	CELL_7_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 7. 1'b 0: This cell is not dead 1'b 1: This cell is dead
5	R	CELL_6_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 6. 1'b 0: This cell is not dead 1'b 1: This cell is dead
4	R	CELL_5_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 5. 1'b 0: This cell is not dead 1'b 1: This cell is dead
3	R	CELL_4_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 4. 1'b 0: This cell is not dead 1'b 1: This cell is dead
2	R	CELL_3_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 3. 1'b 0: This cell is not dead 1'b 1: This cell is dead
1	R	CELL_2_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 2. 1'b 0: This cell is not dead 1'b 1: This cell is dead
0	R	CELL_1_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 1. 1'b 0: This cell is not dead 1'b 1: This cell is dead

**RD_SOVDEAD_STS (2Dh)****Format:** Unsigned binary

The RD_SOVDEAD_STS command reports whether an SOV or a dead condition has been detected on cell 17-24.

Bits	Access	Bit Name	Default	Description
15	R	CELL_24_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 24. 1'b 0: This cell is not dead 1'b 1: This cell is dead
14	R	CELL_23_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 23. 1'b 0: This cell is not dead 1'b 1: This cell is dead
13	R	CELL_22_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 22. 1'b 0: This cell is not dead 1'b 1: This cell is dead
12	R	CELL_21_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 21. 1'b 0: This cell is not dead 1'b 1: This cell is dead
11	R	CELL_20_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 20. 1'b 0: This cell is not dead 1'b 1: This cell is dead
10	R	CELL_19_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 19. 1'b 0: This cell is not dead 1'b 1: This cell is dead
9	R	CELL_18_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 18. 1'b 0: This cell is not dead 1'b 1: This cell is dead
8	R	CELL_17_DEAD_RT_STS	1'b 0	Reports whether a dead condition has been detected on cell 17. 1'b 0: This cell is not dead 1'b 1: This cell is dead
7	R	CELL_24_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 24. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
6	R	CELL_23_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 23. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
5	R	CELL_22_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 22. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
4	R	CELL_21_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 21. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
3	R	CELL_20_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 20. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV



2	R	CELL_19_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 19. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
1	R	CELL_18_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 18. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV
0	R	CELL_17_SOV_RT_STS	1'b 0	Reports whether an SOV condition has been detected on cell 17. 1'b 0: This cell is not SOV 1'b 1: This cell is SOV

RD_CELL_MSMT (2Eh)**Format:** Unsigned binary

The RD_CELL_MSMT command controls reports whether a mismatch condition has been detected on cell 1-16.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 16. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
14	R	CELL_15_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 15. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
13	R	CELL_14_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 14. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
12	R	CELL_13_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 13. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
11	R	CELL_12_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 12. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
10	R	CELL_11_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 11. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
9	R	CELL_10_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 10. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
8	R	CELL_9_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 9. 1'b 0: No mismatch 1'b 1: A mismatch has been detected



7	R	CELL_8_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 8. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
6	R	CELL_7_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 7. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
5	R	CELL_6_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 6. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
4	R	CELL_5_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 5. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
3	R	CELL_4_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 4. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
2	R	CELL_3_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 3. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
1	R	CELL_2_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 2. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
0	R	CELL_1_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 1. 1'b 0: No mismatch 1'b 1: A mismatch has been detected

RD_CELLH_MSMT (2Fh)**Format:** Unsigned binary

The RD_CELLH_MSMT command controls reports whether a mismatch condition has been detected on cell 17-24.

Bits	Access	Bit Name	Default	Description
15:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7	R	CELL_24_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 24. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
6	R	CELL_23_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 23. 1'b 0: No mismatch 1'b 1: A mismatch has been detected



5	R	CELL_22_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 22. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
4	R	CELL_21_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 21. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
3	R	CELL_20_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 20. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
2	R	CELL_19_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 19. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
1	R	CELL_18_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 18. 1'b 0: No mismatch 1'b 1: A mismatch has been detected
0	R	CELL_17_MSMT_RT_STS	1'b 0	Reports whether a mismatch condition has been detected on cell 17. 1'b 0: No mismatch 1'b 1: A mismatch has been detected

RD_RT_STS (30h)**Format:** Unsigned binary

The RD_RT_STS command reports the real-time digital die temperature status, real-time VDD/3V3/5V UV status, real-time VTOP UV/OV/SUV/SOV status, the real-time DSG/CHG MOSFET dead status. It also reports the latched ADC self-diagnostic UV/OV status, fuse fault, and whether VTOP is lower than VTOP_UV_SLEEP_THR.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R	TOP_SLEEP_UV_STS	1'b 0	Reports the VTOP voltage is below than VTOP_UV_SLEEP_THR. If VTOP < VTOP_UV_SLEEP_THR, then IC goes to sleep mode with the VTOP slotA is disabled, this register should be kept as 1. 1'b 0: VTOP ≥ VTOP_UV_SLEEP_THR 1'b 1: VTOP < VTOP_UV_SLEEP_THR
12	R	FUSE_FAULT_STS	1'b 0	Reports whether the fuse fault has been detected or the FUSE pin has detected high voltage, which is the latched status. 1'b 0: Not detected 1'b 1: Detected
11	R	SELF_TEST_STS_UV	1'b 0	Reports whether the ADC self-test UV has been detected 1'b 0: Not detected 1'b 1: Detected


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R	DIE_TEMP_HOT_RT_STS	1'b 0	Reports the real-time digital die temperature status, based on the result of the latest comparison between the digital die temperature reading and the associated hot threshold. 1'b 0: The die temperature is within its normal range 1'b 1: The digital die temperature is too hot
9	R	SELF_TEST_STS_OV	1'b 0	Reports whether the ADC self-test OV has been detected. 1'b 0: Not detected 1'b 1: Detected
8	R	V5_UV_RT_STS	1'b 0	Reports the real-time 5V UV status, based on the result of the latest comparison between the 5V ADC reading and the 5V UV threshold. 1'b 0: Not detected 1'b 1: Detected
7	R	V3V3_UV_RT_STS	1'b 0	Reports the real-time 3V3 UV status, based on the result of the latest comparison between the 3V3 ADC reading and the 3V3 UV threshold. 1'b 0: Not detected 1'b 1: Detected
6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R	CHGFET_DEAD_RT_STS	1'b 0	Reports whether the CHG MOSFET dead has been detected in real-time. 1'b 0: Not detected 1'b 1: Detected
4	R	DSGFET_DEAD_RT_STS	1'b 0	Reports whether the DSG MOSFET dead has been detected in real-time. 1'b 0: Not detected 1'b 1: Detected
3	R	VTOP_SUV_RT_STS	1'b 0	Reports the real-time VTOP SUV status, based on the result of the latest comparison between the VTOP ADC reading and the VTOP SUV threshold. Depending on the settings hysteresis might be applied to the threshold used in comparison. 1'b 0: Not detected 1'b 1: Detected
2	R	VTOP_SOV_RT_STS	1'b 0	Reports the real-time VTOP SOV status, based on the result of the latest comparison between the VTOP ADC reading and the VTOP SOV threshold. Depending on the settings hysteresis might be applied to the threshold used in comparison. 1'b 0: Not detected 1'b 1: Detected
1	R	VTOP_OV_RT_STS	1'b 0	Reports the real-time result of the VTOP vs. OV threshold comparison. Depending on the settings, a hysteresis may be applied to the threshold for comparison. Depending on the settings hysteresis might be applied to the threshold used in comparison. 1'b 0: $VTOP \leq OV$ threshold 1'b 1: $VTOP > OV$ threshold



0	R	VTOP_UV_RT_STS	1'b 0	Reports the real-time result of the VTOP vs. UV threshold comparison. Depending on the settings, a hysteresis may be applied to the threshold for comparison. Depending on the settings hysteresis might be applied to the threshold used in comparison. 1'b 0: VTOP $\geq$ UV threshold 1'b 1: VTOP < UV threshold
---	---	----------------	-------	---

CELL_MSMT_STS (31h)**Format:** Unsigned binary

The CELL_MSMT_STS command reports the voltage difference between the cell with the lowest voltage and the highest, it also reports the cell number with the lowest voltage.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:10	R	CELL_MSMT_LOWE R	5'b 00000	Reports the cell with the lowest voltage. If the interrupt is disabled, this bit indicates the real-time status; otherwise, this bit is latched at the interrupt and returns to the real-time status after being cleared. 5'b 00001: Cell 1 5'b 00010: Cell 2 ... 5'b 10111: Cell 23 5'b 11000: Cell 24
9:0	R	CELL_MSMT_DELTA	10'b 000000000 0	Reports the voltage difference between the cell with the lowest voltage and the highest. If the interrupt is disabled, this bit indicates the real-time status; otherwise, this bit is latched at the interrupt and returns to the real-time status after being cleared. LSB: 4.8828mV Range: 0mV to 4995.1044mV Value (mV) = Setting x 4.8828

PACK_SOV (32h)**Format:** Unsigned binary

The PACK_SOV command configures VTOP SOV threshold, and VTOP SOV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:12	R/W (can lock to read- only)	VTOP_SOV_DG	4'b 0000	Sets the VTOP SOV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: An SOV condition is reported as soon as the SOV threshold is reached 4'b 0001: An SOV condition is reported when the current and previous protection readings violate the VTOP SOV threshold ... 4'b 1111: An SOV condition is reported when a total of 16 protection cycles violate the VTOP SOV threshold These bits allow OTP
11:0	R/W (can lock to read- only)	VTOP_SOV_LIMIT	12'b 110000000 000	Configures the battery pack SOV threshold LSB: 29.297mV Range: 0mV to 119971.215mV Value (mV) = Setting x 29.297 These bits allow MTP

**PACK_SUV (33h)****Format:** Unsigned binary

The PACK_SUV command configures VTOP SUV threshold, and VTOP SUV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:12	R/W (can lock to read-only)	VTOP_SUV_DG	4'b 0000	Sets the VTOP SUV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: An SUV condition is reported as soon as the SUV threshold is reached 4'b 0001: An SUV condition is reported when the current and previous protection readings violate the VTOP SUV threshold ... 4'b 1111: An SUV condition is reported when a total of 16 protection cycles violate the VTOP SUV threshold These bits allow OTP
11:0	R/W (can lock to read-only)	VTOP_SUV_LIMIT	12'b 010101010 101	Configures the battery pack SUV threshold LSB: 29.297mV Range: 0mV to 119971.215mV Value (mV) = Setting x 29.297 These bits allow MTP

CELL_SOV (34h)**Format:** Unsigned binary

The CELL_SOV command configures cell SOV threshold, cell SOV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13:10	R/W (can lock to read-only)	CELL_SOV_DG	4'b 0000	Sets the cell SOV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: An SOV condition is reported as soon as the SOV threshold is reached 4'b 0001: An SOV condition is reported when the current and previous protection readings violate the cell SOV threshold ... 4'b 1111: An SOV condition is reported when a total of 16 protection cycles violate the cell SOV threshold These bits allow OTP
9:0	R/W (can lock to read-only)	CELL_SOV	10'b 001110010 0	Configures the cell SOV threshold. LSB: 4.8828mV Range: 0mV to 4995.1044mV Value (mV) = Setting x 4.8828 These bits allow MTP

CELL_HYST (35h)**Format:** Unsigned binary

The CELL_HYST command configures cell UV threshold hysteresis, cell UV recovery logic, cell OV threshold hysteresis, and cell OV recovery logic.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.



11:8	R/W	CELL_OV_HYST	4'b 1010	Sets the cell OV threshold hysteresis. LSB: 19.53125mV Range: 0mV to 292.96875mV Value (mV) = Setting x 19.53125 These bits allow OTP
7:4	R/W	CELL_UV_HYST	4'b 1010	Sets the cell UV threshold hysteresis. LSB: 19.53125mV Range: 0mV to 292.96875mV Value (mV) = Setting x 19.53125 These bits allow OTP
3:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R/W (can lock to read-only)	CELL_OV_LOGIC_SEL	1'b 0	Selects which logic is used for status recovery from cell OV conditions. 1'b 0: The status recovers by following logic below: • The cell is below (OV threshold - hysteresis) 1'b 1: The status recovers by following either logic below: • The cell is below the OV threshold and the PACKP voltage is below (VTOP -1.6V) • The cell is below the OV threshold and the battery pack current status is discharge. • The cell is below (UV threshold - hysteresis) This bit allows OTP
0	R/W (can lock to read-only)	CELL_UV_LOGIC_SEL	1'b 0	Selects which logic is used for status recovery from cell UV conditions. 1'b 0: The status recovers by following logic below: • The cell exceeds (UV threshold + hysteresis) 1'b 1: The status recovers by following either logic below: • The cell exceeds the UV threshold and the PACKP voltage exceeds (VTOP + 160mV) • The cell exceeds the UV threshold, and the battery pack current status is charging. • The cell exceeds (UV threshold + hysteresis) This bit allows OTP

PACK_UV_OV (36h)**Format:** Unsigned binary

The PACK_UV_OV command configures VTOP UV/OV threshold hysteresis.

Bits	Access	Bit Name	Default	Description
15:10	R/W (can lock to read-only)	VTOP_OV_HYST	6'b 100000	Sets the battery pack OV threshold hysteresis. LSB: 117.19mV Range: 0mV to 7382.97mV Value (mV) = Setting x 117.19 These bits allow MTP
9:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7:2	R/W (can lock to read-only)	VTOP_UV_HYST	6'b 100000	Sets the battery pack UV threshold hysteresis. LSB: 117.19mV Range: 0mV to 7382.97mV Value (mV) = Setting x 117.19 These bits allow MTP
1:0	R	RESERVED	N/A	Reserved. Do not change this register value.

**CELL_UV (37h)****Format:** Unsigned binary

The CELL_UV command configures cell UV threshold and cell UV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11:8	R/W (can lock to read-only)	CELL_UV_DG	4'b 0000	Sets the cell UV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: A UV condition is reported as soon as the UV threshold is reached 4'b 0001: A UV condition is reported when the current and previous protection readings violate the cell UV threshold ... 4'b 1111: A UV condition is reported when a total of 16 protection cycles violate the cell UV threshold These bits allow OTP
7:0	R/W (can lock to read-only)	CELL_UV	8'b 10011000	Configures the cell UV threshold. LSB: 19.53125mV Range: 0mV to 4980.46875mV Value (mV) = Setting x 19.53125 These bits allow MTP

CELL_OV (38h)**Format:** Unsigned binary

The CELL_OV command configures cell OV threshold and cell OV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13:10	R/W (can lock to read-only)	CELL_OV_DG	4'b 0000	Sets the cell OV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: An OV condition is reported as soon as the OV threshold is reached 4'b 0001: An OV condition is reported when the current and previous protection readings violate the cell OV threshold ... 4'b 1111: An OV condition is reported when a total of 16 protection cycles violate the cell OV threshold These bits allow OTP
9:0	R/W (can lock to read-only)	CELL_OV	10'b 001110010 0	Configures the cell OV threshold. LSB: 4.8828mV Range: 0mV to 4995.1044mV Value (mV) = Setting x 4.8828 These bits allow MTP

PACK_UV (39h)**Format:** Unsigned binary

The PACK_UV command configures VTOP UV threshold and VTOP UV deglitch delay.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:12	R/W (can lock to read-only)	VTOP_UV_DG	4'b 0000	Sets the VTOP UV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: A UV condition is reported as soon as the UV threshold is reached 4'b 0001: A UV condition is reported when the current and previous protection readings violate the VTOP UV threshold ... 4'b 1111: A UV condition is reported when a total of 16 protection cycles violate the VTOP UV threshold These bits allow OTP
11:0	R/W (can lock to read-only)	VTOP_UV_LIMIT	12'b 011001100 110	Configures the battery pack UV threshold. LSB: 29.297mV Range: 0mV to 119971.215mV Value (mV) = Setting x 29.297 These bits allow MTP

PACK_OV (3Ah)
Format: Unsigned binary

The PACK_OV command configures VTOP OV threshold and VTOP OV deglitch delay.

Bits	Access	Bit Name	Default	Description
15:12	R/W (can lock to read-only)	VTOP_OV_DG	4'b 0000	Sets the VTOP OV deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: An OV condition is reported as soon as the OV threshold is reached 4'b 0001: An OV condition is reported when the current and previous protection readings violate the VTOP OV threshold ... 4'b 1111: An OV condition is reported when a total of 16 protection cycles violate the VTOP OV threshold These bits allow OTP
11:0	R/W (can lock to read-only)	VTOP_OV_LIMIT	12'b 101110000 101	Configures the battery pack OV threshold. LSB: 29.297mV Range: 0mV to 119971.215mV Value (mV) = Setting x 29.297 These bits allow MTP

CELL_DEAD_THR (3Bh)
Format: Unsigned binary

The CELL_DEAD_THR command configures cell dead threshold and cell dead deglitch delay.

Bits	Access	Bit Name	Default	Description
15:11	R	RESERVED	N/A	Reserved. Do not change this register value.
10:7	R/W (can lock to read-only)	CELL_DEAD_DG	4'b 0000	Sets the dead cell deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: A dead cell condition is reported as soon as the dead cell threshold is reached 4'b 0001: A dead cell condition is reported when the current and previous protection readings violate the dead cell threshold ... 4'b 1111: A dead cell condition is reported when a total of 16 protection cycles violate the dead cell threshold



				These bits allow OTP
6:0	R/W (can lock to read-only)	CELL_DEAD_LIMIT	7'b 1101000	Configurates dead cell threshold. LSB: 19.53125mV Range: 0mV to 2480.46875mV Value (mV) = Setting x 19.53125 These bits allow MTP

CELL_MSMT (3Ch)**Format:** Unsigned binary

The CELL_MSMT command configures cell mismatch threshold, cell mismatch deglitch delay, and the minimum detection threshold for cell mismatch.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:9	R/W (can lock to read-only)	CELL_MSMT_MIN	6'b 000000	Sets the minimum detection threshold for cell mismatch. When the maximum cell is lower than this level, the cell mismatch detection is forced to disable. LSB: 39.0625 mV Range: 2500 mV to 4960.9375 mV Value (mV) = Setting x 39.0625 + 2500 These bits allow OTP
8:5	R/W (can lock to read-only)	CELL_MSMT_DG	4'b 0000	Sets the mismatched cell deglitch delay, which can be set between 1 reading (no deglitch) and 16 readings. 4'b 0000: A mismatched cell condition is reported as soon as the mismatched cell threshold is reached 4'b 0001: A mismatched cell condition is reported when the current and previous protection readings violate the mismatched cell threshold ... 4'b 1111: A mismatched cell condition is reported when a total of 16 protection cycles violate the mismatched cell threshold These bits allow OTP
4:0	R/W (can lock to read-only)	CELL_MSMT_TH	5'b 00010	Sets the mismatched cell threshold. LSB: 39.0625mV Range: 0mV to 1210.9375mV Value (mV) = Setting x 39.0625 These bits allow OTP

LOG_STS (3Dh)**Format:** Unsigned binary

The LOG_STS command reports the statuses of cell dead, die temperature too hot, cell mismatch, VTOP SOV/SUV, DSG/CHG dead, cell SOV, ADC self-diagnostic, and LDO UV.

Bits	Access	Bit Name	Default	Description
15	R	DIE_TEMP_ANA_EVENT	1'b 0	Obtains the status from analog circuit recordings after a hot die temperature event. This bit cannot be read in real time, since digital logic is shutdown when this event occurs. This bit is cleared by DIE_TEMP_ANA_CLEAR. 1'b 0: The die temperature is within its normal range 1'b 1: The analog die temperature is too hot
14:9	R	RESERVED	N/A	Reserved. Do not change this register value.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

8	R	LDO_UV_LOG_STS	1'b 0	Reports whether a LDO UV event has been detected. This status is not cleared by shutting down. It can be cleared with LDO_UV_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
7	R	ADC_FAIL_LOG_STS	1'b 0	Reports whether an ADC fail event has been detected. This status is not cleared by shutting down. It can be cleared with ADC_FAIL_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
6	R	CHGFET_DEAD_LOG_STS	1'b 0	Reports whether a CHG MOSFET dead event has been detected. This status is not cleared by shutting down. It must be cleared with CHGFET_DEAD_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
5	R	DSGFET_DEAD_LOG_STS	1'b 0	Reports whether a DSG MOSFET dead event has been detected. This status is not cleared by shutting down. It must be cleared with DSGFET_DEAD_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
4	R	VTOP_SUV_LOG_STS	1'b 0	Reports whether a VTOP SUV event has been detected. This status is not cleared by shutting down. It must be cleared with VTOP_SUV_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
3	R	VTOP_SOV_LOG_STS	1'b 0	Reports whether a VTOP SOV event has been detected. This status is not cleared by shutting down. It can be cleared with VTOP_SOV_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
2	R	CELL_MSMT_LOG_STS	1'b 0	Reports whether a cell mismatch event has been detected. This status is not cleared by shutting down. It can be cleared with CELL_MSMT_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
1	R	CELL_DEAD_LOG_STS	1'b 0	Reports whether a dead cell event has been detected. This status is not cleared by shutting down. It must be cleared with CELL_DEAD_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected
0	R	CELL_SOV_LOG_STS	1'b 0	Reports whether a cell SOV event has been detected. This status is not cleared by shutting down. It can be cleared with CELL_SOV_DET_CLEAR or WDT reset or power down. 1'b 0: Not detected 1'b 1: Detected

**LOG_STS_CLR (3Eh)****Format:** Unsigned binary

The LOG_STS_CLR command controls the bit-clear of status, including cell dead, analog die temperature too hot, cell mismatch, VTOP SUV/SOV, DSG/CHG MOSFET dead, cell SOV, ADC self-diagnostic, and LDO UV.

Bits	Access	Bit Name	Default	Description
15	W	DIE_TEMP_ANA_CLEAR	1'b 0	Write this bit to 1 to clear the analog die temperature too hot status. This is a self-clearing bit.
14:9	R	RESERVED	N/A	Reserved. Do not change this register value.
8	W	LDO_UV_DET_CLEAR	1'b 0	Write this bit to 1 to clear the LDO UV status. This is a self-clearing bit.
7	W	ADC_FAIL_DET_CLEAR	1'b 0	Write this bit to 1 to clear the ADC fail status. This is a self-clearing bit.
6	W	CHGFET_DEAD_DET_CLEAR	1'b 0	Write this bit to 1 to clear the CHG MOSFET dead status. This is a self-clearing bit.
5	W	DSGFET_DEAD_DET_CLEAR	1'b 0	Write this bit to 1 to clear the DSG MOSFET dead status. This is a self-clearing bit.
4	W	VTOP_SUV_DET_CLEAR	1'b 0	Write this bit to 1 to clear the VTOP SUV status. This is a self-clearing bit.
3	W	VTOP_SOV_DET_CLEAR	1'b 0	Write this bit to 1 to clear the VTOP SOV status. This is a self-clearing bit.
2	W	CELL_MSMT_DET_CLEAR	1'b 0	Write this bit to 1 to clear the cell mismatch status. This is a self-clearing bit.
1	W	CELL_DEAD_DET_CLEAR	1'b 0	Write this bit to 1 to clear the dead cell status. This is a self-clearing bit.
0	W	CELL_SOV_DET_CLEAR	1'b 0	Write this bit to 1 to clear the cell SOV status. This is a self-clearing bit.

BAL_STS (3Fh)**Format:** Unsigned binary

The BAL_STS command controls enable/disable of skipping auto balance under different conditions: open wire detected, IC in safe or standby mode, die too hot, discharging or charging current, it also reports the number of executed balancing cycles, and whether it's ongoing.

Bits	Access	Bit Name	Default	Description
15:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12	R	AUTO_BAL_SKIPPED_AOPW	1'b 0	When true, balancing cycles were skipped due to at least one wire is detected as open during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False 1'b 1: True
11	R	AUTO_BAL_SKIPPED_SAFE	1'b 0	When true, balancing cycles were skipped due to the IC goes into safe mode during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False 1'b 1: True
10	R	AUTO_BAL_SKIPPED_HOT	1'b 0	When true, balancing cycles were skipped due to the die temperature being too hot during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False



				1'b 1: True
9	R	AUTO_BAL_SKIPPE D_DISCHARGE	1'b 0	When true, balancing cycles were skipped due to a detected discharge current during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False 1'b 1: True
8	R	AUTO_BAL_SKIPPE D_STANDBY	1'b 0	When true, balancing cycles were skipped due to a detected standby current during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False 1'b 1: True
7	R	AUTO_BAL_SKIPPE D_CHARGE	1'b 0	When true, balancing cycles were skipped due to a detected charge current during repetitive auto balance mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). 1'b 0: False 1'b 1: True
6:1	R	BALANCING_COUNT _STS	6'b 000000	Shows the number of balancing cycles that are actually performed. To be valid, BAL_DONE_STS must be set to true. This register will get updated in manual balance mode and auto balance mode with AUTO_BAL_ALWAYS = 0. If any balancing cycle is skipped, the reason can be verified by reading AUTO_BAL_SKIPPED_HOT, AUTO_BAL_SKIPPED_DISCHARGE, AUTO_BAL_SKIPPED_STANDBY, or AUTO_BAL_SKIPPED_CHARGE. This register is not updated when AUTO_BAL_ALWAYS is enabled.
0	R	BALANCING_ACTIVE	1'b 0	Reports whether cell-balancing is currently ongoing. 1'b 0: False 1'b 1: True

BAL_CTRL (40h)**Format:** Unsigned binary

The BAL_CTRL command controls enable/disable of cell-balancing, reports the completed cell-balancing, and cell-balancing error.

Bits	Access	Bit Name	Default	Description
15:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R	BAL_DONE_STS	1'b 0	Reports whether cell-balancing is completed. 1'b 0: False 1'b 1: True
0	R/W	BALANCE_GO	1'b 0	Controls the start for both manual and automatic cell-balancing. When set to on, balancing begins. This bit is invalid under the always auto run mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 1). 1'b 0: Off 1'b 1: On

BAL_CFG (41h)**Format:** Unsigned binary

The BAL_CFG command configures the control mode to initiate auto cell-balancing, cell-balancing mode, the number of repetitions for each execution of the balancing list, it controls enable/disable of the synchronous execution of cell balancing and HR ADC scan.



Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R/W	BAL_ODD_ON_HRS CAN	1'b 0	This register is used for internal balancing MOSFET check in the initialization. When enabled, the odd cells internal balancing MOSFET is on during the HR ADC scan. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	BAL_EVEN_ON_HRS CAN	1'b 0	This register is used for internal balancing MOSFET check in the initialization. When enabled, the even cells internal balancing MOSFET is on during the HR ADC scan. 1'b 0: Disabled 1'b 1: Enabled
7:3	R/W (can lock to read- only)	BAL_REPETITION	5'b 11111	Sets the number of repetitions for each execution of the balancing list. When set to 31 repetitions, 32 balancing cycles are executed. LSB: 1 repetition Range: 0 to 31 repetitions Value (repetitions) = Setting These bits allow OTP
2	R/W (can lock to read- only)	AUTO_BAL_ALWAYS	1'b 0	This register is only used when automatic cell-balancing is enabled. When disabled, auto cell-balancing will use BAL_REPETITION to control how many iterations the device repeats. The host MCU need set BALANCE_GO from 0 to 1 to initiate the balance. Once the repetition time is reached or the balancing list is empty, the BAL_DONE_STS will report to 1. When enabled, the auto cell-balancing will always active. The balancing will start automatically if the balancing list is not empty. The BALANCE_GO is invalid under the always run mode. The balancing done interrupt won't be triggered under this mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read- only)	BALANCE_MODE_R EG	1'b 0	Controls the cell-balancing mode. 1'b 0: Manual cell-balancing 1'b 1: Automatic cell-balancing This bit allows OTP
0	R/W (can lock to read- only)	BALANCE_MODE_C TRL	1'b 0	Sets the control mode to initiate automatic cell-balancing. This bit is only valid in repetitive auto run mode (BALANCE_MODE_REG = 1 and AUTO_BAL_ALWAYS = 0). This feature is only available on I2C versions. 1'b 0: Register control (BALANCE_GO) 1'b 1: GPIO4 control (the direction of GPIO4 is set as input, and the input type is set as digital input). When set to high, balancing begins. This bit allows OTP

BAL_THR (42h)**Format:** Unsigned binary

The BAL_THR command controls enable/disable of automatic cell-balancing in safe mode, and auto balancing when charge current exceeds charge standby threshold, and auto balancing when current within standby comparator current, and whether suspend automatic cell-balancing if the silicon digital die

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

temperature is too hot. It also configures the qualifying minimum cell voltage to run auto cell-balancing, and auto cell-balancing threshold.

Bits	Access	Bit Name	Default	Description
15:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12:10	R/W (can lock to read-only)	BAL_MSM_TH	3'b 000	The cell-balancing threshold used by the automatic cell-balancing algorithm. LSB: 9.765 mV Offset: 9.765 mV Range: 9.765 mV to 78.12 mV Value (mV) = Setting x 9.765 + 9.765 These bits allow MTP
9	R/W (can lock to read-only)	STOP_ON_HOT	1'b 1	This register is used only when automatic cell-balancing is enabled. 1'b 0: Don't suspend automatic cell-balancing if the silicon digital die temperature is too hot 1'b 1: Suspend automatic cell-balancing if the silicon digital die temperature is too hot This bit allows OTP
8	R/W (can lock to read-only)	ABAL_ON_SAFE	1'b 1	Enables automatic cell-balancing in safe mode. This register is used only when automatic cell-balancing is enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	ABAL_ON_STBY	1'b 1	Enables automatic cell-balancing when the current is within $\pm$ standby comparator current threshold. This register is used only when automatic cell-balancing is enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	ABAL_ON_CHARGE	1'b 1	Enables automatic cell-balancing when charge current has exceeded the charge standby threshold. This register is used only when automatic cell-balancing is enabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
5:0	R/W (can lock to read-only)	CELL_BAL_MIN	6'b 000000	Sets the qualifying minimum cell voltage to run automatic cell-balancing. When a cell is below this level, it is excluded from the balancing list, while other qualifying cells could be balanced if they meet applicable criteria. LSB: 39.0625 mV Range: 2500 mV to 4960.9375 mV Value (mV) = Setting x 39.0625 + 2500 These bits allow MTP

BAL_LIST (43h)

Format: Unsigned binary

The BAL_LIST command controls enable/disable of cell1-16's balancing.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15	R/W	CELL_16_TO_BALANCE	1'b 0	When true, cell 16 will be balanced during the next balancing session. When false, cell 16 is skipped. 1'b 0: False 1'b 1: True
14	R/W	CELL_15_TO_BALANCE	1'b 0	When true, cell 15 will be balanced during the next balancing session. When false, cell 15 is skipped. 1'b 0: False 1'b 1: True
13	R/W	CELL_14_TO_BALANCE	1'b 0	When true, cell 14 will be balanced during the next balancing session. When false, cell 14 is skipped. 1'b 0: False 1'b 1: True
12	R/W	CELL_13_TO_BALANCE	1'b 0	When true, cell 13 will be balanced during the next balancing session. When false, cell 13 is skipped. 1'b 0: False 1'b 1: True
11	R/W	CELL_12_TO_BALANCE	1'b 0	When true, cell 12 will be balanced during the next balancing session. When false, cell 12 is skipped. 1'b 0: False 1'b 1: True
10	R/W	CELL_11_TO_BALANCE	1'b 0	When true, cell 11 will be balanced during the next balancing session. When false, cell 11 is skipped. 1'b 0: False 1'b 1: True
9	R/W	CELL_10_TO_BALANCE	1'b 0	When true, cell 10 will be balanced during the next balancing session. When false, cell 10 is skipped. 1'b 0: False 1'b 1: True
8	R/W	CELL_9_TO_BALANCE	1'b 0	When true, cell 9 will be balanced during the next balancing session. When false, cell 9 is skipped. 1'b 0: False 1'b 1: True
7	R/W	CELL_8_TO_BALANCE	1'b 0	When true, cell 8 will be balanced during the next balancing session. When false, cell 8 is skipped. 1'b 0: False 1'b 1: True
6	R/W	CELL_7_TO_BALANCE	1'b 0	When true, cell 7 will be balanced during the next balancing session. When false, cell 7 is skipped. 1'b 0: False 1'b 1: True
5	R/W	CELL_6_TO_BALANCE	1'b 0	When true, cell 6 will be balanced during the next balancing session. When false, cell 6 is skipped. 1'b 0: False 1'b 1: True
4	R/W	CELL_5_TO_BALANCE	1'b 0	When true, cell 5 will be balanced during the next balancing session. When false, cell 5 is skipped. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

3	R/W	CELL_4_TO_BALANCE	1'b 0	When true, cell 4 will be balanced during the next balancing session. When false, cell 4 is skipped. 1'b 0: False 1'b 1: True
2	R/W	CELL_3_TO_BALANCE	1'b 0	When true, cell 3 will be balanced during the next balancing session. When false, cell 3 is skipped. 1'b 0: False 1'b 1: True
1	R/W	CELL_2_TO_BALANCE	1'b 0	When true, cell 2 will be balanced during the next balancing session. When false, cell 2 is skipped. 1'b 0: False 1'b 1: True
0	R/W	CELL_1_TO_BALANCE	1'b 0	When true, cell 1 will be balanced during the next balancing session. When false, cell 1 is skipped. 1'b 0: False 1'b 1: True

BAL_LISTH (44h)
Format: Unsigned binary

The BAL_LISTH command controls enable/disable of cell18-24's balancing.

Bits	Access	Bit Name	Default	Description
15	R/W	CELL_24_TO_BALANCE	1'b 0	When true, cell 24 will be balanced during the next balancing session. When false, cell 24 is skipped. 1'b 0: False 1'b 1: True
14	R/W	CELL_23_TO_BALANCE	1'b 0	When true, cell 23 will be balanced during the next balancing session. When false, cell 23 is skipped. 1'b 0: False 1'b 1: True
13	R/W	CELL_22_TO_BALANCE	1'b 0	When true, cell 22 will be balanced during the next balancing session. When false, cell 22 is skipped. 1'b 0: False 1'b 1: True
12	R/W	CELL_21_TO_BALANCE	1'b 0	When true, cell 21 will be balanced during the next balancing session. When false, cell 21 is skipped. 1'b 0: False 1'b 1: True
11	R/W	CELL_20_TO_BALANCE	1'b 0	When true, cell 20 will be balanced during the next balancing session. When false, cell 20 is skipped. 1'b 0: False 1'b 1: True
10	R/W	CELL_19_TO_BALANCE	1'b 0	When true, cell 19 will be balanced during the next balancing session. When false, cell 19 is skipped. 1'b 0: False 1'b 1: True
9	R/W	CELL_18_TO_BALANCE	1'b 0	When true, cell 18 will be balanced during the next balancing session. When false, cell 18 is skipped. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

8	R/W	CELL_17_TO_BALANCE	1'b 0	When true, cell 17 will be balanced during the next balancing session. When false, cell 17 is skipped. 1'b 0: False 1'b 1: True
7:0	R	RESERVED	N/A	Reserved. Do not change this register value.

RD_NTC_SUT (45h)
Format: Unsigned binary

The RD_NTC_SUT command reports the NTC1-10 cell serious under temperature DSG real-time status.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R	NTC10_CELL_SUT_RT_STS	1'b 0	Reports the NTC10 cell serious under temperature DSG real-time status. 1'b 0: The NTC10 cell temperature is within its normal range 1'b 1: The NTC10 cell temperature is serious under temperature
8	R	NTC9_CELL_SUT_RT_STS	1'b 0	Reports the NTC9 cell serious under temperature DSG real-time status. 1'b 0: The NTC9 cell temperature is within its normal range 1'b 1: The NTC9 cell temperature is serious under temperature
7	R	NTC8_CELL_SUT_RT_STS	1'b 0	Reports the NTC8 cell serious under temperature DSG real-time status. 1'b 0: The NTC8 cell temperature is within its normal range 1'b 1: The NTC8 cell temperature is serious under temperature
6	R	NTC7_CELL_SUT_RT_STS	1'b 0	Reports the NTC7 cell serious under temperature DSG real-time status. 1'b 0: The NTC7 cell temperature is within its normal range 1'b 1: The NTC7 cell temperature is serious under temperature
5	R	NTC6_CELL_SUT_RT_STS	1'b 0	Reports the NTC6 cell serious under temperature DSG real-time status. 1'b 0: The NTC6 cell temperature is within its normal range 1'b 1: The NTC6 cell temperature is serious under temperature
4	R	NTC5_CELL_SUT_RT_STS	1'b 0	Reports the NTC5 cell serious under temperature DSG real-time status. 1'b 0: The NTC5 cell temperature is within its normal range 1'b 1: The NTC5 cell temperature is serious under temperature
3	R	NTC4_CELL_SUT_RT_STS	1'b 0	Reports the NTC4 cell serious under temperature DSG real-time status. 1'b 0: The NTC4 cell temperature is within its normal range 1'b 1: The NTC4 cell temperature is serious under temperature
2	R	NTC3_CELL_SUT_RT_STS	1'b 0	Reports the NTC3 cell serious under temperature DSG real-time status. 1'b 0: The NTC3 cell temperature is within its normal range 1'b 1: The NTC3 cell temperature is serious under temperature
1	R	NTC2_CELL_SUT_RT_STS	1'b 0	Reports the NTC2 cell serious under temperature DSG real-time status. 1'b 0: The NTC2 cell temperature is within its normal range 1'b 1: The NTC2 cell temperature is serious under temperature



0	R	NTC1_CELL_SUT_R T_STS	1'b 0	Reports the NTC1 cell serious under temperature DSG real-time status. 1'b 0: The NTC1 cell temperature is within its normal range 1'b 1: The NTC1 cell temperature is serious under temperature
---	---	--------------------------	-------	---

RD_NTC_SOT (46h)**Format:** Unsigned binary

The RD_NTC_SOT command controls enable/disable of reports the NTC1-10 cell serious over temperature DSG real-time status.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R	NTC10_CELL_SOT_ RT_STS	1'b 0	Reports the NTC10 cell serious over temperature DSG real-time status. 1'b 0: The NTC10 cell temperature is within its normal range 1'b 1: The NTC10 cell temperature is too serious over temperature
8	R	NTC9_CELL_SOT_R T_STS	1'b 0	Reports the NTC9 cell serious over temperature DSG real-time status. 1'b 0: The NTC9 cell temperature is within its normal range 1'b 1: The NTC9 cell temperature is too serious over temperature
7	R	NTC8_CELL_SOT_R T_STS	1'b 0	Reports the NTC8 cell serious over temperature DSG real-time status. 1'b 0: The NTC8 cell temperature is within its normal range 1'b 1: The NTC8 cell temperature is too serious over temperature
6	R	NTC7_CELL_SOT_R T_STS	1'b 0	Reports the NTC7 cell serious over temperature DSG real-time status. 1'b 0: The NTC7 cell temperature is within its normal range 1'b 1: The NTC7 cell temperature is too serious over temperature
5	R	NTC6_CELL_SOT_R T_STS	1'b 0	Reports the NTC6 cell serious over temperature DSG real-time status. 1'b 0: The NTC6 cell temperature is within its normal range 1'b 1: The NTC6 cell temperature is too serious over temperature
4	R	NTC5_CELL_SOT_R T_STS	1'b 0	Reports the NTC5 cell serious over temperature DSG real-time status. 1'b 0: The NTC5 cell temperature is within its normal range 1'b 1: The NTC5 cell temperature is too serious over temperature
3	R	NTC4_CELL_SOT_R T_STS	1'b 0	Reports the NTC4 cell serious over temperature DSG real-time status. 1'b 0: The NTC4 cell temperature is within its normal range 1'b 1: The NTC4 cell temperature is too serious over temperature
2	R	NTC3_CELL_SOT_R T_STS	1'b 0	Reports the NTC3 cell serious over temperature DSG real-time status. 1'b 0: The NTC3 cell temperature is within its normal range 1'b 1: The NTC3 cell temperature is too serious over temperature
1	R	NTC2_CELL_SOT_R T_STS	1'b 0	Reports the NTC2 cell serious over temperature DSG real-time status. 1'b 0: The NTC2 cell temperature is within its normal range 1'b 1: The NTC2 cell temperature is too serious over temperature



0	R	NTC1_CELL_SOT_RT_STS	1'b 0	Reports the NTC1 cell serious over temperature DSG real-time status. 1'b 0: The NTC1 cell temperature is within its normal range 1'b 1: The NTC1 cell temperature is too serious over temperature
---	---	----------------------	-------	---

RD_NTC_UT (47h)**Format:** Unsigned binary

The RD_NTC_UT command controls enable/disable of reports the NTC1-10 cell under temperature CHG real-time status.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R	NTC10_CELL_UT_RT_STS	1'b 0	Reports the NTC10 cell under temperature CHG real-time status. 1'b 0: The NTC10 cell temperature is within its normal range 1'b 1: The NTC10 cell temperature is under temperature
8	R	NTC9_CELL_UT_RT_STS	1'b 0	Reports the NTC9 cell under temperature CHG real-time status. 1'b 0: The NTC9 cell temperature is within its normal range 1'b 1: The NTC9 cell temperature is under temperature
7	R	NTC8_CELL_UT_RT_STS	1'b 0	Reports the NTC8 cell under temperature CHG real-time status. 1'b 0: The NTC8 cell temperature is within its normal range 1'b 1: The NTC8 cell temperature is under temperature
6	R	NTC7_CELL_UT_RT_STS	1'b 0	Reports the NTC7 cell under temperature CHG real-time status. 1'b 0: The NTC7 cell temperature is within its normal range 1'b 1: The NTC7 cell temperature is under temperature
5	R	NTC6_CELL_UT_RT_STS	1'b 0	Reports the NTC6 cell under temperature CHG real-time status. 1'b 0: The NTC6 cell temperature is within its normal range 1'b 1: The NTC6 cell temperature is under temperature
4	R	NTC5_CELL_UT_RT_STS	1'b 0	Reports the NTC5 cell under temperature CHG real-time status. 1'b 0: The NTC5 cell temperature is within its normal range 1'b 1: The NTC5 cell temperature is under temperature
3	R	NTC4_CELL_UT_RT_STS	1'b 0	Reports the NTC4 cell under temperature CHG real-time status. 1'b 0: The NTC4 cell temperature is within its normal range 1'b 1: The NTC4 cell temperature is under temperature
2	R	NTC3_CELL_UT_RT_STS	1'b 0	Reports the NTC3 cell under temperature CHG real-time status. 1'b 0: The NTC3 cell temperature is within its normal range 1'b 1: The NTC3 cell temperature is under temperature
1	R	NTC2_CELL_UT_RT_STS	1'b 0	Reports the NTC2 cell under temperature CHG real-time status. 1'b 0: The NTC2 cell temperature is within its normal range 1'b 1: The NTC2 cell temperature is under temperature
0	R	NTC1_CELL_UT_RT_STS	1'b 0	Reports the NTC1 cell under temperature CHG real-time status. 1'b 0: The NTC1 cell temperature is within its normal range 1'b 1: The NTC1 cell temperature is under temperature

RD_NTC_OT (48h)**Format:** Unsigned binary

The RD_NTC_OT command controls enable/disable of reports the NTC1-10 cell over temperature CHG real-time status and NTC1-4 PCB over temperature status.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15	R	NTC4_PCB_MNTR_H OT_STS	1'b 0	Reports the NTC4 over temperature PCB status. 1'b 0: The NTC4 PCB temperature is within its normal range 1'b 1: The NTC4 PCB temperature is over temperature
14	R	NTC3_PCB_MNTR_H OT_STS	1'b 0	Reports the NTC3 over temperature PCB status. 1'b 0: The NTC3 PCB temperature is within its normal range 1'b 1: The NTC3 PCB temperature is over temperature
13	R	NTC2_PCB_MNTR_H OT_STS	1'b 0	Reports the NTC2 over temperature PCB status. 1'b 0: The NTC2 PCB temperature is within its normal range 1'b 1: The NTC2 PCB temperature is over temperature
12	R	NTC1_PCB_MNTR_H OT_STS	1'b 0	Reports the NTC1 over temperature PCB status. 1'b 0: The NTC1 PCB temperature is within its normal range 1'b 1: The NTC1 PCB temperature is over temperature
11:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9	R	NTC10_CELL_OT_RT T_STS	1'b 0	Reports the NTC10 cell over temperature CHG real-time status. 1'b 0: The NTC10 cell temperature is within its normal range 1'b 1: The NTC10 cell temperature is over temperature
8	R	NTC9_CELL_OT_RT _STS	1'b 0	Reports the NTC9 cell over temperature CHG real-time status. 1'b 0: The NTC9 cell temperature is within its normal range 1'b 1: The NTC9 cell temperature is over temperature
7	R	NTC8_CELL_OT_RT _STS	1'b 0	Reports the NTC8 cell over temperature CHG real-time status. 1'b 0: The NTC8 cell temperature is within its normal range 1'b 1: The NTC8 cell temperature is over temperature
6	R	NTC7_CELL_OT_RT _STS	1'b 0	Reports the NTC7 cell over temperature CHG real-time status. 1'b 0: The NTC7 cell temperature is within its normal range 1'b 1: The NTC7 cell temperature is over temperature
5	R	NTC6_CELL_OT_RT _STS	1'b 0	Reports the NTC6 cell over temperature CHG real-time status. 1'b 0: The NTC6 cell temperature is within its normal range 1'b 1: The NTC6 cell temperature is over temperature
4	R	NTC5_CELL_OT_RT _STS	1'b 0	Reports the NTC5 cell over temperature CHG real-time status. 1'b 0: The NTC5 cell temperature is within its normal range 1'b 1: The NTC5 cell temperature is over temperature
3	R	NTC4_CELL_OT_RT _STS	1'b 0	Reports the NTC4 cell over temperature CHG real-time status. 1'b 0: The NTC4 cell temperature is within its normal range 1'b 1: The NTC4 cell temperature is over temperature
2	R	NTC3_CELL_OT_RT _STS	1'b 0	Reports the NTC3 cell over temperature CHG real-time status. 1'b 0: The NTC3 cell temperature is within its normal range 1'b 1: The NTC3 cell temperature is over temperature
1	R	NTC2_CELL_OT_RT _STS	1'b 0	Reports the NTC2 cell over temperature CHG real-time status. 1'b 0: The NTC2 cell temperature is within its normal range 1'b 1: The NTC2 cell temperature is over temperature
0	R	NTC1_CELL_OT_RT _STS	1'b 0	Reports the NTC1 cell over temperature CHG real-time status. 1'b 0: The NTC1 cell temperature is within its normal range 1'b 1: The NTC1 cell temperature is over temperature

**NTC_CFG (49h)****Format:** Unsigned binary

The NTC_CFG command configures NTC1-4 monitor type, and NTC PCB over temperature monitoring deglitch delay.

Bits	Access	Bit Name	Default	Description
15:11	R	RESERVED	N/A	Reserved. Do not change this register value.
10:8	R/W (can lock to read-only)	PCB_MNTR_DG	3'b 001	Sets the NTC PCB over temperature monitoring deglitch delay, which can be set between 1 reading (no deglitch) and 8 readings. 3'b 000: An NTC PCB over temperature is reported as soon as the NTC PCB over temperature threshold is reached 3'b 001: An NTC PCB over temperature is reported when the current and previous protection readings violate the NTC PCB over temperature threshold ... 3'b 111: An NTC PCB over temperature is reported when a total of 8 protection cycles violates the NTC PCB over temperature threshold. These bits allow OTP
7	R/W (can lock to read-only)	NTC4_TYPE_SEL	1'b 1	Configures the NTC4 monitor type. 1'b 0: Cell monitor 1'b 1: PCB monitor This bit allows OTP
6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W (can lock to read-only)	NTC3_TYPE_SEL	1'b 0	Configures the NTC3 monitor type. 1'b 0: Cell monitor 1'b 1: PCB monitor This bit allows OTP
4	R	RESERVED	N/A	Reserved. Do not change this register value.
3	R/W (can lock to read-only)	NTC2_TYPE_SEL	1'b 0	Configures the NTC2 monitor type. 1'b 0: Cell monitor 1'b 1: PCB monitor This bit allows OTP
2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R/W (can lock to read-only)	NTC1_TYPE_SEL	1'b 0	Configures the NTC1 monitor type. 1'b 0: Cell monitor 1'b 1: PCB monitor This bit allows OTP
0	R	RESERVED	N/A	Reserved. Do not change this register value.

NTCC_SOTHR (4Ah)**Format:** Unsigned binary

The NTCC_SOTHR command configures NTC cell discharging serious over temperature monitoring threshold, and NTC cell serious under/over temperature monitoring deglitch delay.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.



13:10	R/W (can lock to read-only)	NTC_CELL_SUTSOT_DG	4'b 0001	Sets the NTC cell serious under/over temperature monitoring deglitch delay, which can be set between 1 reading (no deglitch) and 8 readings. 4'b 0000: An NTC cell serious under/over temperature is reported as soon as the NTC cell serious under/over temperature threshold is reached 4'b 0001: An NTC cell serious under/over temperature is reported when the current and previous protection readings violate the NTC cell serious under/over temperature threshold ... 4'b 1111: An NTC cell serious under/over temperature is reported when a total of 16 protection cycles violate the NTC cell serious under/over temperature threshold These bits allow OTP
9:0	R/W (can lock to read-only)	NTC_CELL_SOT	10'b 010010111 0	Sets the NTC cell discharging serious over temperature monitor voltage threshold. The limit is triggered if the value is equal to or below the set threshold. LSB: 0.09766% Range: 0% to 99.906% Value (%) = Setting x 0.09766 These bits allow MTP

NTCC_SUTHR (4Bh)**Format:** Unsigned binary

The NTCC_SUTHR command configures NTC cell discharging serious under temperature monitor voltage threshold and whether NTC charging colder/hotter operating temperature is lower/higher than discharging.

Bits	Access	Bit Name	Default	Description
15:11	R	RESERVED	N/A	Reserved. Do not change this register value.
10	R/W (can lock to read-only)	NTC_CHG_CURRDG_EN	1'b 0	Enables deglitch time for NTC charge current detection, which is about 1s or 2s depend on ACTIVE_MONITOR_CFG. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9:0	R/W (can lock to read-only)	NTC_CELL_SUT	10'b 101001010 0	Sets the NTC cell discharging serious under temperature monitor voltage threshold. The limit is triggered if the value is equal to or above the set threshold. LSB: 0.09766% Range: 0% to 99.906% Value (%) = Setting x 0.09766 These bits allow MTP

NTCC_OTHR (4Ch)**Format:** Unsigned binary

The NTCC_OTHR command controls configures the cell discharging NTC over temperature monitor voltage threshold and NTC cell under/over temperature monitoring deglitch delay.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.



13:10	R/W (can lock to read-only)	NTC_CELL_UTOT_D G	4'b 0001	Sets the cell NTC under/over temperature monitoring deglitch delay, which can be set between 1 reading (no deglitch) and 8 readings. 4'b 0000: An NTC cell under/over temperature is reported as soon as the NTC cell under/over temperature threshold is reached 4'b 0001: An NTC cell under/over temperature is reported when the current and previous protection readings violate the NTC cell under/over temperature threshold ... 4'b 1111: An NTC cell under/over temperature is reported when a total of 16 protection cycles violate the NTC cell under/over temperature threshold These bits allow OTP
9:0	R/W (can lock to read-only)	NTC_CELL_OT	10'b 010010111 0	Sets the NTC cell charging over temperature monitor voltage threshold. The limit is triggered if the value is equal to or below the set threshold. LSB: 0.09766% Range: 0% to 99.906% Value (%) = Setting x 0.09766 These bits allow MTP

NTCC_UTHR (4Dh)**Format:** Unsigned binary

The NTCC_UTHR command controls configures the NTC cell discharging under temperature monitor voltage threshold and the NTC cell monitor hysteresis.

Bits	Access	Bit Name	Default	Description
15:11	R/W (can lock to read-only)	NTC_CELL_HYST	5'b 10001	Sets the cell NTC monitor hysteresis. LSB: 0.1953% Range: 0% to 6.0543% Value (%) = Setting x 0.1953 These bits allow OTP
10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R/W (can lock to read-only)	NTC_CELL_UT	10'b 101001010 0	Sets the NTC cell charging under temperature monitor voltage threshold. The limit is triggered if the value is equal to or above the set threshold. LSB: 0.09766% Range: 0% to 99.906% Value (%) = Setting x 0.09766 These bits allow MTP

NTCM_OTHr (4Eh)**Format:** Unsigned binary

The NTCM_OTHr command controls enable/disable of the NTC PCB over temperature monitor voltage threshold and NTC PCB monitor hysteresis.

Bits	Access	Bit Name	Default	Description
15:11	R/W (can lock to read-only)	PCB_MNTR_HYST	5'b 10000	Sets the NTC PCB monitor hysteresis. LSB: 0.1953% Range: 0% to 6.0543% Value (%) = Setting x 0.1953


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

				These bits allow OTP
10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R/W (can lock to read-only)	PCB_MNTR_HOT	10'b 001110101 1	Sets the NTC PCB over temperature monitor voltage threshold. The limit is triggered if the value is equal to or below the set threshold. LSB: 0.09766% Range: 0% to 99.906% Value (%) = Setting x 0.09766 These bits allow OTP

DIE_OT (4Fh)
Format: Unsigned binary

The DIE_OT command configures the digital die temperature threshold and the digital die temperature hysteresis.

Bits	Access	Bit Name	Default	Description
15:11	R/W (can lock to read-only)	DIE_TEMP_HYST	5'b 10101	Sets the digital die temperature hysteresis. LSB: 0.474°C Range: 0°C to 14.694°C Value (°C) = Setting x 0.474 These bits allow OTP
10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R/W (can lock to read-only)	DIE_TEMP_HOT	10'b 101110001 1	Sets the digital die temperature threshold. LSB: 0.474°C Offset: - 269.12°C Range: - 269.12°C to 215.782°C Value (°C) = Setting x 0.474 - 269.12 These bits allow OTP

RD_VA3P3 (51h)
Format: Unsigned binary

The RD_VA3P3 command reports the latest ADC reading for the 3V3 regulator.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R	V3V3_VALUE	10'b 000000000 0	Reports the latest ADC reading for the 3V3 regulator. Value (mV) = Reading x 6.4453

RD_VA5 (52h)
Format: Unsigned binary

The RD_VA5 command reports the latest ADC reading for the 5V regulator.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R	V5_VALUE	10'b 000000000 0	Reports the latest ADC reading for the 3V3 regulator. Value (mV) = Reading x 6.4453

**RD_VASELF (53h)****Format:** Two's complement

The RD_VASELF command reports the latest 10-bit ADC reading for the ADC self-test.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R	SELF_TEST_VALUE	10'b 000000000 0	Reports the latest 10-bit complement-signed ADC reading for the ADC self-test. The value should be automatically switched alternately between 1.2V and 0V Value (mV) = Reading x 3.2227

RD_OPENL (54h)**Format:** Unsigned binary

The RD_OPENL command reports whether C0-C15 has as an open wire after a detection has been executed.

Bits	Access	Bit Name	Default	Description
15	R	WIRE_C15_OPEN_R T_STS	1'b 0	Reports whether C15 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
14	R	WIRE_C14_OPEN_R T_STS	1'b 0	Reports whether C14 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
13	R	WIRE_C13_OPEN_R T_STS	1'b 0	Reports whether C13 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
12	R	WIRE_C12_OPEN_R T_STS	1'b 0	Reports whether C12 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
11	R	WIRE_C11_OPEN_R T_STS	1'b 0	Reports whether C11 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
10	R	WIRE_C10_OPEN_R T_STS	1'b 0	Reports whether C10 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
9	R	WIRE_C9_OPEN_RT _STS	1'b 0	Reports whether C9 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

8	R	WIRE_C8_OPEN_RT_STS	1'b 0	Reports whether C8 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
7	R	WIRE_C7_OPEN_RT_STS	1'b 0	Reports whether C7 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
6	R	WIRE_C6_OPEN_RT_STS	1'b 0	Reports whether C6 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
5	R	WIRE_C5_OPEN_RT_STS	1'b 0	Reports whether C5 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
4	R	WIRE_C4_OPEN_RT_STS	1'b 0	Reports whether C4 has an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
3	R	WIRE_C3_OPEN_RT_STS	1'b 0	Reports whether C3 has an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
2	R	WIRE_C2_OPEN_RT_STS	1'b 0	Reports whether C2 has an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
1	R	WIRE_C1_OPEN_RT_STS	1'b 0	Reports whether C1 has an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
0	R	WIRE_C0_OPEN_RT_STS	1'b 0	Reports whether C0 has an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected

RD_OPENH_CTL (55h)
Format: Unsigned binary

The OPW_CTL_RDH(RD_OPENH) command controls enable/disable of open-wire detection and enable/disable of open-wire detection when the device is powered on. The RD_OPENH command reports whether C16-C24 has as an open wire after a detection has been executed.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	OPEN_WIRE_PON	1'b 0	Enables open-wire detection when the device is powered on (leaving shutdown mode). OPEN_WIRE_EN should also be set to enable, thus, this bit can operate correctly. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R/W (can lock to read-only)	OPEN_WIRE_EN	1'b 1	Enables open-wire detection. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
11:9	R	RESERVED	N/A	Reserved. Do not change this register value.
8	R	WIRE_C24_OPEN_R T_STS	1'b 0	Reports whether C24 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
7	R	WIRE_C23_OPEN_R T_STS	1'b 0	Reports whether C23 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
6	R	WIRE_C22_OPEN_R T_STS	1'b 0	Reports whether C22 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
5	R	WIRE_C21_OPEN_R T_STS	1'b 0	Reports whether C21 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
4	R	WIRE_C20_OPEN_R T_STS	1'b 0	Reports whether C20 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
3	R	WIRE_C19_OPEN_R T_STS	1'b 0	Reports whether C19 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
2	R	WIRE_C18_OPEN_R T_STS	1'b 0	Reports whether C18 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected



1	R	WIRE_C17_OPEN_RT_STS	1'b 0	Reports whether C17 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected
0	R	WIRE_C16_OPEN_RT_STS	1'b 0	Reports whether C16 has as an open wire after a detection has been executed. The cell must be enabled from CELL_S_CTRL for this flag to function. 1'b 0: Not detected 1'b 1: Detected

OPW_GO (56h)**Format:** Unsigned binary

The OPW_GO command reports whether open-wire detection is complete, an error has occurred during open-wire detection. It also controls enable/disable of open wire detection.

Bits	Access	Bit Name	Default	Description
15:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R	OPEN_WIRE_DONE_STS	1'b 0	Reports whether open-wire detection is complete. When true, open-wire detection is complete. Write 0 to OPEN_WIRE_GO to clear this flag. 1'b 0: False 1'b 1: True
0	R/W	OPEN_WIRE_GO	1'b 0	Write 1 to schedule an open wire check, and write 0 to clear the done bit of open wire detection. 1'b 0: False 1'b 1: True

OPEN_CFG (57h)**Format:** Unsigned binary

The OPEN_CFG command configures the length of each pull together, pull apart, pull-up and pull-down phase, the open wire low, high, difference threshold used in the detection sequence.

Bits	Access	Bit Name	Default	Description
15:12	R/W (can lock to read-only)	OPW_CHECK_DTH	4'b 0000	Sets the open wire difference threshold used in the detection sequence. LSB: 312.5mV Offset: 312.5mV Range: 312.5mV to 5000mV Value (mV) = Setting x 312.5 + 312.5 These bits allow OTP
11:8	R/W (can lock to read-only)	OPW_CHECK_LOW_TH	4'b 0011	Sets the open wire low threshold used in the detection sequence. It is the threshold to judge if either the upper wire or the lower wire is open during the pull together process. LSB: 39.0625mV Range: 0mV to 585.9375mV Value (mV) = Setting x 39.0625 These bits allow OTP
7:3	R/W (can lock to read-only)	OPW_CHECK_HIGH_TH	5'b 00000	Sets the open wire high threshold used in the detection sequence. It is the threshold to judge if either the upper wire or the lower wire is open during the pull apart process. LSB: 156.25 mV



				Range: 0 mV to 4843.75mV Value (mV) = Setting x 156.25 These bits allow OTP
2:0	R/W (can lock to read-only)	OPW_CHECK_LEN	3'b 000	Sets the length of each pull together, pull apart, pull-up and pull-down phase. LSB: 2ms Offset: 2ms Range: 2ms to 16ms Value (ms) = Setting x 2 + 2 These bits allow OTP

NVM_CRC_GO (58h)**Format:** Unsigned binary

The NVM_CRC_GO command controls enable/disable of NVM CRC function and scheduling an NVM CRC. It also reports the NVM CRC check completion and the errors in the outcome NVM CRC check.

Bits	Access	Bit Name	Default	Description
15:7	R	RESERVED	N/A	Reserved. Do not change this register value.
6	R	NVM_CRC_OUTCOME	1'b 0	Reports the errors in the outcome NVM CRC check. The result of the previous check is retained until NVM_CRC_DO is set to 1 (from 0). Then NVM_CRC_OUTCOME remains at 0 until the check is completed and a new result is available. 1'b 0: No errors detected 1'b 1: One or more error(s) detected
5	R	NVMCHK_DONE_STS	1'b 0	When true, NVM CRC check is complete. This bit can be cleared by setting NVM_CRC_DO = 0. 1'b 0: False 1'b 1: True
4:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R/W	NVM_CRC_DO	1'b 0	Write 0 to this bit to clear the results. Write 1 to this bit to schedule an NVM CRC.
0	R/W (can lock to read-only)	NVM_CRC_EN	1'b 1	Enables the NVM CRC. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

V3P3_UV (59h)**Format:** Unsigned binary

The V3P3_UV command configures the 3V3 UV threshold.

Bits	Access	Bit Name	Default	Description
15:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7:0	R/W	V3V3_UV_LIMIT	8'b 11101000	Sets the threshold for 3V3 UV conditions, which is applied to the ADC reading reported in 3V3_VALUE. LSB: 12.890625mV Range: 0mV to 3287.11mV Value (mV) = Setting x 12.890625 These bits allow OTP

**V5_UV (5Bh)****Format:** Unsigned binary

The V5_UV command configures the 5V UV threshold.

Bits	Access	Bit Name	Default	Description
15:9	R	RESERVED	N/A	Reserved. Do not change this register value.
8:0	R/W (can lock to read-only)	V5_UV_LIMIT	9'b 101011101	Sets the threshold for 5V UV conditions, which is applied to the ADC reading reported in 5V_VALUE. LSB: 12.890625mV Range: 0mV to 6587.11mV Value (mV) = Setting x 12.890625 These bits allow OTP

FT_STS0 (5Ch)**Format:** Unsigned binary

The FT_STS0 command reports whether OC/SC is active, and indicates the fuse fault is triggered by cell SOV/dead/mismatch, VTOP SUV/SOV, DSG/CHG MOSFET dead, or LDO UV.

Bits	Access	Bit Name	Default	Description
15	R	LDO_UV_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a LDO UV event. This status is a latched status, which is cleared by LDO_UV_FAULT_CLEAR. 1'b 0: False 1'b 1: True
14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R	CHGFET_DEAD_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a CHG MOSFET dead event. This status is a latched status, which is cleared by CHGFET_DEAD_FAULT_CLEAR. 1'b 0: False 1'b 1: True
12	R	DSGFET_DEAD_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a DSG MOSFET dead event. This status is a latched status, which is cleared by DSGFET_DEAD_FAULT_CLEAR. 1'b 0: False 1'b 1: True
11	R	VTOP_SUV_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a VTOP SUV event. This status is a latched status, which is cleared by VTOP_SUV_FAULT_CLEAR. 1'b 0: False 1'b 1: True
10	R	VTOP_SOV_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a VTOP SOV event. This status is a latched status, which is cleared by VTOP_SOV_FAULT_CLEAR. 1'b 0: False 1'b 1: True
9	R	CELL_MSMT_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a cell mismatch event. This status is a latched status, which is cleared by CELL_MSMT_FAULT_CLEAR. 1'b 0: False 1'b 1: True



8	R	CELL_DEAD_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a cell dead event. This status is a latched status, which is cleared by CELL_DEAD_FAULT_CLEAR. 1'b 0: False 1'b 1: True
7	R	CELL_SOV_FUSE_STS	1'b 0	Reports whether the fuse fault is triggered by a cell SOV event. This status is a latched status, which is cleared by CELL_SOV_FAULT_CLEAR. 1'b 0: False 1'b 1: True
6:1	R	RESERVED	N/A	Reserved. Do not change this register value.
0	R	O_RECOVERY_BUS_Y	1'b 0	When true, reports that either charge over-current, discharge short-circuit, or over-current recovery is active. 1'b 0: False 1'b 1: True

FT_STS1 (5Dh)**Format:** Unsigned binary

The FT_STS1 command controls indicates the fault is triggered by cell UV/OV, VTOP UV/OV, NTC cell charging/discharging cold and hot, DSG/CHG turn-on failure, discharge OC1/OC2/SC, charge OC, discharge OC1/OC2 auto recovery failure, and charge OC auto recovery failure.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R	OC_DCHG_RECOVERY_FAILED	1'b 0	Reports that the either discharge OC 2 fault or discharge OC 1 fault has failed automatic recovery. In the case of a discharge OC2 or OC 1 fault failed recovery, this bit remains 1 until the fault has been manually cleared by OC2_DCHG_FAULT_CLR or OC1_DCHG_FAULT_CLR, or until the fault has been cleared through detection of charge current when DSG_OC2_EXIT_CHG_CURR or DSG_OC1_EXIT_CHG_CURR =1, or detection of charge voltage when DSG_OC2_EXIT_CHG_VMID or DSG_OC1_EXIT_VMID =1. 1'b 0: False 1'b 1: True
13	R	OC_CHG_RECOVERY_FAILED	1'b 0	Reports that the charge OC fault has failed automatic recovery. In the case of a charge OC fault failed recovery, this bit remains 1 until the DSG FET has successfully turned-on after recovery retry or until the fault has been manually cleared by DSG_DRIVER_FAULT_CLR. until the fault has been manually cleared by OC_CHG_FAULT_CLR or until the fault has been cleared through detection of discharge current when CHG_OC_EXIT_DSG_CURR = 1 or detection of discharge voltage when CHG_OC_EXIT_DSG_VPACKP = 1. 1'b 0: False 1'b 1: True



12	R	OC_CHG_FAULT_STS	1'b 0	Reports whether a charge over-current (OC) condition has triggered a fault. In the case of a charge OC fault, this bit remains 1 until the CHG FET has successfully turned-on after recovery retry and remains 1 during the charge driver cool period or until the fault has been manually cleared by OC_CHG_FAULT_CLR. 1'b 0: False 1'b 1: True
11	R	OC2_DCHG_FAULT_STS	1'b 0	Reports whether a discharge short-circuit (SC) or over-current (OC) 2 condition has triggered a fault. In the case of a discharge SC or OC 2 faults, this bit remains 1 until the DSG FET has successfully turned-on after recovery retry and remains 1 during the discharge driver cool period or until the fault has been manually cleared by OC2_DCHG_FAULT_CLR. 1'b 0: False 1'b 1: True
10	R	OC1_DCHG_FAULT_STS	1'b 0	Reports whether a discharge short-circuit (SC) or over-current (OC) 2 or OC 1 condition has triggered a fault. In the case of a discharge SC or OC 2 or OC1 faults, this bit remains 1 until the DSG FET has successfully turned-on after recovery retry and remains 1 during the discharge driver cool period or until the fault has been manually cleared by OC1_DCHG_FAULT_CLR. 1'b 0: False 1'b 1: True
9	R	CHG_DRIVER_FAULT_STS	1'b 0	Reports whether a CHG driver turn-on condition is triggering a fault. In the case of a CHG FET driver fault, this bit remains 1 until the CHG FET has successfully turned-on after recovery retry or until the fault has been manually cleared by CHG_DRIVER_FAULT_CLR. 1'b 0: False 1'b 1: True
8	R	DSG_DRIVER_FAULT_STS	1'b 0	Reports whether a DSG driver turn-on condition is triggering a fault. In the case of a DSG FET driver fault, this bit remains 1 until the DSG FET has successfully turned-on after recovery retry or until the fault has been manually cleared by DSG_DRIVER_FAULT_CLR. 1'b 0: False 1'b 1: True
7	R	NTC_CELL_DSG_COLD_FAULT_STS	1'b 0	Reports whether an NTC cell monitoring fault triggered due to the NTC temperature exceeds DSG under temperature limit. 1'b 0: False 1'b 1: True
6	R	NTC_CELL_DSG_HOT_FAULT_STS	1'b 0	Reports whether an NTC cell monitoring fault triggered due to the NTC temperature exceeds DSG over temperature limit. 1'b 0: False 1'b 1: True
5	R	NTC_CELL_CHG_COLD_FAULT_STS	1'b 0	Reports whether an NTC cell monitoring fault triggered due to the NTC temperature exceeds CHG under temperature limit. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

4	R	NTC_CELL_CHG_HO T_FAULT_STS	1'b 0	Reports whether an NTC cell monitoring fault triggered due to the NTC temperature exceeds CHG over temperature limit. 1'b 0: False 1'b 1: True
3	R	VTOP_OV_FAULT_S TS	1'b 0	Reports whether a VTOP OV condition is triggering a fault. 1'b 0: False 1'b 1: True
2	R	VTOP_UV_FAULT_S TS	1'b 0	Reports whether a VTOP UV condition is triggering a fault. 1'b 0: False 1'b 1: True
1	R	CELL_OV_FAULT_S TS	1'b 0	Reports whether a cell OV condition is triggering a fault. 1'b 0: False 1'b 1: True
0	R	CELL_UV_FAULT_ST S	1'b 0	Reports whether a cell UV condition is triggering a fault. 1'b 0: False 1'b 1: True

FT_STS2 (5Eh)
Format: Unsigned binary

The FT_STS2 command indicates the fuse fault is triggered by ADC self-diagnostic. It also indicates fault triggered by NVM CRC, NTC PCB too hot, die temperature too hot, open-wire, discharge SC, discharge SC auto recovery failure, cell SOV/dead/mismatch, VTOP SUV/SOV, ADC self-diagnostic, LDO UV, DSG/CHG MOSFET driver auto recovery failure.

Bits	Access	Bit Name	Default	Description
15	R	OPEN_WIRE_FAULT _STS	1'b 0	Reports whether an open-wire condition is triggering a fault. 1'b 0: False 1'b 1: True
14	R	ADC_FAIL_FUSE_ST S	1'b 0	Reports whether the fuse fault is triggered by an ADC fail event. This status is a latched status, which is cleared by ADC_FAIL_FAULT_CLEAR. 1'b 0: False 1'b 1: True
13	R	LDO_FAULT_STS	1'b 0	Reports whether a 3V3, 5V, or VDD UV is triggering a fault. An LDO fault can be triggered by either ADC monitoring or the analog UV comparator. The 3.3V, 5V and VDD analog comparator is always enabled, so when LDO_FAULT_EN is enabled, it can trigger a fault even when ADC 3.3V UV monitoring (V3V3_EN) and ADC 5V UV Monitoring (V5_EN) are disabled. 1'b 0: False 1'b 1: True
12	R	ADC_FAIL_FAULT_S TS	1'b 0	Reports whether an ADC fail condition is triggering a fault. This status is a latched status, which is cleared by ADC_FAIL_FAULT_CLEAR. 1'b 0: False 1'b 1: True
11	R	VTOP_SUV_FAULT_ STS	1'b 0	Reports whether a VTOP SUV condition is triggering a fault. 1'b 0: False 1'b 1: True



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R	VTOP_SOV_FAULT_STS	1'b 0	Reports whether a VTOP SOV condition is triggering a fault. 1'b 0: False 1'b 1: True
9	R	CELL_MSMT_FAULT_STS	1'b 0	Reports whether a cell mismatch condition is triggering a fault. 1'b 0: False 1'b 1: True
8	R	CELL_DEAD_FAULT_STS	1'b 0	Reports whether a dead cell condition is triggering a fault. 1'b 0: False 1'b 1: True
7	R	CELL_SOV_FAULT_STS	1'b 0	Reports whether a cell SOV condition is triggering a fault. 1'b 0: False 1'b 1: True
6	R	DSG_FET_DRV_RECOVERY_FAILED	1'b 0	Reports that the DSG FET driver fault has failed automatic recovery. In the case of a DSG FET driver fault failed recovery, this bit remains 1 until the fault has been manually cleared by DSG_DRIVER_FAULT_CLR, or until the fault has been cleared through detection of charge current when DSG_DRIVER_FAIL_EXIT_CHG_CURR = 1. 1'b 0: False 1'b 1: True
5	R	CHG_FET_DRV_RECOVERY_FAILED	1'b 0	Reports that the CHG FET driver fault has failed automatic recovery. In the case of a CHG FET driver fault failed recovery, this bit remains 1 until the fault has been manually cleared by CHG_DRIVER_FAULT_CLR, or until the fault has been cleared through detection of discharge current when CHG_DRIVER_FAIL_EXIT_DSG_CURR = 1. 1'b 0: False 1'b 1: True
4	R	SC_DCHG_RECOVERY_FAILED	1'b 0	Reports that the discharge SC fault has failed automatic recovery. In the case of a discharge SC fault failed recovery, this bit remains 1 until the fault has been manually cleared by SC_DCHG_FAULT_CLR, or until the fault has been cleared through detection of charge current when DSG_SC_EXIT_CHG_CURR = 1, or detection of charge voltage when DSG_SC_EXIT_CHG_VMID = 1. 1'b 0: False 1'b 1: True
3	R	SC_DCHG_FAULT_STS	1'b 0	Reports whether a discharge short-circuit (SC) condition has triggered a fault. In the case of a discharge SC fault, this bit remains 1 until the DSG FET has successfully turned-on after recovery retry or until the fault has been manually cleared by SC_DCHG_FAULT_CLR. 1'b 0: False 1'b 1: True
2	R	DIE_TEMP_FAULT_STS	1'b 0	Reports whether a die temperature condition is triggering a fault. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

1	R	PCB_MNTR_FAULT_STS	1'b 0	Reports whether an NTC PCB monitor condition is triggering a fault. 1'b 0: False 1'b 1: True
0	R	NVM_CRC_FAULT_STS	1'b 0	Reports whether an NVM CRC condition is triggering a fault. This bit can be cleared with NVM_CRC_CLR. 1'b 0: False 1'b 1: True

FT_CLR (5Fh)
Format: Unsigned binary

The FT_CLR command controls the manual bit-clear of faults, including cell UV/OV, VTOP UV/OV, NTC cell discharging/charging cold and hot, DSG/CHG MOSFET turn-on failure, discharge OC1/OC2, and charge OC.

Bits	Access	Bit Name	Default	Description
15:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12	W	OC_CHG_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a charge OC fault and charge OC fault retry counter. This is a self-clearing register.
11	W	OC2_DCHG_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a discharge OC 2 fault and discharge OC 2 fault retry counter. This is a self-clearing register.
10	W	OC1_DCHG_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a discharge OC 1 fault and discharge OC 1 fault retry counter. This is a self-clearing register.
9	W	CHG_DRIVER_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a CHG FET driver fault and CHG FET driver fault retry counter. This is a self-clearing register.
8	W	DSG_DRIVER_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a DSG FET driver fault and DSG FET driver fault retry counter. This is a self-clearing register.
7	W	NTC_CELL_DSG_COLD_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear an NTC cell under temperature DSG fault. This is a self-clearing register.
6	W	NTC_CELL_DSG_HOT_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear an NTC cell over temperature DSG fault. This is a self-clearing register.
5	W	NTC_CELL_CHG_COLD_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear an NTC cell under temperature CHG fault. This is a self-clearing register.
4	W	NTC_CELL_CHG_HOT_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear an NTC cell over temperature CHG fault. This is a self-clearing register.
3	W	VTOP_OV_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a battery pack OV fault. This is a self-clearing register.
2	W	VTOP_UV_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a battery pack UV fault. This is a self-clearing register.
1	W	CELL_OV_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a cell OV fault. This is a self-clearing register.
0	W	CELL_UV_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a cell UV fault. This is a self-clearing register.

FT2_CLR (60h)
Format: Unsigned binary

The FT2_CLR command controls the manual bit-clear of faults, including NVM CRC, NTC PCB too hot, die temperature too hot, open-wire, discharge SC and discharge SC fault retry counter, cell


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

SOV/dead/mismatch, VTOP SUV/SOV, DSG/CHG MOSFET dead, ADC self-diagnostic failure, and LDO UV.

Bits	Access	Bit Name	Default	Description
15	W	LDO_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear 3.3V and 5V UV faults. This is a self-clearing register.
14	W	ADC_FAIL_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear an ADC fail fault. This is a self-clearing register.
13	W	CHGFET_DEAD_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear a CHG MOSFET dead fault. This is a self-clearing register.
12	W	DSGFET_DEAD_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear a DSG MOSFET dead fault. This is a self-clearing register.
11	W	VTOP_SUV_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear a battery pack SUV fault. This is a self-clearing register.
10	W	VTOP_SOV_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear a battery pack SOV fault. This is a self-clearing register.
9	W	CELL_MSMT_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a mismatched cell fault. This is a self-clearing register.
8	W	CELL_DEAD_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a dead cell fault. This is a self-clearing register.
7	W	CELL_SOV_FAULT_CLEAR	1'b 0	Write 1 to this bit to manually clear a cell SOV fault. This is a self-clearing register.
6:5	R	RESERVED	N/A	Reserved. Do not change this register value.
4	W	SC_DCHG_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a discharge SC fault and discharge SC fault retry counter. This is a self-clearing register.
3	W	OPEN_WIRE_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a cell open wire fault. This is a self-clearing register.
2	W	DIE_TEMP_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear a die temperature fault. This is a self-clearing register.
1	W	PCB_MNTR_FAULT_CLR	1'b 0	Write 1 to this bit to manually clear an NTC PCB monitor fault. This is a self-clearing register.
0	W	NVM_CRC_CLR	1'b 0	Write 1 to this bit to manually clear an NVM CRC fault. This is a self-clearing register.

FT_ACT_CFG (61h)

Format: Unsigned binary

The FT_ACT_CFG command configures MOSFET behavior when fault occurs, fault includes NTC cell discharging/charging cold and hot, cell UV/OV, VTOP UV/OV, discharge OC1/OC2/SC, charge OC, and DSG/CHG MOSFET turn-on failure.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	SC_DSG_FAULT_ACTION	1'b 1	Sets the MOSFET behavior when DSG SC fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP



12	R/W (can lock to read-only)	OC_CHG_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when CHG OC fault happened. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
11	R/W (can lock to read-only)	OC2_DCHG_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when DSG OC 2 fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
10	R/W (can lock to read-only)	OC1_DCHG_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when DSG OC 1 fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
9	R/W (can lock to read-only)	CHG_DRIVER_FAULT_ACTION	1'b 1	Sets the MOSFET behavior when CHG driver fault happened. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
8	R/W (can lock to read-only)	DSG_DRIVER_FAULT_ACTION	1'b 1	Sets the MOSFET behavior when DSG driver fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
7	R/W (can lock to read-only)	NTC_CELL_DSG_COLD_ACTION	1'b 0	Sets the MOSFET behavior when NTC cell under temperature DSG fault happened. Standby region can be considered as charge or discharge depending on CELL_COLD_STBY_MODE setting. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
6	R/W (can lock to read-only)	NTC_CELL_DSG_HOT_ACTION	1'b 0	Sets the MOSFET behavior when NTC cell over temperature DSG fault happened. Standby region can be considered as charge or discharge depending on CELL_HOT_STBY_MODE setting. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
5	R/W (can lock to read-only)	NTC_CELL_CHG_COLD_ACTION	1'b 0	Sets the MOSFET behavior when NTC cell under temperature CHG fault happened. Standby region can be considered as charge or discharge depending on CELL_COLD_STBY_MODE setting. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
4	R/W (can lock to read-only)	NTC_CELL_CHG_HOT_ACTION	1'b 0	Sets the MOSFET behavior when NTC cell over temperature CHG fault happened. Standby region can be considered as charge or discharge depending on CELL_HOT_STBY_MODE setting. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP



3	R/W (can lock to read-only)	VTOP_OV_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when battery pack OV fault happened. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
2	R/W (can lock to read-only)	VTOP_UV_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when battery pack UV fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
1	R/W (can lock to read-only)	CELL_OV_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when cell OV fault happened. 1'b 0: Only turn off CHG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP
0	R/W (can lock to read-only)	CELL_UV_FAULT_ACTION	1'b 0	Sets the MOSFET behavior when cell UV fault happened. 1'b 0: Only turn off DSG MOSFET 1'b 1: Turn off CHG and DSG MOSFET This bit allows OTP

FT_REC_EN (62h)**Format:** Unsigned binary

The FT_REC_EN command controls enable/disable of recovery for fault, fault includes open-wire, die temperature too hot, discharge OC1/OC2/SC, charge OC, NTC cell discharging and charging cold/hot, NTC PCB too hot, cell UV/OV and VTOP UV/OV.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R/W (can lock to read-only)	OPEN_WIRE_REC	1'b 0	Enables automatic recovery for open wire fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
13	R/W (can lock to read-only)	DIE_TEMP_FAULT_REC	1'b 0	Enables automatic recovery for die temperature conditions (when the temperature drops). 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
12	R/W (can lock to read-only)	PCB_MNTR_REC	1'b 0	Enables automatic recovery from an NTC PCB monitor type fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
11	R/W (can lock to read-only)	SC_DCHG_REC	1'b 0	Enables an automatic recovery attempt for discharge SC conditions. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

10	R/W (can lock to read-only)	OC_CHG_REC	1'b 0	Enables an automatic recovery attempt for charge OC conditions. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
9	R/W (can lock to read-only)	OC2_DCHG_REC	1'b 0	Enables an automatic recovery attempt for discharge OC 2 conditions. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
8	R/W (can lock to read-only)	OC1_DCHG_REC	1'b 0	Enables an automatic recovery attempt for discharge OC 1 conditions. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
7	R/W (can lock to read-only)	NTC_CELL_DSG_CO LD_REC	1'b 0	Enables automatic recovery for NTC cell under temperature DSG conditions (when the NTC temperature back in range). 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
6	R/W (can lock to read-only)	NTC_CELL_DSG_HO T_REC	1'b 0	Enables automatic recovery for NTC cell over temperature DSG conditions (when the NTC temperature back in range). 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
5	R/W (can lock to read-only)	NTC_CELL_CHG_CO LD_REC	1'b 0	Enables automatic recovery for NTC cell under temperature CHG conditions (when the NTC temperature back in range). 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
4	R/W (can lock to read-only)	NTC_CELL_CHG_HO T_REC	1'b 0	Enables automatic recovery for NTC cell over temperature CHG conditions (when the NTC temperature back in range). 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
3	R/W (can lock to read-only)	VTOP_OV_REC	1'b 0	Enables automatic recovery for battery pack OV fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
2	R/W (can lock to read-only)	VTOP_UV_REC	1'b 0	Enables automatic recovery for battery pack UV fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP



1	R/W (can lock to read-only)	CELL_OV_REC	1'b 0	Enables automatic recovery for cell OV fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
0	R/W (can lock to read-only)	CELL_UV_REC	1'b 0	Enables automatic recovery for cell UV fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP

FT_REC_CFG0 (63h)**Format:** Unsigned binary

The FT_REC_CFG0 command controls enable/disable of auto recovery for fault, fault includes DSG/CHG MOSFET turn-on failure. It also configures DSG/CHG MOSFET cool down period and DSG/CHG MOSFET turn-on failure retry attempts.

Bits	Access	Bit Name	Default	Description
15:14	R/W (can lock to read-only)	CHG_DRIVER_RETRY	2'b 00	Sets the number of CHG MOSFET driver fault retry attempts. 2'b 00: 1 time 2'b 01: 2 times 2'b 10: 3 times 2'b 11: Infinite These bits allow MTP
13:10	R/W (can lock to read-only)	CHG_DRIVER_COOL	4'b 0000	Sets the time used for CHG MOSFET driver fault hiccup (cool-down) period. LSB: 1s Offset: 1s Range: 1s to 16s Value (s) = Setting + 1 These bits allow MTP
9:8	R/W (can lock to read-only)	DSG_DRIVER_RETRY	2'b 00	Sets the number of DSG MOSFET driver fault retry attempts. 2'b 00: 1 time 2'b 01: 2 times 2'b 10: 3 times 2'b 11: Infinite These bits allow MTP
7:4	R/W (can lock to read-only)	DSG_DRIVER_COOL	4'b 0000	Sets the time used for DSG MOSFET driver fault hiccup (cool-down) period. LSB: 1s Offset: 1s Range: 1s to 16s Value (s) = Setting + 1 These bits allow MTP
3:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R/W (can lock to read-only)	CHG_DRIVER_REC	1'b 0	Enables automatic recovery for CHG driver fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP



0	R/W (can lock to read-only)	DSG_DRIVER_REC	1'b 0	Enables automatic recovery for DSG driver fault. 1'b 0: Manual recovery 1'b 1: Automatic recovery attempt This bit allows OTP
---	--------------------------------	----------------	-------	--

FT_REC_CFG1 (64h)**Format:** Unsigned binary

The FT_REC_CFG1 command controls enable/disable of SC removal detection during OC recovery, GPIO1 as cell UV indicator, GPIO2 as cell OV/open-wire indicator. It also configures DSG OC1 cool-down time and retry attempts, DSG OC2 cool-down time and retry attempts, and CHG OC cool-down time and retry attempts.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	SBD_GPIO	1'b 0	When enabled, the GPIO1 as the cell UV indicator and the GPIO2 as the cell OV or open wire indicator. This setting is only valid when GPIO1/2_IO = 0 and GPIO1/2_FSEL = 1. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	FORCE_SS_ON	1'b 0	When enabled, the DSG can be forced to soft start when it is the first MOSFET to turn on. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R	RESERVED	N/A	Reserved. Do not change this register value.
11:10	R/W (can lock to read-only)	OC_CHG_RETRY	2'b 00	Sets the number of CHG OC fault retry attempts. 2'b 00: 1 time 2'b 01: 2 times 2'b 10: 3 times 2'b 11: Infinite These bits allow MTP
9:8	R/W (can lock to read-only)	OC_CHG_COOL	2'b 00	Sets the CHG OC cool-down time. 2'b 00: 100ms 2'b 01: 200ms 2'b 10: 500ms 2'b 11: 1s These bits allow MTP
7:6	R/W (can lock to read-only)	OC2_DCHG_RETRY	2'b 00	Sets the number of DSG OC 2 fault retry attempts. 2'b 00: 1 time 2'b 01: 2 times 2'b 10: 3 times 2'b 11: Infinite These bits allow MTP



5:4	R/W (can lock to read-only)	OC2_DCHG_COOL	2'b 00	Sets the DSG OC 2 cool-down time. 2'b 00: 100ms 2'b 01: 200ms 2'b 10: 500ms 2'b 11: 1s These bits allow MTP
3:2	R/W (can lock to read-only)	OC1_DCHG_RETRY	2'b 00	Sets the number of DSG OC 1 fault retry attempts. 2'b 00: 1 time 2'b 01: 2 times 2'b 10: 3 times 2'b 11: Infinite These bits allow MTP
1:0	R/W (can lock to read-only)	OC1_DCHG_COOL	2'b 00	Sets the DSG OC 1 cool-down time. 2'b 00: 100ms 2'b 01: 200ms 2'b 10: 500ms 2'b 11: 1s These bits allow MTP

FT_REC_CFG2 (65h)**Format:** Unsigned binary

The FT_REC_CFG2 command controls enable/disable of recovery scenario for fault. If discharging current exists, the following faults will not be triggered: cell/VTOP OV, charge OC, CHG MOSFET turn-on failure, NTC cell cold and hot. If charging current exists, the following faults will not be triggered: cell/VTOP UV, discharge OC1/OC2, discharge SC, DSG MOSFET turn-on failure. If VMID>VTOP+160mV, the following faults will not be triggered: DSG OC1/OC2/SC. If PACKP<VTOP - 1.6V, charge OC will not be triggered.

Bits	Access	Bit Name	Default	Description
15	R/W (can lock to read-only)	DSG_SC_EXIT_CHG_VMID	1'b 0	Enables the battery pack voltage range checking for DSG SC fault exit. If enabled, when the MID voltage exceeds (VTOP + 160mV), the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
14	R/W (can lock to read-only)	CHG_OC_EXIT_DSG_VPACKP	1'b 0	Enables the battery pack voltage range checking for CHG OC fault exit. If enabled, when the PACKP voltage falls below (VTOP -1.6V), the fault can be self-cleared when the CHG OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
13	R/W (can lock to read-only)	DSG_OC2_EXIT_CHG_VMID	1'b 0	Enables the battery pack voltage range checking for DSG OC 2 fault exit. If enabled, when the MID voltage exceeds (VTOP + 160mV), the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



12	R/W (can lock to read-only)	DSG_OC1_EXIT_VMI D	1'b 0	Enables the battery pack voltage range checking for DSG OC 1 fault exit. If enabled, when the MID voltage exceeds (V_{TOP} + 160mV), the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
11	R/W (can lock to read-only)	NTC_COLD_EXIT_B Y_CURR	1'b 0	Enables the battery pack current range checking for NTC UT fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared even NTC voltage exceeds NTC cell UT threshold. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
10	R/W (can lock to read-only)	NTC_HOT_EXIT_BY_ CURR	1'b 0	Enables the battery pack current range checking for NTC OT fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared even NTC voltage falls below NTC cell OT threshold. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
9	R/W (can lock to read-only)	CHG_DRIVER_FAIL_ EXIT_DSG_CURR	1'b 0	Enables the battery pack current range checking for CHG MOSFET driver fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared when the CHG MOSFET driver fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
8	R/W (can lock to read-only)	DSG_DRIVER_FAIL_ EXIT_CHG_CURR	1'b 0	Enables the battery pack current range checking for DSG MOSFET driver fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared when the DSG MOSFET driver fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
7	R/W (can lock to read-only)	DSG_SC_EXIT_CHG_ CURR	1'b 0	Enables the battery pack current range checking for DSG SC fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
6	R/W (can lock to read-only)	CHG_OC_EXIT_DSG_ CURR	1'b 0	Enables the battery pack current range checking for CHG OC fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared when the CHG OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

5	R/W (can lock to read-only)	DSG_OC2_EXIT_CHG_CURR	1'b 0	Enables the battery pack current range checking for DSG OC 2 fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
4	R/W (can lock to read-only)	DSG_OC1_EXIT_CHG_CURR	1'b 0	Enables the battery pack current range checking for DSG OC 1 fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared when the DSG SC/OC fault recovery has failed. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
3	R/W (can lock to read-only)	VTOP_UV_EXIT_CHG_CURR	1'b 0	Enables the battery pack current range checking for VTOP UV fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared even VTOP voltage falls below the VTOP UV threshold. The VTOP UV fault only will be triggered when VTOP voltage falls below the VTOP UV threshold and battery pack current status is discharge or standby. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W (can lock to read-only)	VTOP_OV_EXIT_DS G_CURR	1'b 0	Enables the battery pack current range checking for VTOP OV fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared even VTOP voltage exceeds the VTOP OV threshold. The VTOP OV fault only will be triggered when VTOP voltage exceeds the VTOP OV threshold and battery pack current status is charge or standby. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read-only)	CELL_UV_EXIT_CHG_CURR	1'b 0	Enables the battery pack current range checking for cell UV fault exit. If enabled, when the battery pack current status is in charge state, the fault can be self-cleared even cell voltage falls below the cell UV threshold. The cell UV fault only will be triggered when cell voltage falls below the cell UV threshold and battery pack current status is discharge or standby. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

0	R/W (can lock to read-only)	CELL_OV_EXIT_DSG_CURR	1'b 0	Enables the battery pack current range checking for cell OV fault exit. If enabled, when the battery pack current status is in discharge state, the fault can be self-cleared even cell voltage exceeds the cell OV threshold. The cell OV fault only will be triggered when cell voltage exceeds the cell OV threshold and battery pack current status is charge or standby. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
---	--------------------------------	-----------------------	-------	--

RD_CCIRQL (66h)
Format: Unsigned binary

The RD_CCIRQL command returns the accumulated coulomb counter reading for the LSB.

Bits	Access	Bit Name	Default	Description
15:0	R	CC_ACC_LSBS	16'b 000000000 0000000	Accumulated coulomb counter reading for the LSB. This content is latched from the CC_RT_ACC registers when a Coulomb counting accumulation has been completed. The reading is complement-signed and 28 bits long (combined with CC_ACC_MSBS). The value ranges from -134217728 to +134217727.

RD_CCIRQH (67h)
Format: Unsigned binary

The RD_CCIRQH command returns the accumulated Coulomb counter reading for the MSB.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11:0	R	CC_ACC_MSBS	12'b 000000000 000	Accumulated Coulomb counter reading for the MSB. This content is latched from the CC_RT_ACC registers when a Coulomb counting accumulation has been completed. The reading is complement-signed and 28 bits long (combined with CC_ACC_LSBS). The value ranges from -134217728 to +134217727. The value can be calculated with the following equation: $\text{Value}(\text{m}\Omega \times \text{A} \times \text{s}) = \text{Reading} / 32768 / 5$

RD_CCACCQL (68h)
Format: Unsigned binary

The RD_CCACCQL command returns the real-time accumulated Coulomb counter reading for the LSB.

Bits	Access	Bit Name	Default	Description
15:0	R	CC_RT_ACC_LSBS	16'b 000000000 0000000	Returns the real-time accumulated Coulomb counter reading for the LSB. This register is constantly updated when Coulomb counting is active. The reading is complement-signed and 28 bits long (combining with CC_RT_ACC_MSBS). The value ranges from -134217728 to +134217727.

RD_CCACCQH (69h)
Format: Unsigned binary

The RD_CCACCQH command configures Returns the real-time accumulated Coulomb counter reading for the MSB.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.



11:0	R	CC_RT_ACC_MSBS	12'b 000000000 000	Returns the real-time accumulated Coulomb counter reading for the MSB. This register is constantly updated when Coulomb counting is active. The reading is complement-signed and 28 bits long (combining with CC_RT_ACC_LSBS). The value ranges from -134217728 to +134217727. The value can be calculated with the following equation: Value(mΩ x A x s) = Reading / 32768 / 5
------	---	----------------	--------------------------	--

RD_VTOP (6Ah)**Format:** Unsigned binary

The RD_VTOP command returns the VTOP high-resolution voltage reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	VTOP_V	15'b 000000000 0000000	Returns the VTOP high-resolution voltage reading. Voltage (mV) = Reading x 120000 / 32768

RD_ITOP (6Bh)**Format:** Two's complement

The RD_ITOP command returns the battery pack current reading (synchronous to VTOP_V).

Bits	Access	Bit Name	Default	Description
15:0	R	VTOP_I	16'b 000000000 00000000	Returns the battery pack current reading (synchronous to VTOP_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL1 (6Ch)**Format:** Two's complement

The RD_VCELL1 command returns the cell 1 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_1_V	16'b 000000000 00000000	Returns the cell 1 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL1 (6Dh)**Format:** Two's complement

The RD_ICELL1 command returns the cell 1 battery pack current reading (synchronous with CELL_1_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_1_I_SYNC	16'b 000000000 00000000	Returns the cell 1 battery pack current reading (synchronous with CELL_1_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL2 (6Eh)**Format:** Two's complement

The RD_VCELL2 command returns the cell 2 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_2_V	16'b 000000000 00000000	Returns the cell 2 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

**RD_ICELL2 (6Fh)****Format:** Two's complement

The RD_ICELL2 command returns the cell 2 battery pack current reading (synchronous with CELL_2_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_2_I_SYNC	16'b 000000000 0000000	Returns the cell 2 battery pack current reading (synchronous with CELL_2_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL3 (70h)**Format:** Two's complement

The RD_VCELL3 command returns the cell 3 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_3_V	16'b 000000000 0000000	Returns the cell 3 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL3 (71h)**Format:** Two's complement

The RD_ICELL3 command returns the cell 3 battery pack current reading (synchronous with CELL_3_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_3_I_SYNC	16'b 000000000 0000000	Returns the cell 3 battery pack current reading (synchronous with CELL_3_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL4 (72h)**Format:** Two's complement

The RD_VCELL4 command returns the cell 4 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_4_V	16'b 000000000 0000000	Returns the cell 4 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL4 (73h)**Format:** Two's complement

The RD_ICELL4 command returns the cell 4 battery pack current reading (synchronous with CELL_4_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_4_I_SYNC	16'b 000000000 0000000	Returns the cell 4 battery pack current reading (synchronous with CELL_4_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL5 (74h)**Format:** Two's complement

The RD_VCELL5 command returns the cell 5 high-resolution reading.



Bits	Access	Bit Name	Default	Description
15:0	R	CELL_5_V	16'b 000000000 0000000	Returns the cell 5 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL5 (75h)**Format:** Two's complement

The RD_ICELL5 command returns the cell 5 battery pack current reading (synchronous with CELL_5_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_5_I_SYNC	16'b 000000000 0000000	Returns the cell 5 battery pack current reading (synchronous with CELL_5_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL6 (76h)**Format:** Two's complement

The RD_VCELL6 command returns the cell 6 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_6_V	16'b 000000000 0000000	Returns the cell 6 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL6 (77h)**Format:** Two's complement

The RD_ICELL6 command returns the cell 6 battery pack current reading (synchronous with CELL_6_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_6_I_SYNC	16'b 000000000 0000000	Returns the cell 6 battery pack current reading (synchronous with CELL_6_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL7 (78h)**Format:** Two's complement

The RD_VCELL7 command returns the cell 7 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_7_V	16'b 000000000 0000000	Returns the cell 7 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL7 (79h)**Format:** Two's complement

The RD_ICELL7 command returns the cell 7 battery pack current reading (synchronous with CELL_7_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_7_I_SYNC	16'b 000000000 0000000	Returns the cell 7 battery pack current reading (synchronous with CELL_7_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

**RD_VCELL8 (7Ah)****Format:** Two's complement

The RD_VCELL8 command returns the cell 8 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_8_V	16'b 000000000 0000000	Returns the cell 8 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL8 (7Bh)**Format:** Two's complement

The RD_ICELL8 command returns the cell 8 battery pack current reading (synchronous with CELL_8_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_8_I_SYNC	16'b 000000000 0000000	Returns the cell 8 battery pack current reading (synchronous with CELL_8_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL9 (7Ch)**Format:** Two's complement

The RD_VCELL9 command returns the cell 9 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_9_V	16'b 000000000 0000000	Returns the cell 9 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL9 (7Dh)**Format:** Two's complement

The RD_ICELL9 command returns the cell 9 battery pack current reading (synchronous with CELL_9_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_9_I_SYNC	16'b 000000000 0000000	Returns the cell 9 battery pack current reading (synchronous with CELL_9_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL10 (7Eh)**Format:** Two's complement

The RD_VCELL10 command returns the cell 10 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_10_V	16'b 000000000 0000000	Returns the cell 10 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL10 (7Fh)**Format:** Two's complement

The RD_ICELL10 command returns the cell 10 battery pack current reading (synchronous with CELL_10_V).


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_10_I_SYNC	16'b 000000000 00000000	Returns the cell 10 battery pack current reading (synchronous with CELL_10_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL11 (80h)
Format: Two's complement

The RD_VCELL11 command returns the cell 11 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_11_V	16'b 000000000 00000000	Returns the cell 11 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL11 (81h)
Format: Two's complement

The RD_ICELL11 command returns the cell 11 battery pack current reading (synchronous with CELL_11_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_11_I_SYNC	16'b 000000000 00000000	Returns the cell 11 battery pack current reading (synchronous with CELL_11_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL12 (82h)
Format: Two's complement

The RD_VCELL12 command returns the cell 12 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_12_V	16'b 000000000 00000000	Returns the cell 12 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL12 (83h)
Format: Two's complement

The RD_ICELL12 command returns the cell 12 battery pack current reading (synchronous with CELL_12_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_12_I_SYNC	16'b 000000000 00000000	Returns the cell 12 battery pack current reading (synchronous with CELL_12_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL13 (84h)
Format: Two's complement

The RD_VCELL13 command returns the cell 13 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_13_V	16'b 000000000 00000000	Returns the cell 13 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

**RD_ICELL13 (85h)****Format:** Two's complement

The RD_ICELL13 command returns the cell 13 battery pack current reading (synchronous with CELL_6_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_13_I_SYNC	16'b 000000000 00000000	Returns the cell 13 battery pack current reading (synchronous with CELL_13_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL14 (86h)**Format:** Two's complement

The RD_VCELL14 command returns the cell 14 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_14_V	16'b 000000000 00000000	Returns the cell 14 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL14 (87h)**Format:** Two's complement

The RD_ICELL14 command returns the cell 14 battery pack current reading (synchronous with CELL_14_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_14_I_SYNC	16'b 000000000 00000000	Returns the cell 14 battery pack current reading (synchronous with CELL_14_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL15 (88h)**Format:** Two's complement

The RD_VCELL15 command returns the cell 15 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_15_V	16'b 000000000 00000000	Returns the cell 15 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL15 (89h)**Format:** Two's complement

The RD_ICELL15 command returns the cell 15 battery pack current reading (synchronous with CELL_15_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_15_I_SYNC	16'b 000000000 00000000	Returns the cell 15 battery pack current reading (synchronous with CELL_15_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

**RD_VCELL16 (8Ah)****Format:** Two's complement

The RD_VCELL16 command returns the cell 16 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_16_V	16'b 000000000 0000000	Returns the cell 16 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL16 (8Bh)**Format:** Two's complement

The RD_ICELL16 command returns the cell 16 battery pack current reading (synchronous with CELL_16_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_16_I_SYNC	16'b 000000000 0000000	Returns the cell 16 battery pack current reading (synchronous with CELL_16_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL17 (8Ch)**Format:** Two's complement

The RD_VCELL17 command returns the cell 17 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_17_V	16'b 000000000 0000000	Returns the cell 17 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL17 (8Dh)**Format:** Two's complement

The RD_ICELL17 command returns the cell 17 battery pack current reading (synchronous with CELL_17_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_17_I_SYNC	16'b 000000000 0000000	Returns the cell 17 battery pack current reading (synchronous with CELL_17_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL18 (8Eh)**Format:** Two's complement

The RD_VCELL18 command returns the cell 18 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_18_V	16'b 000000000 0000000	Returns the cell 18 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL18 (8Fh)**Format:** Two's complement

The RD_ICELL18 command returns the cell 18 battery pack current reading (synchronous with CELL_18_V).


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_18_I_SYNC	16'b 000000000 0000000	Returns the cell 18 battery pack current reading (synchronous with CELL_18_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL19 (90h)
Format: Two's complement

The RD_VCELL19 command returns the cell 19 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_19_V	16'b 000000000 0000000	Returns the cell 19 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL19 (91h)
Format: Two's complement

The RD_ICELL19 command returns the cell 19 battery pack current reading (synchronous with CELL_19_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_19_I_SYNC	16'b 000000000 0000000	Returns the cell 19 battery pack current reading (synchronous with CELL_19_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL20 (92h)
Format: Two's complement

The RD_VCELL20 command returns the cell 20 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_20_V	16'b 000000000 0000000	Returns the cell 20 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL20 (93h)
Format: Two's complement

The RD_ICELL20 command returns the cell 20 battery pack current reading (synchronous with CELL_20_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_20_I_SYNC	16'b 000000000 0000000	Returns the cell 20 battery pack current reading (synchronous with CELL_20_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL21 (94h)
Format: Two's complement

The RD_VCELL21 command returns the cell 21 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_21_V	16'b 000000000 0000000	Returns the cell 21 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

**RD_ICELL21 (95h)****Format:** Two's complement

The RD_ICELL21 command returns the cell 21 battery pack current reading (synchronous with CELL_21_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_21_I_SYNC	16'b 000000000 00000000	Returns the cell 21 battery pack current reading (synchronous with CELL_21_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL22 (96h)**Format:** Two's complement

The RD_VCELL22 command returns the cell 22 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_22_V	16'b 000000000 00000000	Returns the cell 22 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL22 (97h)**Format:** Two's complement

The RD_ICELL22 command returns the cell 22 battery pack current reading (synchronous with CELL_22_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_22_I_SYNC	16'b 000000000 00000000	Returns the cell 22 battery pack current reading (synchronous with CELL_22_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VCELL23 (98h)**Format:** Two's complement

The RD_VCELL23 command returns the cell 23 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_23_V	16'b 000000000 00000000	Returns the cell 23 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL23 (99h)**Format:** Two's complement

The RD_ICELL23 command returns the cell 23 battery pack current reading (synchronous with CELL_23_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_23_I_SYNC	16'b 000000000 00000000	Returns the cell 23 battery pack current reading (synchronous with CELL_23_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

**RD_VCELL24 (9Ah)****Format:** Two's complement

The RD_VCELL24 command returns the cell 24 high-resolution reading.

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_24_V	16'b 000000000 0000000	Returns the cell 24 high-resolution reading. Voltage (mV) = Reading x 5000 / 32768

RD_ICELL24 (9Bh)**Format:** Two's complement

The RD_ICELL24 command returns the cell 24 battery pack current reading (synchronous with CELL_24_V).

Bits	Access	Bit Name	Default	Description
15:0	R	CELL_24_I_SYNC	16'b 000000000 0000000	Returns the cell 24 battery pack current reading (synchronous with CELL_24_V). The reading is 16-bit complement-signed. Rsense is the external current-sense resistor, and its unit is mΩ. Current (A) = Reading x 100 / 32768 / Rsense

RD_VNTC1 (9Ch)**Format:** Unsigned binary

The RD_VNTC1 command returns the NTC1 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC1_HIRES_V	15'b 000000000 000000	Returns the NTC1 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VNTC2 (9Dh)**Format:** Unsigned binary

The RD_VNTC2 command returns the NTC2 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC2_HIRES_V	15'b 000000000 000000	Returns the NTC2 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VNTC3 (9Eh)**Format:** Unsigned binary

The RD_VNTC3 command returns the NTC3 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC3_HIRES_V	15'b 000000000 000000	Returns the NTC3 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

**RD_VNTC4 (9Fh)****Format:** Unsigned binary

The RD_VNTC4 command returns the NTC4 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC4_HIRES_V	15'b 000000000 000000	Returns the NTC4 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VNTC5 (A0h)**Format:** Unsigned binary

The RD_VNTC5 command returns the NTC5 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC5_HIRES_V	15'b 000000000 000000	Returns the NTC5 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VNTC6 (A1h)**Format:** Unsigned binary

The RD_VNTC6 command returns the NTC6 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	NTC6_HIRES_V	15'b 000000000 000000	Returns the NTC6 high-resolution voltage ADC reading. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VGPIO1 (A2h)**Format:** Unsigned binary

The RD_VGPIO1 command returns the GPIO1 or NTC7 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	GPIO1_VOLTAGE	15'b 000000000 000000	Returns the GPIO1 high-resolution voltage ADC reading when GPIO1_NTC7 = 0. Voltage (mV) = Reading x 3300 / 32768 Returns the NTC7 high-resolution voltage ADC reading when GPIO1_NTC7 = 1. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VGPIO2 (A3h)**Format:** Unsigned binary

The RD_VGPIO2 command returns the GPIO2 or NTC8 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.



14:0	R	GPIO2_VOLTAGE	15'b 000000000 000000	Returns the GPIO2 high-resolution voltage ADC reading when GPIO2_NTC8 = 0. Voltage (mV) = Reading x 3300 / 32768 Returns the NTC8 high-resolution voltage ADC reading when GPIO2_NTC8 = 1. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768
------	---	---------------	-----------------------------	--

RD_VGPIO3 (A4h)**Format:** Unsigned binary

The RD_VGPIO3 command returns the GPIO3 or NTC9 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	GPIO3_VOLTAGE	15'b 000000000 000000	Returns the GPIO3 high-resolution voltage ADC reading when GPIO3_NTC9 = 0. Voltage (mV) = Reading x 3300 / 32768 Returns the NTC9 high-resolution voltage ADC reading when GPIO3_NTC9 = 1. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_VGPIO4 (A5h)**Format:** Unsigned binary

The RD_VGPIO4 command controls returns the GPIO4 or NTC10 high-resolution voltage ADC reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	GPIO4_VOLTAGE	15'b 000000000 000000	Returns the GPIO4 high-resolution voltage ADC reading when GPIO4_NTC10 = 0. Voltage (mV) = Reading x 3300 / 32768 Returns the NTC10 high-resolution voltage ADC reading when GPIO4_NTC10 = 1. This reading is ratiometric to NTCB, which has a 3.3V nominal value. Value (%NTCB) = Reading x 100 / 32768

RD_TDIE (A6h)**Format:** Unsigned binary

The RD_TDIE command controls returns die temperature high-resolution reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	DIE_T_VOLTAGE	15'b 000000100 001101	Returns the die temperature high-resolution reading, which is proportional to the internal die temperature. The temperature can be calculated with the following equation: $T (^{\circ}\text{C}) = \text{Reading} \times 0.01481 - 269.12$

RD_VPACKP (A7h)**Format:** Unsigned binary

The RD_VPACKP command returns the PACKP high-resolution voltage reading.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:0	R	VPACK_V	15'b 000000000 000000	Returns the PACKP high-resolution voltage reading. Voltage (mV) = Reading x 120000 / 32768

**ADC_CTRL (A8h)****Format:** Unsigned binary

The ADC_CTRL command the status of the feature command scheduler, internal scheduler, the real-time status of auto open-wire, and the high-resolution voltage scan status. It also controls enable/disable of a high-resolution scan for all the selected channels.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R	AOPW_STS	1'b 0	This field reports the status of auto open-wire. It is a real time state. 1'b 0: Auto open-wire is not arranged 1'b 1: Auto open-wire is arranged
12:11	R	VADC_FSM	2'b 00	This field reports the status of the internal scheduler. 2'b 00: Idle 2'b 01: Refreshing voltage protection reading 2'b 10: Running High resolution VADC Scan 2'b 11: High resolution scan waiting (initiated) during voltage protection
10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:8	R	VADC_SCHEDULER	2'b 00	This field reports the status of the feature command scheduler. 2'b 00: Free 2'b 01: Executing voltage ADC scan 2'b 10: Executing cell-balancing 2'b 11: Executing open-wire detection
7:2	R	RESERVED	N/A	Reserved. Do not change this register value.
1	R	SCAN_DONE_STS	1'b 0	Reports whether the high-resolution voltage scan status is complete (true) or not (false). 1'b 0: False 1'b 1: True
0	R/W	ADC_SCAN_GO	1'b 0	Write 1 to this bit to start a high-resolution scan for all the selected channels.

ADC_CFG (A9h)**Format:** Unsigned binary

The ADC_CFG command controls enable/disable of the HR ADC scan continuous mode, the HR ADC scan continuous mode in safe mode, and the ESS mode. It also configures the HR ADC update rate during active, fault and ESS mode and during safe and standby mode. It disables the hardware protection and temperature compensation re-calculated once every 15 intermediate timeslots.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R/W (can lock to read-only)	DISABLE_HWP_SLOT	1'b 0	When this bit set to 1, the hardware protection is disabled and temperature compensation re-calculated once every 15 intermediate timeslots. When toggled from 0 to 1, the hardware protection may run one more time to clear any fault indicating status bits. This bit is valid in any mode, however, it is only recommended to set when ESS_MODE_EN = 1.



13	R/W (can lock to read-only)	ESS_MODE_EN	1'b 0	Enables the ESS mode. In ESS mode, the AFE will remain in active mode always, it cannot go to safe, sleep, standby or fault states. 1'b 0: Disabled 1'b 1: Enabled
12:11	R	RESERVED	N/A	Reserved. Do not change this register value.
10:8	R/W (can lock to read-only)	ADC_SCAN_SAFE	3'b 000	Selects the HR ADC update rate during safe and standby mode. 3'b 000: Generate a command 0 intermediate timeslots after the last HR scan complete 3'b 001: Generate a command 2 intermediate timeslots after the last HR scan complete 3'b 010: Generate a command 4 intermediate timeslots after the last HR scan complete 3'b 011: Generate a command 8 intermediate timeslots after the last HR scan complete 3'b 100: Generate a command 16 intermediate timeslots after the last HR scan complete 3'b 101: Generate a command 32 intermediate timeslots after the last HR scan complete 3'b 110: Generate a command 48 intermediate timeslots after the last HR scan complete 3'b 111: Generate a command 64 intermediate timeslots after the last HR scan complete
7:5	R/W (can lock to read-only)	ADC_SCAN_ACTIVE	3'b 000	Selects the HR ADC update rate during active, fault and ESS mode. 3'b 000: Generate a command 0 intermediate timeslots after the last HR scan complete 3'b 001: Generate a command 1 intermediate timeslots after the last HR scan complete 3'b 010: Generate a command 2 intermediate timeslots after the last HR scan complete 3'b 011: Generate a command 4 intermediate timeslots after the last HR scan complete 3'b 100: Generate a command 8 intermediate timeslots after the last HR scan complete 3'b 101: Generate a command 16 intermediate timeslots after the last HR scan complete 3'b 110: Generate a command 20 intermediate timeslots after the last HR scan complete 3'b 111: Generate a command 24 intermediate timeslots after the last HR scan complete
4	R/W (can lock to read-only)	ADC_SCAN_CONT_SAFE_EN	1'b 0	Enables the HR ADC scan continuous mode in safe mode when ADC_SCAN_CONT_EN = 1. 1'b 0: Disabled 1'b 1: Enabled
3	R/W (can lock to read-only)	ADC_SCAN_CONT_EN	1'b 0	Enables the HR ADC scan continuous mode. The ADC scan done interrupt should be triggered once the full channels ADC reading have been updated. 1'b 0: Disabled 1'b 1: Enabled
2:0	R	RESERVED	N/A	Reserved. Do not change this register value.

**PCB_GAIN_CAL (AAh)****Format:** Two's complement

The PCB_GAIN_CAL command configures reflow gain calibration compensating for post reflow bandgap shift. It controls enable/disable of reflow calibration for all cell voltage reading, PACKP and VTOP, current ADC and Coulomb counter reading, and all other channels. It also controls enable/disable of ADC cell measurement compensation for the voltage drop caused by the cell input resistor.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13	R/W (can lock to read-only)	CELL_1K_COMP	1'b 0	Enables ADC cell measurement compensation for the voltage drop caused by the cell input resistor, R_CELL_FILTER. This configuration should be enabled when the input filter resistance exceeds 500Ω. For R_CELL_FILTER values below 500Ω, this should be disabled. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
12	R	RESERVED	N/A	Reserved. Do not change this register value.
11	R/W (can lock to read-only)	PCB_GC_I_CTRL	1'b 0	Enables control to apply reflow calibration for current ADC and Coulomb counter reading. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
10	R/W (can lock to read-only)	PCB_GC_VOTHER_CTRL	1'b 0	Enables control to apply reflow calibration for all other channels (except NTCx because ADC reference used is not the bandgap). 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
9	R/W (can lock to read-only)	PCB_GC_VT_PK_CTRL	1'b 0	Enables control to apply reflow calibration for PACKP and VTOP. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
8	R/W (can lock to read-only)	PCB_GC_CELL_CTRL	1'b 0	Enables control to apply reflow calibration for all cell voltage reading. 1'b 0: Disabled 1'b 1: Enabled This bit allows MTP
7:0	R/W (can lock to read-only)	PCB_GAIN_CAL	8'b 00000000	Reflow gain calibration compensating for post reflow bandgap shift. This gain is common for all channels. Specific registers can be used to enable this correction on specific group of ADC channels, including PCB_GC_CELL_CTRL, PCB_GC_VT_PK_CTRL, PCB_GC_VOTHER_CTRL, and PCB_GC_I_CTRL, which use the reference, excluding NTC. LSB: 0.0127% Offset: -1.6256% Range: -1.6256% to 1.6129% Gain (%) = Setting x 0.0127 These bits allow MTP

**TRIMG_IPCB (ABh)****Format:** Two's complement

The TRIMG_IPCB command configures current-sense PCB gain correction to compensate for sense resistors and SMT variation.

Bits	Access	Bit Name	Default	Description
15:10	R	RESERVED	N/A	Reserved. Do not change this register value.
9:0	R/W (can lock to read-only)	I_PCB_GAIN_VALUE	10'b 000000000 0	Current-sense PCB gain correction can compensate for sense resistors and SMT variation. The correction is applied to both Coulomb counting and synchronous current ADC readings. It is not applied to short-circuit or over-current detection. The setting is 10-bit complement-signed. LSB: 0.0244% Offset: 100% Range: 87.5316% to 112.4684% Gain (%) = Setting x 0.0244 + 100 These bits allow MTP

HR_SCAN1 (ACh)**Format:** Unsigned binary

The HR_SCAN1 command controls enable/disable of cell 1-16's high-resolution voltages.

Bits	Access	Bit Name	Default	Description
15	R/W	CELL_16_VI_READ_EN	1'b 0	Enables the cell 16 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 16 is also provided. 1'b 0: Disabled 1'b 1: Enabled
14	R/W	CELL_15_VI_READ_EN	1'b 0	Enables the cell 15 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 15 is also provided. 1'b 0: Disabled 1'b 1: Enabled
13	R/W	CELL_14_VI_READ_EN	1'b 0	Enables the cell 14 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 14 is also provided. 1'b 0: Disabled 1'b 1: Enabled
12	R/W	CELL_13_VI_READ_EN	1'b 0	Enables the cell 13 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 13 is also provided. 1'b 0: Disabled 1'b 1: Enabled
11	R/W	CELL_12_VI_READ_EN	1'b 0	Enables the cell 12 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 12 is also provided. 1'b 0: Disabled 1'b 1: Enabled



10	R/W	CELL_11_VI_READ_EN	1'b 0	Enables the cell 11 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 11 is also provided. 1'b 0: Disabled 1'b 1: Enabled
9	R/W	CELL_10_VI_READ_EN	1'b 0	Enables the cell 10 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 10 is also provided. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	CELL_9_VI_READ_EN	1'b 0	Enables the cell 9 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 9 is also provided. 1'b 0: Disabled 1'b 1: Enabled
7	R/W	CELL_8_VI_READ_EN	1'b 0	Enables the cell 8 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 8 is also provided. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	CELL_7_VI_READ_EN	1'b 0	Enables the cell 7 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 7 is also provided. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	CELL_6_VI_READ_EN	1'b 0	Enables the cell 6 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 6 is also provided. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	CELL_5_VI_READ_EN	1'b 0	Enables the cell 5 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 5 is also provided. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	CELL_4_VI_READ_EN	1'b 0	Enables the cell 4 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 4 is also provided. 1'b 0: Disabled 1'b 1: Enabled



2	R/W	CELL_3_VI_READ_EN	1'b 0	Enables the cell 3 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 3 is also provided. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	CELL_2_VI_READ_EN	1'b 0	Enables the cell 2 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 2 is also provided. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	CELL_1_VI_READ_EN	1'b 0	Enables the cell 1 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 1 is also provided. 1'b 0: Disabled 1'b 1: Enabled

HR_SCAN2 (ADh)**Format:** Unsigned binary

The HR_SCAN2 command controls enable/disable of cell 17-24, die temperature, VTOP, PACKP's high-resolution voltages. It also controls enable/disable of battery pack current reading synchronous to each cell and VTOP reading during HR ADC scan.

Bits	Access	Bit Name	Default	Description
15:13	R	RESERVED	N/A	Reserved. Do not change this register value.
12	R/W	VTOP_SYNC_EN	1'b 0	Enables the battery pack current reading that is synchronous with the VTOP reading to be updated during the high-resolution voltage ADC scan. When disabled, the synchronous VTOP current reading cannot be updated. 1'b 0: Disabled 1'b 1: Enabled
11	R/W	SCAN_PACKP_EN	1'b 0	Enables the PACKP reading to be updated during the high-resolution voltage ADC scan. 1'b 0: Disabled 1'b 1: Enabled
10	R/W	SCAN_VTOP_EN	1'b 0	Enables the VTOP reading to be updated during the high-resolution voltage ADC scan. 1'b 0: Disabled 1'b 1: Enabled
9	R/W	CELL_SYNC_EN	1'b 0	Enables the battery pack current reading synchronous to each cell to be updated during the high-resolution voltage ADC scan, following the enable setting of each individual cell. When disabled, no synchronous cell current reading can be updated. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	SCAN_DIE_T	1'b 0	Enables the die temperature to be updated during the high-resolution voltage ADC scan. 1'b 0: Disabled 1'b 1: Enabled



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

7	R/W	CELL_24_VI_READ_EN	1'b 0	Enables the cell 24 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 24 is also provided. 1'b 0: Disabled 1'b 1: Enabled
6	R/W	CELL_23_VI_READ_EN	1'b 0	Enables the cell 23 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 23 is also provided. 1'b 0: Disabled 1'b 1: Enabled
5	R/W	CELL_22_VI_READ_EN	1'b 0	Enables the cell 22 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 22 is also provided. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	CELL_21_VI_READ_EN	1'b 0	Enables the cell 21 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 21 is also provided. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	CELL_20_VI_READ_EN	1'b 0	Enables the cell 20 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 20 is also provided. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	CELL_19_VI_READ_EN	1'b 0	Enables the cell 19 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 19 is also provided. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	CELL_18_VI_READ_EN	1'b 0	Enables the cell 18 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 18 is also provided. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	CELL_17_VI_READ_EN	1'b 0	Enables the cell 17 high-resolution voltage. The cell must be enabled from CELL_S_CTRL for this flag to function. If CELL_SYNC_EN is enabled, the current reading synchronous with cell 17 is also provided. 1'b 0: Disabled 1'b 1: Enabled

**HR_SCAN3 (AEh)****Format:** Unsigned binary

The HR_SCAN3 command controls enable/disable of NTC1-6's high-resolution voltage scan. It also controls enable/disable of GPIO1 or NTC9, GPIO2 or NTC10, GPIO3 or NTC11, GPIO4 or NTC12's high-resolution voltage scan.

Bits	Access	Bit Name	Default	Description
15:12	R	RESERVED	N/A	Reserved. Do not change this register value.
11	R/W	GPIO4_READ_EN	1'b 0	Enables the GPIO4 or NTC10 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
10	R/W	GPIO3_READ_EN	1'b 0	Enables the GPIO3 or NTC9 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
9	R/W	GPIO2_READ_EN	1'b 0	Enables the GPIO2 or NTC8 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
8	R/W	GPIO1_READ_EN	1'b 0	Enables the GPIO1 or NTC7 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
7:6	R	RESERVED	N/A	Reserved. Do not change this register value.
5	R/W	NTC6_READ_EN	1'b 0	Enables the NTC6 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
4	R/W	NTC5_READ_EN	1'b 0	Enables the NTC5 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
3	R/W	NTC4_READ_EN	1'b 0	Enables the NTC4 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
2	R/W	NTC3_READ_EN	1'b 0	Enables the NTC3 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
1	R/W	NTC2_READ_EN	1'b 0	Enables the NTC2 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled
0	R/W	NTC1_READ_EN	1'b 0	Enables the NTC1 high-resolution voltage scan. 1'b 0: Disabled 1'b 1: Enabled

CC_CTRL (AFh)**Format:** Unsigned binary

The CC_CTRL command reports Coulomb Counting(CC) status: error, completion, and the number of accumulations that have been executed. It also reports whether CC accumulation data is valid, conversion was aborted and AFE transitioned to fault or safe mode. Besides, it controls enable/disable of CC.



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:8	R	CC_INT_CNT_STS	7'b 0000000	Shows the number of accumulations that have been executed. The range can be between 0 and 64. When Coulomb counting conversion is ongoing, this reading is stable only when CC_DONE = true and CC_B2B_ACC_CTRL = disabled, or if a fault interrupted an ongoing Coulomb counting conversion. This scenario sets CC_DONE and CC_ABORTED_FAULT to true, even if CC_B2B_ACC_CTRL = enabled.
7	R	RESERVED	N/A	Reserved. Do not change this register value.
6	R	CC_ACC_DATA_VALID	1'b 0	Reports whether Coulomb counting accumulation data is valid (true) or invalid (false). 1'b 0: False 1'b 1: True
5	R	CC_ABORTED_SAFE	1'b 0	When true, the most recent Coulomb counting conversion was aborted because a flag was detected and the AFE transitioned to safe mode. When a conversion is aborted, the accumulation result is still accessible in CC_RT_ACC, bits[15:0] and in CC_RT_ACC, bits[27:16] (instead of CC_IRQ, bits[15:0] and CC_IRQ, bits[27:16] like in a successfully completed conversion scenario), and the real accumulation can be accessed in CC_INT_CNT_STS. 1'b 0: False 1'b 1: True
4	R	CC_ABORTED_FAULT	1'b 0	When true, the most recent Coulomb counting conversion was aborted because a flag was detected and the AFE transitioned to fault mode. When a conversion is aborted, the accumulation result is still accessible in CC_RT_ACC, bits[15:0] and in CC_RT_ACC, bits[27:16] (instead of CC_IRQ, bits[15:0] and CC_IRQ, bits[27:16] like in a successfully completed conversion scenario), and the real accumulation can be accessed in CC_INT_CNT_STS. 1'b 0: False 1'b 1: True
3	R	CC_ERROR_STS	1'b 0	Reports whether a Coulomb counting state machine error has been detected. 1'b 0: Not detected 1'b 1: Detected
2	R	CC_DONE	1'b 0	Reports whether Coulomb counting accumulation has completed (true) or not (false). 1'b 0: False 1'b 1: True
1	R	CC_STATUS	1'b 0	Reports whether the Coulomb counter is running. 1'b 0: False 1'b 1: True
0	R/W (can lock to read-only)	CC_EN	1'b 0	Enables Coulomb counting accumulation. If Coulomb counting is enabled, the device cannot be in standby mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP

**CC_CFG (B0h)****Format:** Unsigned binary

The CC_CFG command controls enable/disable of CC in safe and fault mode, CC operating in power-save mode, back-to-back accumulation mode. It configures CC sample waiting time and integration length.

Bits	Access	Bit Name	Default	Description
15:14	R	RESERVED	N/A	Reserved. Do not change this register value.
13:8	R/W (can lock to read-only)	CC_INT_CNT	6'b 000000	Sets the Coulomb counter integration length. The length is set as the number of current ADC samples (24.5ms each), and a length contains 64 current ADC samples. LSB: 24.5ms Offset: 24.5ms Range: 24.5ms to 1568ms Value (ms) = Setting x 24.5 + 24.5 These bits allow OTP
7	R	RESERVED	N/A	Reserved. Do not change this register value.
6:4	R/W (can lock to read-only)	CC_SAMPLE_TIME_SEL	3'b 000	Sets the Coulomb counter sample waiting time when CC_SAMPLE_MODE = 1 and CC_B2B_ACC_CTRL = 1. 3'b 000: 98ms 3'b 001: 196ms 3'b 010: 294ms 3'b 011: 392ms 3'b 100: 588ms 3'b 101: 784ms 3'b 110: 1176ms 3'b 111: 1568ms These bits allow OTP
3	R/W (can lock to read-only)	CC_B2B_ACC_CTRL	1'b 0	Controls back-to-back accumulation mode. When enabled, a new Coulomb counting conversion automatically restarts as soon as the current count is completed. With back-to-back conversions enabled, the CC_RT_ACC_LSBS and CC_RT_ACC_MSBS registers are automatically cleared and do not report the progress of the new accumulation. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
2	R/W (can lock to read-only)	CC_EN_FAULT	1'b 0	Enables Coulomb counting conversions can occur in fault mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1	R/W (can lock to read-only)	CC_EN_SAFE	1'b 0	Enables Coulomb counting conversions can occur in safe mode. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

0	R/W (can lock to read-only)	CC_SAMPLE_MODE	1'b 0	Enables the Coulomb counter to operate in power-save mode, the conversions can work as an active-wait- active mode. When CC_SAMPLE_MODE = 1 and CC_B2B_ACC_CTRL = 1, the Coulomb counting runs CC_INT_CNT time, then waits for CC_SAMPLE_TIME_SEL time, and then continues. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
---	--------------------------------	----------------	-------	---

COMM_CFG (B1h)
Format: Unsigned binary

The COMM_CFG command configures the device address, type of communication bus, high level threshold level for I2C. It controls enable/disable of the use of CRC over the communication protocol, SDA/SCL input glitch filters and SDA output delay.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14:8	R/W (can lock to read-only)	DEVICE_ADD	7'b 0000000	Sets the configurable device address, only even address available. These bits allow MTP
7:3	R	RESERVED	N/A	Reserved. Do not change this register value.
2	R/W (can lock to read-only)	USE_COMM_CRC	1'b 1	Enables the use of CRC over the communication protocol. 1'b 0: Disabled 1'b 1: Enabled This bit allows OTP
1:0	R	RESERVED	N/A	Reserved. Do not change this register value.

COMM_CTRL (B2h)
Format: Unsigned binary

The COMM_CTRL command configures the size of the read back in words before appending a CRC. It also reports the last write had a bad CRC and was discarded.

Bits	Access	Bit Name	Default	Description
15	R	COMM_CRC_BAD_WRITE	1'b 0	When true, the last write had a bad CRC and was discarded. 1'b 0: False 1'b 1: True
14:7	R	RESERVED	N/A	Reserved. Do not change this register value.
6:0	R/W	XFR_NUM_RD_WORDS	7'b 0000000	When CRC is enabled, this register defines the size of the read back in words before appending a CRC. After each read transaction, this bit self-resets back to 1 word. This setting cannot be 0. LSB: 1 word Range: 1 word to 127 words Value (words) = Setting

SILC_INFO2 (B5h)
Format: Unsigned binary

The SILC_INFO2 command supplies the chip information.


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Bits	Access	Bit Name	Default	Description
15:4	R	RESERVED	N/A	Reserved. Do not change this register value.
3:0	R/W (can lock to read-only)	MODEL	4'b 1001	Hold the identifier of the specific P/N in the BMS family: 0x1 = MPS MP3718 These bits allow OTP

NVM_STATUS (BEh)
Format: Unsigned binary

The NVM_STATUS command reports the thermometer coded program status of NVM memory.

Bits	Access	Bit Name	Default	Description
15:7	R	RESERVED	N/A	Reserved. Do not change this register value.
6:0	R	NVM_STATUS	7'b 0000000	Thermometer coded program status of NVM memory. 7'b xxx1xxx: MTP page 1 has been fully programmed, allows the NVM backed registers to be loaded and use CRC if enabled 7'b xx1xxxx: MTP page 2 has been fully programmed, allows the NVM backed registers to be loaded and use CRC if enabled 7'b x1xxxxx: CAL page 1 has been fully programmed, allows the CAL backed registers to be loaded and use CRC if enabled 7'b 1xxxxxx: CAL page 2 has been fully programmed, allows the CAL backed registers to be loaded and use CRC if enabled CAL and NVM sections can be programmed in any order relative to each other.

NVM_CRC_STATUS (BFh)
Format: Unsigned binary

The NVM_CRC_STATUS command reports the errors in the outcome NVM CRC check.

Bits	Access	Bit Name	Default	Description
15	R	NVM_CRC_ERROR	1'b 0	Reports the errors in the outcome NVM CRC check. The result of the previous check is retained until NVM_CRC_DO is set to 1 (from 0). Then NVM_CRC_OUTCOME remains at 0 until the check is completed and a new result is available. 1'b 0: No errors detected 1'b 1: One or more error(s) detected
14:0	R	RESERVED	N/A	Reserved. Do not change this register value.

NVM_STORE_CMD (C0h)
Format: Unsigned binary

The NVM_STORE_CMD command reports the errors in the outcome NVM CRC check.

Bits	Access	Bit Name	Default	Description
15	R	STORE_IN_PROGRESS	1'b 0	Reports whether the store function is in progress. 1'b 0: Storage of register data to the MTP memory is not in progress 1'b 1: Storage of register data to the MTP memory is in progress
14:10	R	RESERVED	N/A	Reserved. Do not change this register value.



9:7	R/W	STORE_NVM_BURN_CONTROL	3'b 000	Indicates that the page that will be burned or restored. This bit will reset to 0 when operation is complete. 3'b 000: MTP Burn or Restore Depending on State 3'b 011: CAL Burn or Restore Depending on State (Reserved, nvm_start enabled in DIGITAL TEST MODE ONLY) 3'b 001, 010, 100 - 111: RESERVED Read/Write/Self Clears
6:4	R	RESERVED	N/A	Reserved. Do not change this register value.
3	R/W	STORE_NVM_CMD	1'b 0	Store all MTP registers and the MTP CRC code in the NVM if STORE_CMD_ACCESS_CODE has been configured with the correct code. This is a self-clearing register. 1'b 0: No operation (default) 1'b 1: Start operation
2:0	R	RESERVED	N/A	Reserved. Do not change this register value.

STORE_CMD_ACCESS_CODE (C1h)**Format:** Unsigned binary

The STORE_CMD_ACCESS_CODE command controls the enable/disable of store command.

Bits	Access	Bit Name	Default	Description
15:0	R/W	STORE_CMD_ACCESS_CODE	16'b 000000000 0000000	Before enabling the STORE_NVM_CMD command, 0xBAF4 must be written to this register to allow for the store command. This bit should be cleared once storage is complete.

OTP_READ_CMD (C2h)**Format:** Unsigned binary

The OTP_READ_CMD command reports whether the store function is complete.

Bits	Access	Bit Name	Default	Description
15	R	RESERVED	N/A	Reserved. Do not change this register value.
14	R	STORE_DONE	1'b 0	When true, the store function has completed. 1'b 0: False 1'b 1: True
13:0	R	RESERVED	N/A	Reserved. Do not change this register value.

AUTO_BAL_LIST (EDh)**Format:** Unsigned binary

AUTO_BAL_LIST command reports whether cell1-16 are executed during the next balancing session.

Bits	Access	Bit Name	Default	Description
15	R	CELL_16_AUTO_BALANCE	1'b 0	When true, cell 16 will be auto balanced during the next balancing session. When false, cell 16 is skipped. 1'b 0: False 1'b 1: True
14	R	CELL_15_AUTO_BALANCE	1'b 0	When true, cell 15 will be auto balanced during the next balancing session. When false, cell 15 is skipped. 1'b 0: False 1'b 1: True
13	R	CELL_14_AUTO_BALANCE	1'b 0	When true, cell 14 will be auto balanced during the next balancing session. When false, cell 14 is skipped. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

12	R	CELL_13_AUTO_BALANCE	1'b 0	When true, cell 13 will be auto balanced during the next balancing session. When false, cell 13 is skipped. 1'b 0: False 1'b 1: True
11	R	CELL_12_AUTO_BALANCE	1'b 0	When true, cell 12 will be auto balanced during the next balancing session. When false, cell 12 is skipped. 1'b 0: False 1'b 1: True
10	R	CELL_11_AUTO_BALANCE	1'b 0	When true, cell 11 will be auto balanced during the next balancing session. When false, cell 11 is skipped. 1'b 0: False 1'b 1: True
9	R	CELL_10_AUTO_BALANCE	1'b 0	When true, cell 10 will be auto balanced during the next balancing session. When false, cell 10 is skipped. 1'b 0: False 1'b 1: True
8	R	CELL_9_AUTO_BALANCE	1'b 0	When true, cell 9 will be auto balanced during the next balancing session. When false, cell 9 is skipped. 1'b 0: False 1'b 1: True
7	R	CELL_8_AUTO_BALANCE	1'b 0	When true, cell 8 will be auto balanced during the next balancing session. When false, cell 8 is skipped. 1'b 0: False 1'b 1: True
6	R	CELL_7_AUTO_BALANCE	1'b 0	When true, cell 7 will be auto balanced during the next balancing session. When false, cell 7 is skipped. 1'b 0: False 1'b 1: True
5	R	CELL_6_AUTO_BALANCE	1'b 0	When true, cell 6 will be auto balanced during the next balancing session. When false, cell 6 is skipped. 1'b 0: False 1'b 1: True
4	R	CELL_5_AUTO_BALANCE	1'b 0	When true, cell 5 will be auto balanced during the next balancing session. When false, cell 5 is skipped. 1'b 0: False 1'b 1: True
3	R	CELL_4_AUTO_BALANCE	1'b 0	When true, cell 4 will be auto balanced during the next balancing session. When false, cell 4 is skipped. 1'b 0: False 1'b 1: True
2	R	CELL_3_AUTO_BALANCE	1'b 0	When true, cell 3 will be auto balanced during the next balancing session. When false, cell 3 is skipped. 1'b 0: False 1'b 1: True
1	R	CELL_2_AUTO_BALANCE	1'b 0	When true, cell 2 will be auto balanced during the next balancing session. When false, cell 2 is skipped. 1'b 0: False 1'b 1: True


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

0	R	CELL_1_AUTO_BALANCE	1'b 0	When true, cell 1 will be auto balanced during the next balancing session. When false, cell 1 is skipped. 1'b 0: False 1'b 1: True
---	---	---------------------	-------	--

AUTO_BALH_LIST (EEh)
Format: Unsigned binary

AUTO_BALH_LIST command reports whether cell17-24 are executed during the next balancing session.

Bits	Access	Bit Name	Default	Description
15:8	R	RESERVED	N/A	Reserved. Do not change this register value.
7	R	CELL_24_AUTO_BALANCE	1'b 0	When true, cell 24 will be auto balanced during the next balancing session. When false, cell 24 is skipped. 1'b 0: False 1'b 1: True
6	R	CELL_23_AUTO_BALANCE	1'b 0	When true, cell 23 will be auto balanced during the next balancing session. When false, cell 23 is skipped. 1'b 0: False 1'b 1: True
5	R	CELL_22_AUTO_BALANCE	1'b 0	When true, cell 22 will be auto balanced during the next balancing session. When false, cell 21 is skipped. 1'b 0: False 1'b 1: True
4	R	CELL_21_AUTO_BALANCE	1'b 0	When true, cell 21 will be auto balanced during the next balancing session. When false, cell 21 is skipped. 1'b 0: False 1'b 1: True
3	R	CELL_20_AUTO_BALANCE	1'b 0	When true, cell 20 will be auto balanced during the next balancing session. When false, cell 20 is skipped. 1'b 0: False 1'b 1: True
2	R	CELL_19_AUTO_BALANCE	1'b 0	When true, cell 19 will be auto balanced during the next balancing session. When false, cell 19 is skipped. 1'b 0: False 1'b 1: True
1	R	CELL_18_AUTO_BALANCE	1'b 0	When true, cell 18 will be auto balanced during the next balancing session. When false, cell 18 is skipped. 1'b 0: False 1'b 1: True
0	R	CELL_17_AUTO_BALANCE	1'b 0	When true, cell 17 will be auto balanced during the next balancing session. When false, cell 17 is skipped. 1'b 0: False 1'b 1: True



LOCK REGISTER MAP

The MP3718 supports the configuration register lock function. This prevents critical safety settings from changing due to unforeseen circumstances.

All of the bits in an address that are not populated by a register should be treated as reserved. The value of these reserved bits may be 0 or 1, and must be set to 0 when there are write operations changing other bits in the same address. Setting these reserved bits to 0 does not change the value of the reserved bits.

Once the lock bit is set to 1, it cannot be set to 0 unless the MP3718 is reset.

All the locking bits support the MTP and the default value is 0. If the lock bit is MTP to 1, the corresponding setting register cannot be modified anymore. Before the MTP command, setting the locking bit to 1 should be considered carefully.

Table 8 shows the lock register map.

Table 8: Lock Register Map

Register Name, Location and Bit Position for Locking	Locked Address Name and Location	Locked Field Bit Position and Name
VTOP_UV_SLEEP_EN_LOCK: B8h, bit [15]	PWR_CFG: 05h	Bits [11:0]: VTOP_UV_SLEEP_THR
	VTOP_UV_SLEEP_CFG: 23h	Bit [5]: VTOP_UV_SLEEP_EN
GPIO3_LOCK: B8h, bit [14]	GPIO_OUT: 0Bh	Bit [2]: GPIO3_O
	GPIO_CFG: 0Ch	Bit [10]: GPIO3_PUP
		Bit [9]: GPIO3_TYPE
WDT_CFG_LOCK: B8h, bit [13]	WDT_CFG: 09h	Bit [3]: GPIO3_FSEL
		Bits [15:9]: WDT_BITE_CFG
		Bits [8:2]: WDT_BARK_CFG
PINS_CFG_LOCK: B8h, bit [12]	PINS_CFG: 0Dh	Bit [0]: WDT_COM_CTRL
		Bit [9]: NTC6_CFG_WDT
		Bit [8]: GPIO_LV_CFG
		Bit [7]: NTC5_CFG_XALERT
		Bit [6]: WDT_RST_EN
		Bit [5]: WDT_RPT
		Bit [0]: ALERT_POL



GPIO_CFG_LOCK: B8h, bit [11]	GPIO_CFG: 0Ch	Bit [14]: GPIO4_PUP
		Bit [13]: GPIO4_TYPE
		Bit [12]: GPIO4_IO
		Bit [10]: GPIO3_PUP
		Bit [9]: GPIO3_TYPE
		Bit [8]: GPIO3_IO
		Bit [6]: GPIO2_PUP
		Bit [5]: GPIO2_TYPE
		Bit [4]: GPIO2_IO
		Bit [2]: GPIO1_PUP
		Bit [1]: GPIO1_TYPE
		Bit [0]: GPIO1_IO
	PINS_CFG: 0Dh	Bit [4]: GPIO4_FSEL
		Bit [3]: GPIO3_FSEL
		Bit [2]: GPIO2_FSEL
		Bit [1]: GPIO1_FSEL
LC_CFG_LOCK_1: B8h, bit [10]	LOAD_CHARGER_CFG: 04h	Bit [4]: MID_CMP_DG_EN
LC_CFG_LOCK_0: B8h, bit [9]	LOAD_CHARGER_CFG: 04h	Bits [15:14]: LD_RML_RETRY_N
		Bits [13:11]: LD_RML_RETRY_DELAY
		Bits[10:9]:SHORT_CIRCUIT_REMOVAL_DETECT_TIME
		Bit [8]: PACKP_2V_TIMER_ENABLE
		Bits [7:5]: T_PLUGIN_PRECHARGE_MAX
FUSE_SLEEP_CFG_LOCK: B8h, bit [8]	PWR_CFG: 05h	Bit [6]: FORCE_SLEEP_BY_FUSE
SLOTA_LOCK: B8h, bit [7]	PWR_CFG: 05h	Bit [3]: ACTIVE_MONITOR_CFG
		Bit [1]: SAFE_SCOC_EN
		Bit [0]: SAFE_MONITOR_EN
	STB_CFG: 06h	Bits [12:8]: STBY_ENTER_DGL
		Bit [7]: STBY_MONITOR_EN
		Bits [6:4]: STBY_MONITOR_CFG
		Bit [0]: STBY_STATE_EN
LDO_SHDN_CFG_LOCK: B8h, bit [6]	PWR_CFG: 05h	Bit [10]: V5_SHDN_EN
		Bit [7]: V3P3_SHDN_EN
FET_BD_CFG_LOCK: B8h, bit [5]	FET_CFG: 08h	Bit [11]: FET_BD_EN
SAFE_WAKEUP_CFG_LOCK: B8h, bit [4]	PWR_CFG: 05h	Bit [4]: SAFE_WAKEUP_EN
ACT_CFG_LOCK_2: B8h, bit [3]	FET_MODE: 07h	Bit [12]: LOW_SIDE_FET_MODE
		Bit [2]: FT_STATE_SEL
ACT_CFG_LOCK_1: B8h, bit [2]	FET_MODE: 07h	Bit [1]: FET_CFG
		Bit [0]: FET_SRC
ACT_CFG_LOCK_0: B8h, bit [1]	AFE_CFG_STATUS: 02h	Bits [9:8]: ACTIVE_CTRL



CELL_S_CTRL_LOCK: B8h, bit [0]	CELLS_CTRL: 00h	Bit [13]: CELL_23_MONITOR_SKIP
		Bit [12]: CELL_22_MONITOR_SKIP
		Bit [11]: CELL_21_MONITOR_SKIP
		Bit [10]: CELL_20_MONITOR_SKIP
		Bit [9]: CELL_19_MONITOR_SKIP
		Bit [8]: CELL_18_MONITOR_SKIP
		Bits [4:0]: CELL_S_CTRL
	CELL_MONITOR_SKIP: 01h	Bit [15]: CELL_17_MONITOR_SKIP
		Bit [14]: CELL_16_MONITOR_SKIP
		Bit [13]: CELL_15_MONITOR_SKIP
		Bit [12]: CELL_14_MONITOR_SKIP
		Bit [11]: CELL_13_MONITOR_SKIP
		Bit [10]: CELL_12_MONITOR_SKIP
		Bit [9]: CELL_11_MONITOR_SKIP
		Bit [8]: CELL_10_MONITOR_SKIP
		Bit [7]: CELL_9_MONITOR_SKIP
		Bit [6]: CELL_8_MONITOR_SKIP
		Bit [5]: CELL_7_MONITOR_SKIP
		Bit [4]: CELL_6_MONITOR_SKIP
		Bit [3]: CELL_5_MONITOR_SKIP
		Bit [2]: CELL_4_MONITOR_SKIP
		Bit [1]: CELL_3_MONITOR_SKIP
		Bit [0]: CELL_2_MONITOR_SKIP
CHGFET_DEAD_CFG_LOCK: B9h, bit [15]	PRO_EN_CTRL1: 0Eh	Bit [13]: CHGFET_DEAD_EN
	FAULT_EN2: 1Fh	Bit [13]: CHGFET_DEAD_FUSE_EN
	FET_DEAD_LIM: 20h	Bit [13:12]: CHGFET_DEAD_CURR_RNG
		Bits [11:7]: DSGFET_DEAD_CURR_LIM
	FET_DEAD_DEG: 21h	Bit [13]: CHGFET_DEAD_DG_RNG
DSGFET_DEAD_CFG_LOCK: B9h, bit [14]	PRO_EN_CTRL1: 0Eh	Bit [12]: DSGFET_DEAD_EN
	FAULT_EN2: 1Fh	Bit [12]: DSGFET_DEAD_FUSE_EN
	FET_DEAD_LIM: 20h	Bits [6:5]: DSGFET_DEAD_CURR_RNG
		Bits [4:0]: DSGFET_DEAD_CURR_LIM
	FET_DEAD_DEG: 21h	Bits [6]: DSGFET_DEAD_DGL
		Bits [5:0]: CHGFET_DEAD_CURR_LIM
CELL_MSMT_CFG_LOCK: B9h, bit [13]	PRO_EN_CTRL1: 0Eh	Bit [8]: CELL_MSMT_EN
	FAULT_EN1: 1Eh	Bit [15]: CELL_MSMT_FAULT_EN
	FAULT_EN2: 1Fh	Bit [11]: CELL_MSMT_FUSE_EN
	CELL_MSMT: 3Ch	Bits [14:9]: CELL_MSMT_MIN
		Bits [8:5]: CELL_MSMT_DG
		Bits [4:0]: CELL_MSMT_TH

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

CELL_DEAD_CFG_LOCK: B9h, bit [12]	PRO_EN_CTRL1: 0Eh	Bit [7]: CELL_DEAD_EN
	FAULT_EN1: 1Eh	Bit [14]: CELL_DEAD_FAULT_EN
	FAULT_EN2: 1Fh	Bit [7]: CELL_DEAD_FUSE_EN
	CELL_DEAD_THR: 3Bh	Bits [10:7]: CELL_DEAD_DG
		Bits [6:0]: CELL_DEAD_LIMIT
CELL_SOV_CFG_LOCK: B9h, bit [11]	PRO_EN_CTRL1: 0Eh	Bit [6]: CELL_SOV_EN
	FAULT_EN2: 1Fh	Bit [8]: CELL_SOV_FUSE_EN
		Bit [0]: CELL_SOV_FAULT_EN
	CELL_SOV: 34h	Bits [13:10]: CELL_SOV_DG
		Bits [9:0]: CELL_SOV
CELL_OV_CFG_LOCK: B9h, bit [10]	PRO_EN_CTRL1: 0Eh	Bit [5]: CELL_OV_EN
	FAULT_EN1: 1Eh	Bit [1]: CELL_OV_FAULT_EN
	CELL_HYST: 35h	Bit [1]: CELL_OV_LOGIC_SEL
	CELL_OV: 38h	Bits [13:10]: CELL_OV_DG
		Bits [9:0]: CELL_OV
	FT_ACT_CFG: 61h	Bit [1]: CELL_OV_FAULT_ACTION
	FT_REC_EN: 62h	Bit [1]: CELL_OV_REC
	FT_REC_CFG2: 65h	Bit [0]: CELL_OV_EXIT_DSG_CURR
CELL_UV_CFG_LOCK: B9h, bit [9]	PRO_EN_CTRL1: 0Eh	Bit [4]: CELL_UV_EN
	FAULT_EN1: 1Eh	Bit [0]: CELL_UV_FAULT_EN
	CELL_HYST: 35h	Bit [0]: CELL_UV_LOGIC_SEL
	CELL_UV: 37h	Bits [11:8]: CELL_UV_DG
		Bits [7:0]: CELL_UV
	FT_ACT_CFG: 61h	Bit [0]: CELL_UV_FAULT_ACTION
	FT_REC_EN: 62h	Bit [0]: CELL_UV_REC
	FT_REC_CFG2: 65h	Bit [1]: CELL_UV_EXIT_CHG_CURR
SC_DCHG_LOCK: B9h, bit [8]	PRO_EN_CTRL1: 0Eh	Bit [3]: SC_DCHG_EN_CTRL
	FAULT_EN1: 1Eh	Bit [13]: SC_DCHG_FAULT_EN
	DSGSC_CFG: 27h	Bits [14:8]: SC_DCHG_DG
		Bit [5]: SC_DCHG_RNG
		Bits [4:0]: SC_DCHG_LIM
	FT_ACT_CFG: 61h	Bit [13]: SC_DSG_FAULT_ACTION
	FT_REC_EN: 62h	Bit [11]: SC_DCHG_REC
	FT_REC_CFG2: 65h	Bit [15]: DSG_SC_EXIT_CHG_VMID
		Bit [7]: DSG_SC_EXIT_CHG_CURR

**PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE**

OC_CHG_LOCK: B9h, bit [7]	PRO_EN_CTRL1: 0Eh	Bit [2]: OC_CHG_EN_CTRL
	FAULT_EN1: 1Eh	Bit [12]: OC_CHG_FAULT_EN
	CHGOC_DEG: 26h	Bit [14]: OC_CHG_DGL_RNG
		Bits [13:8]: OC_CHG_DG
		Bit [5]: OC_CHG_RNG
		Bits [4:0]: OC_CHG_LIM
	FT_ACT_CFG: 61h	Bit [12]: OC_CHG_FAULT_ACTION
	FT_REC_EN: 62h	Bit [10]: OC_CHG_REC
	FT_REC_CFG1: 64h	Bits [11:10]: OC_CHG_RETRY
		Bits [9:8]: OC_CHG_COOL
OC2_DCHG_LOCK: B9h, bit [6]	PRO_EN_CTRL1: 0Eh	Bit [14]: CHG_OC_EXIT_DSG_VPACKP
		Bit [6]: CHG_OC_EXIT_DSG_CURR
	FAULT_EN1: 1Eh	Bit [11]: OC2_DCHG_FAULT_EN
	DSGOC_LIM: 24h	Bit [13]: OC2_DCHG_RNG
		Bits [12:8]: OC2_DCHG_LIM
	DSGOC_DEG: 25h	Bit [14]: OC2_DCHG_DGL_RNG
		Bits [13:8]: OC2_DCHG_DGL
	FT_ACT_CFG: 61h	Bit [11]: OC2_DCHG_FAULT_ACTION
	FT_REC_EN: 62h	Bit [9]: OC2_DCHG_REC
	FT_REC_CFG1: 64h	Bits [7:6]: OC2_DCHG_RETRY
		Bits [5:4]: OC2_DCHG_COOL
OC1_DCHG_LOCK: B9h, bit [5]	FT_REC_CFG2: 65h	Bit [13]: DSG_OC2_EXIT_CHG_VMID
		Bit [5]: DSG_OC2_EXIT_CHG_CURR
	PRO_EN_CTRL1: 0Eh	Bit [0]: OC1_DCHG_EN_CTRL
	FAULT_EN1: 1Eh	Bit [10]: OC1_DCHG_FAULT_EN
	DSGOC_LIM: 24h	Bit [5]: OC1_DCHG_RNG
		Bits [4:0]: OC1_DCHG_LIM
	DSGOC_DEG: 25h	Bit [6]: OC1_DCHG_DGL_RNG
		Bits [5:0]: OC1_DCHG_DGL
	FT_ACT_CFG: 61h	Bit [10]: OC1_DCHG_FAULT_ACTION
	FT_REC_EN: 62h	Bit [8]: OC1_DCHG_REC
	FT_REC_CFG1: 64h	Bits [3:2]: OC1_DCHG_RETRY
		Bits [1:0]: OC1_DCHG_COOL
	FT_REC_CFG2: 65h	Bit [12]: DSG_OC1_EXIT_VMID
		Bit [4]: DSG_OC1_EXIT_CHG_CURR



NTC_CFG_LOCK: B9h, bit [4]	GPIO_CFG: 0Ch	Bit [15]: GPIO4_NTC10
		Bit [11]: GPIO3_NTC7
		Bit [7]: GPIO2_NTC6
		Bit [3]: GPIO1_NTC5
	PRO_EN_CTRL2: 0Fh	Bit [15]: NTC10_CELL_EN
		Bit [14]: NTC9_CELL_EN
		Bit [13]: NTC8_CELL_EN
		Bit [12]: NTC7_CELL_EN
		Bit [11]: NTC6_CELL_EN
		Bit [10]: NTC5_CELL_EN
		Bit [9]: NTC4_EN
		Bit [8]: NTC3_EN
		Bit [7]: NTC2_EN
		Bit [6]: NTC1_EN
	NTC_CFG: 49h	Bit [7]: NTC4_TYPE_SEL
		Bit [5]: NTC3_TYPE_SEL
		Bit [3]: NTC2_TYPE_SEL
		Bit [1]: NTC1_TYPE_SEL
	NTCC_SOTHR: 4Ah	Bits [13:10]: NTC_CELL_SUTSOT_DG
		Bits [9:0]: NTC_CELL_SOT
	NTCC_SUTHR: 4Bh	Bit [11]: NTC_CHG_COLDER
		Bit [10]: NTC_CHG_HOTTER
	NTCC_OTHr: 4Ch	Bits [9:0]: NTC_CELL_SUT
		Bits [13:10]: NTC_CELL_UTOT_DG
NTC_ADC_CFG_LOCK: B9h, bit [3]	FT_REC_CFG2: 65h	Bits [9:0]: NTC_CELL_OT
		Bits [15:11]: NTC_CELL_HYST
ADC_CFG_LOCK: B9h, bit [2]	FET_CFG: 08h	Bits [9:0]: NTC_CELL_UT
		Bit [11]: NTC_COLD_EXIT_BY_CURR
CHGFET_SOFT_LOCK: B9h, bit [1]	FET_CFG: 08h	Bit [10]: NTC_HOT_EXIT_BY_CURR
		Bit [13]: NTCB_DYNAMIC_ON
DSGFET_SOFT_LOCK: B9h, bit [0]	FET_CFG: 08h	Bit [12]: ADC_DYN_ON
		Bit [13]: ESS_MODE_EN
		Bit [15]: CHG_SOFTON_EN
		Bits [14:12]: CHG_SOFTON_PUP
ADC_SCAN_LOCK: BAh, bit [15]	ADC_CFG: A9h	Bit [10]: DSG_SOFTON_EN
		Bits [9:5]: DSG_SS_TURN_ON_TIMEOUT
		Bit [4]: DSGSS_CURRENT_RNG
		Bits [3:0]: DSGSS_CURRENT
		Bit [14]: DISABLE_HWP_SLOT
REC_CFG_LOCK: BAh, bit [14]	FT_REC_CFG1: 64h	Bits [10:8]: ADC_SCAN_SAFE
		Bits [7:5]: ADC_SCAN_ACTIVE
		Bit [4]: ADC_SCAN_CONT_SAFE_EN
		Bit [3]: ADC_SCAN_CONT_EN
		Bit [15]: SBD_GPIO



DSG_DRIVER_CFG_LOCK: BAh, bit [13]	FT_ACT_CFG: 61h	Bit [9]: CHG_DRIVER_FAULT_ACTION
		Bit [8]: DSG_DRIVER_FAULT_ACTION
	FT_REC_CFG0: 63h	Bits [15:14]: CHG_DRIVER_RETRY
		Bits [13:10]: CHG_DRIVER_COOL
		Bits [9:8]: DSG_DRIVER_RETRY
		Bits [7:4]: DSG_DRIVER_COOL
		Bit [1]: CHG_DRIVER_REC
		Bit [0]: DSG_DRIVER_REC
NVM_CRC_LOCK: BAh, bit [12]	FT_REC_CFG2: 65h	Bit[9]: CHG_DRIVER_FAIL_EXIT_DSG_CURR
		Bit[8]: DSG_DRIVER_FAIL_EXIT_CHG_CURR
	FAULT_EN2: 1Fh	Bit [6]: NVM_FAULT_EN
OPW_CFG_LOCK: BAh, bit [11]	NVM_CRC_GO: 58h	Bit [0]: NVM_CRC_EN
	PRO_EN_CTRL1: 0Eh	Bit [15]: AOPW_EN_DSG
	PRO_EN_CTRL1: 0Eh	Bit [14]: AOPW_EN_CHG
	FAULT_EN2: 1Fh	Bit [5]: OPEN_WIRE_FAULT_EN
	RD_OPENH_CTL: 55h	Bit [10]: OPEN_WIRE_PON
		Bit [8]: OPEN_WIRE_EN
	OPEN_CFG: 57h	Bits [15:12]: OPW_CHECK_DTH
		Bits [11:8]: OPW_CHECK_LOW_TH
		Bits [7:3]: OPW_CHECK_HIGH_TH
		Bits [2:0]: OPW_CHECK_LEN
LDO_CFG_LOCK: BAh, bit [10]	FT_REC_EN: 62h	Bit [14]: OPEN_WIRE_REC
	PRO_EN_CTRL2: 0Fh	Bit [5]: V3V3_EN
	FAULT_EN2: 1Fh	Bit [15]: LDO_UV_FUSE_EN
ADC_SELF_LOCK: BAh, bit [9]		Bit [4]: LDO_FAULT_EN
	PRO_EN_CTRL1: 0Eh	Bit [10]: ADC_SELF_TEST_EN
	FAULT_EN2: 1Fh	Bit [14]: ADC_FAIL_FUSE_EN
VTOP_SUV_CFG_LOCK: BAh, bit [8]		Bit [3]: ADC_FAIL_FAULT_EN
	PRO_EN_CTRL2: 0Fh	Bit [3]: VTOP_SUV_EN
	FAULT_EN2: 1Fh	Bit [10]: VTOP_SUV_FUSE_EN
		Bit [2]: VTOP_SUV_FAULT_EN
VTOP_SOV_CFG_LOCK: BAh, bit [7]	PACK_SUV: 33h	Bits [15:12]: VTOP_SUV_DG
		Bits [11:0]: VTOP_SUV_LIMIT
	PRO_EN_CTRL2: 0Fh	Bit [2]: VTOP_SOV_EN
	FAULT_EN2: 1Fh	Bit [9]: VTOP_SOV_FUSE_EN
DIE_CFG_LOCK: BAh, bit [6]		Bit [1]: VTOP_SOV_FAULT_EN
	PACK_SOV: 32h	Bits [15:12]: VTOP_SOV_DG
		Bits [11:0]: VTOP_SOV_LIMIT
	PRO_EN_CTRL1: 0Eh	Bit [14]: DIE_TEMP_DIG_EN
	FAULT_EN1: 1Eh	Bit [9]: DIE_TEMP_DIG_FAULT_EN
	DIE_OT: 4Fh	Bits [15:11]: DIE_TEMP_HYST
		Bits [9:0]: DIE_TEMP_HOT
	FT_REC_EN: 62h	Bit [13]: DIE_TEMP_FAULT_REC



NTC_PCB_LOCK: BAh, bit [5]	FAULT_EN1: 1Eh	Bit [8]: PCB_MNTR_FAULT_EN
	NTC_CFG: 49h	Bits [10:8]: PCB_MNTR_DG
	NTCM_OTHR: 4Eh	Bits [15:11]: PCB_MNTR_HYST
	FT_REC_EN: 62h	Bits [9:0]: PCB_MNTR_HOT
NTC_DSG_LOCK: BAh, bit [4]	FAULT_EN1: 1Eh	Bit [12]: PCB_MNTR_REC
		Bit [7]: NTC_CELL_COLD_DSG_FAULT_EN
	FT_ACT_CFG: 61h	Bit [6]: NTC_CELL_HOT_DSG_FAULT_EN
		Bit [7]: NTC_CELL_DSG_COLD_ACTION
NTC_CHG_LOCK: BAh, bit [3]	FT_REC_EN: 62h	Bit [6]: NTC_CELL_DSG_HOT_ACTION
		Bit [7]: NTC_CELL_DSG_COLD_REC
	FAULT_EN1: 1Eh	Bit [6]: NTC_CELL_DSG_HOT_REC
		Bit [5]: NTC_CELL_COLD_CHG_FAULT_EN
VTOP_OV_CFG_LOCK: BAh, bit [2]	FT_ACT_CFG: 61h	Bit [4]: NTC_CELL_HOT_CHG_FAULT_EN
		Bit [5]: NTC_CELL_CHG_COLD_ACTION
	FT_REC_EN: 62h	Bit [4]: NTC_CELL_CHG_HOT_ACTION
		Bit [5]: NTC_CELL_CHG_COLD_REC
	PACK_OV: 3Ah	Bit [4]: NTC_CELL_CHG_HOT_REC
		Bit [1]: VTOP_OV_EN
	PRO_EN_CTRL2: 0Fh	Bit [3]: VTOP_OV_FAULT_EN_CTRL
VTOP_UV_CFG_LOCK: BAh, bit [1]	FAULT_EN1: 1Eh	Bits [15:10]: VTOP_OV_HYST
	PACK_UV_OV: 36h	Bits [15:12]: VTOP_OV_DG
	PACK_UV: 39h	Bits [11:0]: VTOP_OV_LIMIT
		Bit [3]: VTOP_OV_FAULT_ACTION
	FT_ACT_CFG: 61h	Bit [3]: VTOP_OV_REC
	FT_REC_EN: 62h	Bit [2]: VTOP_OV_EXIT_DSG_CURR
	FT_REC_CFG2: 65h	Bit [2]: VTOP_UV_EN
	PRO_EN_CTRL2: 0Fh	Bit [2]: VTOP_UV_FAULT_EN_CTRL
INT_CFG_LOCK: BAh, bit [0]	FAULT_EN1: 1Eh	Bits [7:2]: VTOP_UV_HYST
	PACK_UV_OV: 36h	Bits [15:12]: VTOP_UV_DG
	PACK_UV: 39h	Bits [11:0]: VTOP_UV_LIMIT
		Bit [2]: VTOP_UV_FAULT_ACTION
GPIO4_LOCK: BBh, bit [10]	FT_ACT_CFG: 61h	Bit [2]: VTOP_UV_REC
	FT_REC_EN: 62h	Bit [3]: VTOP_UV_EXIT_CHG_CURR
	FT_REC_CFG2: 65h	Bit [15]: INT_ALERT_CTRL
FET_LVL_LOCK: BBh, bit [9]	INT0_EN: 16h	Bit [3]: GPIO4_O
	GPIO_OUT: 0Bh	Bit [13]: GPIO4_TYPE
	GPIO_CFG: 0Ch	Bit [14]: GPIO4_PUP
GAIN_IPCB_LOCK: BBh, bit [8]	PINS_CFG: 0Dh	Bit [4]: GPIO4_FSEL
CAL_LOCK: BBh, bit [7]	PCB_GAIN_CAL: AAh	Bit [15:14]: FET_DRV_LVL
		Bits [9:0]: I_PCB_GAIN_VALUE
		Bit [13]: CELL_1K_COMP
		Bit [11]: PCB_GC_I_CTRL
		Bit [10]: PCB_GC_VOTHER_CTRL
		Bit [9]: PCB_GC_VT_PK_CTRL
		Bit [8]: PCB_GC_CELL_CTRL



		Bits [7:0]: PCB_GAIN_CAL
--	--	--------------------------


MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING
PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

V5_CFG_LOCK: BBh, bit [6]	PRO_EN_CTRL2: 0Fh	Bit [4]: V5_EN
AOPW_CFG_LOCK: BBh, bit [5]	PRO_EN_CTRL1: 0Eh	Bit [15]: AOPW_EN_DSG
		Bit [14]: AOPW_EN_CHG
BAL_CFG_LOCK: BBh, bit [4]	BAL_CFG: 41h	Bits [7:3]: BAL_REPETITION
		Bit [2]: AUTO_BAL_ALWAYS
		Bit [1]: BALANCE_MODE_REG
		Bit [0]: BALANCE_MODE_CTRL (I ² C versions only)
	BAL_THR: 42h	Bits [12:10]: BAL_MSM_TH
		Bit [9]: STOP_ON_HOT
		Bit [8]: ABAL_ON_SAFE
		Bit [7]: ABAL_ON_STBY
		Bit [6]: ABAL_ON_CHARGE
		Bits [5:0]: CELL_BAL_MIN
COMM_CRC_CFG_LOCK: BBh, bit [3]	COMM_CFG: B1h	Bit [2]: USE_COMM_CRC
COMM_ADD_CFG_LOCK: BBh, bit [1]	COMM_CFG: B1h	Bits [14:8]: DEVICE_ADD
CC_CFG_LOCK: BBh, bit [0]	CC_CTRL: AFh	Bit [0]: CC_EN
	CC_CFG: B0h	Bits [13:8]: CC_INT_CNT
		Bits [6:4]: CC_SAMPLE_TIME_SEL
		Bit [3]: CC_B2B_ACC_CTRL
		Bit [2]: CC_EN_FAULT
		Bit [1]: CC_EN_SAFE
		Bit [0]: CC_SAMPLE_MODE



TYPICAL APPLICATION CIRCUITS

Figure 21 shows the typical external components and connections required to interface the MP3718 to the battery pack and the system.

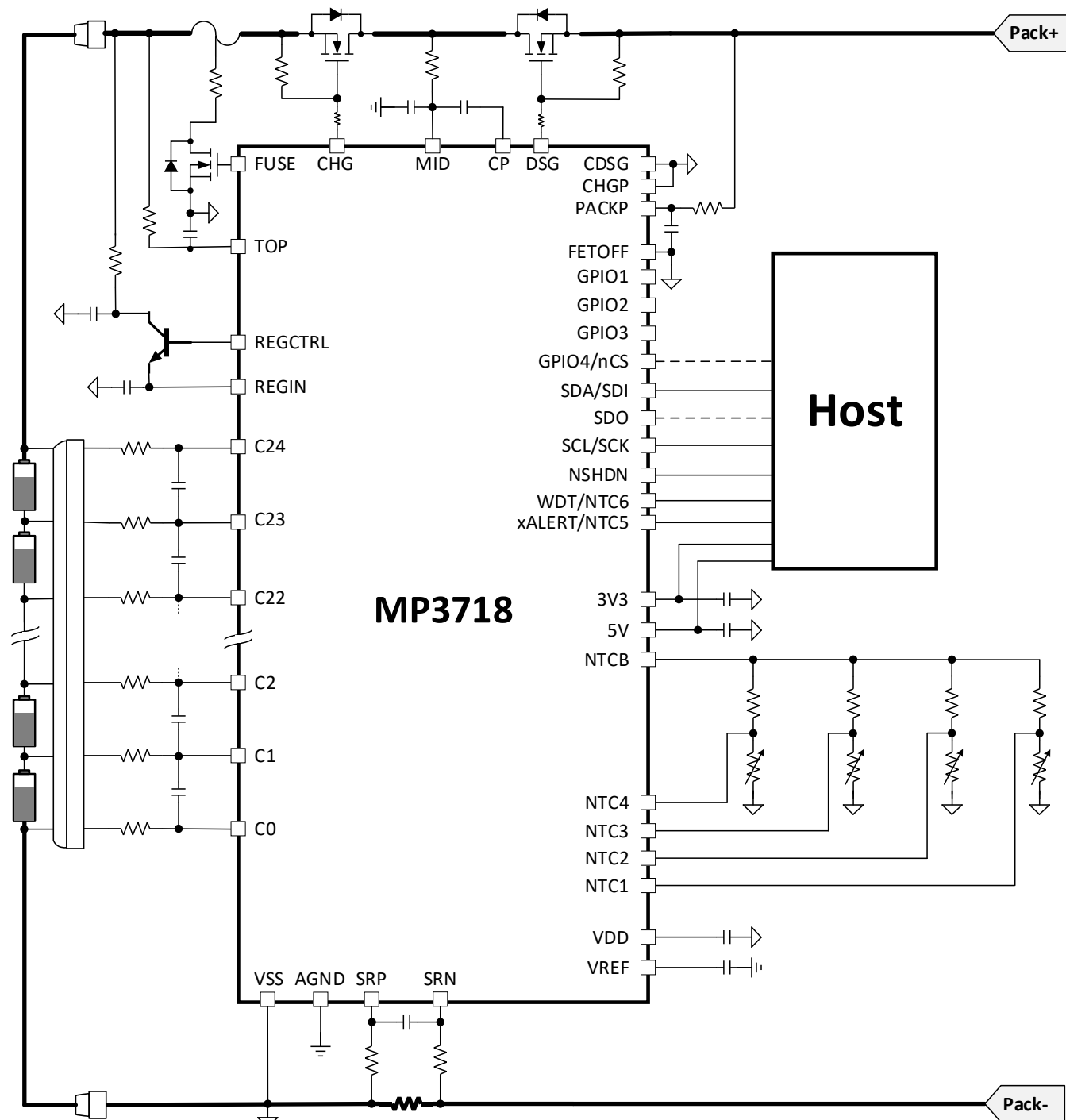


Figure 21: Typical Application Circuit for the MP3718



Figure 22 shows the typical external components and connections required to interface the MP3718 to the battery pack and the separate port system.

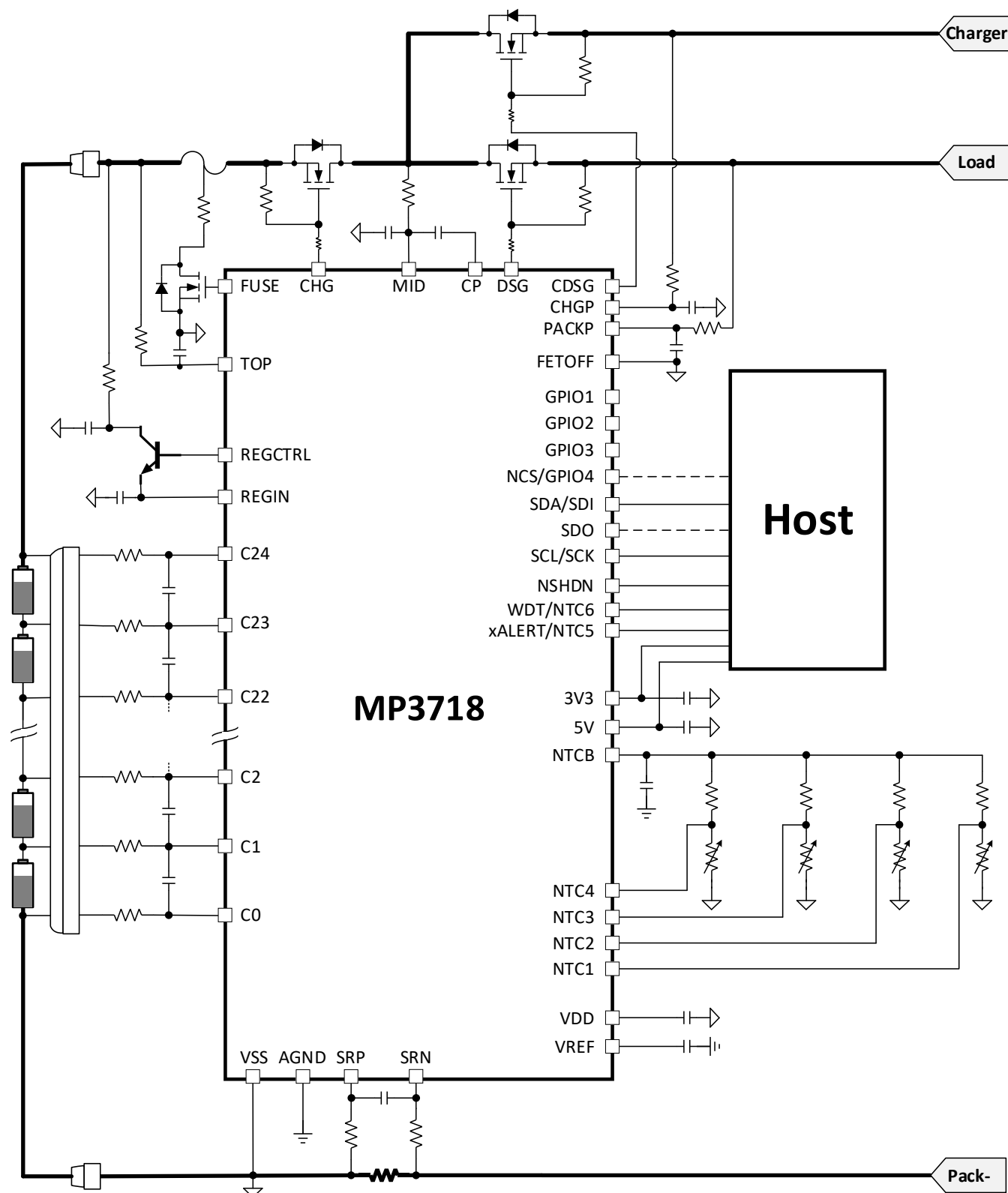


Figure 22: Separate Port Application Circuit for the MP3718

Recommended External Components

Figure 23 shows a detailed view of all the typical recommended external components.

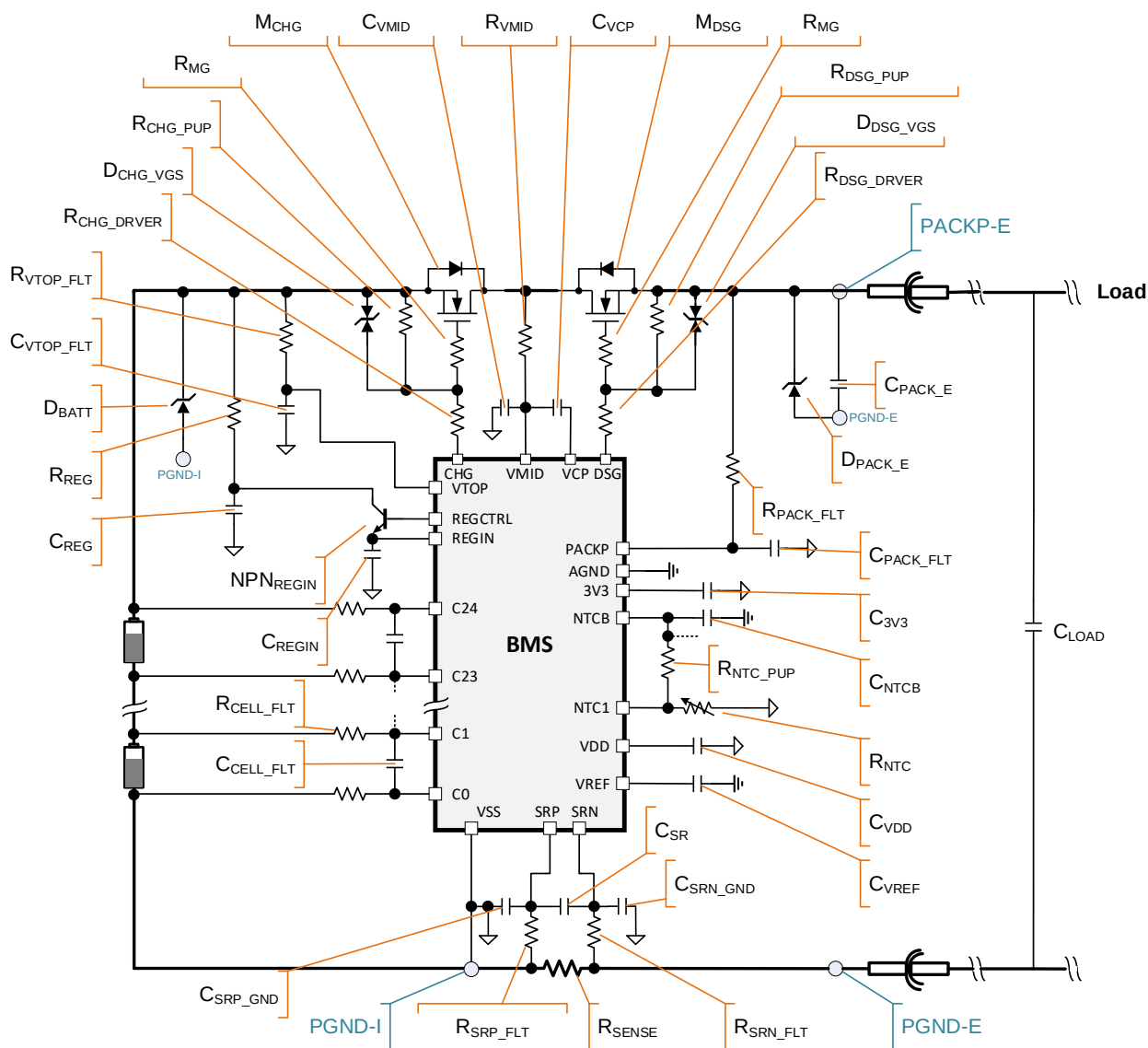


Figure 23: Recommended External Component Diagram

Table 9 provides guidance regarding which parameters should be evaluated when selecting each BOM component (see the Configuration for External or Internal Cell-Balancing section for more details on cell-balancing). Refer to the MP3718 evaluation board datasheet to see each BOM component.

Table 9: Recommended Components

Parameter Identifier	Parameter Description	Component Description	Parameter Comments	Min	Typ	Max	Unit
General							
C _{PACK_E}	Capacitance value	External pack capacitor			47		μF



MP3718 – 4S TO 24S BATTERY MONITOR AND PROTECTOR W/ COULOMB COUNTING

PRELIMINARY SPECIFICATIONS SUBJECT TO CHANGE

Parameter Identifier	Parameter Description	Component Description	Parameter Comments	Min	Typ	Max	Unit
Pin Filtering							
C _{TOP_FLT}	Required capacitance	TOP-filtering capacitor			10		nF
R _{TOP_FLT}	Required resistance	TOP-filtering resistor			1000		Ω
C _{PACK_FLT}	Required capacitance	PACKP-filtering capacitor	If R _{PACK_FLT} increases, C _{PACK_FLT} should be proportionally decreased.		1		nF
R _{PACK_FLT}	Required resistance	PACKP-filtering resistor			1000		Ω
R _{MID}	Required resistance	MID-filtering resistor			100		Ω
Cell Voltage Sensing							
C _{CELL_FLT}	Required capacitance	Cell-filtering capacitor			100		nF
R _{CELL_FLT}	Required resistance	Cell-filtering resistor		20	100		Ω
N-Channel MOSFET Driver							
D _{CHG-VGS}	Recommended Zener diode voltage rating	Diode CHG V _{GS} protection	For a CHG N-channel MOSFET with a ±20V maximum V _{GS} . If the CHG N-channel MOSFET has a smaller V _{GS} , decrease the Zener voltage value accordingly.		±16		V
D _{DSG-VGS}	Recommended Zener diode voltage rating	Diode DSG V _{GS} protection	For a DSG N-channel MOSFET with a ±20V maximum V _{GS} . If the DSG N-channel MOSFET has a smaller V _{GS} , decrease the Zener voltage value accordingly.		±16		V
R _{CHG_PUP}	Recommended resistance	N charge MOSFET pull-up resistor			10		MΩ
R _{DSG_PUP}	Recommended resistance	N discharge MOSFET pull-up resistor			10		MΩ
R _{CHG_DRIVER}	Required resistance	N-channel MOSFET charge driver protection resistor			100		Ω
R _{DSG_DRIVER}	Required resistance	N-channel MOSFET discharge driver protection resistor			100		Ω
R _{MG}	Required resistance	N-channel MOSFET gate protection resistor			100		Ω
Current Sensing							
R _{SRN_FLT}	Recommended resistance	SRN filtering resistor			100		Ω
C _{SRN_GND}	Recommended capacitance	SRN-GND filtering capacitor			100		nF



Parameter Identifier	Parameter Description	Component Description	Parameter Comments	Min	Typ	Max	Unit
C _{SRN_SRP}	SRN-SRP capacitance	SRNP differential filtering capacitor			100		nF
R _{SRP_GND}	Recommended SRP-GND resistance	SRP-GND filter filtering resistor			100		Ω
C _{SRP_GND}	SRP-GND capacitance	SRP-GND filtering capacitor			100		nF
R _{SENSE}	Recommended resistance	Pack current-sense resistor	The min and max values are recommended and are based on the current-sense range.	0.1	2	5	mΩ
Regulators							
C _{3V3}	Required capacitance	3V3 bypass capacitor	Capacitor value for the 3.3V rail.	0.47	1	10	μF
C _{VDD}	VDD required capacitance	VDD bypass capacitor			1		μF
C _{VREF}	Required capacitance	VREF bypass capacitor			1		μF
C _{REGIN}	REGIN input required capacitance	REGIN capacitor	Capacitor value for REGIN.		3.3		μF
C _{REG}	Input regulator required capacitance	Regulator capacitor		3.3			μF
R _{REG}	Regulator-limiting resistor value	Regulator-limiting resistor value			500		Ω
Temperature Sensing							
R _{NTC_PUP}	Typical resistance	NTC pull-up	This is a typical value. Generally, the NTC pull-up value should match the NTC thermistor value at 25°C.		10		kΩ
R _{NTC}	Typical resistance	NTC thermistor	If a smaller resistance is used, the NTCB current limit should be evaluated and compared to the total resistance connected to the NTCB in the worst-case scenario (hot temperatures).		10		kΩ
C _{NTCB}	NTCB capacitance	NTCB capacitor				10	nF

Configuration For External or Internal Cell-Balancing

Cell-balancing can be implemented via the internal balancing MOSFETs for up to 58mA. Higher cell-balancing current is possible with external MOSFETs or BJTs (see Figure 24).

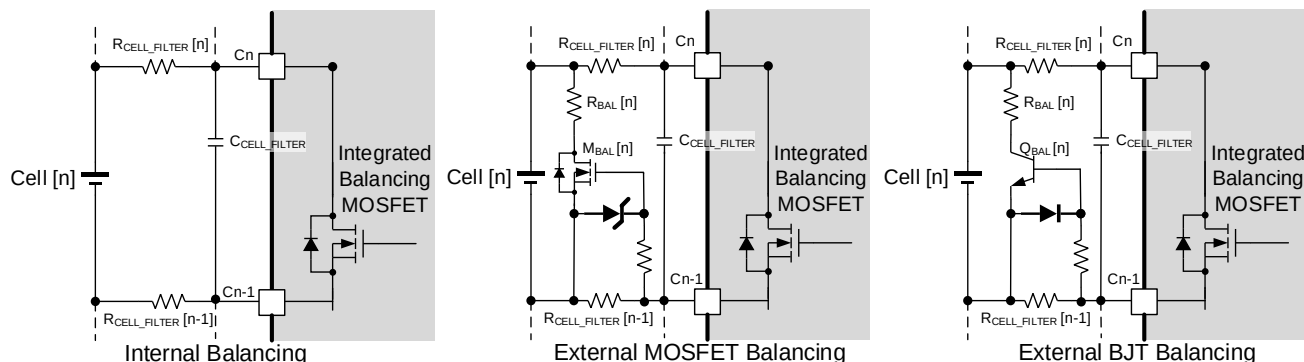


Figure 24: Typical Schematics Internal vs. External Balancing Configurations

Table 10 recommends specific components for internal MOSFET balancing.

Table 10: Recommended Component Selection for Internal MOSFET Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C _{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for internal balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.		100		nF
R _{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for the internal balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.	20	100	200	Ω

The amount of current provided in an internal balancing configuration is easily derived by following the R_{CELL_FILTER} resistor value and considering the integrated MOSFET balancing resistance (R_{DS(ON)_BAL_FET}). For example, for a lithium cell at 4V where R_{CELL_FILTER} = 20Ω, the resistive path allows for 58mA once the integrated balancing resistor MOSFET is enabled.

For higher balancing current, Table 11 and Table 12 provide guidance on selecting the MOSFET or BJT circuit components.

Table 11 recommends specific components for external MOSFET balancing.

Table 11: Recommended Component Selection for External MOSFET Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C _{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for external balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.		10		nF
R _{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for external balancing configuration. If R _{CELL_FILTER} increases, C _{CELL_FILTER} should be proportionally decreased.	800	1000	1100	Ω
R _{BAL}	Required resistance	Cell-balancing current-limiting resistor	Adjust this value to set the balancing current and verify the power dissipation compared to package limit.		43		Ω
V _{TH_MBAL}	MOSFET V _{GS} threshold	External balancing MOSFET M _{BAL}	A higher voltage threshold may prevent the MOSFET from turning on.		1.5	1.8	V

The external balancing MOSFETs should be selected using Table 11. Choose a MOSFET with a sufficiently low V_{GS} threshold. For simplification, assume that the integrated MOSFET-balancing resistance can be ignored. The available V_{GS} that turns on the MOSFET is generated across R_{CELL_FILTER}, meaning its voltage is half of the lithium cell voltage (e.g., 4.2V/2 = 2.1V). In this scenario,



the V_{GS} threshold for M_{BAL} should be below 2.1V with an appropriate safety margin. The safety margin should allow for voltage drops on the integrated MOSFET balancing resistor. This drop can be caused by the following:

- Changes in the operating conditions
- Shifts in the M_{BAL} V_{GS} threshold across operating conditions.
- Drops that develop across the sensing wires due to the balancing current.

Table 12 recommends specific components for external BJT balancing.

Table 12: Recommended Component Selection for External BJT Balancing

Parameter Identifier	Parameter Description	Component Description	Notes	Min	Typ	Max	Unit
C_{CELL_FILTER}	Required capacitance	Cell-filtering capacitor	Capacitance for external balancing configuration. If R_{CELL_FILTER} increases, C_{CELL_FILTER} should be proportionally decreased.		100		nF
R_{CELL_FILTER}	Required resistance	Cell-filtering resistor	Resistance for external balancing configuration. If R_{CELL_FILTER} increases, C_{CELL_FILTER} should be proportionally decreased.	20	100	200	Ω
R_{BAL}	Required resistance	Cell-balancing current-limiting resistor	Adjust this value to set the balancing current and verify the power dissipation compared to package limit.		43		Ω
h_{FE_QBAL}	BJT DC current gain	External balancing BJT Q_{BAL}	A lower h_{FE} may limit the external balancing current.	50			

Because R_{BAL} sets the balancing current and dissipates most of the power, ensure that an appropriate resistor and resistor package are selected, especially for resistors below 100 Ω . For example, if $R_{BAL} = 43\Omega$, there is 410mW of power dissipation when the lithium cell voltage = 4.2V. This means that a package of at least 2512 (6432 Metric) should be used.

It is also important to consider the thermal design of the overall board and enclosure. The MP3718 can simultaneously balance only odd or even cells, meaning 9 cells in a 18-cell system. When balancing is run continuously on 9 cells, 3.69W is dissipated as heat with a 43 Ω R_{BAL} . To keep the MP3718 in the allowed temperature range, consider how this additional heat can be dissipated, as it is added to the amount of heat generated by other components.

SELECTING THE CP CAPACITOR

When multiple MOSFETs in parallel are used for DSG or CHG, the CP capacitance should be selected to avoid causing an excessive voltage drop while the MOSFETs turn on. Table 13 lists values for the CP capacitor depending on the total C_{ISS} that the DSG or CHG MOSFET driver must drive. The total C_{ISS} was obtained from the MOSFET datasheets.

Design the number of parallel DSG MOSFETs to match the number of parallel CHG MOSFETs. If 4 parallel DSG MOSFETs are used, then the individual MOSFET's C_{ISS} should be multiplied by 4, and the resulting C_{ISS} value should be used.

Table 13: CP Capacitor Selection

C_{CP}	Total C_{ISS}
47nF	100nF
68nF	140nF
100nF	220nF



PCB Layout Guidelines

Proper PCB layout is critical to reduce noise, and to optimize the device's accuracy and reliability. For the best results, refer to Figure 25, Figure 26, and Figure 27 follow the guidelines below:

1. Route the ground plane such that it reduces ground noise and prevents stray current (see Figure 25).

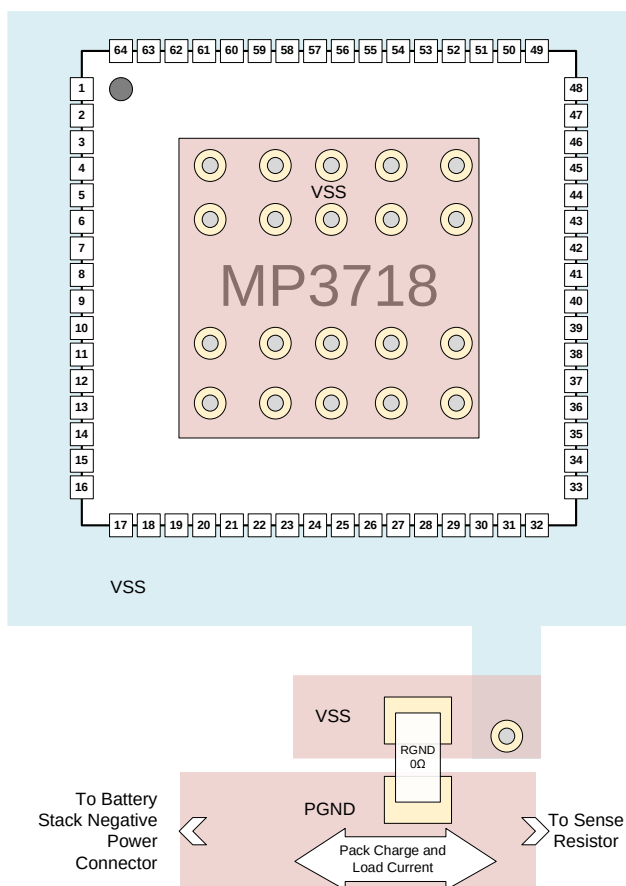


Figure 25: Recommended Layout for AGND

2. Connect the VSS to battery stack negative power connector (PGND) and connect the VSS pin to a dedicated ground plane. Then connect the VSS plane to PGND via a 0Ω resistor.
3. Review the schematic and layout to ensure that each component is connected to the correct ground (power vs. signal). For example, the diodes protecting the different power terminals should be connected to PGND.
4. Consider the expected maximum peak load current, then check the PGND length, width, and thickness to ensure that there is an

appropriate low-resistance path that prevents voltage drops.

5. The REGIN, VDD, VREF, 3V3, and NTCB pins require external decoupling capacitors, which should be placed as close as possible to each pin. Minimize the trace inductance from these pins to VSS (see Figure 26).

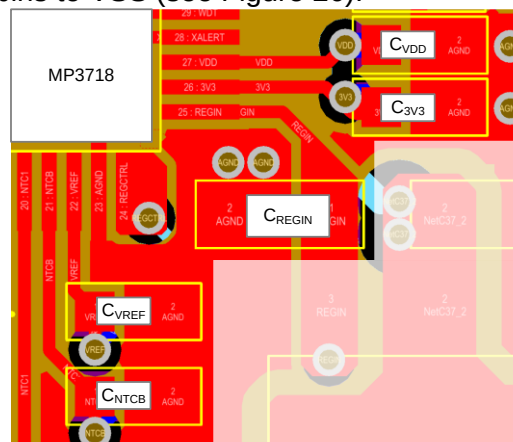


Figure 26: Recommended Layout for Decoupling Capacitors

6. Consider the Kelvin connection at the sense resistor. A sense resistor with a dedicated sensing pad is ideal. A simple, two-terminal sense resistor can be used in a 4-wire sensing configuration (see Figure 27).
7. Route the sensing signals to SRP and SRN in parallel to avoid coupling interfering signals.

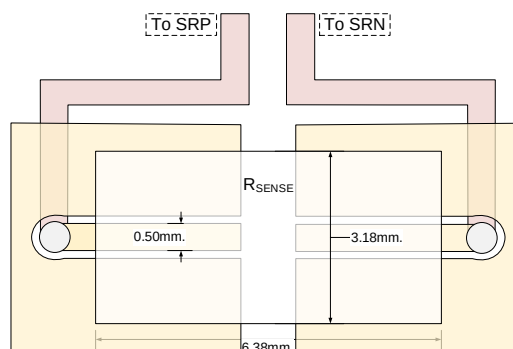


Figure 27: 2512 Sense Resistor using 4-Wire Footprint

8. Connect the temperature sensors (NTCx) to AGND using the star connection technique to avoid contaminating the voltage reading with the resistive voltage drop caused by load or charge ground current.
9. Route the two wires connecting the NTCx pins together in twisted pairs to avoid coupling interfering signals. This is recommended if NTCx is located off-board.



Unused Pins

For a particular application, some pins may not be needed. Table 14 shows how to connect them when these pins are not used.

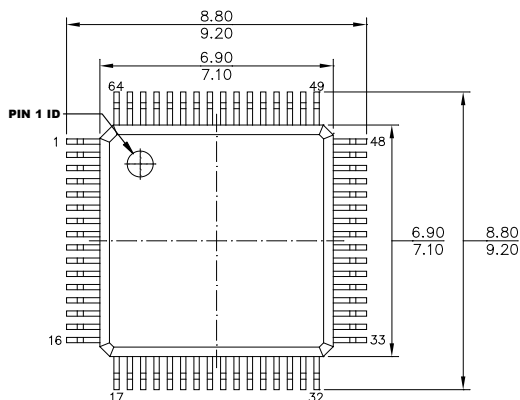
Table 14: Unused pins connection

Pin #	Name	Recommendation
2-17, 60-63	C5 to C24	If less than 24 series cells are used, please directly connect all unused cell channels to the practical maximum cell channel. For example: only 20 cells are used, C24, C23, C22, C21 to C20 should be connected together.
23, 25	SRP, SRN	If the current sense is not required, connect them to VSS.
24, 26- 28	NTC1, NTC2, NTC3, NTC4	The unused NTC channels should be left floating.
28	NTCB	If NTC temperature monitoring is not used, NTCB should connect 10nF to AGND
28	XALERT/N TC6	If not used, this pin should be left floating.
29	WDT/NTC5	If not used, this pin should be left floating.
40	FETOFF	If not used, this pin should be connected to VSS.
41	SCL/SCK	If not used, connect this pin to 3V3 via a 10kΩ resistor.
42	SDA/SDI	If not used, connect this pin to 3V3 via a 10kΩ resistor.
43	SDO	If not used, this pin should be left floating.
44	NCS/GPIO 4	If not used, this pin should be left floating. Note that the unused GPIO4 should be set as 20kΩ pull-up mode by the GPIO_CFG register, otherwise there is an additional power consumption.
45-47	GPIO1, GPIO2, GPIO3	If not used, this pin should be left floating. Note that the unused GPIO should be set as 20kΩ pull-up mode by the GPIO_CFG register, otherwise there is an additional power consumption.
48	FUSE	If not used, this pin should be left floating.
50	PACKP	If the DSG MOSFET driver is not used, and the PACKP vs. TOP comparator is not needed, connect this pin to TOP.
51	DSG	If the HS-FET drivers are not used and active mode is required, connect this pin to PACKP via a 100Ω resistor and a 1nF capacitor. If active mode is not required, float this pin.
56	CP	If the HS-FET drivers are not used and active mode is required, connect this pin to MID via a 47nF CP capacitor. If active mode is not required, float this pin.
57	MID	If the HS-FET drivers are not used, connect this pin to TOP.
58	CHG	If the HS-FET drivers are not used and normal mode is required, connect this pin to TOP via a 100Ω resistor and a 1nF capacitor. If normal mode is not required, float this pin.
53-54	CHGP, CDSG	If the charging port and the discharge port is the same. Connect the PACKP to charging and discharging ports, and connect the CHGP and CDSG to VSS.

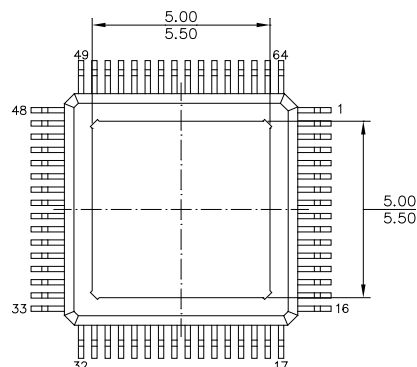


PACKAGE INFORMATION

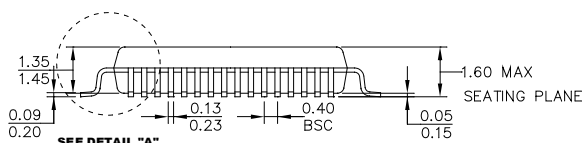
LQFP-64 (7mmx7mm)



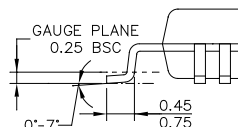
TOP VIEW



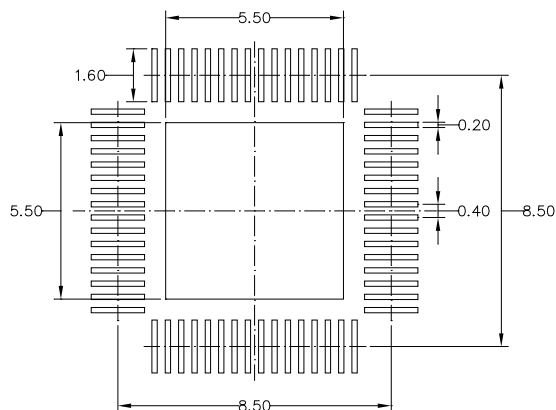
BOTTOM VIEW



SIDE VIEW



DETAIL "A"



RECOMMENDED LAND PATTERN

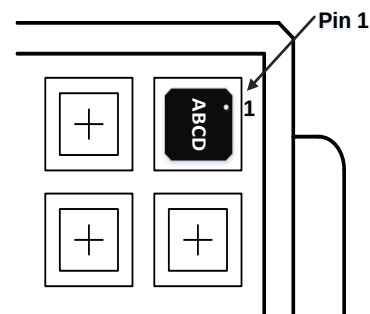
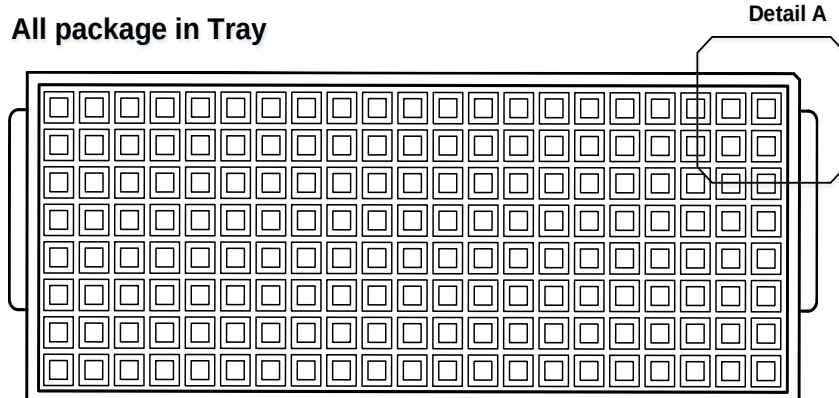
NOTE:

- 1) ALL DIMENSIONS ARE IN MILLIMETERS.
- 2) PACKAGE LENGTH DOES NOT INCLUDE MOLD FLASH, PROTRUSION OR GATE BURR.
- 3) PACKAGE WIDTH DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSION.
- 4) LEAD COPLANARITY (BOTTOM OF LEADS AFTER FORMING) SHALL BE 0.10 MILLIMETERS MAX.
- 5) JEDEC REFERENCE IS MO-143.
- 6) DRAWING IS NOT TO SCALE.



CARRIER INFORMATION

All package in Tray



Detail A

Part Number	Package Description	Quantity/ Reel	Quantity/ Tube	Quantity/ Tray	Reel Diameter	Carrier Tape Width	Carrier Tape Pitch
MP3718DFN- xxxx-T	LQFP-64 (7mmx7mm)	N/A	N/A	250pcs	N/A	N/A	N/A

Notice: The information in this document is subject to change without notice. Please contact MPS for current specifications. Users should warrant and guarantee that third-party Intellectual Property rights are not infringed upon when integrating MPS products into any application. MPS will not assume any legal responsibility for any said applications.